

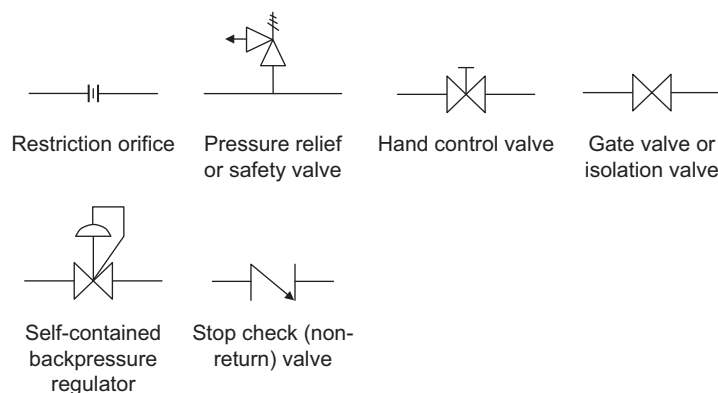


FIG. 5.7 Other common symbols.

Most modern control valves (final control elements) are actuated by electric motors, but older valves are actuated by pneumatic signals using instrument air. Pneumatic actuators are preferred in situations where electronic controllers might cause a process hazard or where electric power is not available or reliable. Pneumatic controllers are also found in many older plants where replacement with electronic controllers has not yet occurred. Motor actuators are used for larger valves, and digital and solenoid actuators are used for valves that switch from open to closed, as often occurs in batch processing. Many newer controllers use a combination of these approaches. For example, a digital signal can be sent to a solenoid that opens or shuts an instrument air line that then actuates a pneumatically driven control valve.

Instrument Lines

The instrument connecting lines are drawn in a manner to distinguish them from the main process lines, as shown in Fig. 5.3. Process lines are drawn as solid lines and are usually drawn thicker. The undefined signal symbol is often used when indicating controllers in a PFD, as the instrument design may not have been specified when the PFD was first drawn.

Failure mode

The direction of the arrow shows the position of the valve on failure of the power supply (Fig. 5.4).

General instrument and controller symbols

General instrument symbols are shown in Fig. 5.5.

Locally mounted means that the controller and display are located out on the plant near the sensing instrument location. *Main panel* means that they are located on a panel in the control room. Except on small plants, most controllers would be mounted in the control room.

Distributed control: Shared display symbols

Symbols for shared displays and programmable logic controllers are shown in Fig. 5.6.

A distributed control system is a system that is functionally integrated but consists of subsystems that may be physically separate and remotely located from one another. A shared display is an operator interface device such as a computer screen or video screen that is used to display process control information from a number of sources at the command of the operator. Most plants built since 1990 (and many older plants) use shared displays instead of instrument panels.

Programmable logic controllers are used to control discrete operations, such as steps in a batch or semicontinuous process, and to program interlock controls that guard against unsafe or uneconomic conditions. For example, a logic controller could be used to ensure that an operator cannot open an air vent line to a vessel unless the feed valves are closed and nitrogen purge is open.

Other common symbols

Other symbols commonly encountered on a P&I diagram are shown in Fig. 5.7.

Type of instrument

This is indicated on the circle representing the instrument controller by a letter code (Table 5.1).

TABLE 5.1 Letter code for instrument symbols (based on ISA-5.1-1984 [R1992])

Initiating or measured variable	First letter	Indicating only	Controllers			Transmitters	Final control element
			Recording	Indicating	Blind		
Analysis (composition)	A	AI	ARC	AIC	AC	AT	AV
Flow rate	F	FI	FRC	FIC	FC	FT	FV
Flow ratio	FF	FFI	FFRC	FFIC	FFC	FFT	FFV
Power	J	JI	JRC	JIC		JT	JV
Level	L	LI	LRC	LIC	LC	LT	LV
Pressure, vacuum	P	PI	PRC	PIC	PC	PT	PV
Pressure differential	PD	PDI	PDRC	PDIC	PDC	PDT	PDV
Quantity	Q	QI	QRC	QIC		QT	QZ
Radiation	R	RI	RRC	RIC	RC	RT	RZ
Temperature	T	TI	TRC	TIC	TC	TT	TV
Temperature differential	TD	TDI	TDRC	TDIC	TDC	TDT	TDV
Weight	W	WI	WRC	WIC	WC	WT	WZ

Notes:

1. The letters C, D, G, M, N, and O are not defined and can be used for any user-specified property.

2. The letter S as second or subsequent letter indicates a switch.

3. The letter Y as second or subsequent letter indicates a relay or a compute function.

4. The letter Z is used for the final control element when this is not a valve.

Consult the standard for the full set of letter codes.

The first letter indicates the property measured; for example, F = flow. Subsequent letters indicate the function; for example,

I = indicating

RC = recorder controller

The letters AH or AL indicate high or low alarms, respectively.

The P&I diagram shows all the components that make up a control loop. For example, Fig. 5.8 shows a field located pressure transmitter connected to a shared display pressure indicator-controller with operator access to adjustments and high and low alarms. The pressure controller sends an electric signal to a fail-closed diaphragm-actuated pressure control valve.

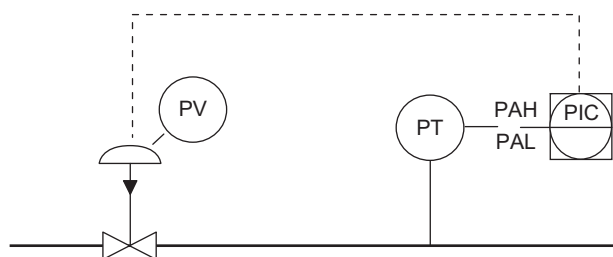


FIG. 5.8 A typical control loop.

5.3 Process instrumentation and control

5.3.1 Instruments

Instruments are provided to monitor the key process variables during plant operation. They may be incorporated in automatic control loops or used for manual monitoring of process operation. In most modern plants, the instruments will be connected to a computer control and data logging system. Instruments monitoring critical process variables will be fitted with automatic alarms to alert the operators to critical and hazardous situations.

Details of process instruments and control equipment can be found in various handbooks: [Green and Southard \(2018\)](#), [Love \(2007\)](#), and [Liptak \(2012\)](#). Reviews of process instruments and control equipment are published periodically in the journals *Chemical Engineering* and *Hydrocarbon Processing*. These reviews give details of instruments and control hardware available commercially. [Table 5.2](#) summarizes some of the more commonly used types of instruments encountered in chemical plants.

TABLE 5.2 Commonly used process instruments

Measured variable	Instrument type	Operating principle
Pressure (gauge)	Differential pressure (DP) cell	Pressure difference causes displacement of a diaphragm. The displacement can be transmitted mechanically to a bellows to register a pneumatic signal or converted to an electrical signal by a strain gauge or by movement of the diaphragm relative to a static capacitor plate. Gauge pressure is measured relative to atmospheric pressure.
Pressure difference	DP cell	As earlier. Pressure difference is measured between two points in the process.
Temperature	Thermocouple	Wires of different metals joined together to form a circuit with one joint hotter than the other will develop an EMF through the Seebeck effect. If one joint is at a reference temperature, the other temperature can be found from the EMF. The reference temperature is usually ambient temperature, which is determined by measuring the electrical resistance of a platinum wire. Different combinations of metal wire are used depending on the temperature range. See Love (2007) for details of thermocouple types.
Volumetric flow	Orifice meter	Flow passes through a restriction orifice. Pressure difference across the orifice is measured with a DP cell. Flow rate is calculated from pressure drop.
Volumetric flow	Venturi meter	Flow passes through a shaped pipe restriction. Pressure difference across the restriction is measured with a DP cell. Flow rate is calculated from pressure drop.
Mass flow	Coriolis meter	Flow through a shaped vibrating pipe loop causes it to twist due to the Coriolis effect. The extent of twist is measured optically. These instruments can be used for multiphase flow, but are expensive, particularly for large flow rates.

TABLE 5.2 Commonly used process instruments—cont'd

Measured variable	Instrument type	Operating principle
Level	DP cell	A DP cell placed between the top and bottom of a vessel can indicate level if there is no internal pressure drop in the vessel.
Level	Capacitance probe	The capacitance between a probe in the center of the vessel and the wall is affected by the dielectric constant of the material between them, and so varies with level.
Interface level	DP cell	A DP cell can determine the interface level between immiscible fluids if they are in a vessel that has an internal weir (so that overall level remains constant).
pH	Glass electrode	The glass electrode and a reference electrode (usually silver/silver chloride) form an electrochemical circuit allowing EMF to be measured.
Composition	Chromatograph	Gas chromatography (GC) can be used to separate simple mixtures and generate a signal through a thermal conductivity detector (TCD) or flame ionization detector (FID). GC methods are difficult to use for online control because the chromatography typically takes a few minutes, but they can be used in cascade control schemes to adjust set points on other controllers.

The process variable that is to be monitored should be measured directly; however, this is often impractical, and some dependent variable that is easier to measure is monitored in its place. For example, in the control of distillation columns, the continuous, online analysis of the overhead product is desirable but is difficult and expensive to achieve reliably, so temperature is often monitored as an indication of composition. The temperature instrument may form part of a control loop controlling, say, reflux flow, with the composition of the overheads checked frequently by automated sampling and online gas chromatograph (GC) analysis.

5.3.2 Instrumentation and control objectives

The primary objectives of the designer when specifying instrumentation and control schemes are:

1. Safe plant operation:
 - a. To keep the process variables within known safe operating limits.
 - b. To detect dangerous situations as they develop and to provide alarms and automatic shut-down systems.
 - c. To provide interlocks and alarms to prevent dangerous operating procedures.
2. Production rate:
 - a. To achieve the design product output.
3. Product quality:
 - a. To maintain the product composition within the specified quality standards.
4. Cost:
 - a. To operate at the lowest production cost, commensurate with the other objectives.
5. Stability:
 - a. To maintain steady, automatic plant operation with minimal operator intervention.

These are not separate objectives and must be considered together. The order in which they are listed is not meant to imply the precedence of any objective over another, other than that of putting safety first. Product quality, production rate, and the cost of production will be dependent on sales requirements. For example, it may be a better strategy to produce a better-quality product at a higher cost.

In a typical chemical processing plant these objectives are achieved by a combination of automatic control, manual monitoring, and laboratory and online analysis.

5.3.3 Automatic control schemes

The detailed design and specification of the automatic control schemes for a large project are usually done by specialists. The basic theory underlying the design and specification of automatic control systems is covered in several texts: Coughanowr and LeBlanc (1994), Shinskey (1984, 1996), Luyben et al. (1999), Henson et al. (1996), Seborg et al. (2017), Love (2007), Chhabra and Rohani (2017), and Green and Southard (2018). The books by Murrill (1988), Shinskey (1996), Kalani (2002), and Love (2007) cover many of the more practical aspects of process control system design and are recommended.

In this chapter only the first step in the specification of the control systems for a process will be considered: the preparation of a preliminary scheme of instrumentation and control, developed from the process flowsheet. This can be drawn up by the process designer based on experience with similar plants and critical assessment of the process requirements. Many of the control loops will be conventional, and a detailed analysis of the system behavior will not be needed, nor justified. Judgment, based on experience, must be used to decide which systems are critical and need detailed analysis and design.

Some examples of typical (conventional) control systems used for the control of specific process variables and unit operations are given in the next section and can be used as a guide in preparing preliminary instrumentation and control schemes.

Guide rules

The following procedure can be used when drawing up preliminary P&I diagrams:

1. Identify and draw in those control loops that are obviously needed for steady plant operation, such as:
 - a. Level controls
 - b. Flow controls
 - c. Pressure controls
 - d. Temperature controls
2. Identify the key process variables that need to be controlled to achieve the specified product quality. Include control loops using direct measurement of the controlled variable, where possible; if not practicable, select a suitable dependent variable.
3. Identify and include those additional control loops required for safe operation not already covered in steps 1 and 2.
4. Decide and show those ancillary instruments needed for the monitoring of the plant operation by the operators and for troubleshooting and plant development. It is well worth including additional connections for instruments that may be needed for future troubleshooting and development, even if the instruments are not installed permanently. These would include extra thermowells, pressure taps, orifice flanges, and sample points.
5. Decide on the location of sample points.
6. Decide on the type of control instrument that will be used, including whether it will be a local instrument or tied into the plant computer control system. Also decide on the type of actuator that can be used, the signal system, and whether the instrument will record data. This step should be done in conjunction with steps 1 to 4.
7. Decide on the alarms and interlocks needed; this should be done in conjunction with step 3 (see [Chapter 10](#)).

In step 1 it is important to remember the following basic rules of process control:

There can only be a single control valve on any given stream between unit operations.

A level controller is needed anywhere where a vapor–liquid or liquid–liquid interface is maintained.

Pressure control is more responsive when the pressure controller actuates a control valve on a vapor stream.

Two operations cannot be controlled at different pressures unless there is a valve or other restriction (or a compressor or pump) between them.

Temperature control is usually achieved by controlling the flow of a utility stream (such as steam or cooling water) or a bypass around an exchanger.

The overall plant material balance is usually set by flow controllers or flow ratio controllers on the process feeds. There cannot be an additional flow controller on an intermediate stream unless there is provision for accumulation (surge), such as an intermediate storage tank.

Some simple examples of control schemes for common unit operations are given in the next section.

5.4 Conventional control schemes

5.4.1 Level control

In any equipment where an interface exists between two phases (e.g., a liquid and a vapor), some means of maintaining the interface at the required level must be provided. This may be incorporated in the design of the equipment, for example, by providing an internal weir, or by automatic control of the flow from the equipment. Fig. 5.9 shows a typical arrangement for the level control at the base of a column. The control valve should be placed on the discharge line from the pump.

5.4.2 Pressure control

Pressure control will be necessary for most systems handling vapor or gas. The method of control depends on the nature of the process. Typical schemes are shown in Figs. 5.10(a)–(d). The scheme shown in Fig. 5.10(a) would not be used where the vented gas was toxic or valuable. In these circumstances the vent should be taken to a vent recovery system, such as a scrubber. The controls shown in Fig. 5.10(b)–(d) are commonly used for controlling the pressure of distillation columns.

In processes that have a high-pressure reaction section and low-pressure separation section, the high-pressure section is usually pressure controlled by expanding the product from the high-pressure section across a control valve. If the process fluid does not change phase, then a more economical scheme is to expand the product through a turbine or turbo-expander and recover shaft work from the expansion.

5.4.3 Flow control

Flow control is usually associated with inventory control in a storage tank or other equipment or with feeds to the process. There must be a reservoir upstream of the control valve to take up the changes in flow rate.

To provide flow control on a compressor or pump running at a fixed speed and supplying a near-constant volume output, a bypass control would be used, as shown in Fig. 5.11(a). The use of variable-speed motors, as shown in

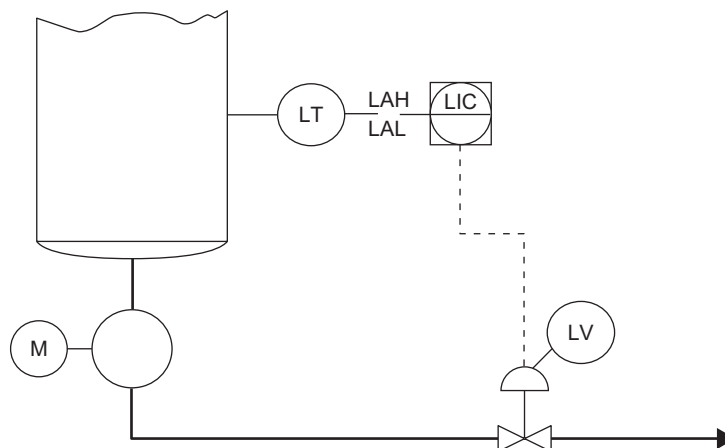


FIG. 5.9 Level control.

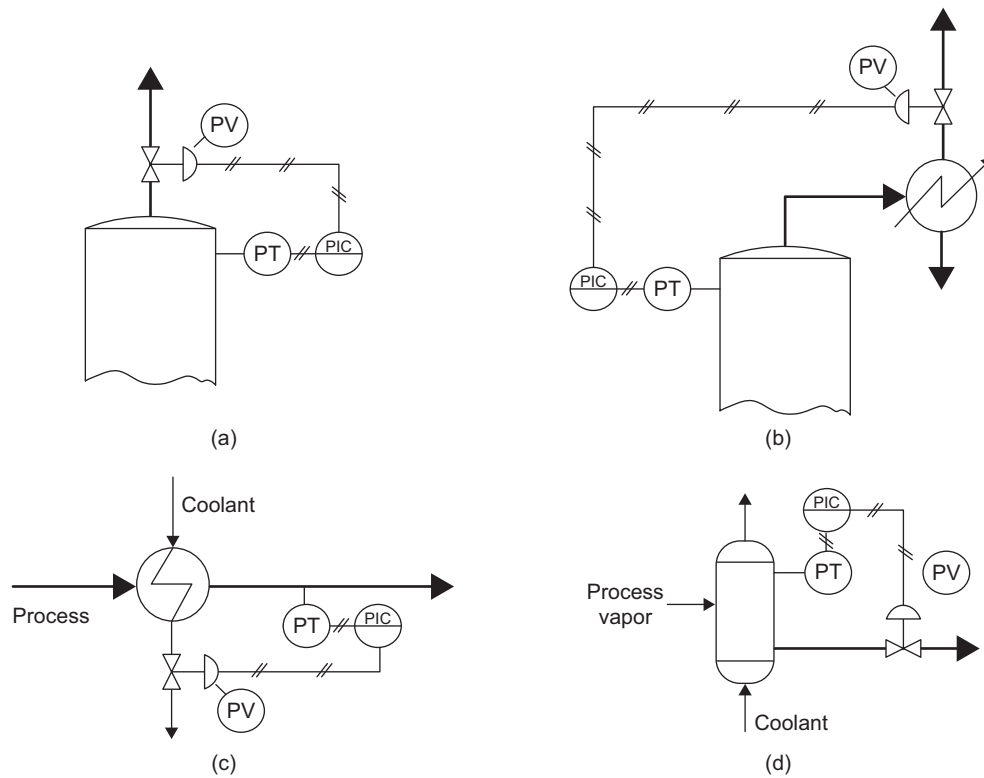


FIG. 5.10 (a) Pressure control by direct venting. (b) Venting of noncondensibles after a condenser. (c) Condenser pressure control by controlling coolant flow. (d) Pressure control of a condenser by varying the heat-transfer area, with the area dependent on liquid level.

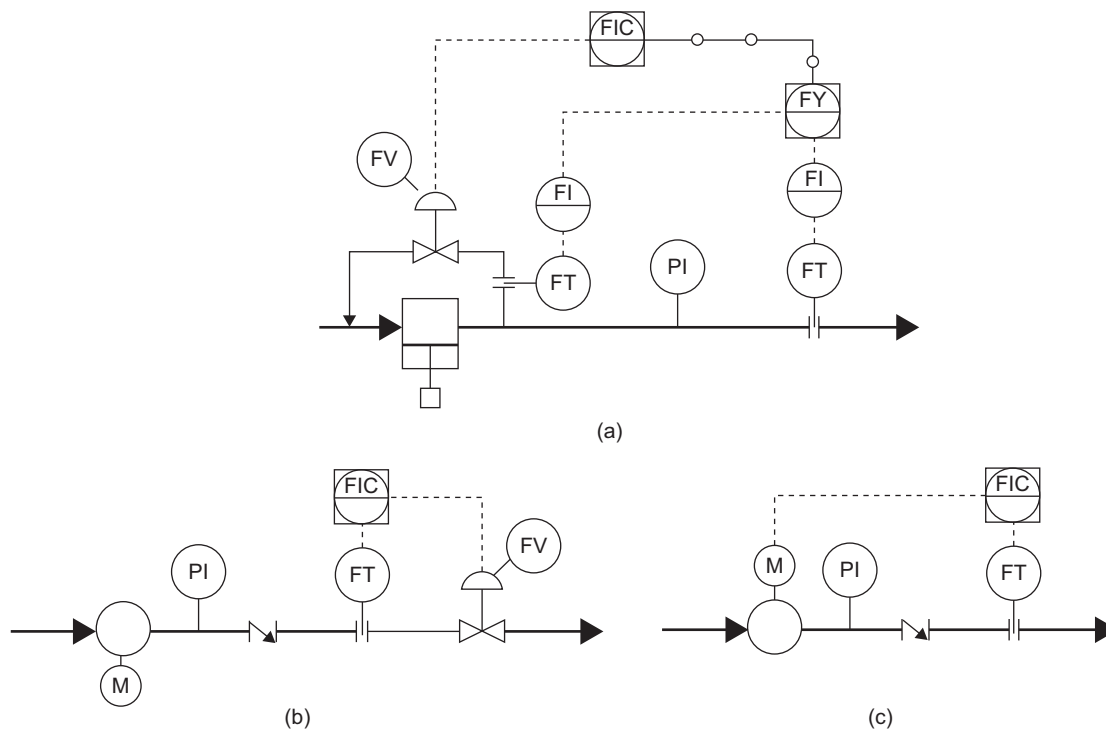


FIG. 5.11 (a) Spill-back flow control for a reciprocating pump. (b) Flow control for a centrifugal pump. (c) Centrifugal pump with variable-speed drive.

Fig. 5.11(c), is more energy efficient than the traditional arrangement shown in Fig. 5.11(b) and is becoming increasingly common; see Hall (2010).

The overall process material balance is usually set by flow controllers on the feed streams. These will often control feeds in ratio to a flow of valuable feed, a solid stream flow (which is difficult to change quickly), or a measured flow of process mixture. Flow rates of small streams are often controlled using special metering pumps that deliver a constant mass flow rate.

The design of pump and control valve systems to assure a desired process flow rate and range of controllability is discussed in more detail in Chapter 20.

5.4.4 Heat exchangers

Fig. 5.12(a) shows the simplest arrangement: the temperature being controlled by varying the flow of the cooling or heating medium.

If the exchange is between two process streams whose flows are fixed, bypass control will have to be used, as shown in Fig. 5.12(b).

For air coolers, the coolant temperature may vary widely on a seasonal (or even hourly) basis. A bypass on the process side can be used, as shown in Fig. 5.12(c), or else a variable-speed motor can be used, as shown in Fig. 5.12(d).

Condenser control

Temperature control is unlikely to be effective for condensers, unless the liquid stream is subcooled. Pressure control is often used, as shown in Fig. 5.10(d), or control can be based on the outlet coolant temperature. Condenser control is discussed in more detail by Kister (2016).

Reboiler and vaporizer control

As with condensers, temperature control is not effective, as the saturated vapor temperature is constant at constant pressure. Level control is often used for vaporizers: the controller controlling the steam supply to the heating surface,

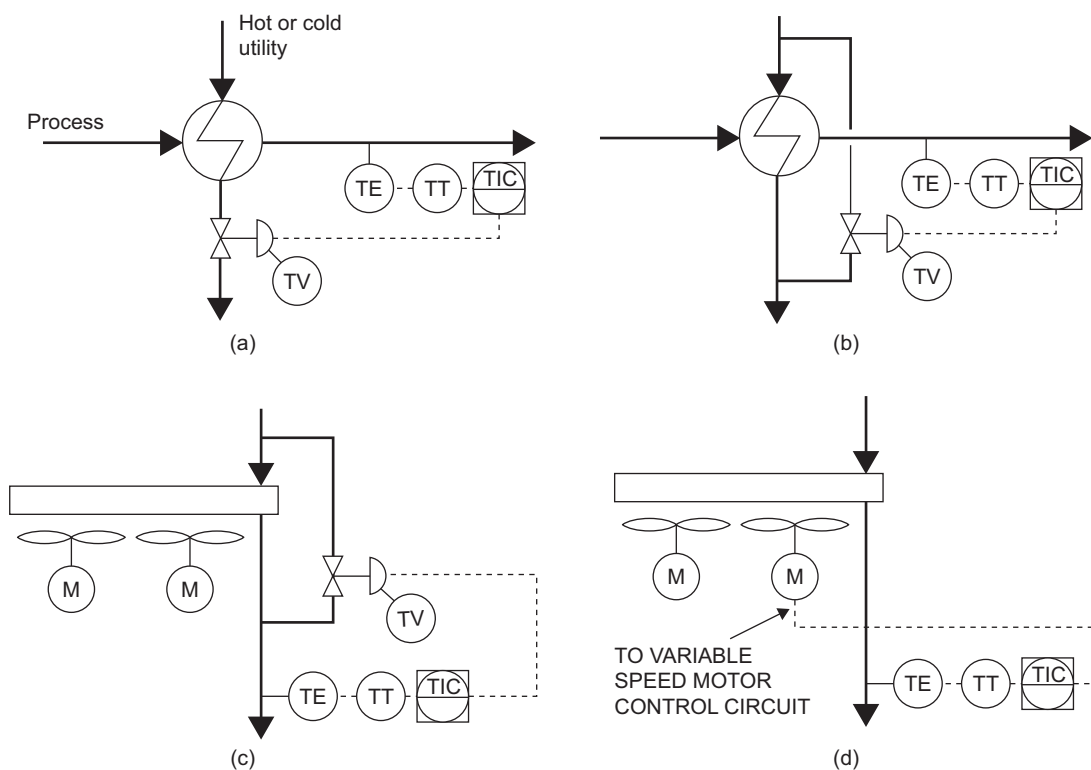


FIG. 5.12 (a) Temperature control of one fluid stream. (b) Bypass control. (c) Air cooler with bypass control. (d) Air cooler with variable-speed drive.

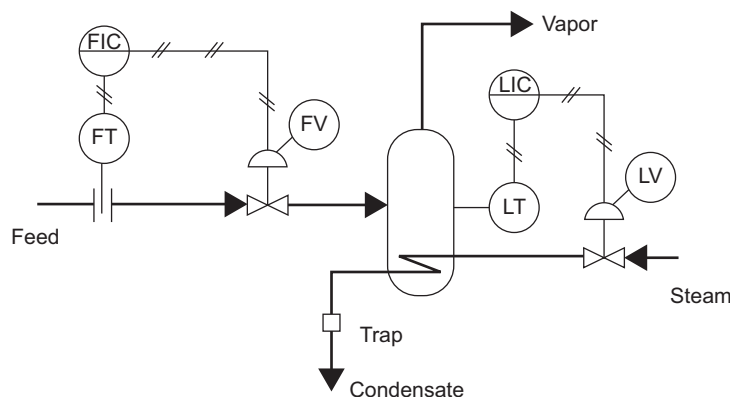


FIG. 5.13 Vaporizer control.

with the liquid feed to the vaporizer on flow control, as shown in Fig. 5.13. An increase in the feed results in an automatic increase in steam to the vaporizer to vaporize the increased flow and maintain a constant level.

Reboiler control systems are selected as part of the general control system for the distillation column and are discussed in Section 5.4.7.

5.4.5 Cascade control

With this arrangement, the output of one controller is used to adjust the set point of another. Cascade control can give smoother control in situations where direct control of the variable would lead to unstable operation. The secondary controller can be used to compensate for any short-term variations in, say, a utility stream flow, which would upset the controlled variable: the primary controller controlling long-term variations. Typical examples are shown in Figs. 5.18 and 5.19.

5.4.6 Ratio control

Ratio control can be used where it is desired to maintain two flows at a constant ratio, for example, reactor feeds or distillation column reflux. A typical scheme for ratio control is shown in Fig. 5.14.

5.4.7 Distillation column control

The primary objective of distillation column control is to maintain the specified composition of the top and bottom products and any side streams, correcting for the effects of disturbances in:

1. Feed flow rate, composition, and temperature
2. Steam or other hot utility supply
3. Cooling water or air cooler conditions
4. Ambient conditions, which can cause cooling of the column shell and changes in internal reflux (see Chapter 17)

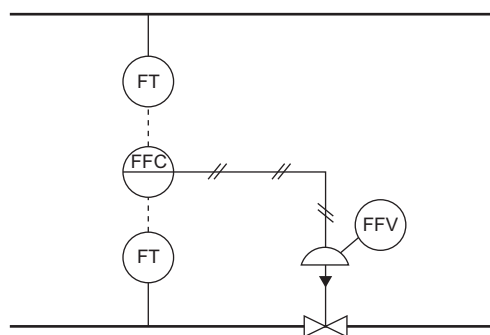


FIG. 5.14 Ratio control.

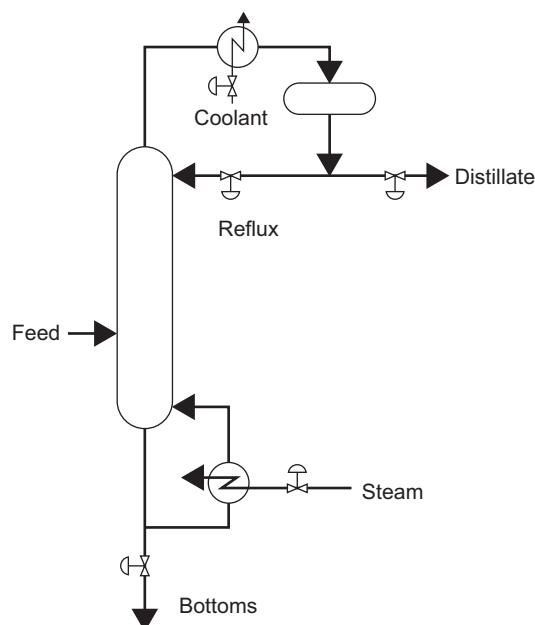


FIG. 5.15 Control valves and degrees of freedom for a simple distillation column.

The feed flow rate is often set by the level controller on a preceding column. It can be independently controlled if the column is fed from a storage or surge tank. Feed temperature is not normally controlled unless a feed pre-heater is used.

In the usual case where the feed rate is set by upstream operations and the column produces a liquid distillate product, there are five control valves, and hence five degrees of freedom; see Fig. 5.15. One degree of freedom is used to set the column pressure, usually by control of the condenser using one of the schemes shown in Fig. 5.10. Column pressure is normally controlled at a constant value, which then sets the vapor inventory in the column. The use of variable pressure control to conserve energy has been discussed by Shinskey (1976). Two degrees of freedom are needed to control the liquid inventories by controlling the vapor–liquid level in the column sump and the reflux drum (or condenser if no reflux drum is used).

The remaining two degrees of freedom can be used to achieve the desired separation, either in terms of product purity or recovery, by adjusting two flow rates. One of these flows is controlled by a flow or flow ratio controller to achieve the desired split between distillate and bottoms, while the other is usually controlled by a column temperature to achieve a desired composition in one of the products. The flow controller cannot be on the distillate or bottoms stream if the designer intends to control composition, as it would then be impossible to maintain product composition if there were changes in feed composition. The temperature controller can, however, control either the distillate or bottoms flow rate. The usual practice is to control a top temperature by varying the reflux ratio or distillate flow rate if the overhead product purity is more important (Fig. 5.16) or to control a bottom temperature by varying the boil-up rate or bottoms flow if bottoms purity is more important (Fig. 5.17).

Control schemes of this type are commonly referred to as *material balance* control schemes, as they achieve the desired product purity by manipulating the column material balance. These schemes are highly robust for processes where the feed flow rate to the column is relatively constant but the composition varies and close control must be maintained on one product composition.

Temperature is usually used as an indication of composition. The temperature sensor should be located at a position in the column where the rate of change of temperature with change in composition of the key component is a maximum; see Parkins (1959). Near the top and bottom of the column the change is usually small. When designing the column, it is a good idea to allow for thermowells on several trays so that the best control point can be found when the column is actually operating. If reliable online composition analyzers are available, they can be incorporated in the control loop, but more complex control equipment will be needed, and composition analyzers are usually used to cascade onto simpler temperature control loops. With multicomponent systems, temperature is not a unique function of composition.

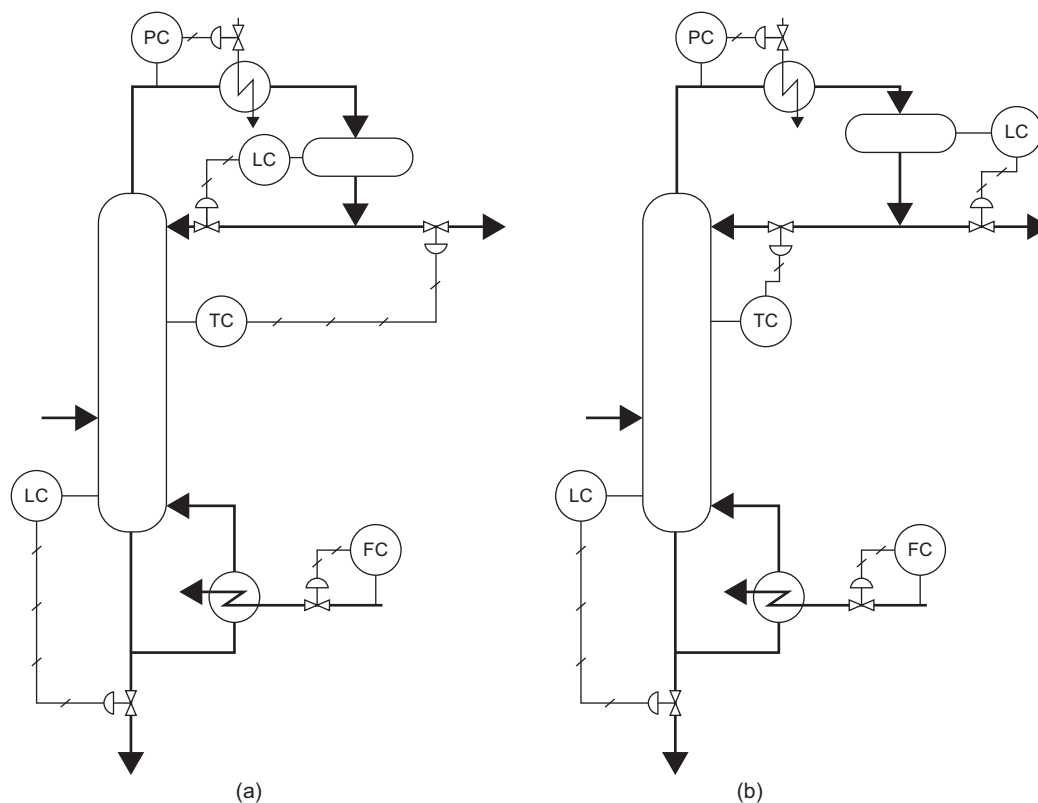


FIG. 5.16 Material balance control schemes for controlling overhead product composition. Flow control on reboiler can be in ratio to feed if feed rate varies. (a) Direct control of distillate by composition. (b) Indirect control of distillate; composition controls reflux.

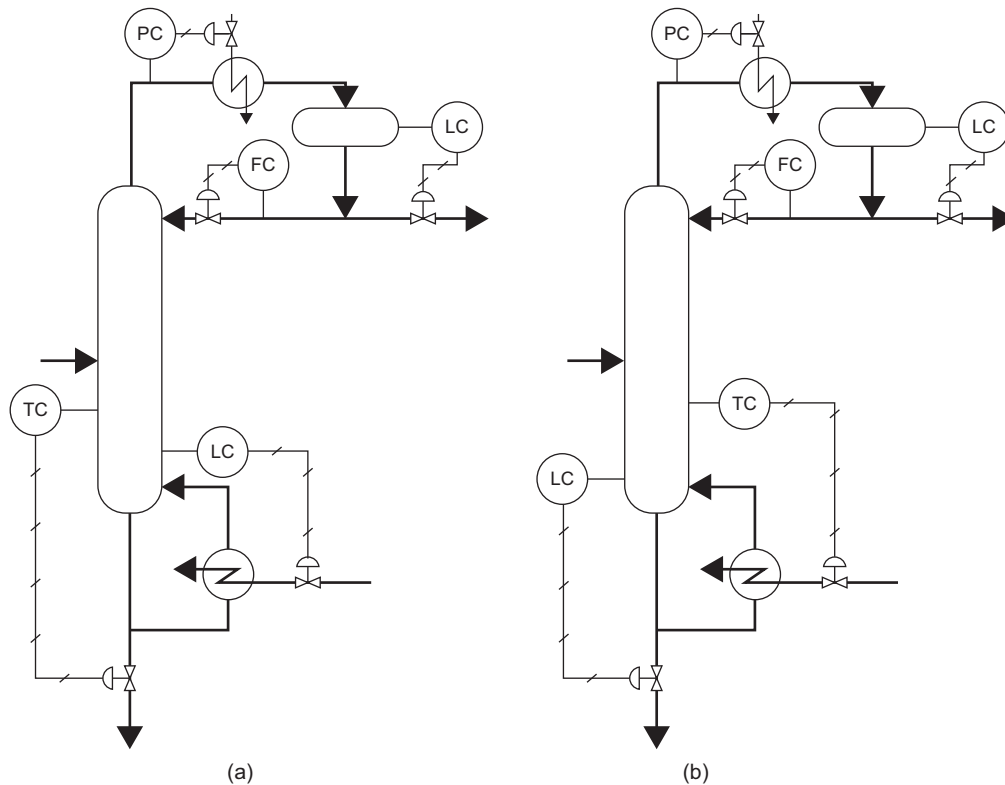


FIG. 5.17 Material balance control schemes for controlling bottoms product composition. Flow control on reflux can be in ratio to feed if feed rate varies. (a) Direct control of bottoms by composition. (b) Indirect control of bottoms; composition controls boil-up.

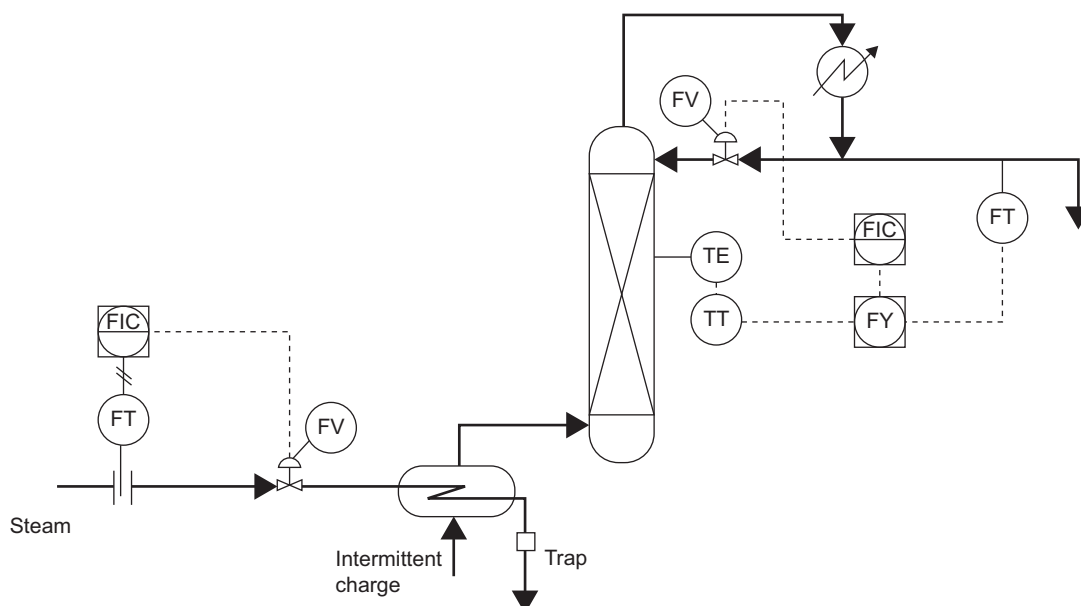


FIG. 5.18 Batch distillation, with reflux flow controlled based on temperature to infer composition.

Flow ratio controllers are sometimes used in distillation control, controlling the reflux or boil-up in ratio to the feed, distillate, or bottoms rate. The same effect can be accomplished using cascade control, with the feed rate adjusting the set point of the flow controller on reflux or boil-up.

Shinskey (1984) has shown that there are 120 ways of connecting the five main pairs of measured and controlled variables in single loops. A variety of control schemes has been devised for distillation column control. Some typical schemes are shown in Figs. 5.16 to 5.18; ancillary control loops and instruments are not shown.

The choice of control scheme may be influenced by many other factors. For example, the control scheme of Fig. 5.17b) controls boil-up by composition and gives the fastest control response to variations in composition of any of the schemes. Kister (1990) discusses the advantages and drawbacks of the material balance control schemes shown in Figs. 5.16 and 5.17.

An older control scheme that is often encountered is similar to Fig. 5.16(b) but has the steam to the reboiler controlled by a temperature in the stripping section of the column. This scheme is known as *temperature-pattern control* or *dual composition control*, and in principle allows both top and bottom compositions to be controlled. The drawback of this scheme is that there is a tendency for the controllers to fight each other, leading to unstable operation.

Distillation column control is discussed in detail by Parkins (1959), Bertrand and Jones (1961), Shinskey (1984), and Buckley et al. (1985).

Additional temperature indicating or recording points should be included up the column for monitoring column performance and for troubleshooting.

5.4.8 Reactor control

The schemes used for reactor control depend on the process and the type of reactor. If a reliable online composition analyzer is available and the reactor dynamics are suitable, the product composition can be monitored continuously and the reactor conditions and feed flows controlled automatically to maintain the desired product composition and yield. More often, the operator is the final link in the control loop, adjusting the controller set points to maintain the product within specification, based on periodic laboratory analyses.

For small stirred-tank reactors, temperature will normally be controlled by regulating the flow of the heating or cooling medium. For larger reactors, temperature is often controlled by recycling a part of the product stream or adding inert material to the feed to act as a heat sink. Pressure is usually held constant. For liquid phase reactors, pressure is often controlled by maintaining a vapor space above the liquid reagents. This space can be pressurized with nitrogen or other suitable gases. Material balance control will be necessary to maintain the correct flow of reactants to the reactor and the flow of products and unreacted materials from the reactor. A typical control scheme for a simple liquid phase reactor is shown in Fig. 5.19. See also Luyben (2007).

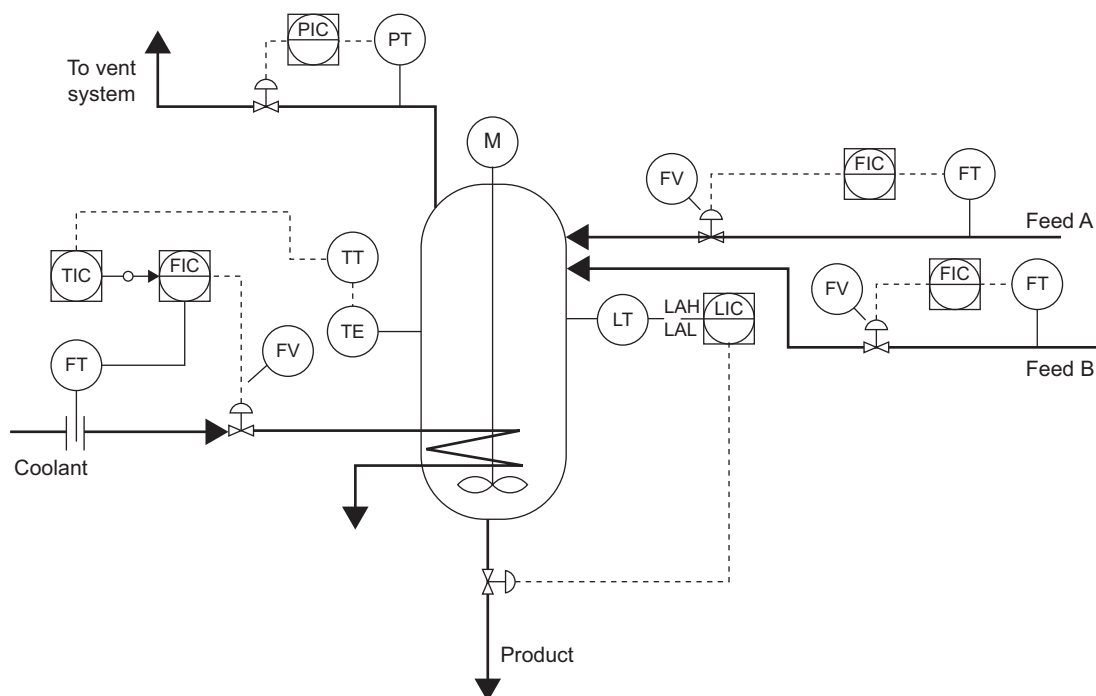


FIG. 5.19 A typical stirred tank reactor control scheme, temperature cascade control of coolant flow, and flow control of reagents.

The control of biological reactions is more complex because it is necessary to maintain many parameters within relatively tight ranges. Control of biological reactors is discussed in Section 15.9.7.

5.5 Alarms, safety trips, and interlocks

Alarms are used to alert operators to serious, and potentially hazardous, deviations in process conditions. Key instruments are fitted with switches and relays or software alarms to operate audible and visual alarms on the control panels and shared display screens. Where delay or lack of response by the operator is likely to lead to the rapid development of a hazardous situation, the instrument would be fitted with a trip system to take action automatically to avert the hazard, such as shutting down pumps, closing valves, and operating emergency systems.

The basic components of an automatic trip system are:

1. A sensor to monitor the control variable and provide an output signal when a preset value is exceeded (the instrument)
2. A link to transfer the signal to the actuator, usually consisting of a system of pneumatic or electric relays
3. An actuator to carry out the required action: close or open a valve, switch off a motor, etc.

A description of some of the equipment (hardware) used is given by [Rasmussen \(1975\)](#).

A safety trip can be incorporated in a control loop, as shown in [Fig. 5.20\(a\)](#). In this system the level control instrument has a built-in software alarm that alerts the operator if the level is too low and a programmed trip set for a level somewhat lower than the alarm level. However, the safe operation of such a system will be dependent on the reliability of the control equipment, and for potentially hazardous situations, it is better practice to specify a separate trip system, such as that shown in [Fig. 5.20\(b\)](#), in which the trip is activated by a separate low-level switch. Provision must be made for the periodic checking of the trip system to ensure that the system operates when needed.

The effective operation of instrumented safety systems depends on the reliable operation of all the components in the system. Because no component is perfectly reliable, designers increase the system reliability by building in redundancy and adding duplicate instruments, switches, relays, etc., so that if one component fails the rest of the system will still operate correctly. More information on the design of safety instrumented systems is given in Section 10.8.

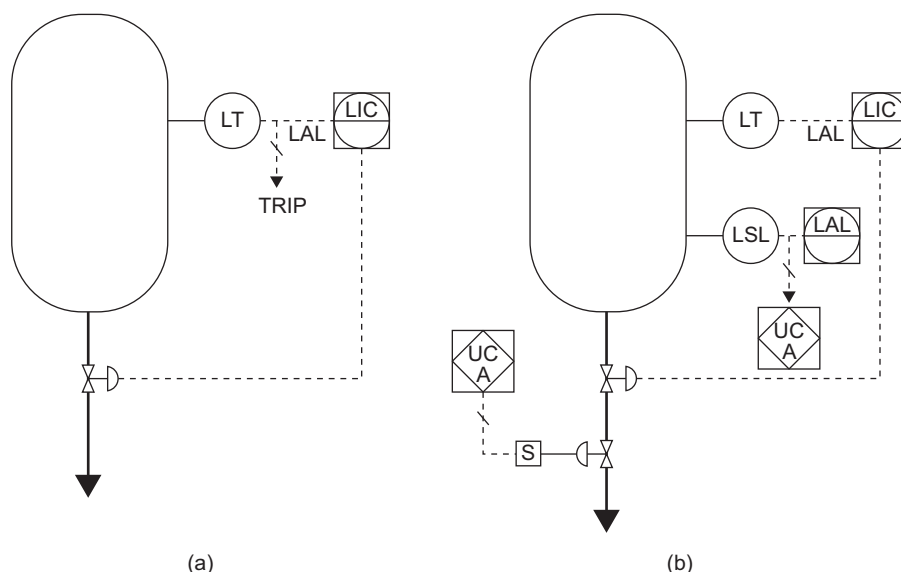


FIG. 5.20 (a) Trip as part of control system. (b) Separate shut-down trip.

Interlocks

Where it is necessary to follow a fixed sequence of operations—for example, during a plant start-up and shut-down, or in batch operations—interlocks are included to prevent operators from departing from the required sequence. They may be incorporated in the control system design, as pneumatic or electric relays, or may be mechanical interlocks. Various proprietary special lock and key systems are also available. In most plants, programmable logic controllers are used, and the interlocks are coded into the control algorithms. Care should be taken to test all of the interlocks in the plant automation during commissioning or whenever changes are made to the plant control and automation.

5.6 Batch process control

Batch processes necessarily involve dynamic variation in process parameters, and so the design of control systems for batch plants is more complex than for plants that operate continuously. In addition to conventional regulatory control functions that maintain temperatures, pressures, flows, and levels at desired values, the designer must consider discrete (on-off) control functions that start and stop operations, as well as the overall recipe or sequence of operations. In a plant that has several batch processes or that makes multiple products, the automation system may also include production planning, batch sequencing, and tracking and logging of batch data for quality control purposes.

The international standard for the design of control systems for batch plants is IEC 61512, which is based on the set of standards developed by the ISA S88 committee. These standards define an architecture for batch process control systems that regulates the flow of information from high-level decisions such as recipe management and production scheduling down to low-level regulatory process control functions. A detailed description of the S88 standards is beyond the scope of this book—for more information the reader should consult ISA 88.00.01 (2010) or the books by [Fleming and Pillai \(1998\)](#), [Parshall and Lamb \(2000\)](#), or [Love \(2007\)](#).

When developing a PFD or P&I diagram for a batch plant, the design team must consider all of the controllers that are needed to regulate the dynamic operation of the plant. The regulatory controllers will control the same variables that would be controlled in a continuous process. Indeed, during some phases of the batch recipe, the regulatory control loops function in the same way as they would for a continuous process. In addition to the regulatory control systems, the designer must then add the discrete control functions that change the set points of the regulatory controllers and activate the opening and closing of isolation valves to start and stop flows to the process. Microprocessor-based programmable logic controllers are usually used to control the sequence of phases that makes up the batch recipe.

5.7 Computer control systems

Almost all process control systems installed on new plants use programmable electronic devices based on microprocessors. These range from simple digitally actuated single-loop controllers that produce a single output signal (single input–single output [SISO] devices) up to complex distributed control systems that carry out control, real-time optimization, and data logging and archiving for multiple process plants across a site or even an enterprise (multiple input–multiple output [MIMO] devices).

The use of microprocessors in controllers allows the controllers to perform more complex control algorithms than could previously be achieved using analog systems based on pneumatic signals. A microprocessor can take input from several instruments and use a sophisticated model to calculate the outputs to multiple actuators. A simple example of a multiple input device is a gas mass-flow controller (Fig. 5.21), in which the gas mass flow is computed based on inputs from temperature, pressure, and flow instruments.

The conventional control schemes described in Section 5.4 mainly make use of SISO controllers, because the schemes were developed for single unit operations. At the unit operation level, the primary focus of process control is usually on safe and stable operation, and it is difficult to take full advantage of the capability of advanced microprocessor-based control systems. When several unit operations are put together to form a process, then the scope for use of MIMO devices increases, particularly when the devices are able to communicate with each other rapidly. The digital control system can then make use of more complex algorithms and models that enable feed-forward control (model-based or multivariable predictive control) and allow data collected from upstream in the process to guide the selection of operating conditions and controller set points for downstream operations. This allows for better response to process dynamics and more rapid operation of batch, cyclic, and other unsteady-state processes. Model-based predictive control is also often used as a means of controlling product quality. This is because devices for measuring product quality typically require analytical procedures that take several minutes to hours to run, making effective real-time feedback control difficult to accomplish.

The use of instruments that log and archive data facilitates remote monitoring of process performance and can improve plant troubleshooting and optimization, as well as providing high-level data for enterprise-wide supply chain management. Increasingly, archived data from plant historian databases is used to develop algorithms that predict safety and maintenance issues and allow more stable and economically efficient operation of the plant. These algorithms can be based on data from multiple sensing instruments and can be either deterministic (developed by regression of equations input by an engineer based on knowledge of engineering fundamentals) or fully empirical (developed by machine learning approaches).

The electronic equipment and systems technology available for process control continues to evolve rapidly. Because of the pace of innovation, industry-wide standards have not been able to keep up, and consequently different manufacturers' systems often use proprietary technology and are often not fully compatible with each other. The implementation of the ISA 50, ISA 100, HART Foundation Fieldbus, and OPC UA standards substantially improved digital communications between control devices, leading to improved control, faster set-up, better reliability through higher redundancy, and even greater distribution of functions between devices.

Wireless systems are beginning to be used in inventory control and maintenance management, but are not widely used yet in plant control. The control systems vendors have overcome problems with interference, signal blocking, and signal loss and have demonstrated robust error checking and transmission protocols. As experience is gained with wireless instrumentation, it is likely to be much more widely adopted in the future, as wireless systems are more convenient to install and can be more robust in the event of incidents such as small fires. A recent survey of wireless control was given by [McKeon-Slattey \(2010\)](#), but this area is currently evolving rapidly.

A detailed treatment of digital technology for process control is beyond the scope of this book. [Kalani \(1988\)](#), [Edgar et al. \(1997\)](#), [Liptak \(2012\)](#), and [Love \(2007\)](#) all provide excellent reviews of the subject. [Mitchell and Law](#)

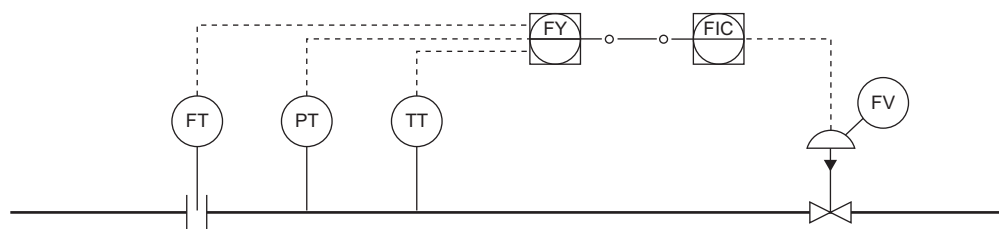
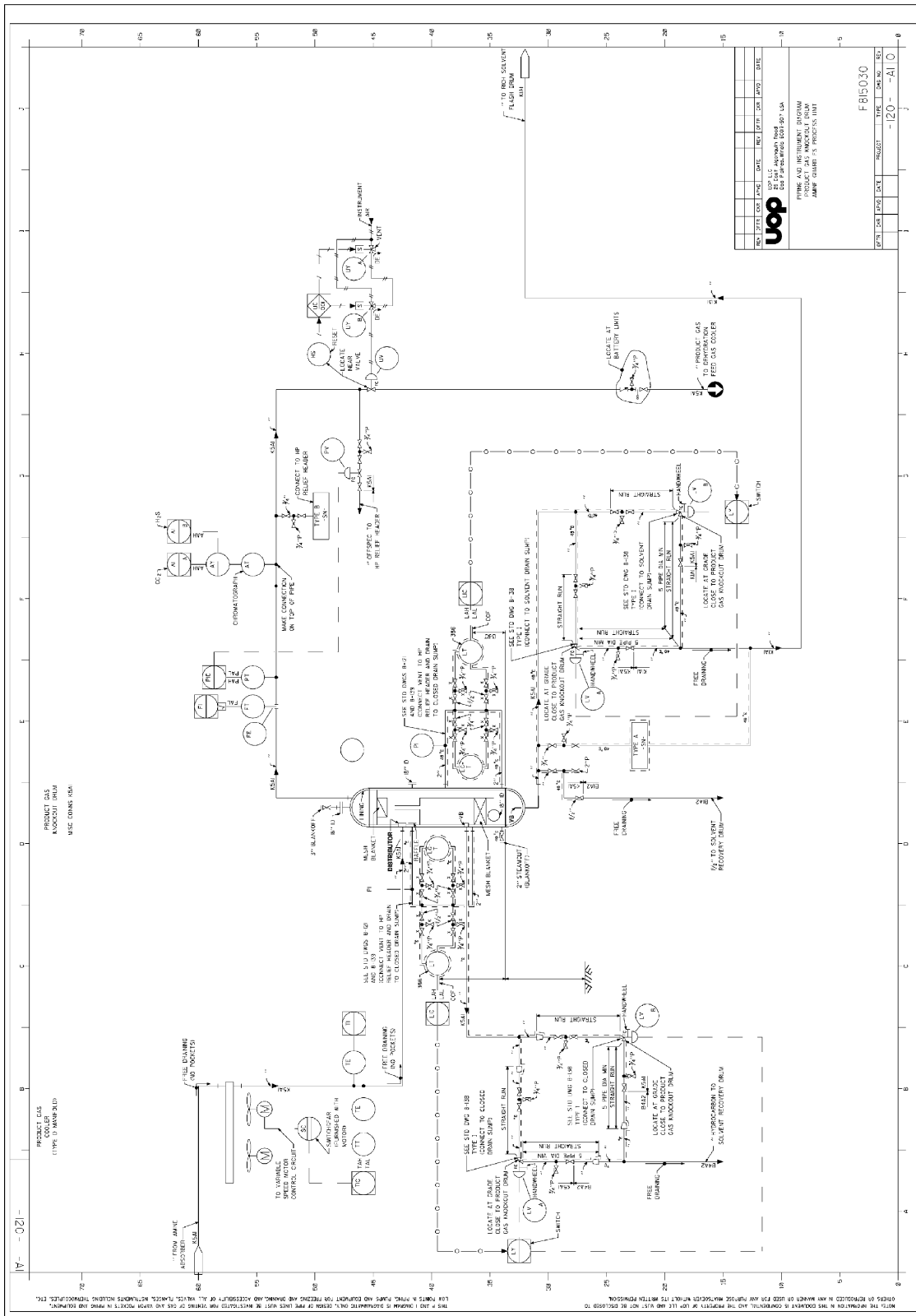


FIG. 5.21 Gas mass flow controller.



(2003) give a good overview of digital bus technologies. Baldi (2016) discusses the cybersecurity defense of industrial control systems. Bagajewicz (2010) gives a comprehensive overview of issues in extracting real-time process data and using them to develop models for advanced control and process performance monitoring and optimization.

5.8 References

- Bagajewicz, M. J. (2010). *Smart process plants: Software and hardware solutions for accurate data and profitable operations*. McGraw-Hill.
- Baldi, M. (2016). Cybersecurity defence for industrial process control systems. *Chemical Engineering*, 123(7), 36.
- Bertrand, L., & Jones, J. B. (1961). Controlling distillation columns. *Chemical Engineering*, 68(Feb. 20), 139. New York.
- Buckley, P. S., Luyben, W. L., & Shunta, J. P. (1985). *Design of distillation column control systems*. Arnold.
- Chhabra, R. P., & Rohani, S. (2017). *Coulson and Richardson's chemical engineering: Volume 3B: Process control* (4th ed.). Butterworth-Heinemann.
- Coughanowr, D. R., & LeBlanc, S. E. (1994). *Process systems analysis and control* (3rd ed.). McGraw-Hill.
- Edgar, T. F., Smith, C. L., Shinskey, F. G., et al. (1997). Process control. In *Perry's chemical engineers handbook* (7th ed.). McGraw-Hill.
- Fleming, D. W., & Pillai, V. (1998). *S88 implementation guide*. McGraw Hill.
- Green, D. W., & Southard, M. Z. (Eds.). (2018). *Perry's chemical engineers' handbook* (9th ed.). McGraw-Hill.
- Hall, J. (2010). Process pump control. *Chemical Engineering*, 117(12), 30.
- Henson, M., Seborg, D. E., & Hempstead, H. (1996). *Nonlinear process control*. Prentice Hall.
- Kalani, G. (1988). *Microprocessor based distributed control systems*. Prentice Hall.
- Kalani, G. (2002). *Industrial process control: Advances and applications*. Butterworth Heinemann.
- Kister, H. Z. (1990). *Distillation operation*. McGraw-Hill.
- Kister, H. Z. (2016). Flooded condenser controls: Principles and troubleshooting. *Chemical Engineering*, 123(1), 37. New York.
- Liptak, B. G. (2012). *Instrument engineers' handbook* (4th ed.). CRC Press.
- Love, J. (2007). *Process automation handbook. a guide to theory and practice*. Springer.
- Luyben, W. L., Tyreus, B. D., & Luyben, M. L. (1999). *Plantwide process control*. McGraw-Hill.
- Luyben, W. L. (2007). *Chemical reactor design and control*. Wiley.
- McKeon-Slattey, M. (2010). The world of wireless. *Chemical Engineering Progress*, 106(2), 6.
- Mitchell, J. A., & Law, G. (2003). Get up to speed on digital buses. *Chemical Engineering*, 110(2). New York, (Feb 1).
- Murrill, P. W. (1988). *Application concepts of process control*. ISA.
- Nayyar, M. L. (Ed.) (2000). *Piping handbook* (7th ed.). McGraw-Hill.
- Parkins, R. (1959). Continuous distillation plant controls. *Chemical Engineering Progress*, 55(July), 60.
- Parshall, J., & Lamb, L. (2000). *Applying S88: Batch control from a user's perspective*. ISA.
- Rasmussen, E. J. (1975). Alarm and shut down devices protect process equipment. *Chemical Engineering*, 82(May 12), 74. New York.
- Seborg, D. E., Edgar, T. F., Mellichamp, D. A., & Doyle, F. J. (2017). *Process dynamics and control* (4th ed.). Prentice Hall.
- Shinskey, F. G. (1976). Energy-conserving control systems for distillation units. *Chemical Engineering Progress*, 72(May), 73.
- Shinskey, F. G. (1984). *Distillation control* (2nd ed.). McGraw-Hill.
- Shinskey, F. G. (1996). *Process control systems* (4th ed.). McGraw-Hill.

American and international standards

- ANSI/ISA-5.1-2009. (2009). *Instrumentation symbols and identification*.
- IEC 61512-1. (1997). *Batch control part 1: Models and terminology*.
- ISA-50.00.01 (R2012). Compatibility of analog signals for electronic industrial process instruments - Formerly ANSI/ISA 50.1-1982 (R1992); formerly ANSI/ISA-50.1-1975 (R1992). Revised 2012.
- ISA-88.00.01-2010. (2010). *Batch control part 1: Models and terminology*.
- ISA-100.11a-2011. (2011). *Wireless systems for industrial automation: Process control and related applications*.

Other standards

- BS 1646:1984. (1984). *Symbolic representation for process measurement control functions and instrumentation*. R2012.
- DIN 2429-2. (1988). *Symbolic representation of pipework components for use on engineering drawings; functional representation*.
- DIN 19227-1. (1993). *Control technology; graphical symbols and identifying letters for process control engineering; symbolic representation for functions*.
- DIN 19227-2. (1991). *Control technology; graphical symbols and identifying letters for process control engineering; representation of details*.
- DIN 28000-4. (2014). *Chemical apparatus - documentation in the life cycle of process plants - part 4: Graphical symbols of valves, pipes and actuators*.

5.9 Problems

5.1. How would you measure:

1. The temperature of a fermentation broth.
2. The mass flow of a gas at high temperature and pressure.
3. The volumetric flow rate of a slurry of diced carrots in water.
4. The level of liquid in a crystallizer.
5. The feed rate of solids to a mixing tank.

- 5.2. a. What alarms would you add to the vaporizer control scheme shown in Fig. 5.13? Indicate whether the alarm would signal high or low conditions, what the alarm would signify, and what operator response would be required in each case.
b. Which alarms should activate a shut-down trip, and which valves should be closed?
- 5.3. Sketch a control scheme for the reactor section shown in Fig. 2.17. The feeds are liquids, and the reactors operate under pressure with inert nitrogen in the vapor space above the reagents. The objective is to achieve full conversion of feed A by the outlet of the last reactor.
- 5.4. A fermentation reactor is charged with a sterile feed of growth media at 35 °C and inoculated with a batch of microorganism. The batch is allowed to grow for 10 days. During the growth period the temperature is maintained at 37 °C by circulating cold water through a jacket on the vessel. Sterile air is sparged into the fermenter to maintain a desired dissolved oxygen concentration. The pH of the fermenter is controlled by periodic addition of a dilute solution of sodium hydroxide. At the end of the growth period the batch is discharged from the reactor to the harvesting section of the process.
a. Sketch a P&I diagram of the reactor and feed section.
b. What pressure would you choose for operation of the fermenter, and how would you control it?
- 5.5. A polymer is produced by the emulsion polymerization of acrylonitrile and methyl methacrylate in a stirred vessel. The monomers and an aqueous solution of catalyst are fed to the polymerization reactor continuously. The product is withdrawn from the base of the vessel as a slurry. Devise a control system for this reactor, and draw up a preliminary piping and instrument diagram. The following points need to be considered:
1. Close control of the reactor temperature is required.
 2. The reactor runs 90% full.
 3. The water and monomers are fed to the reactor separately.
 4. The emulsion is a 30% mixture of monomers in water.
 5. The flow of catalyst will be small compared with the water and monomer flows.
 6. Accurate control of the catalyst flow is essential.
- 5.6. Devise a control system for the distillation column described in Chapter 17, Example 17.2. The flow to the column comes from a storage tank. The product, acetone, is sent to storage and the waste to an effluent pond. It is essential that the specifications on product and waste quality are met.

Materials of construction

KEY LEARNING OBJECTIVES

- Mechanical and chemical properties that must be considered when selecting materials of construction for a chemical plant
- Relative costs of common materials of construction
- Properties of alloys commonly used in engineering
- When to use polymers or ceramic materials

6.1 Introduction

This chapter covers the selection of materials of construction for process equipment and piping. The selection of materials of construction must be made before the capital cost of a process can be estimated, as plant costs can vary significantly with materials selection.

Many factors have to be considered when selecting engineering materials, but for a chemical process plant, the overriding considerations are usually high temperature strength and the ability to resist corrosion. The process designer will be responsible for recommending materials that will be suitable for the process conditions. The process engineer must also consider the requirements of the mechanical design engineer; the material selected must have sufficient strength and be easily worked. The most economical material that satisfies both process and mechanical requirements should be selected; this will be the material that gives the lowest cost over the working life of the plant, allowing for maintenance and replacement. Other factors, such as product contamination and process safety, must also be considered. The mechanical properties that are important in the selection of materials are discussed briefly in this chapter. Several books have been published on the properties of materials and the metal-working processes used in equipment fabrication, and a selection suitable for further study is given in the list of references at the end of this chapter. The mechanical design of process equipment is discussed in [Chapter 14](#).

A detailed discussion of the theoretical aspects of corrosion is not given in this chapter, as this subject is covered comprehensively in several books: [Revie \(2011\)](#), [Fontana \(1986\)](#), [Dillon \(1994\)](#), and [Schweitzer \(1989\)](#). An extensive set of corrosion data for different materials is given by [Craig and Anderson \(1995\)](#).

6.2 Material properties

The most important characteristics to be considered when selecting a material of construction are:

1. Mechanical properties
 - a. Strength: tensile strength
 - b. Stiffness: elastic modulus (Young's modulus)
 - c. Toughness: fracture resistance
 - d. Hardness: wear resistance
 - e. Fatigue resistance
 - f. Creep resistance
2. The effect of high temperature, low temperature, and thermal cycling on the mechanical properties
3. Corrosion resistance
4. Any special properties required, such as thermal conductivity, electrical resistance, and magnetic properties
5. Ease of fabrication: forming, welding, casting ([Table 6.1](#))
6. Availability in standard sizes: plates, sections, and tubes
7. Cost

TABLE 6.1 A guide to the fabrication properties of common metals and alloys

	Machining	Cold working	Hot working	Casting	Welding	Annealing temp. °C
Mild steel	S	S	S	D	S	750
Low alloy steel	S	D	S	D	S	750
Cast iron	S	U	U	S	D/U	—
Stainless steel (18 Cr, 8 Ni)	S	S	S	D	S	1050
Nickel	S	S	S	S	S	1150
Monel	S	S	S	S	S	1100
Copper (deoxidized)	D	S	S	S	D	800
Brass	S	D	S	S	S	700
Aluminum	S	S	S	D	S	550
Dural	S	S	S	—	S	350
Lead	—	S	—	—	S	—
Titanium	S	S	U	U	D	—

S, Satisfactory; D, Difficult, special techniques needed; U, Unsatisfactory.

6.3 Mechanical properties

Typical values of the mechanical properties of the more common materials used in the construction of chemical process equipment are given in [Table 6.2](#).

6.3.1 Tensile strength

The tensile strength (tensile stress) is a measure of the basic strength of a material. It is the maximum stress that the material will withstand, measured by a standard tensile test. The older name for this property, which is more descriptive of the property, was ultimate tensile strength (UTS).

Proof stress is the stress that causes a specified permanent extension, usually 0.1%.

The maximum allowable stress specified by the ASME Boiler and Pressure Vessel (BPV) Code is calculated from these and other material properties at the design temperature, allowing for suitable safety factors. The basis for establishing maximum allowable stress values is discussed in [Chapter 14](#) and is described in detail in the ASME BPV Code Section II Part D, Mandatory Appendix 1.

TABLE 6.2 Mechanical properties of common metals and alloys (typical values at room temperature)

	Tensile strength (N/mm ²)	0.1% proof stress (N/mm ²)	Modulus of elasticity (kN/mm ²)	Brinell hardness	Specific gravity
Mild steel	430	220	210	100–200	7.9
Low-alloy steel	420–660	230–460	210	130–200	7.9
Cast iron	140–170	—	140	150–250	7.2
Stainless steel (18 Cr, 8 Ni)	>540	200	210	160	8.0
Nickel (>99% Ni)	500	130	210	80–150	8.9
Monel	650	170	170	120–250	8.8
Copper (deoxidized)	200	60	110	30–100	8.9
Brass (admiralty)	400–600	130	115	100–200	8.6
Aluminum (>99%)	80–150	—	70	30	2.7
Dural	400	150	70	100	2.7
Lead	30	—	15	5	11.3
Titanium	500	350	110	150	4.5

Note: Tensile stress and proof stress are not the same as the maximum allowable stress permitted by design code. See Tables 6.5 and 6.7 for maximum allowable stress values.

6.3.2 Stiffness

Stiffness is the ability to resist bending and buckling. It is a function of the elastic modulus of the material and the shape of the cross-section of the member (the second moment of area).

6.3.3 Toughness

Toughness is associated with tensile strength, and is a measure of the material's resistance to crack propagation. The crystal structure of ductile materials, such as steel, aluminum, and copper, is such that they stop the propagation of a crack by local yielding at the crack tip. In other materials, such as the cast irons and glass, the structure is such that local yielding does not occur and the materials are brittle. Brittle materials are weak in tension but strong in compression. Under compression any incipient cracks present are closed up. Various techniques have been developed to allow the use of brittle materials in situations where tensile stress would normally occur—for example, the use of prestressed concrete and glass fiber–reinforced plastics in pressure vessel construction.

A detailed discussion of the factors that determine the fracture toughness of materials can be found in the books by the [Institute of Metallurgists \(1960\)](#) and [Boyd \(1970\)](#). [Gordon \(2018\)](#) gives an elementary, but very readable, account of the strength of materials in terms of their macroscopic and microscopic structure.

6.3.4 Hardness

The surface hardness, as measured in a standard test, is an indication of a material's ability to resist wear. This will be an important property if the equipment is being designed to handle abrasive solids or liquids containing suspended solids that are likely to cause erosion.

6.3.5 Fatigue

Fatigue failure is likely to occur in equipment subject to cyclic loading, for example, rotating equipment, such as pumps and compressors, and equipment subjected to temperature or pressure cycling. A comprehensive treatment of this subject is given by [Harris \(1976\)](#).

6.3.6 Creep

Creep is the gradual extension of a material under a steady tensile stress over a prolonged period. It is usually only important at high temperatures, for instance, with steam and gas turbine blades. For a few materials, notably lead, the rate of creep is significant at moderate temperatures. Lead will creep under its own weight at room temperature, and lead linings must be supported at short intervals.

The creep strength of a material is usually reported as the stress to cause rupture in 100,000 hours at the test temperature.

6.3.7 Effect of temperature on the mechanical properties

The tensile strength and elastic modulus of metals decrease with increasing temperature. For example, the tensile strength of mild steel (low-carbon steel, $C < 0.25\%$) is 450 N/mm^2 at 25°C , falling to 210 N/mm^2 at 500°C , and the value of Young's modulus is $200,000 \text{ N/mm}^2$ at 25°C , falling to $150,000 \text{ N/mm}^2$ at 500°C . The ASME BPV Code Section II Part D specifies maximum temperatures for each material. For example, SA-285 plain carbon steel plate cannot be used to construct a pressure vessel that meets the specifications of ASME BPV Code Section VIII Div. 1 with a design temperature greater than 482°C (900°F). Any pressure vessel that is designed for use above this temperature must be made from killed steel or alloy. The maximum allowable stress used in a design is always based on the design temperature. Materials must be chosen that have sufficient strength at the design temperature to give an economic and mechanically feasible wall thickness. The stainless steels are superior in this respect to plain carbon steels.

Creep resistance will be important if the material is subjected to high stresses at elevated temperatures. Special alloys, such as Inconel 600 (UNS N06600) or Incoloy 800 (UNS N08800) (both trademarks of International Nickel Co.) are used for high-temperature equipment such as furnace tubes in environments that do not contain sulfur. The selection of materials for high-temperature applications is discussed by [Day \(1979\)](#) and [Lai \(1990\)](#).

At low temperatures, less than 10°C , metals that are normally ductile can fail in a brittle manner. Serious disasters have occurred through the failure of welded carbon steel vessels at low temperatures. The phenomenon of brittle failure is associated with the crystalline structure of metals. Metals with a body-centered-cubic (bcc) lattice are more liable to brittle failure than those with a face-centered-cubic (fcc) or hexagonal lattice. For low-temperature equipment, such as cryogenic plant and liquefied-gas storages, austenitic stainless steel (fcc) or aluminum alloys (hex) should be specified; see [Wigley \(1978\)](#).

V-notch impact tests, such as the Charpy test, are used to test the susceptibility of materials to brittle failure: see [Wells \(1968\)](#) and ASME BPV Code Sec. VIII Div. 1 Part UG-84.

The brittle fracture of welded structures is a complex phenomenon and is dependent on plate thickness and the residual stresses present after fabrication, as well as the operating temperature. A comprehensive discussion of brittle fracture in steel structures is given by [Boyd \(1970\)](#).

6.4 Corrosion resistance

The conditions that cause corrosion can arise in a variety of ways. For this brief discussion on the selection of materials, it is convenient to classify corrosion into the following categories:

1. General wastage of material – uniform corrosion
2. Galvanic corrosion – dissimilar metals in contact
3. Pitting – localized attack
4. Intergranular corrosion
5. Stress corrosion
6. Erosion – corrosion
7. Corrosion fatigue
8. High-temperature oxidation and sulfidation
9. Hydrogen embrittlement

Metallic corrosion is essentially an electrochemical process. Four components are necessary to set up an electrochemical cell:

1. Anode – the corroding electrode
2. Cathode – the passive, noncorroding electrode

3. The conducting medium (the electrolyte) – the corroding fluid
4. Completion of the electrical circuit through the material

Cathodic areas can arise in many ways:

1. Dissimilar metals
2. Corrosion products
3. Inclusions in the metal, such as slag
4. Less well-aerated areas
5. Areas of differential concentration
6. Differentially strained areas

6.4.1 Uniform corrosion

This term describes the more or less uniform wastage of material by corrosion, with no pitting or other forms of local attack. If the corrosion of a material can be considered to be uniform, the life of the material in service can be predicted from experimentally determined corrosion rates.

Corrosion rates are usually expressed as a penetration rate in inches per year, or mills per year (mpy) (where a mill = 10^{-3} inches). They are also expressed as a weight loss in milligrams per square decimeter per day (mdd). Most of the published data on corrosion rates are in imperial units. In corrosion testing, the corrosion rate is measured by the reduction in weight of a specimen of known area over a fixed period.

$$\text{ipy} = \frac{12w}{tA\rho} \quad (6.1)$$

where w = mass loss in time t , lb

t = time, years

A = surface area, ft²

ρ = density of material, lb/ft³

In SI units 1 ipy = 25 mm per year.

When judging corrosion rates expressed in mdd, it must be remembered that the penetration rate depends on the density of the material. For ferrous metals 100 mdd = 0.02 ipy.

What can be considered an acceptable rate of attack will depend on the cost of the material; the duty, particularly with regard to safety; and the economic life of the plant. For the more commonly used inexpensive materials, such as the carbon and low-alloy steels, a guide to what is considered acceptable is given in Table 6.3. For the more expensive alloys, such as the high-alloy steels, the brasses, and aluminum, the figures given in Table 6.3 should be divided by 2.

If the predicted corrosion rate indicates only short exposures, the design engineer should allow for frequent inspection of the plant and periodic replacement of the affected equipment. This affects process economics in two ways, as it reduces the on-stream factor (number of days of production per year) and increases the maintenance costs. Usually the economic impact of frequent shut-down and replacement is so negative that use of a more expensive alloy with better corrosion resistance can be justified.

Allowances for expected corrosion over the plant life or time between replacements must be added to the minimum vessel wall thicknesses calculated to comply with the ASME BPV Code. These corrosion allowances

TABLE 6.3 Acceptable corrosion rates

	Corrosion rate	
	ipy	mm/y
Completely satisfactory	<0.01	0.25
Use with caution	<0.03	0.75
Use only for short exposures	<0.06	1.5
Completely unsatisfactory	>0.06	1.5

can be economically or mechanically prohibitive if the corrosion rate is high. Guidance on corrosion allowances is given in the ASME BPV Code Sec. VIII Div. 1 Non-mandatory Appendix E. The corrosion allowance should at least equal the expected corrosion loss during the desired life of the vessel.

The corrosion rate will be dependent on the temperature and concentration of the corrosive fluid. An increase in temperature usually results in an increased rate of corrosion, though not always. The rate will depend on other factors that are affected by temperature, such as oxygen solubility.

The effect of concentration can also be complex—for example, the corrosion of mild steel in sulfuric acid, where the rate is unacceptably high in dilute acid and at concentrations above 70%, but is acceptable at intermediate concentrations.

6.4.2 Galvanic corrosion

If dissimilar metals are placed in contact, in an electrolyte, the corrosion rate of the anodic metal will be increased, as the metal lower in the electrochemical series will readily act as a cathode. The galvanic series in sea water for some of the more commonly used metals is shown in Table 6.4. Some metals under certain conditions form a natural protective film, for example, stainless steel in oxidizing environments. This state is denoted by “passive” in the series shown in Table 6.4. Active indicates the absence of the protective film, for example, where the surface of the metal is subject to wear due to moving parts or abrasion by the fluid. Minor shifts in position in the series can be expected in other electrolytes, but the series for sea water is a good indication of the combinations of metals to be avoided. If metals that are widely separated in the galvanic series have to be used together, they should be electrically insulated from each other, breaking the conducting circuit. Alternatively, if sacrificial loss of the anodic material can be accepted, the thickness of this material can be increased to allow for the increased rate of corrosion. The corrosion rate will depend on the relative areas of the anodic and cathodic metals. A high cathode to anode area should be avoided. Sacrificial anodes are used to protect underground steel pipes.

TABLE 6.4 Galvanic series in sea water

Noble end (protected end)	18/8 stainless steel (passive)
	Monel
	Inconel (passive)
	Nickel (passive)
	Copper
	Aluminum bronze (Cu 92%, Al 8%)
	Admiralty brass (Cu 71%, Zn 28%, Sn 1%)
	Nickel (active)
	Inconel (active)
	Lead
	18/8 stainless steel (active)
	Cast iron
	Mild steel
	Aluminum
	Galvanized steel
	Zinc
	Magnesium

6.4.3 Pitting

Pitting is the term given to highly localized corrosion that forms pits in the metal surface. If a material is liable to pitting, penetration can occur prematurely, and corrosion rate data are not a reliable guide to the equipment life.

Pitting can be caused by a variety of circumstances; any situation that causes a localized increase in corrosion rate may result in the formation of a pit. In an aerated medium the oxygen concentration will be lower at the bottom of a pit, and the bottom will be anodic to the surrounding metal, causing increased corrosion and deepening of the pit. A good surface finish will reduce this type of attack. Pitting can also occur if the composition of the metal is not uniform, for example, the presence of slag inclusions in welds. The impingement of bubbles can also cause pitting, as seen when cavitation occurs in pumps, which is an example of erosion-corrosion.

6.4.4 Intergranular corrosion

Intergranular corrosion is the preferential corrosion of material at the grain (crystal) boundaries. Though the loss of material will be small, intergranular corrosion can cause the catastrophic failure of equipment. Intergranular corrosion is a common form of attack on alloys but occurs rarely with pure metals. The attack is usually caused by a differential couple being set up between impurities existing at the grain boundary. Impurities will tend to accumulate at the grain boundaries after heat treatment. The classic example of intergranular corrosion in a chemical plant is the weld decay of unestablished stainless steel. This is caused by the precipitation of chromium carbides at the grain boundaries in a zone adjacent to the weld, where the temperature has been between 500° and 800 °C during welding. Weld decay can be avoided by annealing after welding, if practical (postweld heat treatment) or by using low carbon grades (<0.3% C) or grades stabilized by the addition of titanium or niobium.

6.4.5 Effect of stress

Corrosion rate and the form of attack can be changed if the material is under stress. Generally, the rate of attack will not change significantly within normal design stress values; however, for some combinations of metal, corrosive media, and temperature, the phenomenon called *stress corrosion cracking* can occur. This is the general name given to a form of attack in which cracks are produced that grow rapidly and can cause premature, brittle failure of the metal. The conditions necessary for stress corrosion cracking to occur are:

1. Simultaneous stress and corrosion
2. A specific corrosive substance, in particular, the presence of Cl^- , OH^- , NO_3^- , or NH_4^+ ions

Mild stress can cause cracking; the residual stresses from fabrication and welding are sufficient. For a general discussion of the mechanism of stress corrosion cracking, see [Fontana \(1986\)](#). Some classic examples of stress corrosion cracking are:

- The season cracking of brass cartridge cases
- Caustic embrittlement of steel boilers
- The stress corrosion cracking of stainless steels in the presence of chloride ions

Stress corrosion cracking can be avoided by selecting materials that are not susceptible in the specific corrosion environment or, less certainly, by stress relieving by postweld heat treatment.

Comprehensive tables of materials susceptible to stress corrosion cracking in specific chemicals are given by [Moore \(1979\)](#). Moore's tables are taken from the corrosion data survey published by [Hamner \(1974\)](#). See also ASME BPV Code Sec. II Part D Appendix A-330.

The term *corrosion fatigue* is used to describe the premature failure of materials in corrosive environments caused by cyclic stresses. Even mildly corrosive conditions can markedly reduce the fatigue life of a component. Unlike stress corrosion cracking, corrosion fatigue can occur in any corrosive environment and does not depend on a specific combination of corrosive substance and metal. Materials with a high resistance to corrosion must be specified for critical components subjected to cyclic stresses.

6.4.6 Erosion-corrosion

The term *erosion-corrosion* is used to describe the increased rate of attack caused by a combination of erosion and corrosion. If a fluid stream contains suspended particles, or where there is high velocity or turbulence, erosion will

tend to remove the products of corrosion and any protective film, and the rate of attack will be markedly increased. If erosion is likely to occur, more resistant materials must be specified, or the material surface must be protected in some way. For example, plastic inserts can be used to prevent erosion-corrosion at the inlet to heat exchanger tubes.

6.4.7 High-temperature oxidation and sulfidation

Corrosion is normally associated with aqueous solutions, but oxidation can occur in dry conditions. Carbon steels and low-alloy steels will oxidize rapidly at high temperatures, and their use is limited to temperatures below 480 °C (900 °F).

Chromium is the most effective alloying element to give resistance to oxidation, forming a tenacious oxide film. Chromium alloys should be specified for equipment subject to temperatures above 480 °C in oxidizing atmospheres. For example, type 304L stainless steel (18% Cr) can be used up to 650 °C (1200 °F). For temperatures above 700 °C additional stabilization is needed. Type 347 stainless steel is stabilized with niobium and can be used up to 850 °C. High-nickel alloys can also be used as long as sulfur is not present, and high-chromium-content Ni alloys are used at the highest temperatures. For example, Inconel 600 (15.5% Cr) can be used up to 650 °C (1200 °F) and Incoloy 800 (21% Cr) can be used up to 850 °C (1500 °F).

Sulfur is a common corrosive contaminant in gas processing, oil refining, and energy conversion. In reducing environments sulfur is present as H₂S, which causes sulfidation of metals. The metal chosen must often withstand a sulfidizing environment on one side and an oxidizing environment on the other side, all at high temperature (for example, in a furnace tube). Sulfur can attack the chromium oxide scale that protects the alloy, causing breakaway corrosion, particularly for high-nickel alloys. [Lai \(1990\)](#) gives high-temperature corrosion data for various sulfidizing and mixed-gas environments and recommends the use of high-chromium, high-silicon alloys such as HR-160 in this service.

6.4.8 Hydrogen embrittlement

Hydrogen embrittlement is the name given to the loss of ductility caused by the absorption (and reaction) of hydrogen in a metal. It is of particular importance when specifying steels for use in a hydrogen reforming plant. Alloy steels have a greater resistance to hydrogen embrittlement than the plain carbon steels. A chart showing the suitability of various alloy steels for use in hydrogen atmospheres as a function of hydrogen partial pressure and temperature is given in the corrosion data survey by [Hamner \(1974\)](#). Below 500 °C plain carbon steel can be used.

6.5 Selection for corrosion resistance

To select the correct material of construction, the process environment to which the material will be exposed must be clearly defined. In addition to the main corrosive chemicals present, the following factors must be considered:

1. Temperature – affects corrosion rate and mechanical properties
2. Pressure
3. pH
4. Presence of trace impurities – stress corrosion
5. The amount of aeration – differential oxidation cells
6. Stream velocity and agitation – erosion-corrosion
7. Heat transfer rates – differential temperatures

The conditions that may arise during abnormal operation, such as at start-up and shut-down, must be considered in addition to normal, steady-state operation.

6.5.1 Corrosion charts

The resistance of some commonly used materials to a range of chemicals is shown in Appendix B, which is available in the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793. More comprehensive corrosion data, covering most of the materials used in the construction of a process plant, in a wide range of corrosive media, are given by [Rabald \(1968\)](#), NACE (2012), [Hamner \(1974\)](#), [Green and Southard \(2018\)](#), [Lai \(1990\)](#), and [Schweitzer \(1989, 2004, 2006\)](#). The 12-volume *Dechema Corrosion Handbook* is an extensive

guide to the interaction of corrosive media with materials (Kreysa and Schütze, 2007). The ASM *Handbook of Corrosion Data* also has extensive data (Craig and Anderson, 1995).

These corrosion guides can be used for the preliminary screening of materials that are likely to be suitable, but the fact that published data indicate that a material is satisfactory cannot be taken as a guarantee that it will be suitable for the process environment being considered. Slight changes in the process conditions, or the presence of unsuspected trace impurities, can markedly change the rate of attack or the nature of the corrosion. The guides will, however, clearly show those materials that are manifestly unsuitable. Judgment, based on experience with the materials in similar process environments, must be used when assessing published corrosion data.

Pilot plant tests and laboratory corrosion tests under simulated plant conditions will help in the selection of suitable materials if actual plant experience is not available. Preliminary tests can be carried out by inserting coupons of different materials into an apparatus that is known to resist corrosion and not cause galvanic effects before testing plant components. This reduces the likelihood of component failure and possible release of chemicals during testing. Care is needed in the interpretation of laboratory tests.

The advice of the technical service department of the company supplying the materials should also be sought.

6.6 Material costs

An indication of the cost of some commonly used metals is given in Table 6.5. The actual cost of metals and alloys will fluctuate quite widely, depending on movements in the world metal exchanges.

Current metals prices can be found at:

www.steelonthenet.com Free site with monthly carbon steel prices

www.steelbb.com Steel business briefing – subscription site with weekly carbon steel and stainless steel prices

www.steelweek.com Subscription site with weekly international prices

<http://metalprices.com> Great alloy calculator – 3-month-old prices are free; current prices are subscription only

The quantity of a material used will depend on the material density and strength (maximum allowable stress), and these must be taken into account when comparing material costs. Moore (1970) compares costs by calculating a cost rating factor defined by the equation:

$$\text{Cost rating} = \frac{C \times \rho}{\sigma_d} \quad (6.2)$$

where C = cost per unit mass, \$/kg

ρ = density, kg/m³

σ_d = maximum allowable stress, N/mm²

TABLE 6.5 Relative cost of metals (November 2010)

Metal	Type or grade	Price (\$/lb)	Max allowable stress (ksi = 1000 psi)	Relative cost rating
Carbon steel	A285	0.37	12.9	1
Austenitic stainless steel	304	1.156	20	2.0
	316	1.721	20	3.0
Copper	C10400	3.83	6.7	22.8
Aluminum alloy	AO3560	1.0789	8.6	1.5
Nickel	99% Ni	9.861	10	39.2
Incoloy	N08800	3.733	20	6.7
Monel	N04400	7.76	18.7	16.4
Titanium	R50250	3.35	10	6.8

Note: The maximum allowable stress values are at 40 °C (100 °F) and are taken from ASME BPV Code Sec. II Part D. The code should be consulted for values at other temperatures. Several other grades exist for most of the materials listed.

Cost ratings relative to the rating for mild steel (low carbon) are shown in Table 6.5 for November 2010 prices. Materials with a relatively high maximum allowable stress, such as stainless and low-alloy steels, can be used more efficiently than carbon steel. Note that the simplified formula given in Equation 6.2 does not take into account different corrosion allowances for the different materials.

The relative cost of equipment made from different materials will depend on the cost of fabrication, as well as the basic cost of the material. Unless a particular material requires special fabrication techniques, the relative cost of the finished equipment will be lower than the relative bare material cost. For example, the purchased cost of a stainless steel storage tank will typically be 2 to 3 times the cost of the same tank in carbon steel, whereas the relative cost of the metals is typically from 3 to 8.

If the corrosion rate is uniform, the optimum material can be selected by calculating the annual costs for the possible candidate materials. The annual cost will depend on the predicted life, calculated from the corrosion rate, and the purchased cost of the equipment. In a given situation, it may prove more economic to install a cheaper material with a high corrosion rate and replace it frequently rather than select a more resistant but more expensive material. This strategy would only be considered for relatively simple equipment with low fabrication costs and where premature failure would not cause a serious hazard. For example, carbon steel could be specified for an aqueous effluent line in place of stainless steel, accepting the probable need for replacement. The pipe wall thickness would be monitored *in situ* frequently, to determine when replacement was needed.

The more expensive, corrosion-resistant alloys are often used as a cladding on carbon steel. If a thick plate is needed for structural strength, as for pressure vessels, the use of clad materials can substantially reduce the cost. The design requirements for pressure vessels with cladding or applied internal linings are given in ASME BPV Code Sec. VIII Div. 1 Part UCL.

6.7 Contamination

With some processes, the prevention of the contamination of a process stream, or a product, by certain metals or the products of corrosion overrides any other considerations when selecting suitable materials. For instance, in textile processes, stainless steel or aluminum is often used in preference to carbon steel, which would be quite suitable except that any slight rusting will mark the textiles (iron staining).

With processes that use catalysts, care must be taken to select materials that will not cause contamination and poisoning of the catalyst.

Stainless steels are preferred in biological processes to reduce contamination by corrosion products. Copper alloys were used historically, but have been shown to inhibit fermentation. Most fermentation plants are built from 316 stainless steel, but 304 stainless steel is widely used in food processing applications.

Some other examples that illustrate the need to consider the effect of contamination by trace quantities of other materials are:

1. For equipment handling acetylene, the pure metals, or alloys containing copper, silver, mercury or gold, must be avoided to prevent the formation of explosive acetylides.
2. The presence of trace quantities of mercury in a process stream can cause the catastrophic failure of brass heat exchanger tubes as a result of the formation of a mercury–copper amalgam. Incidents have occurred where the contamination has come from unsuspected sources, such as the failure of mercury-in-steel thermometers.
3. In the Flixborough disaster, there was evidence that the stress corrosion cracking of a stainless steel pipe had been caused by zinc contamination from galvanized wire supporting lagging.

6.7.1 Surface finish

In industries such as the food, pharmaceutical, biochemical, and textile industries, the surface finish of the material is as important as the choice of material to avoid contamination.

Stainless steel is widely used, and the surfaces, inside and out, are given a high finish by abrasive blasting and mechanical polishing. This is done for the purposes of hygiene, to prevent material adhering to the surface, and to aid cleaning and sterilization. The surface finishes required in food processing are discussed by Timperley (1984) and Jowitt (1980).

A good surface finish is important in textile fiber processing to prevent the fibers from snagging.

6.8 Commonly used materials of construction

The general mechanical properties, corrosion resistance, and typical areas of use of some of the materials commonly used in the construction of chemical plant are given in this section. The values given are for a typical, representative grade of the material or alloy. The alloys used in chemical plant construction are known by a variety of trade names, and code numbers are designated in the various national standards. With the exception of the stainless steels, no attempt has been made in this book to classify the alloys discussed by using one or other of the national standards; the commonly used, generic names for the alloys have been used. For the full details of the properties and compositions of the different grades available in a particular class of alloy, and the designated code numbers, reference should be made to the appropriate national code, to the various handbooks, or to manufacturers' literature. See, for example, ASME BPV Code Sec. II Part D for a full listing of materials properties and ASME BPV Code Sec. VIII Div.1 for material-specific fabrication guidelines.

The U.S. trade names and codes are given by [Green and Southard \(2018\)](#). Comprehensive reviews of the engineering materials used for chemical and process plants can be found in the books by [Evans \(1980\)](#) and [Hansen and Puyear \(1996\)](#).

6.8.1 Iron and steel

Low-carbon steel (mild steel) is the most commonly used engineering material. It is cheap, is available in a wide range of standard forms and sizes, and can be easily worked and welded. It has good tensile strength and ductility.

The carbon steels and iron are not resistant to corrosion, except in certain specific environments, such as concentrated sulfuric acid and the caustic alkalis. They are suitable for use with most organic solvents, except chlorinated solvents, but traces of corrosion products may cause discoloration. Mild steel is susceptible to stress-corrosion cracking in certain environments.

The corrosion resistance of the low-alloy steels (less than 5% of alloying elements), where the alloying elements are added to improve the mechanical strength and not for corrosion resistance, is not significantly different from that of the plain carbon steels. Low-alloy steels are widely used in plants that process hydrocarbons at moderate temperatures, for example, in oil refining.

A comprehensive reference covering the properties and application of steels, including the stainless steels, is the book by [Llewellyn and Hudd \(1998\)](#). The use of carbon steel in the construction of chemical plants is discussed by [Clark \(1970\)](#).

The high-silicon irons (14% to 15% Si) have a high resistance to mineral acids, except hydrofluoric acid. They are particularly suitable for use with sulfuric acid at all concentrations and temperatures. They are, however, very brittle.

6.8.2 Stainless steel

The stainless steels are the most frequently used corrosion-resistant materials in the chemical industry.

To impart corrosion resistance, the chromium content must be above 12%, and the higher the chromium content, the more resistant the alloy is to corrosion in oxidizing conditions. Nickel is added to improve the corrosion resistance in nonoxidizing environments.

Types

A wide range of stainless steels is available, with compositions tailored to give the properties required for specific applications. They can be divided into three broad classes according to their microstructure:

1. Ferritic: 13% to 20% Cr, <0.1% C, with no nickel
2. Austenitic: 18% to 20% Cr, >7% Ni
3. Martensitic: 12% to 10% Cr, 0.2% to 0.4% C, up to 2% Ni

The uniform structure of austenite (fcc, with the carbides in solution) is the structure desired for corrosion resistance, and it is these grades that are widely used in the chemical industry. The compositions of the main grades of austenitic steels are shown in [Table 6.6](#). Their properties are discussed next.

TABLE 6.6 Commonly used grades of austenitic stainless steel

Specification no.	Composition %							
AISI No.	C max	Si Max	Mn Max	Cr Range	Ni Range	Mo Range	Ti	Nb
304	0.08	—	2.00	17.5 20.0	8.0 11.0	—	—	—
304L	0.03	1.00	2.00	17.5 20.0	8.0 12.0	—	—	—
321	0.12	1.00	2.00	17.0 20.0	9.0 12.0	—	4 × C	—
347	0.08	1.00	2.00	17.0 20.0	9.0 13.0	—	—	10 × C
316	0.08	1.00	2.00	16.0 18.0	10.0 14.0	2.0 3.0	—	—
316L	0.03	1.0	2.0	16.0 18.0	10.0 14.0	2.0 3.0	—	—
309	0.20	—	—	22.0 24.0	12.0 15.0	—	—	—
310	0.25	—	—	24.0 26.0	19.0 22.0	—	—	—

S and P 0.045% all grades.

AISI, American Iron and Steel Institute.

- Type 304 (the so-called *18/8 stainless steels*): The most generally used stainless steel. It contains the minimum Cr and Ni that give a stable austenitic structure. The carbon content is low enough for heat treatment not to be normally needed with thin sections to prevent weld decay (see Section 6.4.4).
- Type 304L: Low-carbon version of type 304 (<0.03% C) used for thicker welded sections, where carbide precipitation would occur with type 304.
- Type 321: A version of 304 stabilized with titanium to prevent carbide precipitation during welding. It has a slightly higher strength than 304L and is more suitable for high-temperature use.
- Type 347: Stabilized with niobium.
- Type 316: In this alloy, molybdenum is added to improve the corrosion resistance in reducing conditions, such as in dilute sulfuric acid, and, in particular, to solutions containing chlorides.
- Type 316L: A low-carbon version of type 316, which should be specified if welding or heat treatment is liable to cause carbide precipitation in type 316.
- Types 309/310: Alloys with a high chromium content to give greater resistance to oxidation at high temperatures. Alloys with greater than 25% Cr are susceptible to embrittlement due to sigma phase formation at temperatures above 500 °C. Sigma phase is an intermetallic compound, FeCr. The formation of the sigma phase in austenitic stainless steels is discussed by [Hills and Harries \(1960\)](#).

Mechanical properties

The austenitic stainless steels have greater strength than the plain carbon steels, particularly at elevated temperatures ([Table 6.7](#)).

As was mentioned in Section 6.3.7, the austenitic stainless steels, unlike the plain carbon steels, do not become brittle at low temperatures. It should be noted that the thermal conductivity of stainless steel is significantly lower than that of mild steel.

TABLE 6.7 Comparative strength of carbon steel and stainless steel

Temperature °F		100	300	500	700	900
Maximum allowable stress (1000 psi)	Carbon steel (A285 plate)	12.9	12.9	12.9	11.5	5.9
	Stainless steel (304L plate)	16.7	16.7	14.7	13.5	11.9

Maximum allowable stress values from ASME BPV Code Sec. II Part D.

Typical values at 100 °C are:	type 304 (18/8)	mild steel
	16 W/m°C	60 W/m°C

Austenitic stainless steels are nonmagnetic in the annealed condition.

General corrosion resistance

The higher the alloying content, the better the corrosion resistance over a wide range of conditions, strongly oxidizing to reducing, but with a higher cost. A ranking in order of increasing corrosion resistance, taking type 304 as 1, is given here:

304	304L	321	316	316L	310
1.0	1.1	1.1	1.25	1.3	1.6

Intergranular corrosion (weld decay) and stress corrosion cracking are problems associated with the use of stainless steels and must be considered when selecting types suitable for use in a particular environment. Stress corrosion cracking in stainless steels can be caused by a few parts per million (ppm) of chloride ions (see Section 6.4.5).

In general, stainless steels are used for corrosion resistance when oxidizing conditions exist. Special types, or other high-nickel alloys, should be specified if reducing conditions are likely to occur. The properties, corrosion resistance, and uses of the various grades of stainless steel are discussed fully by [Peckner and Bernstein \(1977\)](#). A comprehensive discussion of the corrosion resistance of stainless steels is given in [Sedriks \(1996\)](#). Stress corrosion cracking in stainless steels is discussed by [Turner \(1989\)](#).

High-alloy-content stainless steels

Super-austenitic, high-nickel stainless steels, containing between 29% and 30% nickel and 20% chromium, have a good resistance to acids and acid chlorides. They are more expensive than the lower-alloy content, 300 series, of austenitic stainless steels.

Duplex and super-duplex stainless steels contain high percentages of chromium. They are called *duplex* because their structure is a mixture of the austenitic and ferritic phases. They have better corrosion resistance than the austenitic stainless steels and are less susceptible to stress corrosion cracking. The chromium content of duplex stainless steels is around 20%, and around 25% in the super-duplex grades. The super-duplex steels were developed for use in aggressive offshore environments.

The duplex range of stainless steels can be readily cast, wrought, and machined. Problems can occur in welding as a result of the need to keep the correct balance of ferrite and austenite in the weld area, but this can be overcome using the correct welding materials and procedures.

The cost of the duplex grades is comparable with the 316 steels. Super-duplex is around 50% higher than the cost of duplex.

The selection and properties of duplex stainless steels are discussed by [Bendall and Guha \(1990\)](#), and [Warde \(1991\)](#).

6.8.3 Nickel

Nickel has good mechanical properties and is easily worked. The pure metal (>99%) is not generally used for chemical plants; instead, its alloys are preferred for most applications. The main use is for equipment handling

caustic alkalis at temperatures above that at which carbon steel could be used: above 70 °C. Nickel is not subject to corrosion cracking like stainless steel.

6.8.4 Monel

Monel, the classic nickel–copper alloy with the metals in the ratio 2:1, is probably, after the stainless steels, the most commonly used alloy for chemical plants. It is easily worked and has good mechanical properties up to 500 °C. It is more expensive than stainless steel but is not susceptible to stress-corrosion cracking in chloride solutions. Monel has good resistance to dilute mineral acids and can be used in reducing conditions, where the stainless steels would be unsuitable. It may be used for equipment handling alkalis, organic acids and salts, and sea water.

6.8.5 Inconel and Incoloy

Inconel (typically 76% Ni, 7% Fe, 15% Cr) is used primarily for acid resistance at high temperatures. It maintains its strength at elevated temperature and is resistant to furnace gases, if sulfur-free. It is not suitable for use in sulfidizing environments. Nickel alloys with higher chromium content such as Incoloy 800 (21% Cr) and RA-33 (25% Cr) have better oxidation resistance at higher temperatures.

6.8.6 The Hastelloys

The trade name Hastelloy covers a range of nickel, chromium, molybdenum, and iron alloys that were developed for corrosion resistance to strong mineral acids, particularly HCl. The corrosion resistance, and use, of the two main grades, Hastelloy B (65% Ni, 28% Mo, 6% Fe) and Hastelloy C (54% Ni, 17% Mo, 15% Cr, 5% Fe), are discussed in papers by [Weisert \(1952a,b\)](#).

6.8.7 Copper and copper alloys

Pure copper is not widely used for chemical equipment. It has been used traditionally in the food industry, particularly in brewing. Copper is a relatively soft, very easily worked metal, and is used extensively for small-bore pipes and tubes.

The main alloys of copper are the brasses, alloyed with zinc, and the bronzes, alloyed with tin. Other so-called bronzes are the aluminum bronzes and the silicon bronzes.

Copper is attacked by mineral acids except cold, dilute, unaerated sulfuric acid. It is resistant to caustic alkalis, except ammonia, and to many organic acids and salts. The brasses and bronzes have a similar corrosion resistance to the pure metal. Their main use in the chemical industry is for valves and other small fittings and for heat exchanger tubes and tube sheets. If brass is used, a grade must be selected that is resistant to dezincification.

The cupro–nickel alloys (70% Cu) have good resistance to corrosion-erosion and are used for heat exchanger tubes, particularly where sea water is used as a coolant.

6.8.8 Aluminum and its alloys

Pure aluminum lacks mechanical strength but has higher resistance to corrosion than its alloys. The main structural alloys used are the Duralumin (Dural) range of aluminum–copper alloys (typical composition 4% Cu, with 0.5% Mg), which have a tensile strength equivalent to that of mild steel. The pure metal can be used as a cladding on Dural plates to combine the corrosion resistance of the pure metal with the strength of the alloy. The corrosion resistance of aluminum is due to the formation of a thin oxide film (as with the stainless steels). It is therefore most suitable for use in strong oxidizing conditions. It is attacked by mineral acids and by alkalis but is suitable for concentrated nitric acid (greater than 80%). It is widely used in the textile and food industries, where the use of mild steel would cause contamination. It is also used for the storage and distribution of demineralized water.

6.8.9 Lead

Lead was one of the traditional materials of construction for chemical plants but has now, because of its price, been largely replaced by other materials, particularly plastics. It is a soft, ductile material and is mainly used in the form of sheets (as linings) or pipe. It has a good resistance to acids, particularly sulfuric.

6.8.10 Titanium

Titanium is now used quite widely in the chemical industry, mainly for its resistance to halide solutions, including sea water and wet chlorine. It is rapidly attacked by dry chlorine, but the presence of as low a concentration of moisture as 0.01% will prevent attack. Like the stainless steels, titanium depends for its resistance on the formation of an oxide film. Titanium is also used in other halide services, for example, in liquid phase oxidation processes, such as the manufacture of terephthalic acid, that use bromide as a catalyst or promoter.

Alloying with palladium (0.15%) significantly improves the corrosion resistance, particularly to HCl. Titanium is being increasingly used for heat exchangers, replacing cupro–nickel for use with sea water.

The use of titanium for corrosion resistance is discussed by [Deily \(1997\)](#).

6.8.11 Tantalum

The corrosion resistance of tantalum is similar to that of glass, and it has been called a *metallic glass*. It is expensive, about five times the cost of stainless steel, and is used for special applications, where glass or a glass lining would not be suitable. Tantalum plugs are used to repair glass-lined equipment.

The use of tantalum as a material of construction in the chemical industry is discussed by [Fensom and Clark \(1984\)](#) and [Rowe \(1994, 1999\)](#).

6.8.12 Zirconium

Zirconium and zirconium alloys are used in the nuclear industry because of their low neutron absorption cross-section and resistance to hot water at high pressures.

In the chemical industry zirconium is finding use where resistance to hot and boiling acids is required: nitric, sulfuric, and particularly hydrochloric. Its resistance is equivalent to that of tantalum, but zirconium is less expensive, similar in price to high-nickel steel. [Rowe \(1999\)](#) gives a brief review of the properties and use of zirconium for chemical plants.

6.8.13 Silver

Silver linings are used for vessels and equipment handling hydrofluoric acid. It is also used for special applications in the food and pharmaceutical industries, where it is vital to avoid contamination of the product.

6.8.14 Gold

Because of its high cost, gold is rarely used as a material of construction. It is highly resistant to attack by dilute nitric acid and hot concentrated sulfuric acid, but is dissolved by aqua regia (a mixture of concentrated nitric and sulfuric acids). It is attacked by chlorine and bromine, and forms an amalgam with mercury. It has been used as thin plating on condenser tubes and other surfaces.

6.8.15 Platinum

Platinum has a high resistance to oxidation at high temperatures. One of its main uses has been in the form of an alloy with copper in the manufacture of the spinnerets used in synthetic textile spinning processes.

6.9 Plastics as materials of construction for chemical plants

Plastics are being increasingly used as corrosion-resistant materials for chemical plant construction. They are also widely used in food processing and biochemical plants. They can be divided into two broad classes:

1. Thermoplastic materials, which soften with increasing temperature, for example, polyvinyl chloride (PVC) and polyethylene
2. Thermosetting materials, which have a rigid, cross-linked structure, for example, the polyester and epoxy resins

Details of the chemical composition and properties of the wide range of plastics used as engineering material can be found in the books by [Butt and Wright \(1980\)](#), [Evans \(1980\)](#), and [Harper \(2001\)](#).

The biggest use of plastics is for piping; sheets are also used for lining vessels and for fabricated ducting and fan casings. Moldings are used for small items, such as pump impellers, valve parts, and pipe fittings.

The mechanical strength and operating temperature of plastics are low compared with metals. The mechanical strength, and other properties, can be modified by the addition of fillers and plasticizers. When reinforced with glass or carbon fibers, thermosetting plastics can have a strength equivalent to mild steel at low temperatures and are used for pressure vessels and pressure piping. Guidelines for the design of fiber-reinforced plastic pressure vessels are given in the ASME BPV Code Sec. X Part RD. Unlike metals, plastics are flammable. Plastics can be considered to complement metals as corrosion-resistant materials of construction. They generally have good resistance to dilute acids and inorganic salts, but suffer degradation in organic solvents that would not attack metals. Unlike metals, plastics can absorb solvents, causing swelling and softening. The properties and typical areas of use of the main plastics used for chemical plants are reviewed briefly in the following sections. A comprehensive discussion of the use of plastics as corrosion-resistant materials is given in a book by [Fontana \(1986\)](#). Information on the selection of plastics for different applications is also given by [Harper \(2001\)](#). The mechanical properties and relative cost of plastics are given in [Table 6.8](#).

6.9.1 Polyvinyl chloride

PVC is probably the most commonly used thermoplastic material in the chemical industry. Of the available grades, rigid (unplasticized) PVC is the most widely used. It is resistant to most inorganic acids, except strong sulfuric and nitric acids, and inorganic salt solutions. It is unsuitable, because of swelling, for use with most organic solvents. The maximum operating temperature for PVC is low: 60 °C. The use of PVC as a material of construction in chemical engineering is discussed in a series of articles by [Mottram and Lever \(1957\)](#).

6.9.2 Polyolefins

Low-density polyethylene is a relatively cheap, tough, flexible plastic. It has a low softening point and is not suitable for use above about 60 °C. The higher-density polymer (950 kg/m³) is stiffer and can be used at higher temperatures. Polypropylene is a stronger material than the polyethylenes and can be used at temperatures up to 120 °C. The chemical resistance of the polyolefins is similar to that of PVC.

6.9.3 Polytetrafluoroethylene

Polytetrafluoroethylene (PTFE), known under the trade names Teflon and Fluon, is resistant to all chemicals except molten alkalis and fluorine, and can be used at temperatures up to 250 °C. It is a relatively weak material, but its mechanical strength can be improved by the addition of fillers (glass and carbon fibers). It is expensive and difficult to fabricate. PTFE is used extensively for gaskets, gland packings (for example, on valve stems), and demister pads. As a coating, it is used to confer nonstick properties to surfaces such as filter plates. It can also be used as a liner for vessels.

TABLE 6.8 Mechanical properties and relative cost of polymers

Material	Tensile strength (N/mm ²)	Elastic modulus (kN/mm ²)	Density (kg/m ³)	Relative cost
PVC	55	3.5	1400	1.5
Polyethylene (low density)	12	0.2	900	1.0
Polypropylene	35	1.5	900	1.5
PTFE	21	1.0	2100	30.0
GRP polyester	100	7.0	1500	3.0
GRP epoxy	250	14.0	1800	5.0

Approximate cost relative to polyethylene, volumetric basis.

6.9.4 Polyvinylidene fluoride

Polyvinylidene fluoride (PVDF) has properties similar to PTFE but is easier to fabricate. It has good resistance to inorganic acids and alkalis and organic solvents. It is limited to a maximum operating temperature of 140 °C.

6.9.5 Glass-fiber—reinforced plastics

The polyester resins, reinforced with glass fiber, are the most common thermosetting plastics used for chemical plants. Complex shapes can be easily formed using the techniques developed for working with reinforced plastics. Glass-reinforced plastics are relatively strong and have a good resistance to a wide range of chemicals. The mechanical strength depends on the resin used, the form of the reinforcement (chopped mat or cloth), and the ratio of resin to glass.

By using special techniques, in which the reinforcing glass fibers are wound on in the form of a continuous filament, high strength can be obtained, and this method is used to produce pressure vessels.

The polyester resins are resistant to dilute mineral acids, inorganic salts, and many solvents. They are less resistant to alkalis.

Glass-fiber—reinforced epoxy resins are also used for chemical plants but are more expensive than the polyester resins. In general they are resistant to the same range of chemicals as the polyesters, but are more resistant to alkalis.

The chemical resistance of glass-fiber—reinforced plastic (GRP) is dependent on the amount of glass reinforcement used. High ratios of glass to resin give higher mechanical strength but generally lower resistance to some chemicals. The design of chemical plant equipment in GRP is the subject of a book by [Malleson \(1969\)](#); see also [Shaddock \(1971\)](#), [Baines \(1984\)](#), and ASME BPV Code Sec. X.

6.9.6 Rubber

Rubber, particularly in the form of linings for tanks and pipes, has been extensively used in the chemical industry for many years. Natural rubber is most commonly used because of its good resistance to acids (except concentrated nitric acid) and alkalis. It is unsuitable for use with most organic solvents.

Synthetic rubbers are also used for particular applications. Hypalon (trademark, E. I. du Pont de Nemours) has a good resistance to strongly oxidizing chemicals and can be used with nitric acid. It is unsuitable for use with chlorinated solvents. Viton (trademark, E. I. du Pont de Nemours) has a better resistance to solvents, including chlorinated solvents, than other rubbers. Both Hypalon and Viton are expensive compared with other synthetic and natural rubbers.

The use of natural rubber lining is discussed by [Saxman \(1965\)](#) and the chemical resistance of synthetic rubbers by [Evans \(1963\)](#). Butt and Wright (1984) give an authoritative account of the application and uses of rubber and plastic linings and coatings.

6.10 Ceramic materials (silicate materials)

Ceramics are compounds of nonmetallic elements and include the following materials used for chemical plants:

- Glass, the borosilicate glasses (hard glass)
- Stoneware
- Acid-resistant bricks and tiles
- Refractory materials
- Cements and concrete
- Ceramic materials have a cross-linked structure and are therefore brittle.

6.10.1 Glass

Borosilicate glass (known by several trade names, including Pyrex) is used for chemical plants as it is stronger than the soda glass used for general purposes; it is more resistant to thermal shock and chemical attack. Glass equipment is often used in small-scale manufacture of specialty chemicals. Glass can be used up to moderately high temperatures (700 °C) but is not suitable for pressures above atmospheric unless used as a lining.

Glass equipment is available from several specialist manufacturers. Pipes and fittings are produced in a range of sizes, up to 0.5 m. Special equipment, such as heat exchangers, is available and, together with the larger sizes of pipe, can be used to construct distillation and absorption columns. Teflon gaskets are normally used for jointing glass equipment and pipes.

Where failure of the glass could cause injury, pipes and equipment should be protected by external shielding or wrapping with plastic tape. Glass apparatus should allow adequate venting to the atmosphere to handle anticipated relief scenarios without accumulating high pressure.

Glass linings, also known as *glass enamel*, have been used on steel and iron vessels for many years. Borosilicate glass is used, and the thickness of the lining is about 1 mm. The techniques used for glass lining, and the precautions to be taken in the design and fabrication of vessels to ensure a satisfactory lining, are discussed by [Landels and Stout \(1970\)](#) and by the ASME BPV Code Sec. VIII Div. 1, Mandatory Appendix 27. Borosilicate glass is resistant to acids, salts, and organic chemicals. It is attacked by the caustic alkalis and fluorine.

6.10.2 Stoneware

Chemical stoneware is similar to the domestic variety, but of higher quality, stronger, and with a better glaze. It is available in a variety of shapes for pipe runs and columns. As for glass, it is resistant to most chemicals except alkalis and fluorine. The composition and properties of chemical stoneware are discussed by [Holdridge \(1961\)](#). Stoneware and porcelain shapes are used for packing absorption and distillation columns (see [Chapter 17](#)).

6.10.3 Acid-resistant bricks and tiles

High-quality bricks and tiles are used for lining vessels and ditches and to cover floors. The linings are usually backed with a corrosion-resistant membrane of rubber or plastic, placed behind the tiles, and special acid-resistant cements are used for the joints. Brick and tile linings are covered in a book by [Falcke and Lorentz \(1985\)](#).

6.10.4 Refractory materials (refractories)

Refractory bricks and cements are needed for equipment operating at high temperatures, such as fired heaters, high-temperature reactors, and boilers. The refractory bricks in common use are composed of mixtures of silica (SiO_2) and alumina (Al_2O_3). The quality of the bricks is largely determined by the relative amounts of these materials and the firing temperature. Mixtures of silica and alumina form a eutectic composition (94.5% SiO_2 , 1545 °C), and for a high refractoriness under load (the ability to resist distortion at high temperature), the composition must be well removed from the eutectic composition. The highest-quality refractory bricks, for use in load-bearing structures at high temperatures, contain high proportions of silica or alumina. "Silica bricks," containing greater than 98% SiO_2 , are used for general furnace construction. High-alumina bricks, 60% Al_2O_3 , are used for special furnaces where resistance to attack by alkalis is important, such as lime and cement kilns. Fire bricks, typical composition 50% SiO_2 , 40% Al_2O_3 , and the balance CaO and Fe_2O_3 , are used for general furnace construction. Silica can exist in a variety of allotropic forms, and bricks containing a high proportion of silica undergo reversible expansion when heated up to working temperature. The higher the silica content, the greater the expansion, and this must be allowed for in furnace design and operation.

Ordinary fire bricks, fire bricks with a high porosity, and special bricks composed of diatomaceous earths are used for insulating walls.

Full details of the refractory materials used for process and metallurgical furnaces can be found in the books by [Norton \(1968\)](#) and [Lyle \(1947\)](#). Additional information on refractories can be found in the books by [Schacht \(1995, 2004\)](#) and [Routschka and Wuthnow \(2008\)](#).

6.11 Carbon

Impervious carbon, impregnated with chemically resistant resins, is used for specialized equipment, particularly heat exchangers. It has high conductivity and good resistance to most chemicals, except oxidizing acids of concentrations greater than 30%. Carbon tubes can be used in conventional shell and tube exchanger arrangements, or

proprietary designs can be used, in which the fluid channels are formed in blocks of carbon; see Hilland (1960) and Denyer (1991).

6.12 Protective coatings

A wide range of paints and other organic coatings is used for the protection of mild steel structures. Paints are used mainly for protection from atmospheric corrosion. Special chemically resistant paints have been developed for use on chemical process equipment. Chlorinated rubber paints and epoxy-based paints are used. In the application of paints and other coatings, good surface preparation is essential to ensure good adhesion of the paint film or coating.

Brief reviews of the paints used to protect chemical plants are given by Ruff (1984) and Hullcoop (1984).

6.13 Design for corrosion resistance

The life of equipment subjected to corrosive environments can be increased by proper attention to design details. Equipment should be designed to drain freely and completely. The internal surfaces should be smooth and free from crevices where corrosion products and other solids can accumulate. Butt joints should be used in preference to lap joints. The use of dissimilar metals in contact should be avoided, or care taken to ensure that they are effectively insulated to avoid galvanic corrosion. Fluid velocities and turbulence should be high enough to avoid the deposition of solids but not so high as to cause erosion-corrosion. The design and operating procedures should make allowance for changes in the environment to which the materials are exposed. For example, heating and cooling rates should be slow enough to prevent thermal shocks, and care should be taken during maintenance not to damage corrosion-resistant films that have developed during operation.

6.14 References

- Baines, D. (1984). Glass reinforced plastics in the process industries. *Chemical Engineering*, 161(July), 24. London.
- Bendall, K., & Guha, P. (1990). Balancing the cost of corrosion resistance. *Process Industry Journal*, (Mar), 31.
- Boyd, G. M. (1970). *Brittle fracture of steel structures*. Butterworths.
- Butt, L. T., & Wright, D. C. (1980). *Use of polymers in chemical plant construction*. Applied Science.
- Clark, E. E. (1970). Carbon steels for the construction of chemical and allied plant. *Chemical Engineering*, 242(Oct), 312. London.
- Craig, B. D., & Anderson, D. B. (1995). *Handbook of corrosion data*. ASM International.
- Day, M. F. (1979). *Materials for high temperature use, engineering design guide no. 28*. Oxford University Press.
- Deily, J. E. (1997). Use titanium to stand up to corrosives. *Chemical Engineering Progress*, 93(June), 50.
- Denyer, M. (1991). Graphite as a material for heat exchangers. *Processing*, (July), 23.
- Dillon, C. P. (1994). *Corrosion control in the chemical process industries* (2nd ed.). McGraw-Hill.
- Evans, L. S. (1963). The chemical resistance of rubber and plastics. *Rubber and Plastics Age*, 44, 1349.
- Evans, L. S. (1980). *Chemical and process plant: A guide to the selection of engineering materials* (2nd ed.) Hutchinson.
- Falcke, F. K., & Lorentz, G. (Eds.). (1985). *Handbook of acid proof construction*. VCH.
- Fensom, D. H., & Clark, B. (1984). Tantalum: Its uses in the chemical industry. *Chemical Engineering*, 162(Aug), 46. London.
- Fontana, M. G. (1986). *Corrosion engineering* (3rd ed.). McGraw-Hill.
- Gordon, J. E. (2018). *The new science of strong materials* (2nd ed.). Princeton University Press.
- Green, D. W., & Southard, M. Z. (Eds.). (2018). *Perry's chemical engineers' handbook* (9th ed.). McGraw-Hill.
- Hamner, N. E. (1974). *Corrosion data survey* (5th ed.). National Association of Corrosion Engineers.
- Hansen, D. A., & Puyear, K. B. (1996). *Materials selection for hydrocarbon and chemical plants*. Marcel Dekker.
- Harper, C. A. (2001). *Handbook of materials for product design*. McGraw-Hill.
- Harris, W. J. (1976). *The significance of fatigue*. Oxford University Press.
- Hilland, A. (1960). Graphite for heat exchangers. *Chemical and Process Engineering*, 41, 416.
- Hills, R. F., & Harries, D. P. (1960). Sigma phase in austenitic stainless steel. *Chemical and Process Engineering*, 41, 391.
- Holdridge, D. A. (1961). *Ceramics*. *Chemical and Process Engineering*, 42, 405.
- Hullcoop, R. (1984). The great cover up. *Processing*, (April), 13.
- Institute of Metallurgists. (1960). *Toughness and brittleness of metals*. Iliffe.
- Jowitt, R. (Ed.). (1980). *Hygienic design and operation of food plant*. Ellis Horwood.
- Kreysa, G., & Schütze, M. (2007). *Corrosion handbook*. Wiley-VCH.
- Lai, G. Y. (1990). *High temperature corrosion of engineering alloys*. ASM International.
- Landels, H. H., & Stout, E. (1970). Glassed steel equipment: A guide to current technology. *British Chemical Engineering*, 15, 1289.
- Llewellyn, D. T., & Hudd, R. C. (1998). *Steels: Metallurgy and applications* (3rd ed.). Butterworth-Heinemann.
- Lyle, O. (1947). *Efficient use of steam*. HMSO.

- Malleson, J. H. (1969). *Chemical plant design with reinforced plastics*. McGraw-Hill.
- Moore, D. C. (1970). Copper. *Chemical Engineering*, 242(Oct), 326. London.
- Moore, R. E. (1979). Selecting materials to resist corrosive conditions. *Chemical Engineering*, 86(July 30), 91. New York.
- Mottram, S., and Lever, D. A. (1957). Unplasticized P.V.C. as a constructional material in chemical engineering. *The Industrial Chemist*, 33, 62, 123, 177 (in three parts).
- Norton, F. H. (1968). *Refractories* (4th ed.). McGraw-Hill.
- Peckner, D., & Bernstein, I. M. (1977). *Handbook of stainless steels*. McGraw-Hill.
- Rabald, E. (1968). *Corrosion guide* (2nd ed.). Elsevier.
- Revie, R. W. (2011). *Uhlig's corrosion handbook* (3rd ed.). Wiley.
- Routschka, G., & Wuthnow, H. (2008). *Pocket manual of refractory materials*. Vulkan-Verlag.
- Rowe, D. (1994). Tempted by tantalum. *Process Industry Journal*, (March), 37.
- Rowe, D. (1999). Tantalising materials. *Chemical Engineering*, 683(June 24), 19. London.
- Ruff, C. (1984). Paint for plants. *Chemical Engineering*, 409(Dec), 27. London.
- Saxman, T. E. (1965). Natural rubber tank linings. *Materials Protection*, 4(Oct), 43.
- Schacht, C. A. (1995). *Refractory linings: Thermochemical design and applications*. Marcel Dekker.
- Schacht, C. A. (2004). *Refractories handbook*. Marcel Dekker.
- Schweitzer, P. A. (Ed.). (1989). *Corrosion and corrosion protection handbook* (2nd ed.). Marcel Dekker.
- Schweitzer, P. A. (2004). *Corrosion resistance tables* (5th ed.). CRC Press.
- Schweitzer, P. A. (2006). *Corrosion engineering handbook*. CRC Press.
- Sedriks, A. J. (1996). *Corrosion of stainless steels* (2nd ed.). Wiley.
- Shaddock, A. K. (1971). Designing for reinforced plastics. *Chemical Engineering*, 78(Aug. 9), 116. New York.
- Timperley, D. A. (1984). Surface finish and spray cleaning of stainless steel. *Institute of Chemical Engineers Symposium Series*, 84, 31.
- Turner, M. (1989). What every chemical engineer should know about stress corrosion cracking. *Chemical Engineering*, 460(May), 52. London.
- Warde, E. (1991). Which super-duplex? *Chemical Engineering*, 502(Aug. 15), 35. London.
- Weisert, E. D. (1952a). Hastelloy alloy C. *Chemical Engineering*, 59(June), 267. New York.
- Weisert, E. D. (1952b). Hastelloy alloy B. *Chemical Engineering*, 59(July), 314. New York.
- Wells, A. A. (1968). Fracture control of thick steels for pressure vessels. *British Welding Journal*, 15, 221.
- Wigley, D. A. (1978). *Materials for low temperatures, engineering design guide no. 28*. Oxford University Press.

American standards

- ASME Boiler and Pressure Vessel Code Section II part D. Materials Properties.(2019).
- ASME Boiler and Pressure Vessel Code Section VIII Division 1. Rules for the Construction of Pressure Vessels.(2019).
- ASME Boiler and Pressure Vessel Code Section X. Fiber-reinforced Plastic Vessels.(2019).
- NACE TM-01-69. (2012). *Standard guide for laboratory immersion corrosion testing of metals*. National Association of Corrosion Engineers.

Bibliography

- Further reading on materials, materials selection, and equipment fabrication.
- Callister, W. D., & Rethwisch, D. G. (2013). *Materials science and engineering, an introduction* (9th ed.). Wiley.
- Champion, F. A. (1967). *Corrosion testing procedures* (3rd ed.). Chapman Hall.
- Crane, F. A. A., & Charles, J. A. (1997). *Selection and use of engineering materials* (3rd ed.). Butterworths.
- Ewalds, H. L. (1984). *Fracture mechanics*. Arnold.
- Flinn, R. A., & Trojan, P. K. (1990). *Engineering materials and their applications* (4th ed.). Houghton Mifflin.
- Ray, M. S. (1987). *The technology and application of engineering materials*. Prentice Hall.
- Rolfe, S. T. (1987). *Fracture mechanics and fatigue control in structures* (2nd ed.). Prentice Hall.

6.15 Nomenclature

		Dimensions in MLT\$
A	Area	L^2
C	Cost of material	$\$/M$
t	Time	T
w	Mass loss	M
P	Density	ML^{-3}
σ_d	Maximum allowable stress	$ML^{-1} T^{-2}$

6.16 Problems

- 6.1. A pipeline constructed of carbon steel failed after 3 years of operation. On examination it was found that the wall thickness had been reduced by corrosion to about half the original value. The pipeline was constructed of nominal 100 mm (4 in) schedule 40, pipe, inside diameter 102.3 mm (4.026 in), and outside diameter 114.3 mm (4.5 in). Estimate the rate of corrosion in ipy and mm per year.
- 6.2. The pipeline described in Question 6.1 was used to carry wastewater to a hold-up tank. The effluent is not hazardous. A decision has to be made on what material to use to replace the pipe. Three suggestions have been made:
1. Replace with the same schedule carbon steel pipe and accept renewal at 3-year intervals.
 2. Replace with a thicker pipe, schedule 80, outside diameter 114.3 mm (4.5 in) and inside diameter 97.2 mm (3.826 in).
 3. Use stainless steel pipe, which will not corrode.

The estimated cost of the pipes per unit length is schedule 40 carbon steel \$5, schedule 80 carbon steel \$8.3, and stainless steel (304) schedule 40 \$24.8.

- Installation and fittings for all the materials add \$16.5 per unit length.
- The downtime required to replace the pipe does not result in a loss of production.
- If the expected future life of the plant is 7 years, recommend which pipe to use.

- 6.3. Choose a suitable material of construction for the following duties:

1. 98% w/w sulfuric acid at 70 °C.
2. 5% w/w sulfuric acid at 30 °C.
3. 30% w/w hydrochloric acid at 50 °C.
4. 5% aqueous sodium hydroxide solution at 30 °C.
5. Concentrated aqueous sodium hydroxide solution at 50 °C.
6. 5% w/w nitric acid at 30 °C.
7. Boiling concentrated nitric acid.
8. 10% w/w sodium chloride solution.
9. A 5% w/w solution of cuprous chloride in hydrochloric acid.
10. 10% w/w hydrofluoric acid.

In each case, select the material for a 50-mm pipe operating at approximately 2 bar pressure.

- 6.4. Suggest suitable materials of construction for the following applications:

1. A 10,000 m³ storage tank for toluene.
2. A 5.0 m³ tank for storing a 30% w/w aqueous solution of sodium chloride.
3. A 2 m diameter, 20 m high distillation column, distilling acrylonitrile.
4. A 100 m³ storage tank for strong nitric acid.
5. A 500 m³ aqueous waste hold-up tank. The wastewater pH can vary from 1 to 12. The wastewater will also contain traces of organic material.
6. A packed absorption column 0.5 m diameter, 3 m high, absorbing gaseous hydrochloric acid into water. The column will operate at essentially atmospheric pressure.

- 6.5. Aniline is manufactured by the hydrogenation of nitrobenzene in a fluidized bed reactor. The reactor operates at 250 °C and 20 bar. The reactor vessel is approximately 3 m diameter and 9 m high. Suggest suitable materials of construction for this reactor.
- 6.6. Methyl ethyl ketone (MEK) is manufactured by the dehydrogenation of 2-butanol using a shell and tube type reactor. Flue gases are used for heating and pass through the tubes. The flue gases will contain traces of sulfur dioxide. The reaction products include hydrogen. The reaction takes place in the shell at a pressure of 3 bar and temperature of 500 °C. Select suitable materials for the tubes and shell.

- 6.7. In the manufacture of aniline by the hydrogenation of nitrobenzene, the off-gases from the reactor are cooled and the products and unreacted nitrobenzene condensed in a shell and tube exchanger. A typical composition of the condensate is kmol/h: aniline 950, cyclo-hexylamine 10, water 1920, nitrobenzene 40. The gases enter the condenser at 230 °C and leave at 50 °C. The cooling water enters the tubes at 20 °C and leaves at 50 °C. Suggest suitable materials of construction for the shell and the tubes.
- 6.8. A slurry of acrylic polymer particles in water is held in storage tanks before filtering and drying. Plain carbon steel would be a suitable material for the tanks, but it is essential that the polymer does not become contaminated with iron in storage. Suggest some alternative materials of construction for the tanks.
- 6.9. Coal gasification is carried out at 850 °C and 40 atmospheres pressure by reaction of coal with steam and oxygen. The empirical formula of the coal is roughly $\text{CH}_{0.8}\text{S}_{0.013}$. What materials of construction would you recommend for:
- The coal addition system
 - The oxygen injection system
 - The gasification reactor
 - The product gas transfer line

Capital cost estimating

KEY LEARNING OBJECTIVES

- Techniques for estimating the capital cost of a process
- How to estimate the costs of major process equipment
- How to use commercial cost estimating software
- How to update cost data
- How to estimate cost data for international projects
- How to estimate costs for proprietary equipment when data are scarce

7.1 Introduction

Most chemical engineering design projects are carried out to provide information from which estimates of capital and operating costs can be made. Chemical plants are built to make a profit, and an estimate of the investment required is needed before the profitability of a project can be assessed. Cost estimation is a specialized subject and a profession in its own right, but the design engineer must be able to make rough cost estimates to decide between project alternatives and optimize the design.

This chapter introduces the components of capital costs and the techniques used for capital cost estimating. Simple costing methods and some cost data are given, which can be used to make preliminary estimates of capital costs in the early stages of design. Sources of cost data and methods for updating cost estimates are described. The more sophisticated software that is used in industry for preliminary estimating is discussed in Section 7.10.

For a more detailed treatment of the subject, the reader should refer to the numerous specialized texts that have been published on cost estimation. The following books are particularly recommended: [Happle and Jordan \(1975\)](#), [Guthrie \(1974\)](#), [Page \(1996\)](#), [Garrett \(1989\)](#), [Humphreys \(1991, 2005\)](#).

7.2 Components of capital cost

This section introduces the components of project costs.

7.2.1 Fixed capital investment

The fixed capital investment is the total cost of designing, constructing, and installing a plant and the associated modifications needed to prepare the plant site. The fixed capital investment is made up of:

1. The inside battery limits (ISBL) investment – the cost of the plant itself
2. The modifications and improvements that must be made to the site infrastructure, known as *offsite battery limits* (OSBL) investment
3. Engineering and construction costs
4. Contingency charges

ISBL plant costs

The ISBL plant cost includes the cost of procuring and installing all the process equipment that makes up the new plant.

The direct field costs include:

1. All the major process equipment, such as vessels, reactors, columns, furnaces, heat exchangers, coolers, pumps, compressors, motors, fans, turbines, filters, centrifuges, driers, etc., including field fabrication and testing if necessary.
2. Bulk items, such as piping, valves, wiring, instruments, structures, insulation, paint, lube oils, solvents, catalysts, etc.
3. Civil works such as roads, foundations, piling, buildings, sewers, ditches, bunds, etc.
4. Installation labor and supervision.

In addition to the direct field costs there will be indirect field costs, including:

1. Construction costs such as construction equipment rental, temporary construction (rigging, trailers, etc.), temporary water and power, construction workshops, etc.
2. Field expenses and services such as field canteens, specialists' costs, overtime pay, and adverse weather costs.
3. Construction insurance.
4. Labor benefits and burdens (social security, workers compensation, etc.).
5. Miscellaneous overhead items such as agent's fees, legal costs, import duties, special freight costs, local taxes, patent fees or royalties, corporate overheads, etc.

In the early stages of a project it is important to define the ISBL scope carefully, as other project costs are often estimated from ISBL cost. The overall project economics can be badly miscalculated if the ISBL scope is poorly defined. Several methods for estimating ISBL costs are given in the following sections of this chapter.

Offsite costs

Offsite cost or OSBL investment includes the costs of the additions that must be made to the site infrastructure to accommodate adding a new plant or increasing the capacity of an existing plant. Offsite investments may include:

- Electric main substations, transformers, switchgear, and power lines
- Power generation plants, turbine engines, and standby generators
- Boilers, steam mains, condensate lines, boiler feed water treatment plant, and supply pumps
- Cooling towers, circulation pumps, cooling water mains, and cooling water treatment
- Water pipes, water demineralization, waste water treatment plant, site drainage, and sewers
- Air separation plants to provide site nitrogen for inert gas, as well as nitrogen lines
- Driers and blowers for instrument air, as well as instrument air lines
- Pipe bridges, feed, and product pipelines
- Tanker farms, loading facilities, silos, conveyors, docks, warehouses, railroads, and lift trucks
- Laboratories and analytical equipment
- Offices, canteens, changing rooms, and central control rooms
- Workshops and maintenance facilities
- Emergency services, firefighting equipment, fire hydrants, medical facilities, etc.
- Site security, fencing, gatehouses, and landscaping

Offsite investments often involve interactions with utility companies such as electricity or water suppliers. They may be subject to equal or greater scrutiny than ISBL investments because of their impact on the local community through water consumption and discharge, traffic, and other similar factors.

Offsite costs are typically estimated as a proportion of ISBL costs in the early stages of design. Offsite costs are usually in the range from 10% to 100% of ISBL costs, depending on the project scope and its impact on site infrastructure. For typical petrochemical projects, offsite costs are usually between 20% and 50% of ISBL cost, and 40% is usually used as an initial estimate if no details of the site are known. For an established site with well-developed

infrastructure, offsite costs will generally be lower. This is particularly true of sites that have undergone contraction, where some plants have closed, leaving underutilized infrastructure (“brownfield” sites). On the other hand, if the site infrastructure is in need of repair or upgrading to meet new regulations, or if the plant is built on a completely new site (a “greenfield” site), then offsite costs will be higher. Offsite costs are discussed in more detail in Section 7.9.

Once a site has been chosen for the project, the modifications to the site infrastructure that are needed can be designed in detail in the same manner as the ISBL investments. Infrastructure upgrades are usually the first part of a project to be implemented, as they usually need to be commissioned before the plant can begin operation.

Engineering costs

The engineering costs, sometimes referred to as *home office costs* or *contractor charges*, include the costs of detailed design and other engineering services required to carry out the project:

1. Detailed design engineering of process equipment, piping systems, control systems and offsites, plant layout, drafting, cost engineering, scale models, and civil engineering.
2. Procurement of main plant items and bulks.
3. Construction supervision and services.
4. Administrative charges, including engineering supervision, project management, expediting, inspection, travel and living expenses, and home office overheads.
5. Bonding.
6. Contractor’s profit.

Very few operating companies retain a large enough engineering staff to carry out all of these activities internally, except for very small projects. In most cases, one or more of the major engineering contracting firms will be brought in.

Engineering costs are best estimated individually based on project scope, as they are not directly proportional to project size. A rule of thumb for engineering costs is 30% of ISBL plus OSBL cost for smaller projects and 10% of ISBL plus OSBL cost for larger projects. The actual charges paid for real industrial projects vary considerably from customer to customer and are strongly influenced by long-term client–contractor relationships and overall market demand for engineering services. Customers usually have to pay premiums or surcharges if they want to complete a project on an accelerated timeline or if they make a lot of changes once a project is underway.

Contingency charges

Contingency charges are extra costs added into the project budget to allow for variation from the cost estimate. All cost estimates are uncertain (see Section 7.3.1), and the final installed cost of many items is not known until installation has been successfully completed. Apart from errors in the cost estimate, contingency costs also help cover:

- Minor changes in project scope
- Changes in prices (e.g., prices of steel, copper, catalyst, etc.)
- Currency fluctuations
- Labor disputes
- Subcontractor problems
- Other unexpected problems

A contingency charge can be thought of as an additional fee charged by the Engineering, Procurement and Construction (EPC) company to address the risk that the project will run over budget. Contingency charges therefore reduce the likelihood that the contractor will lose money on a fixed-price bid. A minimum contingency charge of 10% of ISBL plus OSBL cost should be used on all projects. If the technology is uncertain, higher contingency charges (up to 50%) are used. Contingency charges are discussed in more detail in Section 9.8.4.

7.2.2 Working capital

In addition to the fixed capital investment that was used to design and construct the plant, the owner needs to invest some capital in maintaining plant operations. The capital that is tied up in maintaining inventories of feeds, products, and spare parts, together with cash on hand and the difference between money owed by customers (accounts receivable) and money owed to suppliers (accounts payable), is termed the *working capital* of the plant. Working capital is required as long as the plant is in operation, but is recovered if the plant is shut down. Working capital is discussed in more detail in the chapter on project finance and economics; see Section 9.2.3.

7.3 Accuracy and purpose of capital cost estimates

The accuracy of an estimate depends on the amount of design detail available, the accuracy of the cost data available, and the time spent on preparing the estimate. In the early stages of a project only an approximate estimate will be justified based on the amount of information available.

7.3.1 AACE International cost estimate classes

The Association for the Advancement of Cost Estimating International (AACE International) is the professional association representing the cost engineering profession in the United States. AACE International classifies capital cost estimates into five types according to their accuracy and purpose:

1. Order of magnitude estimates ("ballpark estimate," "guesstimate" "Class 5 estimate"), accuracy typically $\pm 30\%$ to 50% , usually based on the costs of similar processes and requiring essentially no design information. These are used in initial feasibility studies and for screening purposes.
2. Preliminary ("approximate," "study," "feasibility," "Class 4") estimates, accuracy typically $\pm 30\%$, which are used to make coarse choices between design alternatives. They are based on limited cost data and design detail.
3. Definitive ("authorization," "budgeting," "control," "Class 3") estimates, accuracy typically $\pm 10\%$ to 15% . These are used for the authorization of funds to proceed with the design to the point where an accurate and more detailed estimate can be made. Authorization may also include funds to cover cancellation charges on any long delivery equipment ordered at this stage of the design to avoid delay in the project. In a contracting organization this type of estimate could be used with a large contingency factor to obtain a price for tendering a bid. Normally, however, an accuracy of about $\pm 5\%$ would be needed and a more detailed estimate would be made if time permitted. With experience, and where a company has cost data available from similar projects, estimates of acceptable accuracy can be made at the flowsheet stage of the project. A rough P&I diagram and the approximate sizes of the major items of equipment would also be needed.
4. Detailed estimates ("quotation," "tender," "firm estimate," "contractor's estimate," "Class 2 estimate"), accuracy $\pm 5\%$ to 10% , which are used for project cost control and estimates for fixed-price contracts. Estimates of this quality can be made once the front-end engineering design (FEED) is completed, including a complete (or near-complete) process design, firm quotes for equipment, and a detailed breakdown and estimation of the construction cost. By this stage, the contractor can usually present a list of all the items that must be purchased and can make a firm commitment to the client.
5. Check estimates ("tender," "as-bid," "Class 1 estimate"), accuracy $\pm 5\%$ to 10% . This is based on a completed design and concluded negotiations on procurement of specialized items and long lead-time items.

It is important for design engineers to understand how much information is necessary to arrive at a given level of confidence in an estimate, particularly when working with professional cost engineers. It is not possible to generate a Class 1 or Class 2 estimate without completing the design to a sufficient level of detail. A Class 3 estimate may also require a substantially complete design package unless the company has recently built projects with similar scope. Class 4 and 5 estimates are relatively easy to generate but have high uncertainty, which must be taken into consideration in economic analysis and sensitivity analysis; see Section 9.8.

7.3.2 Development of cost estimates

As a project proceeds from initial concept through detailed design to start-up, costs begin to be accumulated, particularly once procurement and construction get underway (Fig. 7.1a). At the same time, the ability of the design engineer to influence project cost decreases and is minimal by the time construction begins (Fig. 7.1b). There is therefore a strong incentive to try to estimate project costs at as early a stage as possible, even if the design information is incomplete, so that the project can be optimized, evaluated, and abandoned if it is not attractive.

It is difficult to go beyond the accuracy of a Class 4 estimate without making a fairly detailed design of the plant. The cost of preparing the estimate thus becomes the cost of designing the process and sizing and optimizing the main equipment items. To obtain even greater accuracy, the contractor needs to establish the plot plan and plant layout so as to make more accurate estimates of the amount of piping, wiring, and structural steel that will be needed. The cost of preparing an estimate thus increases from about 0.1% of the total project cost for $\pm 30\%$ accuracy to about 3% for a detailed estimate with an accuracy of $\pm 5\%$.

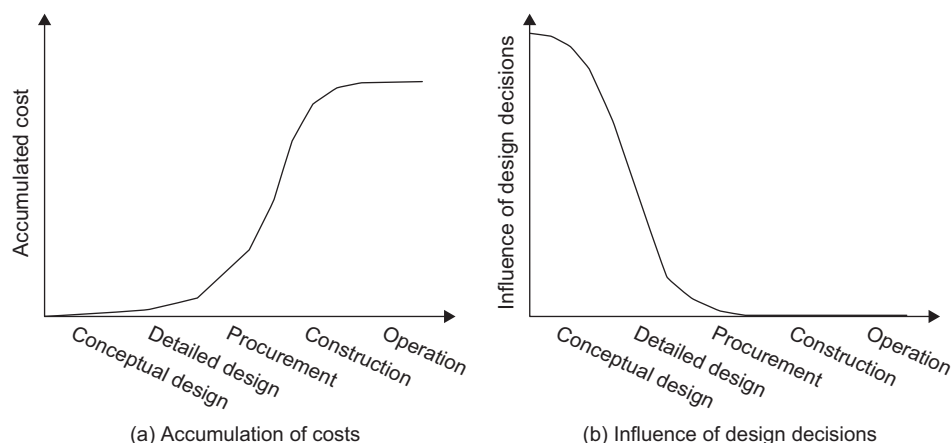


FIG. 7.1 Influence of design decisions on project cost.

7.4 Order-of-magnitude estimates

In the early stages of design, or when carrying out preliminary marketing studies, the design engineer may want to make a quick (Class 5) capital cost estimate without completing a plant design. Several shortcut methods have been developed that allow estimates of total plant cost to be made within $\pm 50\%$ accuracy for preliminary studies. These methods can also be used to provide a rough check on more detailed estimates developed from process equipment costs later in the design of the process.

7.4.1 Cost curve methods

The quickest way to make an order-of-magnitude estimate of plant cost is to scale it from the known cost of an earlier plant that used the same technology or from published data. This requires no design information other than the production rate.

The capital cost of a plant can be related to capacity by the equation

$$C_2 = C_1 \left(\frac{S_2}{S_1} \right)^n \quad (7.1)$$

where C_2 = ISBL capital cost of the plant with capacity S_2

C_1 = ISBL capital cost of the plant with capacity S_1

The exponent n is typically 0.8 to 0.9 for processes that use a lot of mechanical work or gas compression (e.g., methanol, paper pulping, solids-handling plants). For typical petrochemical processes n is usually about 0.7. For small-scale, highly instrumented processes, such as specialty chemical or pharmaceuticals manufacture, n is in the range of 0.4 to 0.5. Averaged across the whole chemical industry, n is about 0.6, and hence Equation 7.1 is commonly referred to as the *six-tenths rule*. The equation can be used with $n = 0.6$ to get a rough estimate of the capital cost if there are not sufficient data available to calculate the index for the particular process. [Estrup \(1972\)](#) gives a critical review of the six-tenths rule. Equation 7.1 is only an approximation, and if sufficient data are available, the relationship is best represented on a log-log plot. [Garrett \(1989\)](#) has published capital cost-plant capacity curves for over 250 processes.

The journal *Hydrocarbon Processing* publishes supplements on refining, petrochemical, and gas processing processes every other year. These supplements are available in print or CD format to subscribers and give approximate capital cost data for various licensed processes, which can be fitted using a rearranged form of Equation 7.1:

$$C_2 = \frac{C_1}{S_1^n} \times S_2^n = a S_2^n \quad (7.2)$$

Values of the parameters a and n for some fuels and commodity chemical processes are given in [Table 7.1](#). Note that these correlations are only valid over a range of plant sizes between S_{lower} and S_{upper} , corresponding to the range over which the data were fitted. The costs in the *Hydrocarbon Processing* supplements are supplied by the technology vendors and are suitable for ballpark estimates only.

TABLE 7.1 Process cost correlations

Process	Licensor	Units	Capacity		<i>a</i>	<i>n</i>
			<i>S_{lower}</i>	<i>S_{upper}</i>		
ABS resin (15% Rubber) by emulsion polymerization	Generic	MMlb/y	50	300	12.146	0.6
Acetic acid by Cativa process	BP	MMlb/y	500	2000	3.474	0.6
Acetic acid by low water methanol carbonylation	Celanese	MMlb/y	500	2000	2.772	0.6
Acrolein by propylene oxidation with Bi/Mo catalyst	Generic	MMlb/y	30	150	6.809	0.6
Adipic acid from phenol	Generic	MMlb/y	300	1000	3.533	0.6
Alkylation (sulfuric acid effluent refrigeration process)	Stratco/DuPont	bpd	4000	20,000	0.160	0.6
Alkylation (HF process)	UOP	bpd	5000	12,000	0.153	0.6
Allyl chloride by propylene chlorination	Generic	MMlb/y	80	250	7.581	0.6
Alpha olefins (full range process)	Chevron Phillips	MMlb/y	400	1200	5.240	0.6
Alpha olefins (full range process)	Shell	MMlb/y	400	1000	8.146	0.6
Benzene by sulfolane extraction	UOP/Shell	MMgal/y	50	200	7.793	0.6
Benzene by toluene hydrodealkylation	Generic	MMgal/y	50	200	7.002	0.6
Benzene reduction by Bensat process	UOP	bpd	8000	15,000	0.0275	0.6
Biodiesel (FAME) from vegetable oil	Generic	MMlb/y	100	500	2.747	0.6
bis-HET by Eastman Glycolysis	Eastman	MMlb/y	50	200	0.500	0.6
BTX Aromatics by Cyclar process	BP/UOP	tpy	200,000	800,000	0.044	0.6
BTX Aromatics by CCR Platforming process	UOP	tpy	200,000	800,000	0.015	0.6
Butadiene by extractive distillation	UOP/BASF	MMlb/y	100	500	5.514	0.6
Butadiene by Oxo-D plus extractive distillation	Texas Petrochem.	MMlb/y	100	500	11.314	0.6
Butene-1 by alhabutol ethylene dimerization	Axens	tpy	5000	30,000	0.0251	0.6
Butene-1 by BP Process	BP	tpy	20,000	80,000	0.169	0.6

Caprolactam from nitration-grade toluene	SNIA BPD S.p.A.	tpy	40,000	120,000	0.321	0.6
Carbon monoxide by steam methane reforming	Generic	MMscf/y	2000	6000	0.363	0.6
Catalytic condensation for gasoline production	UOP	bpd	10,000	30,000	0.222	0.6
Catalytic reforming by CCR platforming process	UOP	bpd	15,000	60,000	0.179	0.6
Coking by Flexicoking, including fluid coking	ExxonMobil	bpd	15,000	40,000	0.343	0.6
Coking by selective yield delayed coking	Foster Wheeler/UOP	bpd	15,000	60,000	0.109	0.68
Copolymer polypropylene by INNOVENE	BP	MMlb/y	300	900	3.430	0.6
Copolymer polypropylene by Unipol	Dow	MMlb/y	300	900	3.641	0.6
Copolymer polypropylene by SPHERIPOL Bulk	Basell	MMlb/y	300	900	3.649	0.6
Copolymer polypropylene by BORSTAR	Borealis	MMlb/y	300	900	4.015	0.6
Crude distillation by D2000	TOTAL/Technip	bpd	150,000	300,000	0.151	0.6
Cumene by Q-Max process	UOP	tpy	150,000	450,000	0.0120	0.6
Cyclic olefin copolymer by Mitsui Process	Mitsui	MMlb/y	60	120	12.243	0.6
Cyclohexane by liq-phase hydrogenation of benzene	Axens	tpy	100,000	300,000	0.0061	0.6
Dewaxing by ISODEWAXING	Chevron Lummus	bpd	6,000	15,000	0.256	0.6
2,6-Dimethylnaphthalene by MeOH alkylation	Exxon Mobil/Kobe	MMlb/y	50	100	7.712	0.6
Dimethyl terephthalate by methanolysis	Generic	MMlb/y	30	80	5.173	0.6
Dimethyl terephthalate by Huels Oxidation	Huels	MMlb/y	300	800	7.511	0.6
Ethanol by ethylene hydration	Generic	Mgal/y	30	90	9.643	0.6
Ethanol (fuel grade) by corn dry milling	Generic	tpy	100,000	300,000	0.0865	0.6
Ethylbenzene by EBOne process	ABB Lummus/UOP	tpy	300,000	700,000	0.0085	0.6

Continued

TABLE 7.1 Process cost correlations—cont'd

Process	Licensor	Units	Capacity		<i>a</i>	<i>n</i>
			<i>S</i> _{lower}	<i>S</i> _{upper}		
Ethylene by ethane cracking	Generic	MMlb/y	500	2000	9.574	0.6
Ethylene by UOP/Hydro MTO process	UOP/INEOS	MMlb/y	500	2000	8.632	0.6
Ethylene: light naphtha cracker (max ethylene)	Generic	MMlb/y	1000	2000	16.411	0.6
Ethylene by ethane/propane cracker	Generic	MMlb/y	1000	2000	7.878	0.6
Ethylene by gas oil cracker	Generic	MMlb/y	1000	2000	17.117	0.6
Ethylene glycol via ethylene oxide hydrolysis	Shell	MMlb/y	500	1000	5.792	0.6
Expandable polystyrene by suspension process	Generic	MMlb/y	50	100	3.466	0.6
Fischer Tropsch process	ExxonMobil	tpy	200,000	700,000	0.476	0.6
Fluid catalytic cracking	KBR	bpd	20,000	60,000	0.210	0.6
Fluid catalytic cracking with power recovery	UOP	bpd	20,000	60,000	0.302	0.6
Gas to liquids by Syntroleum process	Syntroleum	bpd	30,000	100,000	2.279	0.6
Gas sweetening by Amine Guard FS process	UOP	MMscf/d	300	800	0.386	0.6
Gasification by GE gasification process Maya crude	GE Energy	bpd	7000	15,000	0.681	0.6
Gasoline desulfurization, ultra-deep by Prime-G+	Axens	bpd	7000	15,000	0.0420	0.58
Glucose (40% solution) by basic wet corn milling	Generic	MMlb/y	300	800	3.317	0.6
HDPE pellets by BP gas phase process	BP Amoco	MMlb/y	300	700	3.624	0.6
HDPE pellets by Phillips slurry process	Phillips	MMlb/y	300	700	3.370	0.6
HDPE pellets by Zeigler slurry process	Zeigler	MMlb/y	300	700	4.488	0.6
High-impact polystyrene by bulk polymerization	Dow	MMlb/y	70	160	2.970	0.6
Hydrocracking by isocracking	Chevron Lummus	bpd	20,000	45,000	0.221	0.6

Hydrocracking by unicracking (distillate) process	UOP	bpd	20,000	45,000	0.136	0.66
Hydrocracking	Axens	bpd	20,000	45,000	0.198	0.6
Hydrogen by steam methane reforming	Foster Wheeler	MMscf/d	10	50	1.759	0.79
Hydrotreating by Unionfining process	UOP	bpd	10,000	40,000	0.0532	0.68
Isomerization by once-through Penex process	UOP	bpd	8,000	15,000	0.0454	0.6
Isomerization by Penex-Molex process	UOP	bpd	8000	15,000	0.120	0.6
Isophthalic acid by m-xylene oxidation	Generic	MMlb/y	160	300	9.914	0.6
Isoprene via isobutylene carbonylation	IFP	MMlb/y	60	200	10.024	0.6
Isoprene by propylene dimerization and pyrolysis	Generic	MMlb/y	60	200	6.519	0.6
Linear alkylbenzene by PACOL/DeFine/PEP/Detal ⁶	UOP	MMlb/y	100	250	4.896	0.6
Linear alpha olefins	Chevron	MMlb/y	300	700	5.198	0.6
Linear alpha olefins by Linear-1 process	UOP	tpy	200,000	300,000	0.122	0.6
Maleic anhydride by fluid bed process	Generic	MMlb/y	70	150	7.957	0.6
Methacrylic acid by isobutylene oxidation	Generic	MMlb/y	70	150	7.691	0.6
Methanol via steam reforming and synthesis	Davy Process Tech.	tpd	3000	7000	2.775	0.6
m-Xylene by MX Sorbex process	UOP	MMlb/y	150	300	4.326	0.6
Naphthalene by three-stage fractional crystallizer	Generic	MMlb/y	20	50	2.375	0.6
N-Butanol from crude C4s	BASF	MMlb/y	150	300	8.236	0.6
Norbornene by Diels–Alder reaction	Generic	MMlb/y	40	90	7.482	0.6
Pentaerythritol by condensation	Generic	MMlb/y	40	90	6.220	0.6
PET resin chip with comonomer by NG3	DuPont	MMlb/y	150	300	4.755	0.6

Continued

TABLE 7.1 Process cost correlations—cont'd

Process	Licensor	Units	Capacity		<i>a</i>	<i>n</i>
			<i>S</i> _{lower}	<i>S</i> _{upper}		
Phenol from cumene (zeolite catalyst)	UOP/Sunoco	MMlb/y	200	600	6.192	0.6
Phthalic anhydride by catalytic oxidation	Generic	MMlb/y	100	200	7.203	0.6
Polycarbonate by interfacial polymerization	Generic	MMlb/y	70	150	20.680	0.6
Polyethylene terephthalate (melt phase)	Generic	MMlb/y	70	200	5.389	0.6
Polystyrene by bulk polymerization, plug flow	Generic	MMlb/y	70	200	2.551	0.6
Propylene by Oleflex process	UOP	tpy	150,000	350,000	0.0943	0.6
Propylene by metathesis	Generic	MMlb/y	500	1,000	1.899	0.6
Purified terephthalic acid	EniChem/Technimont	MMlb/y	350	700	10.599	0.6
p-Xylene by Isomar and Parex processes	UOP	tpy	300,000	700,000	0.0230	0.6
p-Xylene by Tatoray process	UOP	bpd	12,000	20,000	0.0690	0.6
Refined glycerin by distillation/adsorption	Generic	MMlb/y	30	60	2.878	0.6
Sebacic acid by cyclododecanone route	Sumitomo	MMlb/y	8	16	13.445	0.6
Sorbitol (70%) by continuous hydrogenation	Generic	MMlb/y	50	120	4.444	0.6
Styrene by SMART process	ABB Lummus/UOP	tpy	300,000	700,000	0.0355	0.6
Vinyl acetate by Cativa integrated process	BP	MMlb/y	300	800	7.597	0.6
Vinyl acetate by Celanese Vantage Process	Celanese	MMlb/y	300	800	6.647	0.6
Visbreaking by coil-type visbreaker	Foster Wheeler/UOP	bpd	6,000	15,000	0.278	0.48

Notes:

1. Values of *a* are in January 2006 MM\$ on a USGC basis (Nelson Farrer index = 1961.6, CEPCI index = 478.6).2. *S*_{lower} and *S*_{upper} indicate the bounds of the region over which the correlation can be applied.3. *S* is based on product rate for chemicals, feed rate for fuels.4. If the index *n* is 0.6, then the correlation is an extrapolation around a single cost point.5. Correlations are based on data taken from [Hydrocarbon Processing \(2003, 2004a,b\)](#), except where the licensor is stated as "Generic," in which case the correlations are based on data from Nexant PERP reports (see www.Nexant.com/products for a full list of reports available).

6. PACOL/DeFine/PEP/Detal processes.

Economy of scale

It can be seen from the discussion earlier and the correlations in Table 7.1 that the exponent n is always less than 1.0. One result of this is that larger plants tend to cost less to construct per unit of product produced, an advantage that is known as an *economy of scale*. From Equation 7.2, the capital cost per unit of production of product will be:

$$\frac{C_2}{S_2} = a S_2^{n-1} \quad (7.3)$$

As $n - 1$ is less than zero, the cost per unit of production decreases as S_2 increases. A smaller capital cost per unit of product allows the owners of large plants to price their products more competitively and still recover their capital investment. This creates an incentive for chemical companies to build plants at larger scale.

7.4.2 Step count method

If cost data for a similar process are not available then an order-of-magnitude estimate can sometimes be made by adding contributions for different plant sections or functional units.

Experienced design engineers can often figure out costs of plant sections from historic total plant costs. For example, in many petrochemical processes roughly 20% of ISBL capital cost is in the reactor section and 80% is in the distillation and product purification sections. If the separation and recovery system has a similar size and complexity to that of a known process, then the capital cost can be estimated based on the cost of the known process and the reactor section cost taken as one-quarter of the separation section cost.

An alternative approach is Bridgewater's method, which correlates plant cost to the number of processing steps (Bridgewater & Mumford, 1979). For plants primarily processing liquids and solids:

$$Q \geq 60,000: C = 4320 N \left(\frac{Q}{s} \right)^{0.675} \quad (7.4)$$

$$Q < 60,000: C = 380,000 N \left(\frac{Q}{s} \right)^{0.3} \quad (7.5)$$

where: C = ISBL capital cost in US\$, US Gulf Coast, January 2010 basis (CEPCI = 532.9)

Q = plant capacity in metric tons per year

s = reactor conversion (= mass of desired product per mass fed to the reactor)

N = number of functional units

(Note: The correlations have been updated from the original reference.)

A functional unit includes all the equipment and ancillaries needed for a significant process step or function, such as a reaction, separation, or other major unit operation. Pumping and heat exchange are not normally considered functional units unless they have substantial cost, for example, compressors, refrigeration systems, or process furnaces.

7.4.3 Reverse engineering methods

In some situations, a very rough estimate of capital cost can be backed out from operating costs or product prices.

Payback method

The difference between the sales price and the cost of production is the gross profit of the plant; see Section 8.2.4. If an investor were to build a new plant, then the gross profit minus taxes would have to be sufficient to give a return on the investment made in building the plant. Assuming the plant pays back in 3 to 5 years (average = 4 years), a very rough estimate of the plant cost would be:

$$\text{Plant cost} = 4 \times (\text{gross profit} - \text{taxes}) \quad (7.6)$$

For chemicals, raw materials costs are typically 80% to 90% of the cash cost of production, so the gross profit can be roughly estimated as:

$$\text{Gross profit} = \text{product value} - (1.2 \times \text{raw materials costs}) \quad (7.7)$$

This method obviously can only be used in situations where the designer believes that new plants are being built and generating a reasonable return on capital.

Turnover ratio method

An even simpler (and less accurate) method is to estimate the capital investment based on the turnover ratio. The turnover ratio is the gross annual sales divided by the fixed capital investment. The turnover ratio can vary widely, but a typical value for the chemical industry is between 1.0 and 1.25 (Humphreys, 1991).

TCOP method

For large-scale production of formed or assembled components (>500,000 pieces per year) a rule of thumb is:

$$\text{Total cost of production (TCOP)} = 2 \times \text{raw materials cost} \quad (7.8)$$

The TCOP is equal to the cost of raw materials, plus utility costs, fixed costs, and an annualized capital charge that is typically one-third to one-fifth of the capital cost (see Section 8.2.4, Equation 8.6). Equation 7.8 can thus be used to make a very approximate estimate of plant cost for manufactured products if fixed costs and utilities can be estimated.

Example 7.1

The process for making cyclohexane by saturation of benzene consists of a feed-effluent heat exchanger, a saturation reactor, and a product stabilizer column. Estimate the cost of a plant that produces 200,000 metric tons per year (200 kMTA) of cyclohexane using the correlation in Table 7.1 and Bridgewater's method.

Solution

From Table 7.1, the cost correlation for the Axens process for benzene saturation gives:

$$\begin{aligned} C &= 0.0061(S)^{0.6} \\ &= 0.0061(2 \times 10^5)^{0.6} \\ &= \underline{\$9.2 \text{ MM}} \text{ expressed on a Jan. 2006 USGC basis} \end{aligned}$$

Using Bridgewater's method, we have two functional units (the reactor and product stabilizer—the heat exchanger doesn't count), and assuming that the reactor conversion is 1.0, we can substitute into Equation 7.4:

$$\begin{aligned} C &= 4320 \times 2(Q)^{0.675} \\ &= 4320 \times 2 \times (2 \times 10^5)^{0.675} \\ &= \underline{\$33 \text{ MM}} \text{ expressed on a Jan. 2010 USGC basis} \end{aligned}$$

Note that we have obtained two very different answers. Bridgewater's correlation is known to be only an approximation; however, Table 7.1 is based on data from technology vendors that may be somewhat understated. With the level of information available, it is probably safe to say that the cost is in the range \$10 MM to \$20 MM. Note also that the costs are not on the same time basis. Methods for correcting costs on different time bases will be discussed in Section 7.7.

7.5 Estimating purchased equipment costs

When more design information is available, the cost of a plant can be worked up from the cost of individual items of process equipment. Costs of single pieces of equipment are also often needed for minor revamp and de-bottlenecking projects.

7.5.1 Sources of equipment cost data

The best source of purchased equipment costs is recent data on actual prices paid for similar equipment. Engineers working for engineering, procurement, and construction (EPC) companies (often referred to as *contractors*) have access to large amounts of high-quality data, as these companies carry out many projects globally every year. Engineers working in operating companies may have access to data from recent projects, but unless they work for a large company that carries out many capital projects, they are unlikely to be able to develop and maintain current cost correlations for more than a few basic equipment types. Most large companies recognize the difficulty of making reliable cost estimates and employ a few experienced cost engineering specialists who collect data and work closely with the EPC companies on project budgets.

Actual prices paid for equipment and bulk items may differ substantially from catalog or list prices, depending on the purchasing power of the contractor or client and the urgency of the project. Discounts and surcharges are highly confidential business information and will be closely guarded even within EPC companies.

Those design engineers who are outside the EPC sector and do not have the support of a cost estimating department must rely on cost data from the open literature or use cost estimating software. The most widely used software tools for estimating chemical plant costs are the Aspen Capital Cost Estimator (ACCE) suite of tools licensed by Aspen Technology, Inc., and Cleopatra Enterprise owned by Cost Engineering Consultancy. The cost estimation software within ACCE is built on the ICARUS cost estimation algorithms. ACCE does not use the factorial method, but instead estimates equipment costs, bulk costs, and installation costs from the costs of materials and labor, following the practice used by cost engineers for detailed estimating. The models in ACCE are developed by a team of cost engineers based on data collected from EPC companies and equipment manufacturers. The models are updated annually. The ACCE software is included in the standard Aspen/Hysys academic package and is available in most universities. Cleopatra Enterprise is an integrated project cost management software that supports the entire project life cycle. It uses databases of equipment costs, bulk materials costs, and installation costs that were developed from vendor quotations and historic data and are updated annually. Cleopatra's database allows cost estimates to be built in different levels of detail, supporting the preparation of high-level, semi-detailed, and detailed estimates that cover the full scope of a project. The software also has a variety of features for analyzing and comparing different estimates and scenarios. Both ACCE and Cleopatra can give reasonably good estimates when used properly, and both are described in more detail in Section 7.10.

There is an abundance of equipment cost data and cost correlations in the open literature, but much of it is of very poor quality. The relationship between size and cost given in Equations 7.1 and 7.2 can also be used for equipment if a suitable size parameter is used. If the size range spans several orders of magnitude, then log-log plots usually give a better representation of the relationship than simple equations.

Some of the most reliable information on equipment costs can be found in the professional cost engineering literature. Correlations based on recent data are occasionally published in *Cost Engineering*, which is the journal of AACE International. AACE International also has an excellent website, www.aacei.org, which has cost models that can be used by members. There is also an extensive listing of other web resources for cost estimating at <https://web.aacei.org/resources/publications>. The UK Association of Cost Engineers (ACostE) publishes the journal *The Cost Engineer* and also prints a guide to capital cost estimating (Gerrard, 2000), which gives cost curves for the main types of process equipment based on recent data. The prices are given in British pounds sterling on a UK basis, but this book is useful for making estimates of prices in northwest Europe. The International Cost Engineering Council website (www.icoste.org) provides links to 46 international cost engineering societies, several of which maintain databases of local costs.

Current prices for new and used equipment can be found on resale websites such as www.equipnet.com. It is not always easy to find the exact piece of equipment needed on such sites, but they can be used to give a good indication of the accuracy of cost correlations. A free web-based costing tool is available at www.matche.com. This tool appears to have not been updated since 2003, and the source of the cost correlations given is not clear, so it is not suitable for any use other than undergraduate design projects.

Many cost correlations can be found in chemical engineering textbooks, for example, Douglas (1988), Garrett (1989), Turton et al. (2003), Peters et al. (2003), Ulrich and Vasudevan (2004), and Seider et al. (2016). The references for such correlations should always be checked very carefully. When they are properly referenced, they are often found to be based on data published by Guthrie (1969, 1974) and updated using either cost indices (as described in Section 7.7) or a few recent data points. Guthrie's correlations were reasonably good when published, but there have been substantial changes in the relative contributions of material and fabrication costs of most process equipment since then. Academic authors usually do not have access to sufficient high-quality cost data to be able to make

reliable correlations, and most of the academic correlations predict lower costs than would be obtained using ACCE or other detailed estimating methods. These correlations are adequate for the purposes of university design projects but should not be used in real projects. It is to be hoped that the authors of these publications will benchmark their correlations against ACCE or Cleopatra in future editions, which will improve the accuracy of the correlations and make them more useful to those who do not have access to costing software.

7.5.2 Cost curves for purchased equipment costs

For those design engineers who lack access to reliable cost data or estimating software, the correlations given in Table 7.2 can be used for preliminary estimates. The correlations in Table 7.2 are of the form:

$$C_e = a + b S^n \quad (7.9)$$

where: C_e = purchased equipment cost on a U.S. Gulf Coast basis, Jan. 2010 (CEPCI = 532.9, NF refinery inflation index = 2281.6)

a, b = cost constants in Table 7.2

S = size parameter, units given in Table 7.2

n = exponent for that type of equipment

The correlations in Table 7.2 are only valid between the lower and upper values of S indicated. The prices are all for carbon steel equipment except where noted in the table. Extrapolation to other materials is discussed in Section 7.6.3.

Example 7.2

Estimate the purchased equipment cost of a plain carbon steel shell and tube heat exchanger with area 400 m².

Solution

From Table 7.2, the cost correlation for a shell and tube heat exchanger is:

$$C_e = 28,000 + 54 A^{1.2} \quad (7.9)$$

where A is the area in m² and C_e is on a January 2010 basis. Substituting for the area:

$$C_e = 28,000 + (54 \times 400^{1.2}) = \underline{\underline{\$99,600}}$$

7.5.3 Detailed method of cost estimating

When the method of design and construction of a piece of equipment are known, the cost can be estimated from the costs of the materials, parts, labor, and manufacturer's profit. This is the method preferred by professional cost estimators and procurement managers, as it allows the estimator to obtain an unbiased estimate of the real cost of the equipment, which can then be used in negotiations with the vendor to determine a fair price. This method is also used in many of the correlations in commercial cost estimating programs such as ACCE.

A detailed estimate requires an itemized list of the required parts, an understanding of the fabrication steps, knowledge of the machinery involved (so as to estimate the machine costs), and an understanding of the amount of labor needed for each step. The fabrication method is stated as a work breakdown structure (WBS) to arrive at an accurate estimate of the labor. Machine time costs are usually allocated by taking the annual cost of the machine, including allowances for capital recovery, maintenance, and electric power costs, and then dividing these costs by the number of hours in use to arrive at an hourly machine cost. The overall cost can then be made by summing the component costs, machine costs, and labor costs and adding suitable amounts for supervision, overhead, and manufacturer's profit. A hypothetical example of a cost breakdown structure for a shell and tube heat exchanger is given in Table 7.3.

The detailed method of cost estimating must be followed whenever cost or price data are not available, for example, when making an estimate of the cost of specialized equipment that cannot be found in the literature. For example, a reactor design is usually unique for a particular process but the design can be broken down into

TABLE 7.2 Purchased equipment cost for common plant equipment

Equipment	Units for size, <i>S</i>	<i>S_{lower}</i>	<i>S_{upper}</i>	<i>a</i>	<i>b</i>	<i>n</i>	Note
Agitators and mixers							
Propeller	driver power, kW	5.0	75	17,000	1130	1.05	
Spiral ribbon mixer	driver power, kW	5.0	35	30,800	125	2.0	
Static mixer	liters/s	1.0	50	570	1170	0.4	
Boilers							
Packaged, 15 to 40 bar	kg/h steam	5000	200,000	124,000	10.0	1.0	
Field erected, 10 to 70 bar	kg/h steam	20,000	800,000	130,000	53	0.9	
Centrifuges							
High-speed disk	diameter, m	0.26	0.49	57,000	480,000	0.7	
Atmospheric suspended basket	power, kW	2.0	20	65,000	750	1.5	
Compressors							
Blower	m ³ /h	200	5000	4450	57	0.8	
Centrifugal	driver power, kW	75	30,000	580,000	20,000	0.6	
Reciprocating	driver power, kW	93	16,800	260,000	2700	0.75	
Conveyors							
Belt, 0.5 m wide	length, m	10	500	41,000	730	1.0	
Belt, 1.0 m wide	length, m	10	500	46,000	1320	1.0	
Bucket elevator, 0.5 m bucket	height, m	10	30	17,000	2600	1.0	
Crushers							
Reversible hammer mill	t/h	30	400	68,400	730	1.0	
Pulverizers	kg/h	200	4000	16,000	670	0.5	
Jaw crusher	t/h	100	600	−8000	62,000	0.5	
Gyratory crusher	t/h	200	3000	5000	5100	0.7	
Ball mill	t/h	0.7	60	−23,000	242,000	0.4	
Crystallizers							
Scraped surface crystallizer	length, m	7	280	10,000	13,200	0.8	
Distillation columns							
See pressure vessels, packing, and trays							
Dryers							
Direct contact rotary	area, m ²	11	180	15,000	10,500	0.9	1
Atmospheric tray batch	area, m ²	3.0	20	10,000	7900	0.5	
Spray dryer	evap rate kg/h	400	4,000	410,000	2200	0.7	
Evaporators							
Vertical tube	area, m ²	11	640	330	36,000	0.55	
Agitated Falling film	area, m ²	0.5	12	88,000	65,500	0.75	2

Continued

TABLE 7.2 Purchased equipment cost for common plant equipment—cont'd

Equipment	Units for size, <i>S</i>	<i>S_{lower}</i>	<i>S_{upper}</i>	<i>a</i>	<i>b</i>	<i>n</i>	Note
Exchangers							
U-tube shell and tube	area, m ²	10	1000	28,000	54	1.2	
Floating head shell and tube	area, m ²	10	1000	32,000	70	1.2	
Double pipe	area, m ²	1.0	80	1900	2500	1.0	
Thermosiphon reboiler	area, m ²	10	500	30,400	122	1.1	
U-tube Kettle reboiler	area, m ²	10	500	29,000	400	0.9	
Plate and frame	area, m ²	1.0	500	1600	210	0.95	2
Filters							
Plate and frame	capacity, m ³	0.4	1.4	128,000	89,000	0.5	
Vacuum drum	area, m ²	10	180	−73,000	93,000	0.3	
Furnaces							
Cylindrical	duty, MW	0.2	60	80,000	109,000	0.8	
Box	duty, MW	30	120	43,000	111,000	0.8	
Packings							
304 ss Raschig rings	m ³			0	8000	1.0	
Ceramic Intalox saddles	m ³			0	2000	1.0	
304 ss pall rings	m ³			0	8500	1.0	
PVC structured packing	m ³			0	5500	1.0	
304 ss structured packing	m ³			0	7600	1.0	3
Pressure vessels							
Vertical, cs	shell mass, kg	160	250,000	11,600	34	0.85	4
Horizontal, cs	shell mass, kg	160	50,000	10,200	31	0.85	4
Vertical, 304 ss	shell mass, kg	120	250,000	17,400	79	0.85	4
Horizontal, 304 ss	shell mass, kg	120	50,000	12,800	73	0.85	4
Pumps and drivers							
Single-stage centrifugal	flow liters/s	0.2	126	8000	240	0.9	
Explosion proof motor	power, kW	1.0	2500	−1100	2100	0.6	
Condensing steam turbine	power, kW	100	20,000	−14,000	1900	0.75	
Reactors							
Jacketed, agitated	volume, m ³	0.5	100	61,500	32,500	0.8	2
Jacketed, agitated, glass lined	volume, m ³	0.5	25	12,800	88,200	0.4	
Tanks							
Floating roof	capacity, m ³	100	10,000	113,000	3250	0.65	
Cone roof	capacity, m ³	10	4000	5800	1600	0.7	
Trays							
Sieve trays	diameter, m	0.5	5.0	130	440	1.8	5
Valve trays	diameter, m	0.5	5.0	210	400	1.9	
Bubble cap trays	diameter, m	0.5	5.0	340	640	1.9	

TABLE 7.2 Purchased equipment cost for common plant equipment—cont'd

Equipment	Units for size, S	S_{lower}	S_{upper}	a	b	n	Note
Utilities							
Cooling tower and pumps	flow liters/s	100	10,000	170,000	1500	0.9	6
Packaged mechanical refrigerator	evaporator duty, kW	50	1,500	24,000	3500	0.9	
Water ion exchange plant	flow m^3/h	1	50	14,000	6200	0.75	

Notes:

1. Direct heated.

2. Type 304 stainless steel.

3. With surface area $350 m^2/m^3$.

4. Not including heads, ports, brackets, internals, etc. (see Chapter 14 for how to calculate wall thickness).

5. Cost per tray, based on a stack of 30 trays.

6. Field assembly.

7. All costs are U.S. Gulf Coast basis, Jan. 2010 (CEPCI index = 532.9, NF refinery inflation index = 2281.6).

TABLE 7.3 Hypothetical cost breakdown for fabrication of a heat exchanger

Step	Materials	Parts	Machine	Labor
1. Shell fabrication				
1.1 Shell forming	Steel plate, $L_s \times \pi D_s \times t_s$		Cutting Rolling Welding	2 hr 2 hr 2 hr
1.2 End flange forming	Two steel plate discs, D_f diameter $\times t_f$		Cutting Drilling (for bolt holes)	1 hr each 2 hr each
1.3 Flange attachment			Welding	2 hr each
1.4 Nozzle reinforcement	Two steel plate discs, $2 D_n$ diameter		Cutting/rolling Welding	2 hr 1 hr each
1.5 Nozzle attachment		Two flanged pipes	Welding	1 hr each
2. Head fabrication ($\times 2$) (Note: Both nozzles may be on one head, depending on the exchanger type.)				
2.1 Head forming	Steel plate disc		Cutting Drop forging	1 hr
2.2 Hole cutting			Cutting	4 hr
2.3 End flange forming	Two steel plate discs, D_f diameter $\times t_f$		Cutting Drilling (for bolt holes)	1 hr 2 hr
2.4 Flange attachment			Welding	2 hr
2.5 Nozzle reinforcement	Two steel plate discs, D_n diameter		Cutting/rolling Welding	1 hr each 1 hr each
2.6 Nozzle attachment		Two flanged pipes	Welding	1 hr each
3. Tube bundle fabrication				
3.1 Tube sheet fabrication	Steel plate disc, D_s diameter		Stamping	1 hr
3.2 Baffle cutting	N_{baf} steel plate discs		Stamping	0.5 hr each
3.3 Tie rod cutting	10 to 12 $\times L$, steel rod		Cutting Threading	1 hr 2 hr
3.4 Tube preparation	N_t tubes cut to right length		Cutting Bending	0.25 hr each 0.25 hr each

Continued

TABLE 7.3 Hypothetical cost breakdown for fabrication of a heat exchanger—cont'd

Step	Materials	Parts	Machine	Labor
3.5 Bundle assembly				0.25 hr/tube
3.6 Tube sheet sealing			Welding/rolling	0.25 hr/tube
4. Exchanger assembly				
4.1 Bundle insertion			Large crane	1hr
4.2 Gasket cutting	Two sheets gasket material		Stamping	1 hr
4.3 Head attachment		N_b bolts		2 hr

Notes:

1. L_s = shell length, D_s = shell diameter, t_s = shell thickness, D_f = flange diameter, t_f = flange thickness, D_n = nozzle diameter, N_{baf} = number of baffles, N_t = number of tubes, N_b = number of bolts.

2. Dimensions are not exact, labor hours are approximate, and some steps have been omitted to simplify the example. There could be considerable variation in labor times depending on the exchanger layout and complexity.

3. Costs would also include supervision, testing and inspection, overhead, and manufacturer's margin.

4. The components and fabrication of a shell and tube heat exchanger are discussed in detail in [Chapter 19](#).

standard components (vessels, heat-exchange surfaces, spargers, agitators, etc.), the cost of which can be found in the literature and used to build up an estimate of the reactor cost.

More information on estimating costs from a detailed breakdown of parts and labor is given by [Dysert \(2015\)](#) in the AACE International's training manual ([Hastak, 2015](#)). Breakdowns of the materials and labor components for many types of process equipment are given by [Page \(1996\)](#). [Pikulik and Diaz \(1977\)](#) give a method of costing major equipment items from cost data on the basic components: shells, heads, nozzles, and internal fittings. [Purohit \(1983\)](#) gives a detailed procedure for estimating the cost of heat exchangers.

7.5.4 Use of vendor data in cost estimating

At some point in the development of a design it is always necessary to get a real price quotation from an equipment vendor. Although vendor quotations are always realistic, some caution is needed to ensure that such estimates are valid for the use intended.

A large amount of vendor information is now available online and can easily be found using any of the major search engines or by starting from directories such as www.purchasing.com. Online costs are usually manufacturer's catalog prices for small-order quantities. Large order sizes (as filled by contractors) are often heavily discounted, which is one reason why many operators subcontract plant construction to EPC companies that have greater purchasing power. Items requiring special fabrication, for example, large vessels or compressors, may experience discounts or surcharges depending on the state of the manufacturer's order books and the purchasing power of the customer.

When a vendor is contacted directly, the quality of estimate that they can provide depends very much on the quality of the information that they are given. Often the vendor will ask the customer to provide information on the process conditions and equipment duty, so as to be able to complete their own optimization and ensure that the model or design specified is correct for the purpose. It is very important to include any constraints on materials of construction in the specification provided to the vendor, as materials substitution can cause significant increases in cost; see Section 7.6.3.

As with any purchasing decision, it is always a good idea to contact several vendors and comparison shop for the best price. Care must be taken to be ethical in handling vendors' confidential information, and commercially sensitive information such as pricing should not be shared with other vendors.

7.6 Estimating installed costs: The factorial method

Capital cost estimates for chemical process plants are often based on an estimate of the purchase cost of the major equipment items required for the process, with the other costs being estimated as factors of the equipment cost. The accuracy of this type of estimate will depend on what stage the design has reached at the time the estimate is made

TABLE 7.4 Installation factors proposed by Hand (1958)

Equipment type	Installation factor
Compressors	2.5
Distillation columns	4
Fired heaters	2
Heat exchangers	3.5
Instruments	4
Miscellaneous equipment	2.5
Pressure vessels	4
Pumps	4

and on the reliability of the data available on equipment costs. In the later stages of the project design, when detailed equipment specifications are available and firm quotes have been obtained from vendors, a Class 3 estimate of the capital cost of the project can be made by this method, but generally this method leads to a Class 4 estimate.

7.6.1 Lang factors

Lang (1948) proposed that the ISBL fixed capital cost of a plant is given as a function of the total purchased equipment cost by the equation:

$$C = F \left(\sum C_e \right) \quad (7.10)$$

where: C = total plant ISBL capital cost (including engineering costs).

$\sum C_e$ = total delivered cost of all the major equipment items: reactors, tanks, columns, heat exchangers, furnaces, etc. The delivered cost is the purchased cost plus shipping or delivery costs.

F = an installation factor, later widely known as a *Lang factor*.

Lang originally proposed the following values of F , based on 1940s economics:

$F = 3.1$ for solids processing plants

$F = 4.74$ for fluids processing plants

$F = 3.63$ for mixed fluids-solids processing plants

Hand (1958) suggested that better results are obtained by using different factors for different types of equipment. Examples of the factors proposed by Hand are given in Table 7.4. Hand also observed that this approach should only be used in the earliest stages of process design and in the absence of detailed design information.

Both Lang (1948) and Hand (1958) included home office costs but not offsite costs or contingency in their installation factors, so beware of double counting EPC costs when using this approach. The relative costs of materials and labor have changed substantially from when these factors were developed, and the accuracy of the correlation probably never warranted three significant figures for F . Most practitioners using this method therefore use a Lang factor of 3, 4, or 5, depending on the plant scale (larger plant = smaller factor) and type.

7.6.2 Detailed factorial estimates

Equation 7.10 can be used to make a preliminary estimate once the flowsheet has been drawn up and the main plant equipment has been sized. When more detailed design information is available, the installation factor can be estimated somewhat more rigorously by considering the cost factors that are compounded into the Lang factor individually.

The direct-cost items that are incurred in the construction of a plant in addition to the delivered cost of equipment are:

1. Equipment erection, including foundations and minor structural work
2. Piping, including insulation and painting
3. Electrical, power, and lighting
4. Instruments and automatic process control (APC) systems
5. Process buildings and structures
6. Ancillary buildings, offices, laboratory buildings, and workshops (if not costed separately as offsites)
7. Storage for raw materials and finished product (if not costed separately as offsites)
8. Utilities and provision of plant for steam, water, air, and firefighting services (if not costed separately as offsites)
9. Site preparation

The contribution of each of these items to the total capital cost is calculated by multiplying the total purchased equipment cost by an appropriate factor. As with the basic Lang factor, these factors are best derived from historical cost data for similar processes. Typical values for the factors are given in several references: [Happle and Jordan \(1975\)](#) and [Garrett \(1989\)](#). [Guthrie \(1974\)](#) splits the costs into the material and labor portions and gives separate factors for each.

The accuracy and reliability of an estimate can be further improved by dividing the process into subunits and using factors that depend on the function of the subunits; see [Guthrie \(1969\)](#). In Guthrie's detailed method of cost estimation, the installation, piping, and instrumentation costs for each piece of equipment are costed separately. Detailed costing is only justified if the cost data available are reliable and the design has been taken to the point where all the cost items can be identified and included. [Gerrard \(2000\)](#) gives factors for individual pieces of equipment as a function of equipment cost and complexity of installation.

Typical factors for the components of the capital cost are given in [Table 7.5](#). These can be used to make an approximate estimate of capital cost using equipment cost data published in the literature.

7.6.3 Materials factors

The installation factors given in [Tables 7.4 and 7.5](#) are for plants built from carbon steel. When more exotic materials are used, a materials factor f_m should also be introduced:

TABLE 7.5 Typical factors for estimation of project fixed capital cost

Item	Process type		
	Fluids	Fluids–solids	Solids
1. Major equipment, total purchase cost	C_e	C_e	C_e
f_{er} Equipment erection	0.3	0.5	0.6
f_p Piping	0.8	0.6	0.2
f_i Instrumentation and control	0.3	0.3	0.2
f_{el} Electrical	0.2	0.2	0.15
f_c Civil	0.3	0.3	0.2
f_s Structures and buildings	0.2	0.2	0.1
f_l Lagging and paint	0.1	0.1	0.05
ISBL cost, $C = \Sigma C_e \times$	3.3	3.2	2.5
Offsites (OS)	0.3	0.4	0.4
Design and Engineering (D&E)	0.3	0.25	0.2
Contingency (X)	0.1	0.1	0.1
Total fixed capital cost $C_{FC} = C(1 + OS)(1 + D\&E + X)$			
$= C \times$	1.82	1.89	1.82
$= \Sigma C_e \times$	6.00	6.05	4.55

$$f_m = \frac{\text{purchased cost of item in exotic material}}{\text{purchased cost of item in carbon steel}} \quad (7.11)$$

Note that f_m is not equal to the ratio of the metal prices, as the equipment purchased cost also includes labor costs, overheads, fabricator's profit, and other costs that do not scale directly with metal price. Equation 7.10 can then be expanded for each piece of equipment to give:

$$C = \sum_{i=1}^{i=M} C_{e,i,CS} [(1+f_p)f_m + (f_{er}+f_{el}+f_i+f_c+f_s+f_l)] \quad (7.12)$$

Equation 7.12 should be used when the purchased equipment cost has been found on a carbon steel basis and the designer is estimating the cost for alloy construction. If the purchased equipment cost has been obtained on an alloy basis, then the designer should instead correct the other installation factors so as not to overestimate the cost of installation:

$$C = \sum_{i=1}^{i=M} C_{e,i,A} [(1+f_p) + (f_{er}+f_{el}+f_i+f_c+f_s+f_l) / f_m] \quad (7.13)$$

where: $C_{e,i,CS}$ = purchased equipment cost of equipment i in carbon steel

$C_{e,i,A}$ = purchased equipment cost of equipment i in alloy

M = total number of pieces of equipment

f_p = installation factor for piping

f_{er} = installation factor for equipment erection

f_{el} = installation factor for electrical work

f_i = installation factor for instrumentation and process control

f_c = installation factor for civil engineering work

f_s = installation factor for structures and buildings

f_l = installation factor for lagging, insulation, or paint

Failure to properly correct installation factors for materials of construction is one of the most common sources of error with the factorial method. Typical values of the materials factor for common engineering alloys are given in [Table 7.6](#). A more detailed discussion of the relative cost of materials and sources of material cost factors is given in Section 6.6.

7.6.4 Summary of the factorial method

Many variations of the factorial method are used. The method outlined here can be used with the data given in this chapter to make a quick, approximate estimate of the fixed capital investment needed for a project.

TABLE 7.6 Materials cost factors, f_m , relative to plain carbon steel

Material	f_m
Carbon steel	1.0
Aluminum and bronze	1.07
Cast steel	1.1
304 stainless steel	1.3
316 stainless steel	1.3
321 stainless steel	1.5
Hastelloy C	1.55
Monel	1.65
Nickel and Inconel	1.7

1. Prepare material and energy balances; draw up preliminary flowsheets; size major equipment items and select materials of construction.
2. Estimate the purchased cost of the major equipment items; see Section 7.5.
3. Calculate the ISBL installed capital cost, using the factors given in Table 7.5 and correcting for materials of construction using Equation 7.12 or 7.13 with the materials factors given in Table 7.6.
4. Calculate the OSBL, engineering, and contingency costs using the factors given in Table 7.5.
5. The sum of ISBL, OSBL, engineering, and contingency costs is the fixed capital investment.
6. Estimate the working capital as a percentage of the fixed capital investment; 10% to 20% is typical (or, better, calculate it from the cost of production if this has been estimated—see Section 9.2.3).
7. Add the fixed and working capital to get the total investment required.

Example 7.3

A plant modification has been proposed that will allow recovery of a by-product. The modification consists of adding the following equipment:

Distillation column, height 30 m, diameter 3 m, 50 sieve trays, operating pressure 10 bar

- U-tube heat exchanger, area 60 m²
- Kettle reboiler, area 110 m²
- Horizontal pressure vessel, volume 3 m³, operating pressure 10 bar
- Storage tank, volume 50 m³
- Two centrifugal pumps, flow rate 3.6 m³/h, driver power 500 W
- Three centrifugal pumps, flow rate 2.5 m³/h, driver power 1 kW (two installed plus one spare)

Estimate the installed ISBL capital cost of the modification if the plant is to be built from type 304 stainless steel. Estimate the cost using both Hand's method and the factors given in Table 7.5.

Solution

The first step is to convert the units to those required for the correlations and determine any missing design information. The distillation column can be costed as a combination of a vertical pressure vessel and internals. For both pressure vessels we need to know the wall thickness. The details of how to calculate vessel wall thickness in accordance with the ASME Boiler and Pressure Vessel Code are given in Section 14.5, and the equation to use is Equation 14.13.

The design pressure of the vessels should be 10% above the operating pressure (see Chapter 14), so the design pressure is 11 bar, or roughly 1.1×10^6 N/m². The maximum allowable stress for type 304 stainless steel at 500 °F (260 °C) is 12.9 ksi or roughly 89 N/mm² (see Table 14.2). Assuming the welds will be fully radiographed, the weld efficiency is 1.0. Substituting in Equation 14.13 for the column wall thickness, t_w , then gives:

$$t_w = \frac{1.1 \times 10^6 \times 3}{(2 \times 89 \times 10^6 \times 1.0) - (1.2 \times 1.1 \times 10^6)} \quad (14.13)$$

$$= 0.0187 \text{ m, say } 20 \text{ mm}$$

We can now calculate the shell mass, using the density of 304 stainless steel (= 8000 kg/m³, from Table 6.2).

$$\text{Shell mass} = \pi D_c L_c t_w \rho$$

where: D_c = vessel diameter, m

L_c = vessel length, m

t_w = wall thickness, m

ρ = metal density, kg/m³

So the shell mass for the distillation column is:

$$\text{Shell mass} = \pi \times 3.0 \times 30 \times 0.02 \times 8000 = 46685 \text{ kg}$$

For the horizontal pressure vessel we need to convert the volume into a length and diameter. Assuming that the vessel is a cylinder with $L_c = 2D_c$ then we can follow the same method as for the column and find $t_w = 8$ mm and shell mass = 636 kg.

Using the correlations in Table 7.2, we obtain the following purchased costs for the stainless steel pressure vessels:

$$\text{Distillation column shell, cost} = 17,400 + 79 (46685)^{0.85} = \$753,000$$

$$\text{Horizontal pressure vessel, cost} = 12,800 + 73 (636)^{0.85} = \$30,400$$

For the remaining equipment we obtain the following purchased costs from the correlations in Table 7.2 based on carbon steel construction:

Distillation column trays, cost per tray = $130 + 440(3.0)^{1.8}$	= \$3310
Cost for 50 trays	= \$165,500
U – tube heat exchanger, cost = $28,000 + 54(60)^{1.2}$	= \$35,300
Kettle reboiler, cost = $29,000 + 400(110)^{0.9}$	= \$56,500
Storage tank (conical head), cost = $5800 + 1600(50)^{0.7}$	= \$30,500
Centrifugal pump, $3.6 \text{ m}^3/\text{h} = 1 \text{ L/s}$, SO:	
cost each = $8000 + 240(1.0)^{0.9} = \8240 , cost for two pumps	= \$16,480
driver (electric motor) cost each = $-1100 + 2,100(0.5)^{0.6}$	= \$285
cost for two drivers	= \$570
Centrifugal pump, $2.5 \text{ m}^3/\text{h} = 0.694 \text{ L/s}$, so	
cost each = $8000 + 240(0.694)^{0.9} = \8170 , cost for three	= \$24,520
driver (electric motor) cost each = $-1100 + 2100(1.0)^{0.6}$	= \$1000
cost for three drivers	= \$3000

Note that the pumps and drivers are at the lower end of the range of validity of the cost correlations and the driver cost for the smaller motors looks suspect; however, the pump and motor costs are small compared with the other costs, and the error introduced is therefore negligible given the overall accuracy of $\pm 30\%$.

Following Hand's method, the installed cost of the distillation column is then:

$$C = 4 \times 753,000 = \$3,012,000$$

The cost of the trays can be converted to type 304 stainless steel by multiplying by the appropriate materials factor from Table 7.6, giving:

$$C = 1.3 \times 165,500 = \$215,150$$

This then gives a total cost for the column plus internals of $3,012,000 + 215,150 = \$3,227,150$.

The installed cost of the horizontal pressure vessel is $4 \times 30,400 = \$121,600$.

The installed cost for the exchangers and storage tank in carbon steel construction is:

$$C = 3.5(35,300 + 56,500) + 2.5(30,500) = \$397,550$$

so the cost in type 304 stainless steel is $1.3 \times 397,550 = \$516,800$.

For the pumps, we need to add the cost of the pump and driver before determining the installed cost. Only the cost of the pump needs to be converted to stainless steel. For the first set of pumps:

$$C = 4 \times (570 + (1.3 \times 16,480)) = \$88,000$$

For the second set of pumps only two are installed (the other is a warehouse spare), so the total installed cost is:

$$C = (1.3 \times 8170) + 1000 + (4 \times 2 \times (1000 + (1.3 \times 8170))) = \$105,000$$

The total installed ISBL cost of the plant is then:

$$C = 3,227,150 + 121,600 + 516,800 + 88,000 + 105,000 = \$4,058,550$$

or \$4.1 MM $\pm 30\%$ within the accuracy of the method.

If instead we use the factors given in Table 7.5, then using Equation 7.12, the installed cost for the exchangers, tank, and pumps is equal to:

$$C = (35,300 + 56,500 + 30,500 + 16,480 + 8170)[(1 + 0.8) \times 1.3 + (0.3 + 0.3 + 0.2 + 0.3 + 0.2 + 0.1)]$$

$$C = (155,120)[3.74] = \$580,150$$

The installed cost for the pump drivers (which do not require a materials conversion factor) is:

$$C = (1140 + 6,000)[1 + 0.8 + 0.3 + 0.3 + 0.2 + 0.3 + 0.2 + 0.1]$$

$$C = (7140)[3.2] = \$22,900$$

The installed cost for the pressure vessels can be found using Equation 7.13:

$$C = (753,000 + 30,400)[1 + 0.8 + (0.3 + 0.3 + 0.2 + 0.3 + 0.2 + 0.1) / 1.3]$$

$$C = (783,400)[2.88] = \$2,256,200$$

In addition to this, we require the cost of the trays in stainless steel and the cost of the spare pump and driver:

$$C = 1000 + 1.3(165,500 + 8170) = \$226,800$$

The total installed ISBL cost of the plant is then:

$$C = 580,150 + 22,900 + 2,256,200 + 226,800 = \$3,086,050$$

or \$3.1 MM ± 30% within the accuracy of the method.

Note that although the answers obtained by the two methods are different, the second answer is within the range of accuracy of the first, and the first is very close to the upper end of the range predicted by the second method. Both estimates should be stated as being on a U.S. Gulf Coast basis, January 2010, as this is the basis for the correlations in Table 7.2.

7.7 Cost escalation

All cost estimating methods use historical data and are themselves forecasts of future costs. The prices of the materials of construction and the costs of labor are subject to inflation. Some method has to be used to update old cost data for use in estimating at the design stage and to forecast the future construction cost of the plant.

The method usually used to update historical cost data makes use of published cost indices. These relate present costs to past costs and are based on data for labor, material, and energy costs published in government statistical digests.

$$\text{Cost in year A} = \text{Cost in year B} \times \frac{\text{Cost index in year A}}{\text{Cost index in year B}} \quad (7.14)$$

To get the best estimate, each job should be broken down into its components, and separate indices should be used for labor and materials. It is often more convenient to use the composite indices published for various industries in the trade journals. These are weighted average indices combining the various components of costs in proportions considered typical for the particular industry.

A composite index for the U.S. process plant industry is published monthly in the journal *Chemical Engineering*; this is the Chemical Engineering Plant Cost Index (CEPCI), often referred to as the *CE index*. *Chemical Engineering* also used to publish the Marshall and Swift index (M&S equipment cost index), but that index has been discontinued since 2010.

For oil refinery and petrochemicals projects, the *Oil and Gas Journal* used to publish the Nelson-Farrar Refinery Construction Index (NF index). The index is now available as a subscription service from <https://www.bakerrisk.com/products/nelson-farrar-cost-index/>. The index is updated monthly, and indices for 40 types of equipment are updated quarterly. The Nelson-Farrar index is on a U.S. Gulf Coast basis rather than U.S. average, and is more reliable than the CE index for the types of equipment used in hydrocarbon processing.

The journal *Engineering News Record* publishes a monthly construction cost index. This is based on civil engineering projects and is sometimes used for updating offsite costs. This index has been published since 1904 and is the oldest of all the indices.

For international projects, the journal *Process Engineering* publishes monthly cost indices for several countries, including the United States, United Kingdom, Japan, Australia, and many of the EU countries.

All cost indices should be used with caution and judgment. They do not necessarily relate the true make-up of costs for any particular piece of equipment or plant, nor the effect of supply and demand on prices. The longer

the period over which the correlation is made, the more unreliable the estimate. Between 1970 and 1990 prices rose dramatically. Prices then grew at a more or less steady 2% to 3% per year until 2003, when high demand for fuels projects and high energy prices caused another period of steeper price inflation. Prices then decreased during the 2008–2010 recession before another period of growth driven by major projects associated with the U.S. shale gas expansion and downstream petrochemical investments. At time of writing, the indices have not yet shown any impact of the 2020 COVID-19 pandemic and resulting economic recession and oil price crash; however, it is likely that there will be a substantial decrease in the indices in 2020 and that recovery to 2019 levels will not occur until there is renewed capital investment in the fuels and chemicals sector. The major cost indices are plotted in Fig. 7.2. Fig. 7.3 shows the same data plotted relative to the 1990 value of each index. Fig. 7.3 shows that the NF index is usually a leading indicator compared with the M&S and CE indices, as fuels-sector activity has been a big component of price inflation from 2000 onwards.

To estimate the future cost of a plant, some prediction has to be made of the future annual rate of cost inflation. This can be based on the extrapolation of one of the published indices, tempered by the engineer’s own assessment of what the future may hold. Inflation is difficult to forecast, and allowance for inflation is often included in the contingency charges added to the project cost.

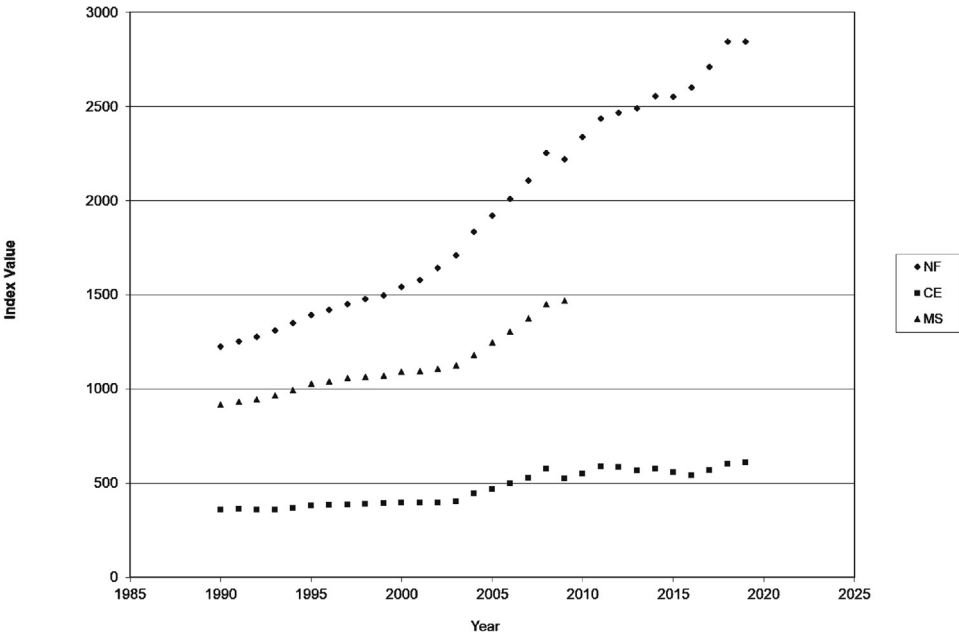


FIG. 7.2 Variation of major cost indices.

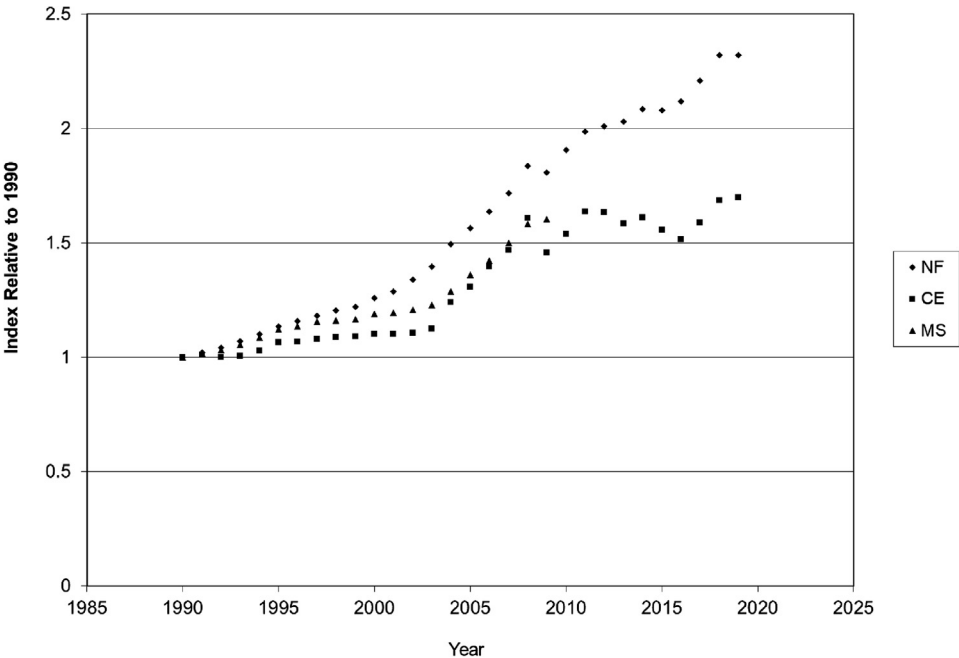


FIG. 7.3 Variation of major cost indices relative to 1990 = 1.0.

Example 7.4

The purchased cost of a shell and tube heat exchanger, carbon shell, 316 stainless steel tubes, heat transfer area 500 m², was \$64,000 in January 2003; estimate the cost in January 2020. Use the CEPCI Equipment Cost Index.

Solution

From Fig. 7.2 (or by looking up the index in *Chemical Engineering*):

Index in 2003 = 402

Index in January 2020 = 596

So, estimated cost in January 2020 = $\$64,000 \times 596 / 402 = \underline{\underline{\$95,000}}$

Example 7.5

The purchased cost of a distillation column was \$136,000 in 2004. Estimate the cost in 2024. Use the Nelson-Farrer index.

Solution

From Fig. 7.2 (or by looking up the index in *Oil and Gas Journal*):

Index in 2004 = 1833.6

The index in 2024 is difficult to extrapolate numerically, as it is not yet clear how deep or long the 2020 recession will be, nor whether we will see another sharp increase afterwards, as experienced in 2015–2018. Looking at Fig. 7.2, it seems likely that the NF index will have a value in the range of 3000 to 3200. A conservative approach would be to make a higher estimate, so let us assume 3200.

So, estimated cost in January 2022 = $\$136,000 \times 3200 / 1833.6 = \underline{\underline{\$237,000}}$

7.8 Location factors

Most plant and equipment cost data are given on a U.S. Gulf Coast (USGC) or northwest Europe (NWE) basis, as these were historically the main centers of the chemical industry, for which the most data were available. The cost of building a plant in any other location will depend on:

- Local fabrication and construction infrastructure
- Local labor availability and cost
- Costs of shipping or transporting equipment to site
- Import duties or other local tariffs
- Currency exchange rates, which affect the relative cost of locally purchased items such as bulk materials, when converted to a conventional pricing basis such as U.S. dollars

These differences are captured in cost estimating by using a location factor:

$$\text{Cost of plant in location A} = \text{cost of plant on USGC} \times LF_A \quad (7.15)$$

where: LF_A = location factor for location A relative to USGC basis

Location factors for international locations are a strong function of currency exchange rates and hence fluctuate with time. Cran (1976a,b), Bridgewater (1979), Soloman (1990), and Gerrard (2000) give location factors for international locations from which this variation can be seen. It can be argued that as a result of globalization, all international installation factors are trending closer to 1.0 (Gerrard, 2000). Location factors within a country are somewhat

TABLE 7.7 Location factors

Country	Region	Location factor
United States	Gulf Coast	1.00
	East Coast	1.04
	West Coast	1.07
	Midwest	1.02
Canada	Ontario	1.00
	Fort McMurray	1.60
Mexico		1.03
Brazil		1.14
China	imported	1.12
	indigenous	0.61
Japan		1.26
Southeast Asia		1.12
Australia		1.21
India		1.02
Middle East		1.07
France		1.13
Germany		1.11
Italy		1.14
Netherlands		1.19
Russia		1.53
United Kingdom		1.02

easier to predict, and Bridgewater (1979) suggested a simple rule of thumb: add 10% for every 1000 miles from the nearest major industrial center.

Table 7.7 gives example location factors. These are based on data from Aspen Richardson's *International Construction Cost Factor Location Manual* (2003). More recent versions of this manual can be found by searching for Richardson Engineering Services at www.aspentech.com. The values in Table 7.7 give costs on a local basis in U.S. dollars. The location factors in Table 7.7 are based on 2003 data and can be updated by dividing by the ratio U.S. dollar/local currency in 2003 and multiplying by the ratio U.S. dollar/local currency in the year of interest. If a cost estimate for a future year is being made, then currency variation will have to be forecasted.

Example 7.6

The cost of constructing a 30,000 metric tons per year (30 kMTA) acrolein plant was estimated as \$80 million (\$80MM) on a 2006 U.S. Gulf Coast basis. What would be the cost in U.S. dollars on a 2006 Germany basis?

Solution

From Table 7.7, the 2003 location factor for Germany was 1.11.

The exchange rate in 2003 averaged about €1 = \$1.15, and in 2006 it averaged about €1 = \$1.35.

The 2006 location factor for Germany is thus $1.11 \times 1.35/1.15 = 1.30$

The cost of building the acrolein plant in Germany in 2006 is $\$80 \text{ MM} \times 1.30 = \underline{\underline{\$104 \text{ MM}}}$

7.9 Estimating off-site capital costs

Improvements to the site infrastructure are almost always needed when a new plant is added to a site or a major expansion is carried out. The cost of such improvements is known as the *off-site or OSBL investment*, as described in Section 7.2.1.

In the early stages of designing a new process, the off-site requirements are usually not precisely known, and an allowance for off-site costs can be made by assuming that they will be a ratio of the ISBL investment. A typical number is 30% to 50% of ISBL investment, depending on the process and site conditions. Table 7.8 gives some guidelines for making approximate estimates of off-site capital costs as a function of plant complexity and site conditions.

TABLE 7.8 Guidelines for estimating approximate OSBL costs as a percentage of ISBL cost

Process complexity	Site condition		
	Existing: underused	Existing: tight capacity	New site
Typical large-volume chemical	30%	40%	40%
Low-volume specialty chemical	20%	40%	50%
High solids-handling requirement	40%	50%	100%

As the design details are established and the requirements for utilities such as steam, electricity, and cooling water are determined, the site requirements can also be determined. If there is insufficient spare capacity in the existing site infrastructure, potential modifications can be designed to accommodate the new plant. For existing sites, the design engineer must always beware of scope creep, in which the new project is used to justify all kinds of (possibly long overdue) improvements to the site infrastructure. If scope creep occurs, excessive off-site costs may make the project economics unattractive.

Many of the off-site items are designed as “packaged” plants or systems that are purchased from specialized suppliers. In some cases, the supplier may even offer an *over-the fence* contract, in which the supplier builds, owns, and operates the off-site plant and contracts to supply the site with the desired utility stream or service. Over-the-fence contracts are widely used for industrial gases such as nitrogen, oxygen, and hydrogen, and most plants also import electricity from the local utility company. Over-the-fence contracts for steam, cooling water, and effluent treatment are less common, but are sometimes used in smaller plants or where several companies share a site.

The question of whether to build a self-contained infrastructure for a plant or contract for off-site services is an example of a *make-or-buy* problem. The over-the-fence price will usually be higher than the cost of producing the utility or service internally, because the supplier needs to make a profit and recover their capital investment. On the other hand, contracting for the service reduces the project capital investment and fixed costs, because the supplier must take on the costs of labor, maintenance, and overheads. The make-or-buy decision is usually made by comparing annualized costs, as described in Section 9.7.2. Correlations for costs of utility plants and other off-sites are given in the sources listed in Section 7.5.

7.10 Computer tools for cost estimating

It is difficult for engineers outside the EPC sector to collect recent cost data from a large set of real projects and maintain accurate and up-to-date cost correlations. Instead, the most common method for making preliminary estimates in industry is to use commercial cost estimating software.

A wide variety of cost estimating programs is available. These include ACCE (Aspen Technology, Inc.), Cleopatra Enterprise (Cost Engineering Consultancy), CostLink/CM (Building Systems Design, Inc.), Cost Track (OnTrack Engineering Ltd.), PRISM Project Estimator (ARES Corp.), Success Estimator (U.S. Cost), Visual Estimator (CPR International Inc.), WinEst (Win Estimator), and others that can be found by searching on the Web or looking at the listings provided by AACE International at www.aacei.org. The discussion in this section will focus on the most widely used programs: Aspen Technology’s ACCE software and Cost Engineering Consultancy’s Cleopatra Enterprise. Both of these programs have academic licenses available for university use and are also available in most chemical companies.

The ACCE and Cleopatra cost estimating tools are simple to use and give quick, defensible estimates without requiring a lot of design data. Design information can be uploaded from major flowsheet simulation programs or entered manually. The programs allow the design to be updated as more information on design details becomes available, so that a more accurate estimate can be developed. Costs can be estimated for a whole plant or for one piece of equipment at a time. Both programs include a large number of equipment types, and these can be designed in a broad range of materials, including U.S., UK, German, and Japanese standard alloys.

Both ACCE and Cleopatra use a combination of mathematical models and expert systems to develop cost estimates. Costs are based on the materials and labor required (following the practice used for detailed estimates) rather than installation factors. If design parameters are not specified by the user, then they are calculated or set to default values by the program. The user should always review the design details carefully to make sure that the default values make sense for the application. If any values are not acceptable, they can be manually adjusted and a more realistic estimate can be generated.

A detailed description of how to run ACCE or Cleopatra is beyond the scope of this book. Both programs have user manuals available from the licensor or as .pdf files downloadable from the “help” button in the program, see, for example, [AspenTech \(2002a,b\)](#). Some of the common issues that arise in using these programs are discussed next. These or similar problems are also faced when using other cost estimating software.

7.10.1 Mapping simulation data

Instructions on loading data from a process simulation are given in the ACCE User’s Guide ([AspenTech, 2002a](#)). When a simulator report file is loaded, ACCE generates a block-flow diagram with each unit operation of the simulation shown as a block. These blocks must then be “mapped” to ICARUS project components (pieces of equipment or bulk items).

Unless the user specifies otherwise, each simulator block is mapped to a default ICARUS project component. The mapping defaults need to be understood properly, as large errors can be introduced if unit operations are mapped incorrectly. The default mapping specifications are given in section 3 of the user’s guide ([AspenTech, 2002a](#)). Some mappings that commonly cause problems include:

1. Reactors: Plug-flow reactor models (PLUG in Hysys and ProII, RPLUG in AspenPlus) are mapped to a packed tower, which is fine for fixed-bed catalytic reactors, but not for other types of plug-flow reactors. All other reactor models (Gibbs, stoichiometric, equilibrium, and yield) are mapped to agitated tank reactors. Reactors that are not suitable for these mappings can be mapped to other ICARUS project components or set up as user models (see later).
2. Heaters, coolers, and heat exchangers: The default mapping for all heat transfer equipment is the floating head heat exchanger. ICARUS contains several different heat exchanger types, including a generic TEMA heat exchanger that can be customized to the other types, as well as fired heater and air cooler components. It is often worthwhile to change the default mapping to the TEMA exchanger to allow the exchangers to be customized in ICARUS.
3. Distillation columns: The simulator column models include not just the column itself but also the reboiler, condenser, overhead receiver drum, and reflux pump (but not bottoms pump). ICARUS has 10 possible configurations to which a column can be mapped. Alternatively, the column can be mapped to a packed or trayed tower, and the ancillary items can be created as separate ICARUS project components.
4. Dummy items: Process simulations often contain models of items that are not actual plant equipment (see [Chapter 4](#)). For example, heat exchangers are sometimes modeled as a series of heaters and coolers linked by a calculator block as a means of checking for internal pinch points or allowing for heat losses to the ambient environment. When the simulation is mapped into ICARUS, dummy items should be excluded from the mapping process. In the previous example, only the heaters should be mapped, so as to avoid double counting the heat transfer area.

The default mapping can be edited by right-clicking on “Project Component Map Specifications” in the Project Basis/Process Design folder. A simulator model can be excluded from the mapping by selecting the item and then selecting “Delete All Mappings.” New mappings can be specified by selecting a simulator item and adding a new mapping.

To map loaded simulator data, click the map button on the toolbar (which maps all items) or right-click on an area or plant item in the process view window (which allows items to be mapped individually). If individual items are selected, then the user is given an option to use simulator data to override the default mapping in the Component

Map Specs file. This is useful for heat exchangers and other equipment where the simulator allows the equipment type to be specified.

The procedure for mapping equipment in Cleopatra is somewhat simpler and is illustrated in Example 7.7.

7.10.2 Design factors in ACCE

All good designs include an appropriate degree of overdesign to allow for uncertainties in the design data and method; see Section 1.6. For some equipment, the design factor or margin is specified by design codes and standards, for example, in the design of pressure vessels, as described in [Chapter 14](#). In other cases, the design engineer must specify the degree of overdesign or margin based on experience, judgment, or company policy.

The equipment sizes calculated by a process simulator will be at the design flow rate unless a higher throughput was specified by the user, and hence include no design margin. The ACCE software adds an “equipment design allowance” to the equipment cost to allow for the design factor that will be introduced when the equipment is designed in detail. The equipment design allowance is based on the process description as follows:

New and unproven process	15%
New process	10%
Redesigned process	7%
Licensed process	5%
Proven process	3%

The process description is entered by right-clicking on “General Specs” in the Project Basis/Basis for Capital Costs folder.

The equipment design allowance is only applied to system-developed costs. If different design margins are needed for different equipment types, then the default should be set to “proven process,” and the equipment can then be oversized appropriately. Design margins can also be added to components using the ACCE custom model tool. Care should be taken to avoid adding more design margin than is necessary.

7.10.3 Pressure vessels

When costing pressure vessels such as reactors and distillation columns, care must be taken to ensure that the wall thickness is adequate. The default method in both ACCE and Cleopatra calculates the wall thickness required based on the ASME Boiler and Pressure Vessel Code Section VIII Division 1 method for the case where the wall thickness is governed by containment of internal pressure (see [Chapter 14](#) for details of this method). If other loads govern the design, then the ACCE software can significantly underestimate the vessel cost. This is particularly important for vessels that operate at pressures below 5 bara, where the required wall thickness is likely to be influenced by dead weight loads and bending moments from the vessel supports, and for tall vessels such as distillation columns and large packed-bed reactors, where combined loading under wind loads may govern the thickness. Similarly, if the vessel is designed under a different section of the Boiler and Pressure Vessel Code, which is usually the case for vessels operated at high pressures, then ACCE can overestimate the vessel cost. It is important to always remember to enter the design pressure and temperature of the vessel, not the operating pressure and temperature.

The best approach to costing pressure vessels using commercial costing software is to enter all of the dimensions after completing the mechanical design of the vessel using the methods given in [Chapter 14](#), or using suitable pressure vessel design software.

7.10.4 Nonstandard components in ACCE

Although ACCE contains over 250 equipment types, many processes require equipment that is not on the list of available project components. Also, in some cases the user will want to specify a certain make or model of equipment that may only be available in discrete sizes (for example, gas turbine engines or large pumps and compressors). In these situations, the nonstandard equipment can be included by setting up an Equipment Model Library (EML). Many companies maintain standard EMLs listing equipment that they often specify.

A new EML can be created by selecting the “Libraries” tab in the palette and opening the folder Cost Libraries/Equipment Model Library. Right-clicking on either of the subfolders then allows the user to create a new EML in the appropriate set of units. Once an EML has been created, equipment items can be added to it. When a new item is added, a dialog box opens in which the user has to specify the sizing or costing method (linear, log-log, semi-log or discrete) and primary sizing parameters. Two costs and sizes must also be entered to establish the cost correlation.

Equipment model libraries are useful for completing an ACCE model of a process that contains nonstandard items. Care must be taken to update the EML costs so that they remain current.

Example 7.7

Estimate the cost of a waste heat boiler designed to produce 4000 lb/h of steam. The exchanger area has been estimated as 1300 ft².

Solution in ACCE

Starting from the ACCE project explorer window (on the far left of the screen), right-click on the Main Area and select Add Project Component (Fig. 7.4a).

Select Process Equipment, then Heat Exchangers (Fig. 7.4b). Select Waste Heat Boiler and enter a name (Fig. 7.4c).

Enter the size parameters and then click the Evaluate button (Fig. 7.4d). This runs the evaluator program and gives the results screen shown in Fig. 7.4e. The purchased equipment cost is \$145,900 on a Jan. 2006 USGC basis. The installed cost is \$196,225. Note that the installed cost is calculated directly by estimating bulk materials and labor rather than using an installation factor.

Solution in Cleopatra Enterprise

Starting from the Database Explorer (top-right menu), create a new document for the estimate (Fig. 7.4f, label A). Select the Estimating tab and then select the Equipment Knowledgebase tab, and in the Level tab select Unit-rates (Fig. 7.4f, label B). Then work through the drop-down menus to select Process Equipment / Special packaged mechanical equipment / Boilers / Steam boiler / Water tube packaged boiler (Fig. 7.4f, label C).

Clicking on the waste heat boiler then opens the menu for that item (Fig. 7.4g) and allows the user to add it to the shopping cart (label A) and open the data entry window by clicking the pyramid plus icon (label B). The equipment data can then be entered (Fig. 7.4h). The flow rate of 4000 lb/h corresponds to a duty of roughly 1220 kW, and we can select 40 bar as the operating pressure (a reasonable value for high-pressure steam). Cleopatra Enterprise then returns an equipment cost of \$119,000 on a Jan. 2018 U.S. basis, and we can add this into an overall estimate by clicking the pyramid button at the bottom right to give the results page shown in Fig. 7.4i.

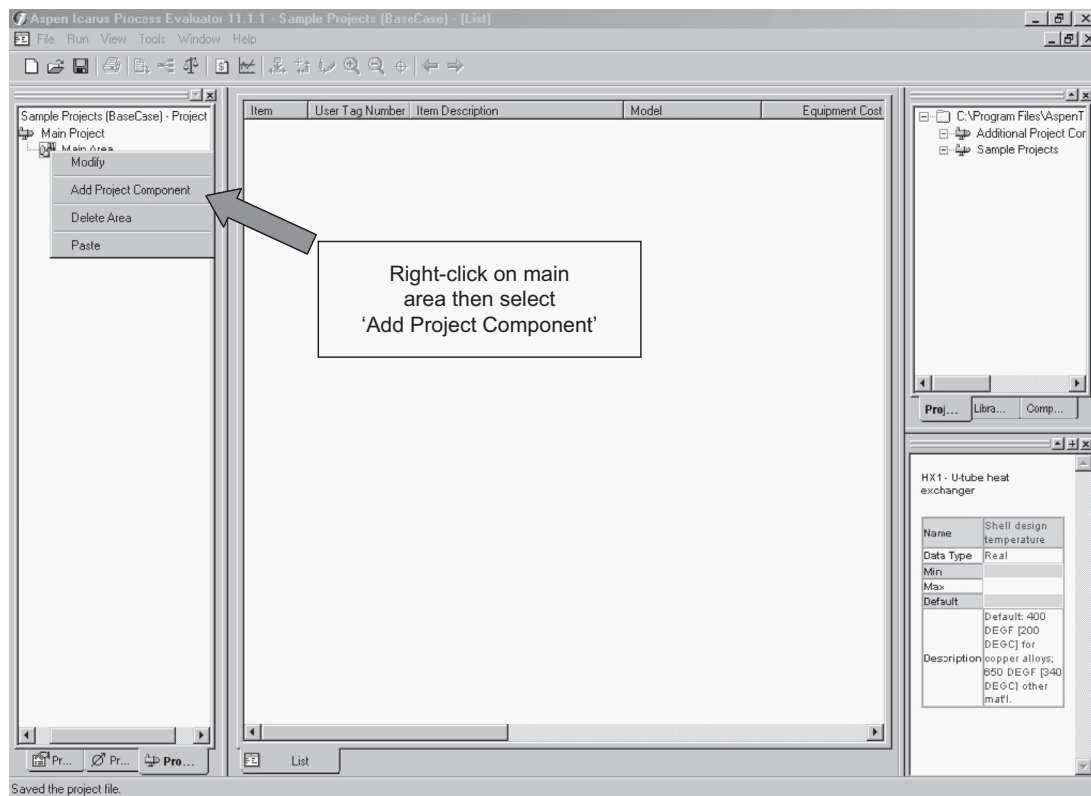
The cost generated by Cleopatra Enterprise is a bare-module cost for the equipment. We would need to enter more details about the plant layout and associated instrumentation and piping to generate an installed cost (or use a suitable Hand factor).

Note that the two programs do not give identical answers. This is a common occurrence when using costing software, as the different programs are based on different data sets. An experienced cost engineer will benchmark the results from the software against recent cost data for equipment they have purchased and tune the costing models to more accurately fit their experience of local costs. Tuning of the cost models is beyond the scope of this book, but is typically covered in the software user manuals.

7.11 Validity of cost estimates

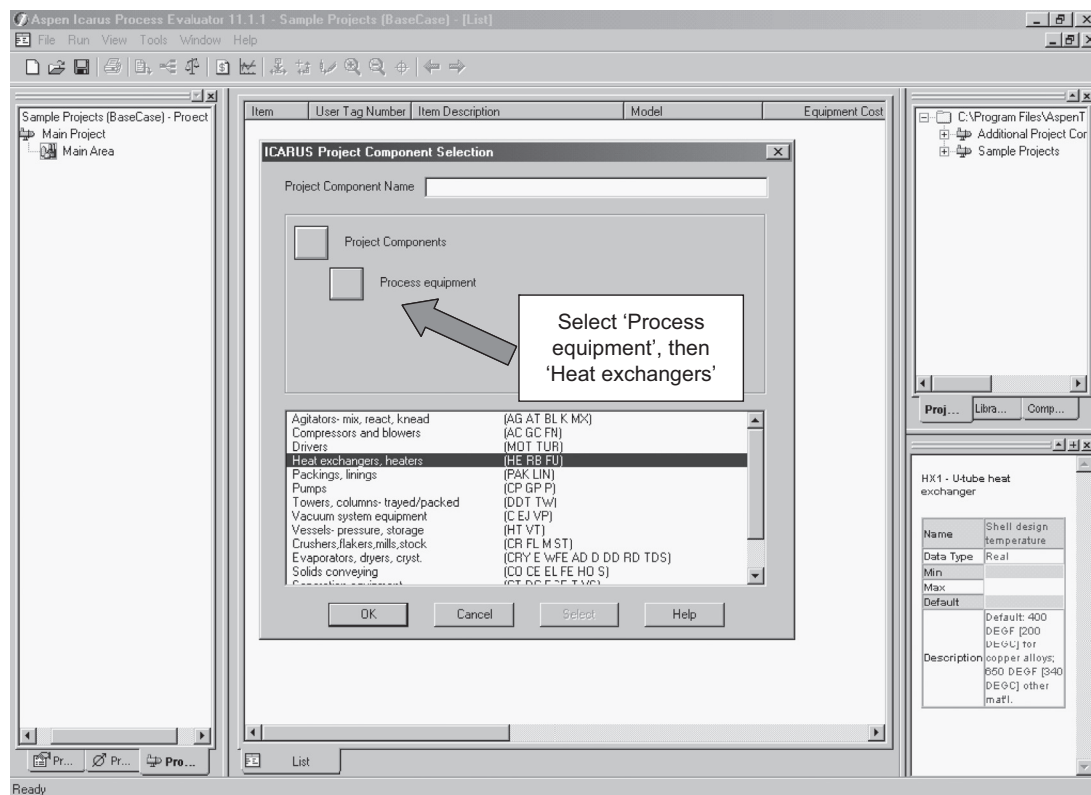
It should always be remembered that cost estimates are only estimates and are subject to error. An estimate should always indicate the margin of error. The error in a cost estimate is primarily determined by the degree of design detail that is available, and even a skilled estimator cannot estimate an accurate cost for an incomplete design.

When more design information has been developed, a professional cost engineer will be able to develop a more accurate estimate. The process design engineer should compare this estimate with the preliminary estimate to gain a better understanding of where the preliminary estimate could have been improved (either through capturing missing plant items or using better costing methods). This will help the design engineer to produce better preliminary estimates in the future.



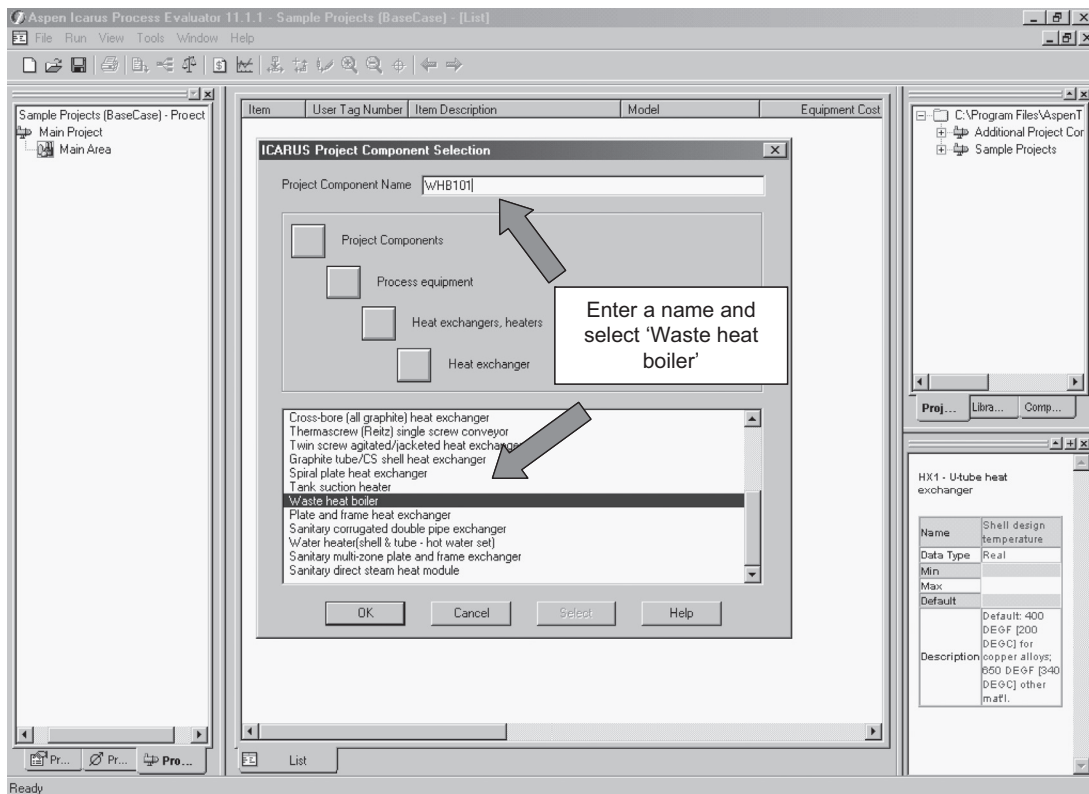
(a)

FIG. 7.4 (a–e) ACCE example. (f–i) Cleopatra Enterprise example.



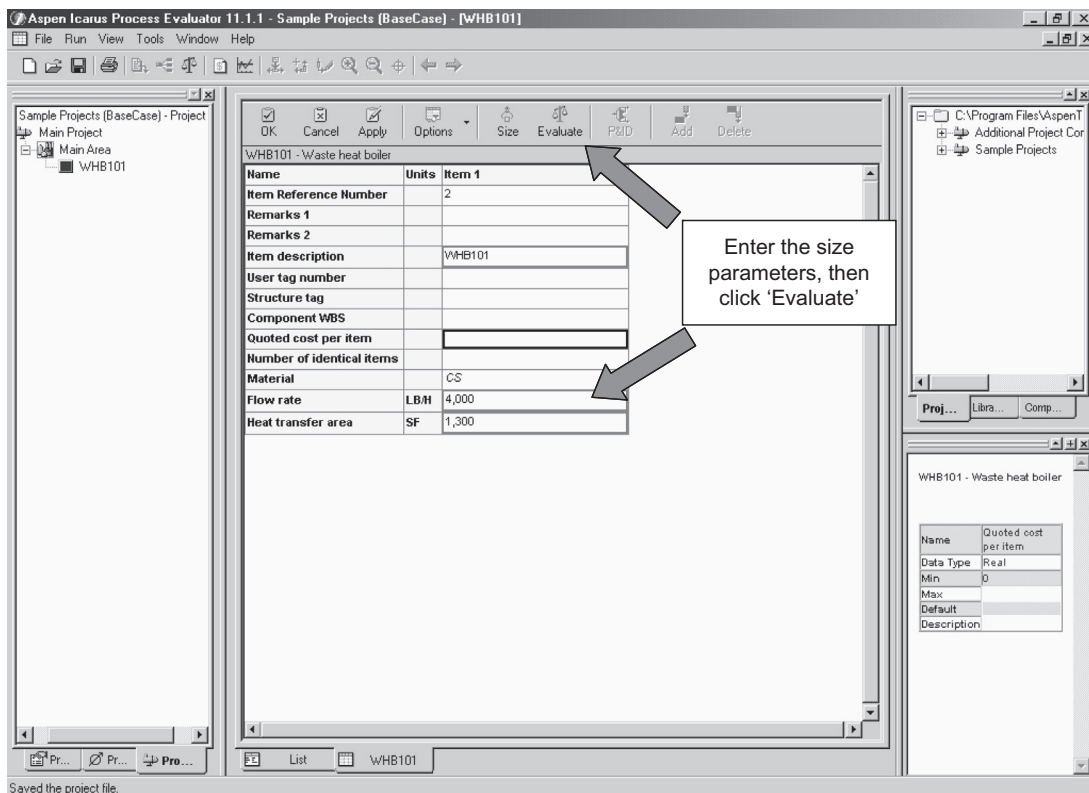
(b)

FIG. 7.4 Cont'd.



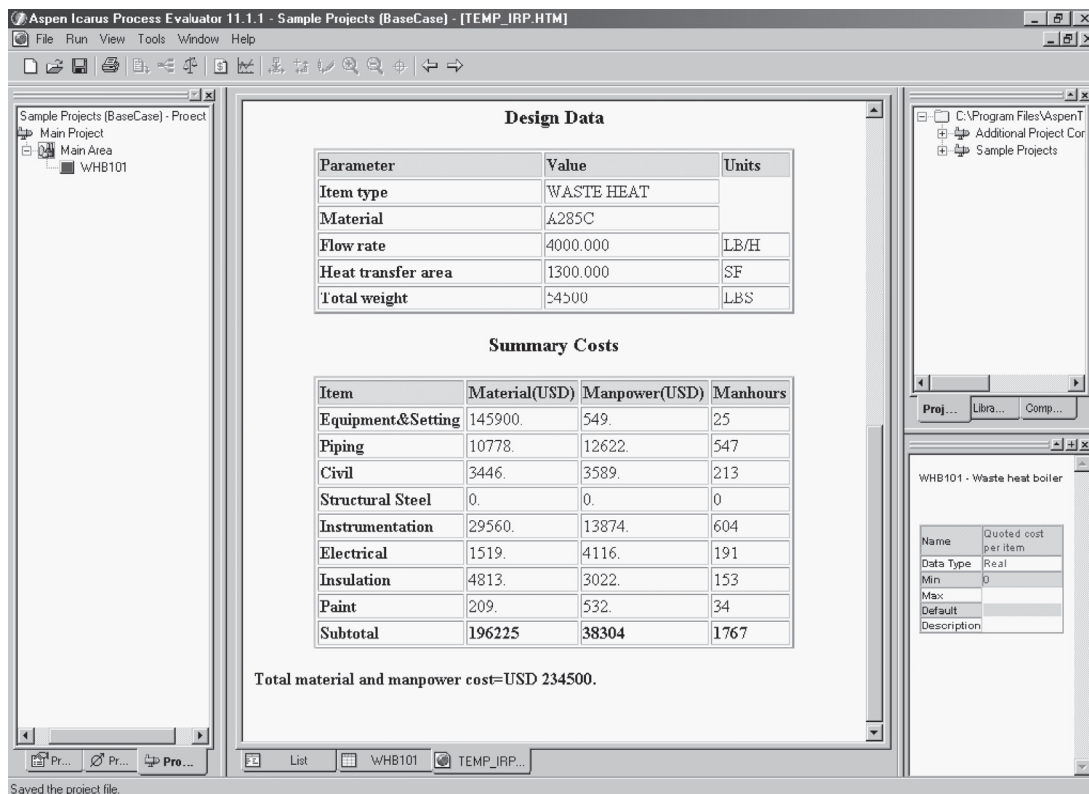
(c)

FIG. 7.4 Cont'd.



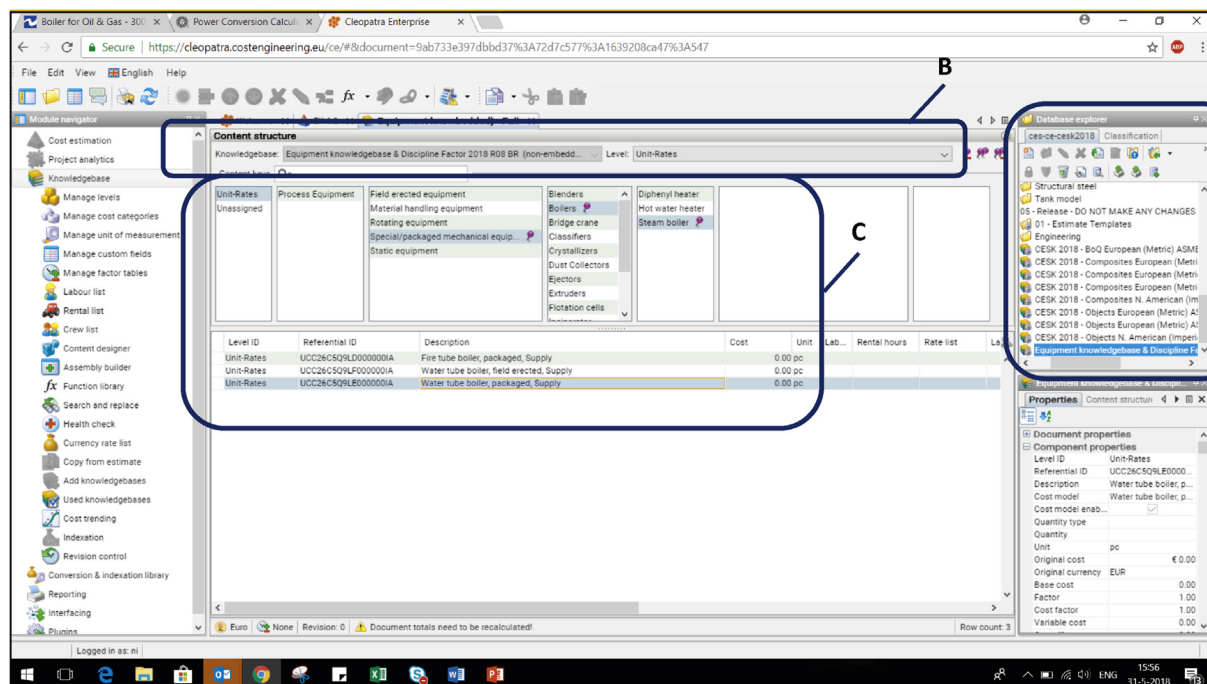
(d)

FIG. 7.4 Cont'd.



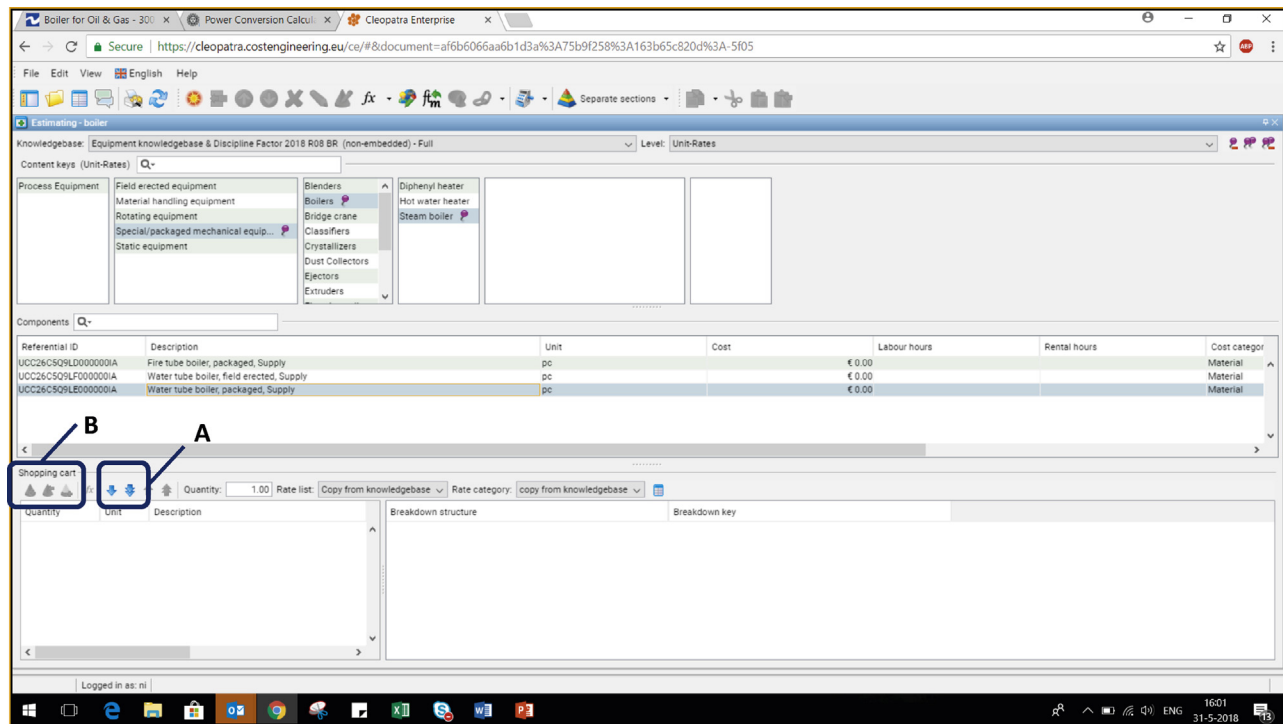
(e)

FIG. 7.4 Cont'd.



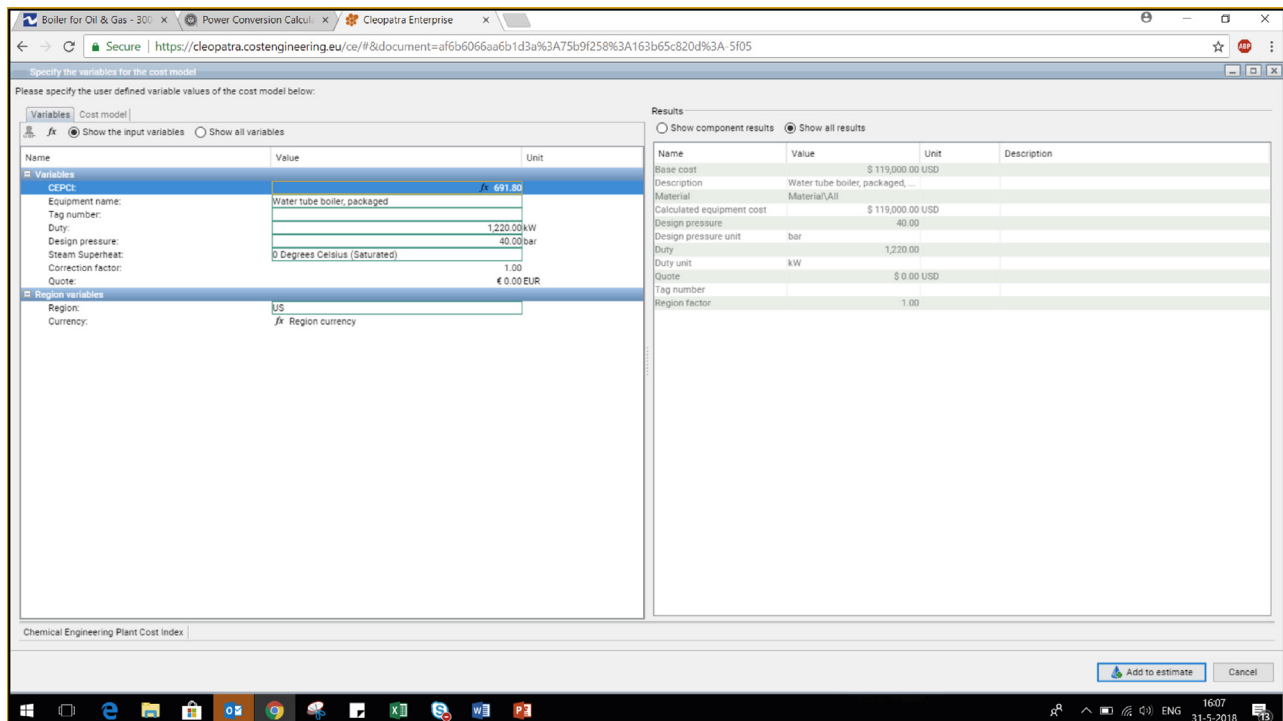
(f)

FIG. 7.4 Cont'd.



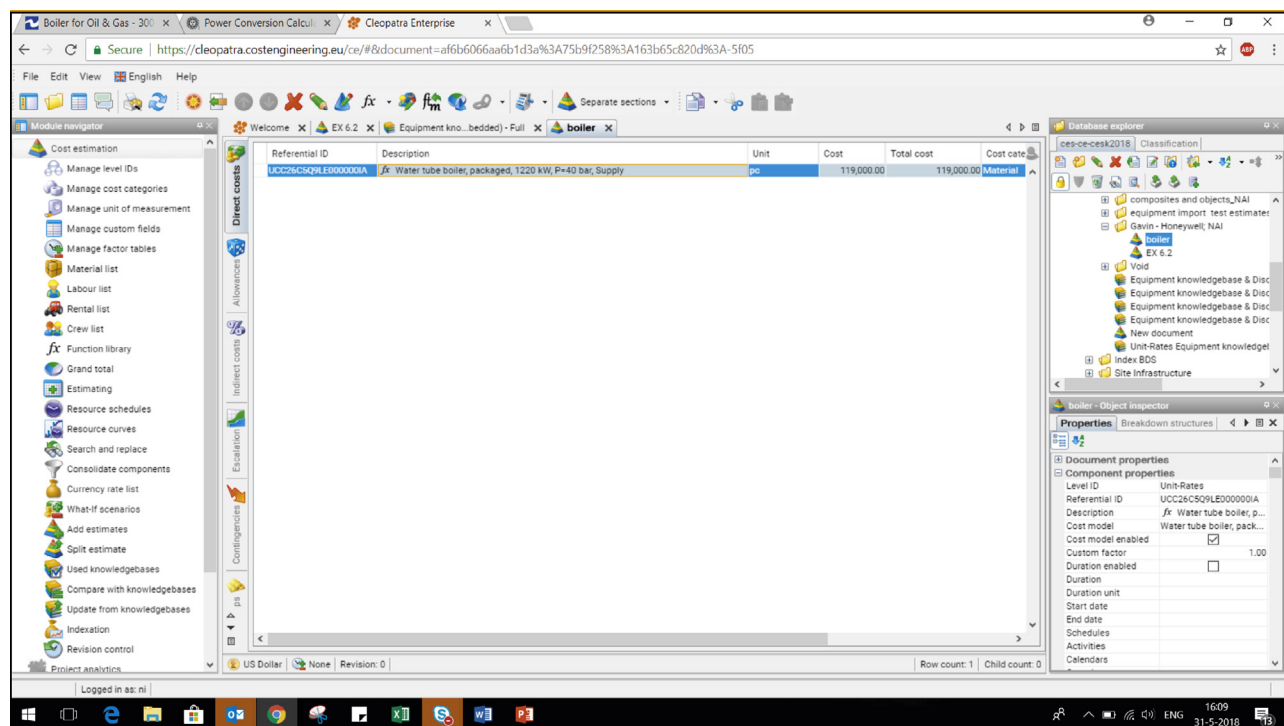
(g)

FIG. 7.4 Cont'd.



(h)

FIG. 7.4 Cont'd.



(i)

FIG. 7.4 Cont'd.

Additional resources for cost estimating are available from the various cost estimating associations: the AACE International (www.aacei.org); the Project Management Institute (www.pmi.org); the UK Association of Cost Engineers (www.acoste.org.uk); and the International Cost Engineering Council (ICEC) (www.icoste.org). The ICEC website has links to cost engineering societies in 46 countries.

Example 7.8

Adipic acid is used in the manufacture of nylon 6,6. It is made by hydrogenation of phenol to a mixture of cyclohexanol and cyclohexanone (known as *KA oil*—ketone and alcohol), followed by oxidation with nitric acid. Estimate the fixed capital cost for a 400,000 metric ton per year (400 kMTA) adipic acid plant located in northeast Asia in 2020.

Solution

The capital cost of the process can be estimated based on historic data using the correlation given in Table 7.1. The correlation is based on the plant capacity in MMlb/y, so we need to convert the capacity: 400 kMTA is equal to 880 MMlb/y:

$$\text{ISBL capital cost} = 3.533 S^{0.6} = 3.533(880)^{0.6} = \$206.5 \text{ MM}$$

The ISBL cost is on a 2006 U.S. Gulf Coast basis, so we need to convert to a 2020 northeast Asia basis. If we look up the location factor in Table 7.7, it is not clear what factor we should use. The location factor for Japan is 1.26, whereas for China it varies from 0.6 to 1.1, depending on the amount of indigenous vs. imported equipment used. Because the exact location of the plant has not yet been specified, we are not able to make a definitive assessment of what the location factor should be. As a first approximation we therefore assume it is 1.0 and note that this should be revisited as part of the sensitivity analysis.

The OSBL capital cost is estimated as 40% of ISBL cost. The engineering cost and contingency are estimated as 10% and 15% of the sum (ISBL + OSBL) cost, respectively, giving a total fixed capital cost on a 2006 basis of \$361.3 MM.

Note that this cost is on a January 2006 basis, as that was the basis of the correlation in Table 7.1. We can update this to a 2020 basis using the CEPCI index. From Fig. 7.2 (or by looking up the index in *Chemical Engineering*):

$$\text{Index in 2006} = 499.6$$

Index for full year 2020 is not yet known, but is likely to be in the range of 570 to 600,

so as a conservative estimate we can assume Index in 2020 = 600.

$$\text{So, estimated total fixed capital cost in 2020} = \$361.3 \text{ MM} \times 600/499.6 = \underline{\underline{\$433.9 \text{ MM}}}.$$

The impact of updating this estimate is further explored in Problems 7.9 and 9.6.

7.12 References

- Aspen Richardson. (2003). *International construction cost factor location manual*. Aspen Technology Inc.
- AspenTech. (2002a). *Aspen ICARUS process evaluator user's guide*. Aspen Technology Inc.
- AspenTech. (2002b). *ICARUS reference: ICARUS evaluation engine (IEE) 30.0*. Aspen Technology Inc.
- Bridgewater, A. V. (1979). International construction cost location factors. *Chemical Engineering*, 86(24), 119.
- Bridgewater, A. V., & Mumford, C. J. (1979). *Waste recycling and pollution control handbook*. George Godwin.
- Cran, J. (1976a). EPE plant cost indices international (1970=100). *Engineering and Process Economics*, 1, 109–112.
- Cran, J. (1976b). EPE plant cost indices international. *Engineering and Process Economics*, 1, 321–323.
- Douglas, J. M. (1988). *Conceptual design of chemical processes*. McGraw-Hill.
- Dysert, L. R. (2015). Chapter 9. In *cost estimating in skills and knowledge of cost engineering* (6th ed.). AACE International. revised.
- Estrup, C. (1972). The history of the six-tenths rule in capital cost estimation. *Brit. Chemical Engineering Proc. Tech*, 17, 213.
- Garrett, D. E. (1989). *Chemical engineering economics*. Van Norstrand Reinhold.
- Gerrard, A. M. (2000). *Guide to capital cost estimation* (4th ed.). Institution of Chemical Engineers.
- Guthrie, K. M. (1969). Capital cost estimating. *Chemical Engineering*, 76(6), 114.
- Guthrie, K. M. (1974). *Process plant estimating, evaluation, and control*. Craftsman Book Co.
- Hand, W. E. (1958). From flow sheet to cost estimate. *Petroleum Refiner*, 37(9), 331.
- Happle, J., & Jordan, D. G. (1975). *Chemical process economics* (2nd ed.). Marcel Dekker.
- Hastak, M. (2015). *Skills and knowledge of cost engineering* (6th ed.). AACE International. revised.
- Humphreys, K. K. (1991). *Jelen's cost and optimization engineering* (3rd ed.). McGraw-Hill.
- Humphreys, K. K. (2005). *Project and cost engineers' handbook* (4th ed.). AACE International.
- Hydrocarbon Processing. (2003). *Petrochemical processes 2003*. Gulf Publishing Co.
- Hydrocarbon Processing. (2004a). *Gas processes 2004*. Gulf Publishing Co.
- Hydrocarbon Processing. (2004b). *Refining processes 2004*. Gulf Publishing Co.
- Lang, H. J. (1948). Simplified approach to preliminary cost estimates. *Chemical Engineering*, 55(6), 112.
- Page, J. S. (1996). *Conceptual cost estimating manual* (2nd ed.). Gulf Publishing Co.
- Peters, M. S., Timmerhaus, K. D., & West, R. E. (2003). *Plant design and economics* (5th ed.). McGraw-Hill.
- Pikulik, A., & Diaz, H. E. (1977). Cost estimating for major process equipment. *Chemical Engineering*, 84(Oct. 10), 106. New York.
- Purohit, G. P. (1983). Estimating the cost of heat exchangers. *Chemical Engineering*, 90(Aug. 22), 56. New York.
- Seider, W. D., Lewin, D. R., Seader, J. D., Widagdo, S., Gani, R., & Ng, K. M. (2016). *Product and process design principles* (4th ed.). Wiley.
- Soloman, G. (1990). Location factors. *The Cost Engineer*, 28(2).
- Turton, R., Shaeiwitz, J. A., Bhattacharyya, D., & Whiting, W. B. (2018). *Analysis, synthesis and design of chemical processes* (5th ed.). Prentice Hall.
- Ulrich, G. D., & Vasudevan, P. T. (2004). *Chemical engineering process design and economics: A practical guide* (2nd ed.). Process Publishing.

7.13 Nomenclature

		Dimensions in \$MLT ⁰
A	Area	L^2
a	Constant	\$
b	Constant	\$
C	Capital cost	\$
C_e	Purchased equipment cost	\$
$C_{e,i,A}$	Purchased cost of equipment i in alloy	\$
$C_{e,i,CS}$	Purchased cost of equipment i in carbon steel	\$
C_1	Capital cost of plant with capacity S_1	\$
C_2	Capital cost of plant with capacity S_2	\$
D_c	Diameter of distillation column	L
D_f	Flange diameter	L
D_n	Nozzle diameter	L

Continued

		Dimensions in \$MLT ⁰
D_s	Shell diameter	L
F	Installation (Lang) factor	—
f_c	Installation factor for civil engineering work	—
f_{el}	Installation factor for electrical work	—
f_{er}	Installation factor for equipment erection	—
f_i	Installation factor for instrumentation and control	—
f_l	Installation factor for lagging, insulation and paint	—
f_m	Materials factor	—
f_p	Installation factor for piping	—
f_s	Installation factor for structures and buildings	—
L_c	Vessel length	L
L_s	Shell length	L
LF_A	Location factor for location A relative to U.S. Gulf Coast basis	—
M	Total number of pieces of equipment	—
N	Number of significant processing steps (functional units)	—
N_b	Number of bolts	—
N_{baf}	Number of baffles	—
N_t	Number of tubes	—
n	Capital cost exponent	—
Q	Plant capacity	MT ⁻¹
S	Plant or equipment capacity	*
S_{lower}	Lower end of capacity range over which correlation is valid	*
S_{upper}	Upper end of capacity range over which correlation is valid	*
S_1	Capacity of plant	1 *
S_2	Capacity of plant	2 *
s	Reactor conversion	—
TCOP	Total cost of production	\$M ⁻¹ or \$T ⁻¹
t	Time, project life in years	T
t_f	Flange thickness	L
t_s	Shell thickness	L
t_w	Vessel wall thickness	L
ρ	Metal density	ML ⁻³

Asterisk (*) indicates that the dimensions are dependent on the type of equipment or process.

Acronyms

AACEI Association for the Advancement of Cost Estimating International
APC Automatic Process Control
CEPCI Chemical Engineering Plant Cost Index, published monthly in *Chemical Engineering*
EPC Engineering, Procurement and Construction
FEED Front End Engineering Design
ISBL Inside Battery Limits, referring to parts of the plant itself
MM Million (abbreviation commonly used in industry in United States)
M&S Marshall and Swift cost index, published monthly in *Chemical Engineering*
NF Nelson-Farrer refinery cost index, published monthly in *Oil and Gas Journal*
NWE Northwest Europe, often used as a location basis in capital costing
OSBL Outside Battery Limits, referring to site infrastructure, also known as off-sites
TCOP Total cost of production, including allowance for capital charges
USGC U.S. Gulf Coast, often used as a location basis in capital costing
WBS Work Breakdown Structure
WC Working Capital

7.14 Problems

- 7.1. Estimate the capital cost of a plant that produces 80,000 metric tons per year of caprolactam.
- 7.2. The process used in the manufacture of aniline from nitrobenzene is described in Appendix F, design problem F.8. The process involves six significant stages:
- Vaporization of the nitrobenzene
 - Hydrogenation of the nitrobenzene
 - Separation of the reactor products by condensation
 - Recovery of crude aniline by distillation
 - Purification of the crude nitrobenzene
 - Recovery of aniline from wastewater streams
 - Estimate the capital cost of a plant to produce 20,000 metric tons per year.
- 7.3. A reactor vessel cost \$365,000 in June 1998; estimate the cost in January 2020.
- 7.4. The cost of a distillation column was \$225,000 in early 1998; estimate the cost in January 2024.
- 7.5. Using the data on equipment costs given in this chapter or commercial cost estimating software, estimate the cost of the following equipment:
1. A shell and tube heat exchanger, heat transfer area 50 m^2 , floating head type, carbon steel shell, stainless steel tubes, operating pressure 25 bar.
 2. A kettle reboiler: heat transfer area 25 m^2 , carbon steel shell and tubes, operating pressure 10 bar.
 3. A horizontal, cylindrical, storage tank, 3 m diameter, 12 m long, used for liquid chlorine at 10 bar, material carbon steel.
 4. A plate column: diameter 2 m, height 25 m, stainless clad vessel, 20 stainless steel sieve plates, operating pressure 5 bar.
- 7.6. Compare the cost of the following types of heat exchangers, each with a heat transfer area of 10 m^2 . Take the construction material as carbon steel.
1. Shell and tube, fixed head.
 2. Double-pipe.
- 7.7. Estimate the cost of the following items of equipment:
1. A packaged boiler to produce 20,000 kg/h of steam at 40 bar.
 2. A centrifugal compressor, driver power 75 kW.
 3. A plate and frame filter press, filtration area 10 m^2 .
 4. A floating roof storage tank, capacity $50,000 \text{ m}^3$.
 5. A cone roof storage tank, capacity $35,000 \text{ m}^3$.
- 7.8. A storage tank is purged continuously with a stream of nitrogen. The purge stream leaving the tank is saturated with the product stored in the tank. A major part of the product lost in the purge could be recovered by

installing a scrubbing tower to absorb the product in a solvent. The solution from the tower could be fed to a stage in the production process and the product and solvent recovered without significant additional cost. A preliminary design of the purge recovery system has been made. It would consist of:

1. A small tower 0.5 m diameter, 4.0 m high, packed with 25 mm ceramic saddles, packed height 3.0 m.
2. A small storage tank for the solution, 5 m³ capacity.
3. The necessary pipe work, pump, and instrumentation.

All the equipment can be constructed from carbon steel.

Using the following data, evaluate how long it would take for the cost savings to pay for the capital cost of installing the recovery system:

1. Cost of product \$5 per lb.
2. Cost of solvent \$0.5 per lb.
3. Additional solvent make-up 10 kg/d.
4. Current loss of product 0.7 kg/h.
5. Anticipated recovery of product 80%.
6. Additional utility costs, negligible.

Other operating costs will be insignificant.

- 7.9. Example 7.8 developed the cost for an adipic acid plant on a northeast Asia January 2006 basis. Estimate the cost of the plant on a 2024 basis if the plant is located in Japan, South Korea, or China. In the case of China, assume that 85% of the plant cost can be sourced indigenously.
- 7.10. The production of methyl ethyl ketone (MEK) is described in Appendix F, Problem F.3. A preliminary design has been made for a plant to produce 10,000 metric tons per year (t/y). The major equipment items required are listed here. The plant operating rate will be 8000 hours per year. Estimate the capital required for this project.
- The plant will be built on an existing site with adequate infrastructure to provide the ancillary requirements of the new plant (no off-site investment is needed).

Major equipment items:

1. Butanol vaporizer: shell and tube heat exchanger, kettle type, heat transfer area 15 m², design pressure 5 bar, material carbon steel.
2. Reactor feed heaters (two): shell and tube, fixed head, heat transfer area 25 m², design pressure 5 bar, material stainless steel.
3. Reactors (three): shell and tube construction, fixed tube sheets, heat transfer area 50 m², design pressure 5 bar, material stainless steel.
4. Condenser: shell and tube heat exchanger, fixed tube sheets, heat transfer area 25 m², design pressure 2 bar, material stainless steel.
5. Absorption column: packed column, diameter 0.5 m, height 6.0 m, packing height 4.5 m, packing 25 mm ceramic saddles, design pressure 2 bar, material carbon steel.
6. Extraction column: packed column, diameter 0.5 m, height 4 m, packed height 3 m, packing 25 mm stainless steel pall rings, design pressure 2 bar, material carbon steel.
7. Solvent recovery column: plate column, diameter 0.6 m, height 6 m, 10 stainless steel sieve plates, design pressure 2 bar, column material carbon steel.
8. Recovery column reboiler: thermosyphon, shell and tube, fixed tube sheets, heat transfer area 4 m², design pressure 2 bar, material carbon steel.
9. Recovery column condenser: double pipe, heat transfer area 1.5 m², design pressure 2 bar, material carbon steel.
10. Solvent cooler: double pipe exchanger, heat transfer area 2 m², material stainless steel.
11. Product purification column: plate column, diameter 1 m², height 20 m, 15 sieve plates, design pressure 2 bar, material stainless steel.
12. Product column reboiler: kettle type, heat transfer area 4 m², design pressure 2 bar, material stainless steel.
13. Product column condenser: shell and tube, floating head, heat transfer area 15 m², design pressure 2 bar, material stainless steel.
14. Feed compressor: centrifugal, rating 750 kW.
15. Butanol storage tank: cone roof, capacity 400 m³, material carbon steel.
16. Solvent storage tank: horizontal, diameter 1.5 m, length 5 m, material carbon steel.
17. Product storage tank: cone roof, capacity 400 m³, material carbon steel.

Estimating revenues and production costs

KEY LEARNING OBJECTIVES

- How to evaluate the revenues and production costs associated with a project
- How to determine prices for feeds, products, by-products, and fuels
- How to estimate fixed and variable production costs
- How to forecast prices over the expected life of a project
- How to summarize production cost information in a standard form

8.1 Introduction

Estimation of the product revenues and costs of production is a key step in determining the profitability of a process. An understanding of the breakdown of production costs is critically important to process optimization, regardless of whether the project is a new grassroots design or a revamp or expansion of an existing plant.

Several companies regularly publish economic analyses of chemical processes. Nexant publishes the Process Evaluation and Research Planning (PERP) reports (www.nexant.com/products). Roughly ten new reports are issued each year, and almost 200 processes have been analyzed. The PERP reports provide estimates of capital and operating costs, usually for two or three process alternatives, as well as an overview of the current market conditions for the product. SRI publishes the Chemical Economics Handbook (CEH) series, which contains 281 reports on a range of commodity and specialty chemicals. The CEH reports provide an overview of production technologies and analyses of several regional markets but do not provide the level of production cost detail given in the PERP reports. Various consulting firms also carry out paid economic studies of “state-of-the-art” technology. Although there are minor variations in methodology, most of these studies estimate production costs using similar assumptions. The conventional terminology and assumptions used in commercial studies will be introduced in the following sections and should be followed when making preliminary economic analyses and when accurate cost information is not available.

8.2 Costs, revenues, and profits

This section introduces the components of project costs and revenues. Costs and revenues for a plant are typically expressed per unit time or per unit of production of main product. The most common measures used for chemicals and bulk materials are \$/year, \$/day, \$/lb, and \$/metric ton, but some industries use other measures, such as \$/kg for high-value, small-production pharmaceutical compounds or \$/bbl for oil refineries.

8.2.1 Variable costs of production

Variable costs of production are costs that are proportional to the plant output or operation rate. These include the costs of:

1. Raw materials consumed by the process.
2. Utilities – fuel burned in process heaters, steam, cooling water, electricity, raw water, instrument air, nitrogen, and other services brought in from elsewhere on the site.
3. Consumables – solvents, acids, bases, inert materials, corrosion inhibitors, additives, catalysts, and adsorbents that require continuous or frequent replacement.
4. Effluent disposal – running costs of treating any solid, liquid, or vapor emissions or waste products from the process.
5. Packaging and shipping – drums, bags, tankers, freight charges, etc.

Variable costs are mainly determined by the choice of feedstock, process chemistry, and plant location, and can usually be reduced by more efficient design or operation of the plant. Methods for estimating variable costs are discussed in [Section 8.4](#).

8.2.2 Fixed costs of production

Fixed production costs are costs that are incurred regardless of the plant operation rate or output. If the plant cuts back its production, these costs are not reduced. Fixed costs are discussed in more detail in [Section 8.5](#). Fixed costs include:

1. Operating labor – see [Section 8.5.1](#).
2. Supervision – usually taken as 25% of operating labor.
3. Direct salary overhead – usually 40% to 60% of operating labor plus supervision. These are nonsalary costs such as employee health insurance and other benefits.
4. Maintenance, which includes both materials and labor, and is typically estimated as 3% to 5% of inside battery limits (ISBL) investment, depending on the expected plant reliability. Plants with more moving equipment or more solids handling usually require higher maintenance.
5. Property taxes and insurance – typically 1% to 2% of ISBL fixed capital.
6. Rent of land (and/or buildings) – typically estimated as 1% to 2% of ISBL plus offsite battery limits (OSBL) investment. Most projects assume land is rented rather than purchased, but in some cases the land is bought and the cost is added to the fixed capital investment and recovered at the end of the plant life.
7. General plant overhead – charges to cover corporate overhead functions such as human resources, research and development (R&D), information technology, finance, etc. These costs can vary widely and are discussed in more detail in [Section 8.5.6](#).
8. Allocated environmental charges to cover superfund payments (see [Chapter 11](#)) – typically 1% of ISBL plus OSBL cost.
9. Running license fees and royalty payments. Royalties are discussed in more detail in [Section 8.5.7](#).
10. Capital charges – these include interest payments due on any debt or loans used to finance the project, but *do not* include expected returns on invested equity capital; see [Section 9.3](#).
11. Sales and marketing costs – in some cases these are considered part of general plant overhead. They can vary from almost zero for some commodities to millions of dollars a year for branded items such as foods, toiletries, drugs, and cosmetics.

Fixed costs should never be neglected, even in the earliest stages of design, as they can have a significant impact on project economics. Very few chemical plants in the United States carry less than \$1 million per year (\$1 MM/y) of fixed costs.

Fixed costs are also a strong disincentive for building small plants. As plant size is increased, labor, supervision, and overhead costs usually do not increase, and hence the fixed cost per pound of product decreases. This, together with economies of scale in capital investment (see [Section 7.4.1](#)), gives larger plants more flexibility to reduce prices and hence force smaller plants out of business during downturns in the business cycle.

Fixed costs are not easily influenced by better design or operation of the plant, other than improvements that allow the plant to be operated safely with a smaller workforce. Fixed costs are more amenable to control at the corporate level than the plant level.

8.2.3 Revenues

The revenues for a project are the incomes earned from sales of main products and by-products.

The production rate of main product is usually specified in the design basis and is determined by the marketing department based on predictions of overall market growth.

By-product revenues

Determining which by-products to recover, purify, and sell is usually more difficult than determining the main product. Some by-products are produced by the main reaction stoichiometry, whereas others are produced from feed impurities or by nonselective reactions.

The decision to recover, purify, and sell; recycle or otherwise attenuate; or dispose of by-products as wastes is an important design optimization problem. A good deal of process design effort is often spent analyzing by-product recovery. Potentially valuable by-products include:

1. Materials produced in stoichiometric quantities by the reactions that form the main product (see Table 8.1 for examples). These stoichiometric by-products are unavoidable unless new chemistry can be found. If these are not recovered as by-products and sold for whatever price they can get, the waste disposal costs will be excessive.
2. Components that are produced in high yield by side reactions. Some examples include propylene, butylenes, and butadiene, all of which are by-products of ethylene from steam cracking of naphtha feed. Orthoxylene and metaxylene are by-products of paraxylene manufacture by catalytic reforming of naphtha.
3. Components formed in high yield from feed impurities. Most sulfur is produced as a by-product of fuels manufacture. Crude oil and natural gas contain sulfur compounds that are converted to H_2S during refining or gas treatment. The H_2S is then converted to elemental sulfur by the Claus process. Mannitol (a valuable hexose) is made from fructose that is present in the glucose feed to the sorbitol process.
4. Components produced in low yield that have high value. Dicyclopentadiene can be recovered from the products of steam naphtha cracking. Acetophenone is recovered as a by-product of phenol manufacture, although it can also be made by oxidation of ethylbenzene or fermentation of cinnamic acid.
5. Degraded consumables such as solvents that have reuse value. Spent sulfuric acid can be neutralized with calcium oxide to form low-value gypsum that is used as fill for roads. Alternatively, more expensive ammonia can be used for the neutralization, forming ammonium sulfate that has a higher by-product value as a fertilizer.

For the by-product to have value, it must meet the specifications for that material, which may entail additional processing costs. The design engineer must therefore assess whether the additional cost of recovering and purifying the by-product is justified by the by-product value and avoided waste disposal cost before deciding whether to value the material as a by-product or as a waste stream.

TABLE 8.1 Some stoichiometric by-products

Feeds	Main product	By-product
cumene + air	phenol	acetone
propylene + ethylbenzene + air	propylene oxide	styrene
ethylene + chlorine	vinyl chloride monomer	HCl
allyl chloride + HOCl + NaOH	epichlorohydrin	NaCl
methane + steam	hydrogen	carbon dioxide
glucose	ethanol (by fermentation)	carbon dioxide
acetone cyanohydrin + methanol + H_2SO_4	methyl methacrylate	ammonium sulfate
sodium chloride + electricity	chlorine	sodium hydroxide

An algorithm for assessing the economic viability of recovering a by-product X is given in Fig. 8.1. Note that it is important to consider not only the cost of purifying the by-product but also whether it can be converted into something more valuable. This would include recycling the by-product within the process if that might be expected to lead to a higher yield of main product or formation of a more valuable by-product. Note also that when analyzing whether to recover a by-product, the value created by recovering the by-product includes not only the revenue from by-product sales but also the avoided by-product disposal cost. If the by-product has fuel value, the fuel value should be subtracted from the revenue instead.

A rule of thumb that can be used for preliminary screening of by-products for large plants is that for by-product recovery to be economically viable, the net benefit must be greater than \$200,000 per year. The net benefit is the by-product revenue plus the avoided waste disposal cost. (This rule of thumb is based on the assumption that recovering by-product is going to add at least one separation to the process, which will cost at least \$0.5MM of capital, or an annualized cost of about \$170,000, as described in Section 9.7.2.).

8.2.4 Margins and profits

Margins

The sum of product and by-product revenues minus raw material costs is known as the gross margin (or sometimes product margin or just margin).

$$\text{Gross margin} = \text{Revenues} - \text{Raw materials costs} \quad (8.1)$$

Gross margin is a useful concept, as raw materials costs are almost always the largest contributor to production costs (typically 80% to 90% of total cost of production). Raw materials and product prices of commodities are often subject to high variability and can be difficult to forecast, but margins suffer less variability if producers are able to pass feedstock price increases on to their customers. Margins are therefore often used in price forecasting, as described in Section 8.3.3.

Note that the gross margin is based on the actual amount of raw materials consumed and is not simply the product price per ton minus the feed price per ton. This is a common source of confusion.

Margins vary widely between different sectors of the chemical industry. For commodities such as bulk petrochemicals and fuels, margins are typically very low (less than 10% of revenues) and may even occasionally be negative. Commodity businesses are usually cyclical because of investment cycles and experience higher margins when supply is short, as described in Section 8.3.1. When a product is tightly regulated (making market entry difficult) or subject to patent protection, then margins can be much higher. For example, margins on food additives, pharmaceutical products, and biomedical implants are typically more than 40% of revenues and often higher than 80% of revenues.

The variable contribution margin is the sum of revenues minus variable costs:

$$\text{Variable contribution margin} = \text{Revenues} - \text{Variable costs of production} \quad (8.2)$$

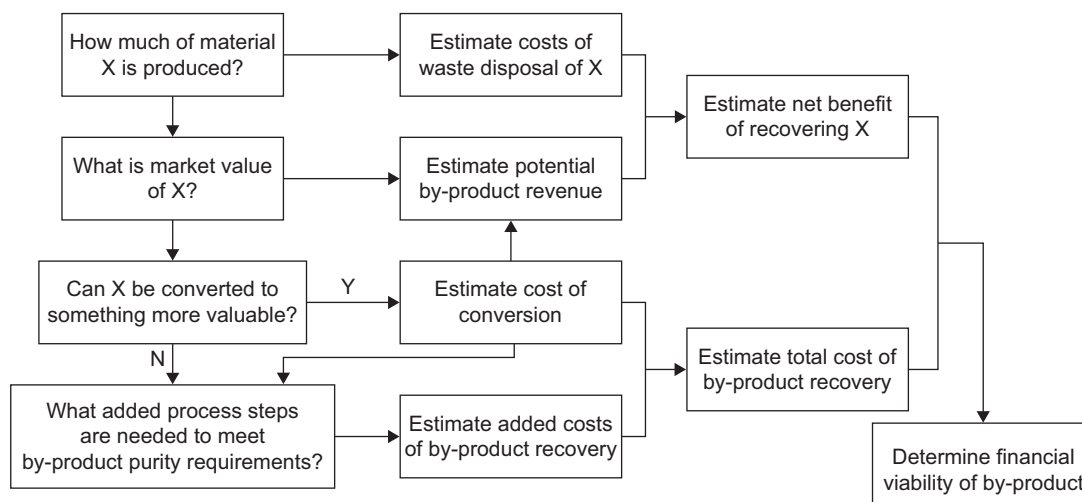


FIG. 8.1 Algorithm for assessing the economic viability of by-product recovery.

Variable contribution margin indicates the profitability of the process on a fixed-cost-free basis. It is often used in batch processing when assessing the impact of adding a new product to an existing plant that is not fully utilized, so that the new product can be accommodated without increasing fixed costs.

Profits

The cash cost of production (CCOP) is the sum of the fixed and variable production costs:

$$\text{CCOP} = \text{VCOP} + \text{FCOP} \quad (8.3)$$

where VCOP = sum of all the variable costs of production minus by-product revenues

FCOP = sum of the fixed costs of production

The CCOP is the cost of making a product, not including any return on the equity capital invested. By convention, by-product revenues are usually taken as a credit and included in the VCOP. This makes it easier to determine the \$/lb cost of producing the main product.

The gross profit is:

$$\text{Gross profit} = \text{Main product revenues} - \text{CCOP} \quad (8.4)$$

Gross profit should not be confused with gross margin, as gross profit includes all the other variable costs in addition to raw materials and also includes fixed costs and by-product revenues.

In some companies gross profit is stated on a plant basis, omitting the general overhead charges and selling costs (selling, general, and administrative, or SG&A, charges: items 7 and 11 in the list of fixed costs in [Section 8.2.2](#)). The SG&A charges are then subtracted from the gross profit at the corporate level to give the operating income reported in the annual income statement.

The profit made by the plant is usually subject to taxation. Different tax codes apply in different countries and locations, and the taxable income may not be the full gross profit. Taxes are discussed in more detail in [Section 9.4](#). The net profit (or cash flow after tax) is the amount left after taxes are paid:

$$\text{Net profit} = \text{gross profit} - \text{taxes} \quad (8.5)$$

The net profit from the project is the money that is available as a return on the initial investments. Methods for evaluating the economic performance of investments are introduced in [Chapter 9](#).

It is sometimes useful to calculate a total cost of production (TCOP), assuming that a plant generates a specified return on investment. In this case, an annual capital charge (ACC) is added to the cash cost of production:

$$\text{TCOP} = \text{CCOP} + \text{ACC} \quad (8.6)$$

Methods for calculating the annual capital charge are discussed in [Section 9.7](#).

8.3 Product and raw material prices

The revenues and variable costs of production are obtained by multiplying the product, feed, or utility flow rates from the flowsheet by the appropriate prices. The difficult step is usually finding good price data. The prices used in economic analysis of a design should reflect the market conditions that the company expects to face over the life of the project. It is therefore necessary to make a long-term forecast of prices. This section discusses the factors that influence prices, where to obtain historic price data, and how to use that data to make forecasts of future prices for use in economic analysis.

The chemical and process industries handle a very broad range of materials, ranging from bulk commodities such as crude oil, naphtha, grains, and basic chemicals to specialized products such as drugs, food additives, scents, and biologically derived molecules. Some of the most basic materials are traded on commodity exchanges, and their price can fluctuate significantly on a daily or even hourly basis. Materials that are supplied in large bulk quantities are usually sold under long-term contracts. The contract price may be pegged to the price of a commodity such as natural gas, corn, or crude oil, or may be subject to periodic renegotiation, typically on a quarterly or annual basis. Materials that are supplied in smaller quantities have a list price specified by the supplier, but the actual price paid by a purchaser is usually the result of a negotiation between the purchaser's procurement department and the supplier's sales department and can depend on the quantity purchased, delivery terms, and other factors. Even finished products such as household chemicals, medicines, and personal care products that are ready for sale to

end users are not immune from price negotiation, as large retailers typically use their control of sales volume to exert price pressure on their suppliers. Companies like Walmart, Costco, and Target can extract significant price concessions from chemicals manufacturers, and hence maximize their own margins.

8.3.1 Pricing fundamentals

The price of any substance is determined by the balance between supply and demand; indeed, in economics the price is the mechanism used by the market to bring supply and demand into dynamic equilibrium.

As the price of any product or service decreases, more customers can afford or are willing to buy the product, and demand for the product increases. We can plot a graph of how much demand there will be for the product (i.e., the quantity that can be sold) as a function of price, which is known as a *demand curve*; Fig. 8.2(a). A demand curve can be thought of as a rank ordering of the customers in terms of the amount they will consume and their willingness to pay.

Similarly, if the price of a product or service increases, more suppliers will be attracted to enter the market, and the quantity of the product that is supplied will increase. We can make a similar plot of the marginal production of the product as a function of price, which is known as a *supply curve*; Fig. 8.2(b). A supply curve can also be thought of as a rank ordering of producers in terms of the amount they can produce and their willingness to sell.

Because the supply and demand curves both relate price to quantity, they can be plotted on the same graph, as in Fig. 8.3. The intersection of the supply and demand curves sets the equilibrium between supply and demand. The intersection point gives the transaction price at which the most marginal consumer agrees to buy and the most marginal producer agrees to sell, as well as setting the volume of the market. At any point to the right of the equilibrium point there is no supplier who can provide product at a price that the next most marginal customer is willing to pay.

The slope of the supply or demand curve is referred to as the elasticity of supply or demand. If the curve is flat and price varies very little with quantity, then the supply or demand is said to be elastic [Fig. 8.4(a)], whereas if the slope is steep and a large change in price makes little difference to demand, then the supply or demand is said to be inelastic [Fig. 8.4(b)]. The elasticity of demand depends on the availability of substitutes, the amount of money

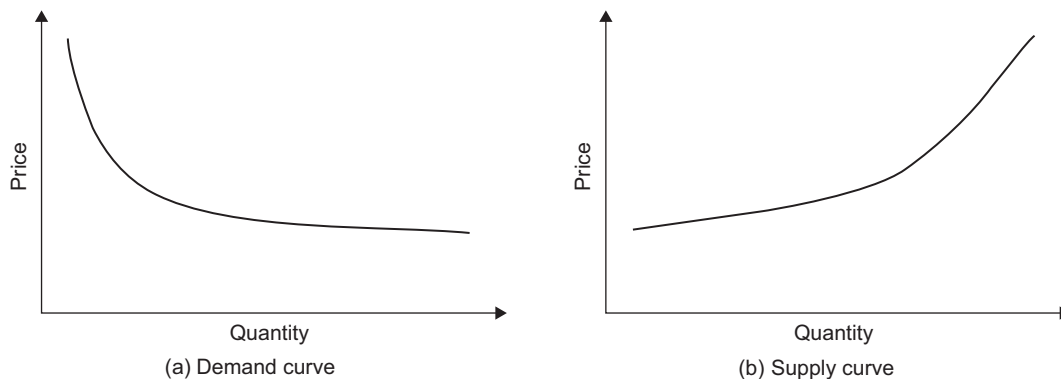


FIG. 8.2 Demand and supply curves.

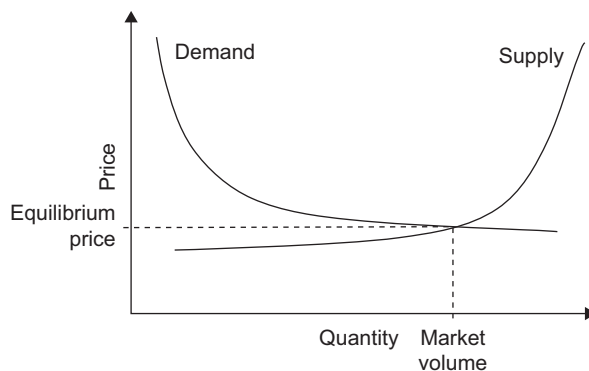


FIG. 8.3 Market equilibrium.

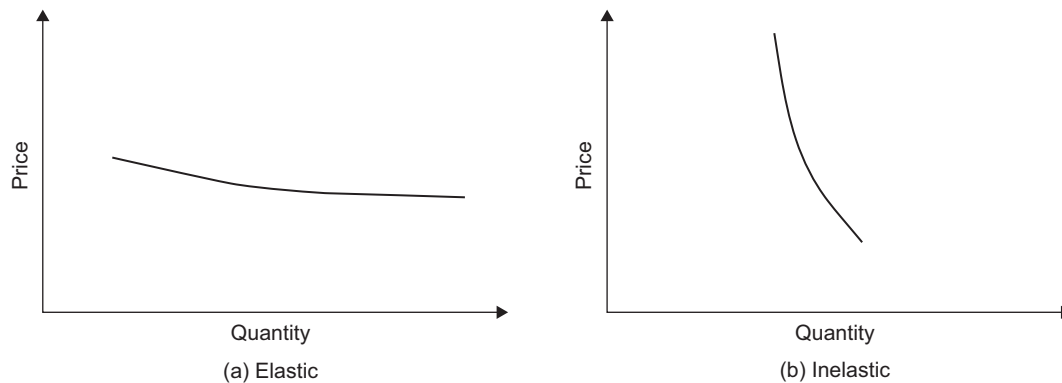


FIG. 8.4 Elasticity of demand.

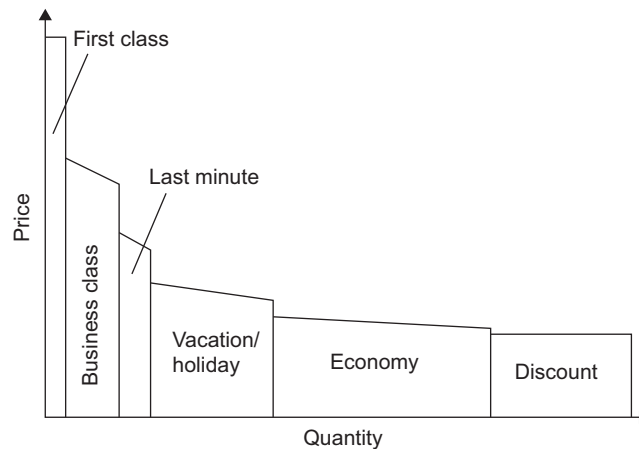


FIG. 8.5 Demand segmentation for air travel.

available to consumers, and the consumers' perception of the marginal utility or value of the product or service. The elasticity of supply depends on how many producers are able to produce an equivalent product or service, how difficult it is to enter the market, and producers' perception of the marginal value of the product or service.

There are several methods that companies can use to maximize the price that they obtain for their products or services. One of the most straightforward is to offer products of different quality aimed at different segments of the market. Each product has its own intersection point on the demand curve and hence can have a different price. For example, Fig. 8.5 is a schematic demand curve for airline travel. By segmenting the customers according to need to travel and willingness to pay, an airline can develop a pricing structure that commands substantial premiums for offering essentially the same service of moving a person from city A to city B. Another familiar example of market segmentation is the garment industry, where retailers develop brands aimed at different demographics so as to maximize the price commanded for relatively small differences in product quality.

Technology and new products can also be used to create barriers to entry into a market and capture a higher price from a defined customer segment. New technologies are generally launched in relatively small quantity at high price, as patent holders are able to control the supply of the product. As patents expire and competitors emerge, the supply increases and price decreases. Products therefore tend to move down the demand curve from left to right with time. Many pharmaceutical products undergo a substantial drop in price when their patents expire, as generic manufacturers are then able to enter the market and increase the supply.

If a supplier gains a monopoly of supply or if a group of suppliers form a cartel, then they can restrict supply and use this as a means of raising prices. In this case, the supply is constrained to a value that corresponds to the target price sought by the suppliers. International trade, competition, and antitrust laws prohibit cartels and impose fines on companies that engage in price fixing; however, cartels tend to be unstable in any case, as there will always be the possibility that the producer can increase their income by allowing demand to grow and segmenting the market to target different consumers. There are well-publicized examples of companies that have benefited from abandoning

cartels and making better use of customer segmentation instead. For example, De Beers now has better financial performance with a 40% market share than it had when it restricted diamond supply and maintained 80% market share in the 1990s.

For chemicals, fuels, foods, and pharmaceutical products, the barriers to entry into a market can create inelasticity in supply. It usually takes several years to design, build, and commission a new chemical plant. The supply curve can be segmented in a similar manner to the demand curve. For example, Fig. 8.6 shows a hypothetical supply curve for a commodity chemical. The lowest-cost producers are typically large, efficient new plants located in regions that have advantageously priced feedstocks. Next are large, efficient plants without feedstock advantage, as these have low utility costs and utilities are usually the next largest cost after feeds. Smaller and older plants carry a higher burden of fixed costs and labor costs, so labor cost advantages are only usually significant for smaller scales of production. The most marginal older plants will be idled or mothballed if they cannot produce product at a price the market is willing to pay.

In most chemical sectors, the overall capacity utilization rate varies between about 75% and 85%, and there is sufficient spare capacity to allow for some plants to be down for maintenance or unexpected interruptions in production. The market equilibrium is therefore at a point where the supply is relatively elastic; however, if the capacity utilization rate increases to greater than 90%, the market equilibrium moves to the inelastic region of the supply curve and prices can become volatile. The market is then much more sensitive to events that cause temporary reductions in supply, such as unexpected shutdowns of large plants. If new plants cannot be brought on stream quickly, prices will spike until additional supply becomes available. This leads to boom and bust cycles, which tend to have the following pattern:

1. High prices create attractive conditions for investment. Several companies commit to building new plants or revamping old plants, adding more capacity than is needed.
2. When the new plants are commissioned, they have the latest technology, lowest cost feeds, and lowest cost of production. The new capacity needs to operate at full production to recoup its investment, so prices fall rapidly as supply increases. Overall, capacity utilization decreases as demand has not yet caught up with the new supply.
3. Lower prices force the least competitive producers to reduce production rates or close older, less efficient plants. Lower prices remove the incentive to add new capacity and cause underinvestment in new plant. A new equilibrium is reached at low price.
4. Demand eventually catches up with capacity, and utilization rates creep up. At some point the elasticity of supply is lost and prices begin to spike again, creating conditions to return to the start of the cycle.

Many commodity chemicals and fuels undergo such periodic oscillations in price. These oscillations can make it difficult to forecast prices for use in economic analysis of new projects, as discussed in the following sections.

8.3.2 Sources of price data

This section describes the most widely used sources of current and historic price data. Some pricing terminology is given in Table 8.2.

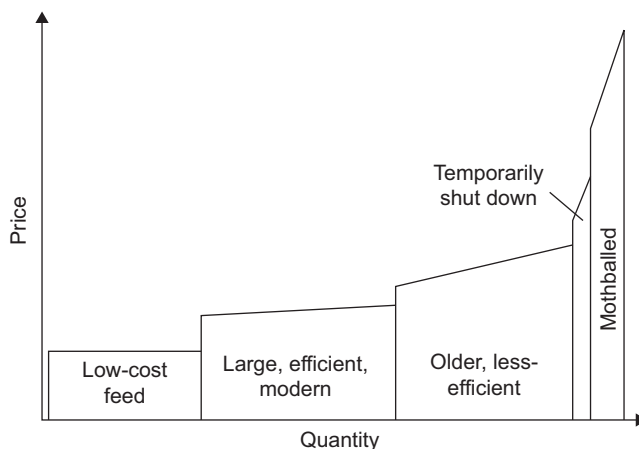


FIG. 8.6 Supply segmentation for a commodity chemical.

TABLE 8.2 Pricing terminology

Abbreviation	Meaning
c.i.f.	Cost, insurance, and freight (does not include customs charges)
dlvd.	Delivered to the end user
f.o.b.	Free on board (loaded at specified supply point, but does not include freight, insurance, or customs duties)
frt. alld.	Freight allowed (f.o.b. plus freight)
dms.	Drums
bgs.	Bags
refy.	Refinery gate
syn.	Synthetic
t.t.	Tank truck
t.c.	Tank car (rail)
t.l.	Truck load
imp.	Imported

Internal company forecasts

In many large companies the marketing or planning department develops official forecasts of prices for use in internal studies. These forecasts sometimes include multiple price scenarios, and projects must be evaluated under every scenario. Company forecasts are occasionally made available to the public. See, for example, the Shell future energy scenarios, which can be downloaded from www.Shell.com/energy-and-innovation/the-energy-future/scenarios. When an officially approved price set exists, the design engineer should use it. The main concern is then ensuring that prices for feeds, products, or consumables that are not part of the standard forecast are put on a consistent basis.

Trade journals

Several journals publish chemicals and fuel prices on a weekly basis:

ICIS Chemical Business Americas — formerly known as *Chemical Marketing Reporter* (ICIS Publications) used to list prices for 757 chemicals with multiple locations and product grades for some. This list was reduced to only 85 compounds in 2006, with most of the remaining set being natural extracts. Data for 80 chemicals, 44 fuels, and 11 base oils is now provided online through the subscription service www.icis.com. At the time of writing this service was very expensive compared with some of the other alternatives listed. ICIS also publishes *ICIS Chemical Business Europe* and *ICIS Chemical Business Asia*, which provide regional price data for a smaller set of compounds.

The *Oil and Gas Journal* (Pennwell) publishes prices for several crude oils and a range of petroleum products on U.S., northwest European, and southeast Asian bases, as well as natural gas prices for the United States. The *Oil and Gas Journal* also publishes formula estimates of gross and net margins for refineries, natural gas liquids recovery, and ethylene plants on a monthly basis.

Chemical Week (IHS) gives spot and contract prices for 22 commodity chemicals in U.S. and northwest European markets.

Consultants

There are many companies that can be hired as consultants to provide economic and marketing information or that allow access to such information on a subscription basis. The information provided generally includes market surveys and technical and economic analyses of competing technologies, as well as price data and forecasts. There is not room here to list all of these companies, but some of the most widely used are:

Purvin and Gertz: Quarterly forecasts of oil, gas, and fuels prices are widely used in the oil industry. They have a 10-year archive of historic data and forecast prices of most fuel products as well as crude oils on U.S., northwest European, Middle Eastern, and Asian bases.

Cambridge Energy Research Associates: Publishes forecasts of crude oil prices based on macroeconomics and industry trends (drilling rates, etc.).

Chemical Market Associates, Inc. (CMAI): Maintains a large archive of historic data and future price forecasts for 70 commodity chemicals, including multiple grades, U.S., northwest European, Middle Eastern, and northeast and southeast Asian. Spot and contract prices are given for some compounds, and in some cases margins are also estimated by formula.

SRI: The *Chemical Economics Handbook* series of reports published by SRI provides overviews of the markets for over 300 compounds. These reports are not updated as frequently as the others, but are useful for less commoditized compounds.

Online brokers and suppliers

A lot of price data are available online from suppliers' websites that can be found through directory sites such as www.chemicalregister.com.

Some caution is needed when using price data from the Web. The prices quoted are generally for spot sales of small-quantity orders, and are thus much higher than the market rates for large order sizes under long-term contract. The prices listed online are also often for higher-quality material such as analytical, laboratory, or United States Pharmacopeia (USP) pharmaceutical grades, which have much higher prices than bulk grades.

Reference books

Prices for some of the more common commodity chemicals are sometimes given in process economics textbooks. These prices are usually single data points rather than forecasts. They are only suitable for undergraduate design projects.

8.3.3 Forecasting prices

In most cases, it will take between 1 and 3 years for a project to go through the phases of design, procurement, and construction before the plant can begin operation. The plant will then operate for the project life of 10 to 20 years. The design engineer thus needs to carry out the economic analysis using prices forecasted over the next 20 or so years rather than the current price when the design is carried out.

For some compounds the only variation in price over time is minor adjustments to allow for inflation. This is the case for some specialty compounds that have relatively high prices and are not subject to competitive pressure (which tends to drive prices down). Prices can also be stable if they are controlled by governments, but this is increasingly rare. In most cases, however, prices are determined largely by feedstock prices, which are ultimately determined by fluctuations in the prices of commodity fuels and chemicals. The prices of these commodities are set by markets in response to variations in supply and demand, and vary widely over time.

Most price forecasts are based on an analysis of historic price data. Several methods are used, as illustrated in Fig. 8.7. The simplest method is to use the current price, Fig. 8.7(a), but this is unsatisfactory for most commodities. Linear regression of past prices is a good method for capturing long-term trends (>10 years), but can give very different results depending on the start date chosen, as shown in Fig. 8.7(b). This method can be very misleading if the data set is too small.

Many commodity prices exhibit cyclic behavior due to the investment cycle, so in some cases nonlinear models can be used, as shown in Fig. 8.7(c). Unfortunately, both the amplitude and the frequency of the price peaks usually vary somewhat erratically, making it difficult to fit the cyclic price behavior with simple wave models or even advanced Fourier transform methods.

A fourth approach, illustrated in Fig. 8.7(d) is to recognize that feed and product prices are usually closely linked, because increases in feed costs are passed on to customers whenever possible via increases in product price. Although feed and product prices may both be variable, the gross margin is subject to much less variation and can be forecasted more reliably. Forecasting of margins is the method used widely in the fuels and petrochemicals industry, as it is much easier to predict the variation in margins than the underlying variation in the prices of crude oil and natural gas. The drawbacks of this method are that it does not work so well when there are multiple routes to the same product, and it involves making assumptions about yields that may not hold true throughout the forecast period. In cases where the gross margin is high, it can be more difficult for the manufacturer to pass on the full impact of feedstock price increases in the form of increased product prices. In such cases, when feed prices rise rapidly, there is a drop in margins while producers wait for the market to absorb the impact of higher prices.

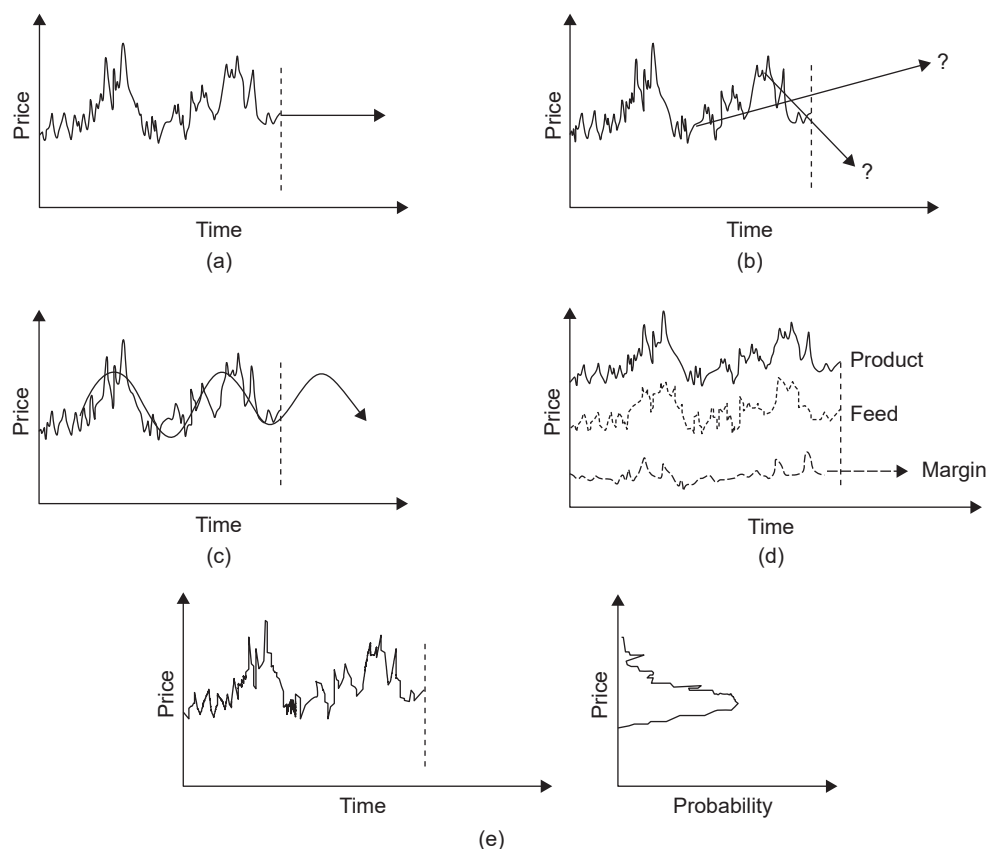


FIG. 8.7 Forecasting commodity prices.

Another method is to model the statistical distribution of the price (or margin), as illustrated in Fig. 8.7 (e). At its simplest, this method involves taking the average price, adjusted for inflation, over a recent period. This method can miss long-term trends in the data, and few prices follow any of the more commonly used distributions. It is, however, useful in combination with sensitivity analysis methods such as Monte Carlo simulation (see Section 9.8.3).

Fig. 8.8 shows North American prices from CMAI data for polyethylene terephthalate resin (PET), which is made from terephthalic acid (TPA), which in turn is made from paraxylene (PX). Several things are apparent from Fig. 8.8:

1. The spot prices of PX and TPA show more volatility than the contract prices, as would be expected.
2. All the prices follow the same broad trends, with a major peak in 1995 and long recovery leading to a second peak in 2006–2008.
3. The sharp peak in PX spot price in 1995 was not passed on to the other prices.

Fig. 8.9 shows the simple margins for TPA-PX and PET-PX over the same period, all based on contract prices. The degree of variation in margins is clearly less than the variation in the base prices. There also appears to be a long-term decline in both TPA and PET margins relative to PX in North America.

A similar examination of feed and product prices along the value chain of a given chemical can usually provide valuable insights into the best method of forecasting. No method is perfect, and anyone capable of accurately predicting commodity prices would be well advised to pursue a more lucrative career than chemical engineering. For process design purposes, it is usually sufficient to show that the prices used for optimization and economic analysis are realistic and consistent with consensus views of the market.

8.3.4 Transfer pricing

If the raw material for plant B is the product of plant A on the same site and owned by the same company, then the price that plant B pays to plant A is known as a *transfer price*. Whenever realistic, transfer prices should be set by open market prices. This reflects the reality that plant A could sell its product on the open market or plant B could similarly buy its feed. Some cases when transfer prices do not match market prices include:

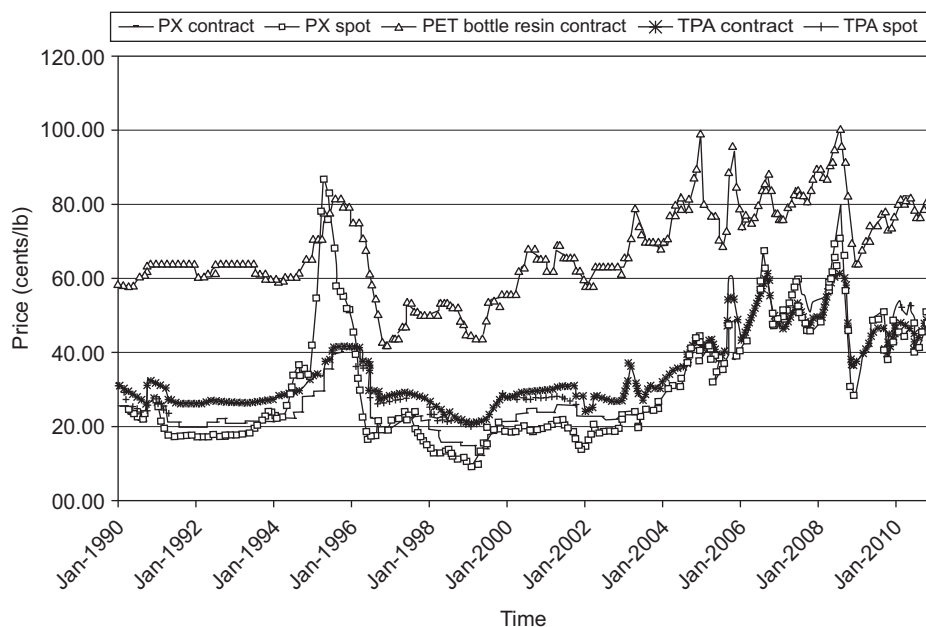


FIG. 8.8 North American prices for the PET value chain.

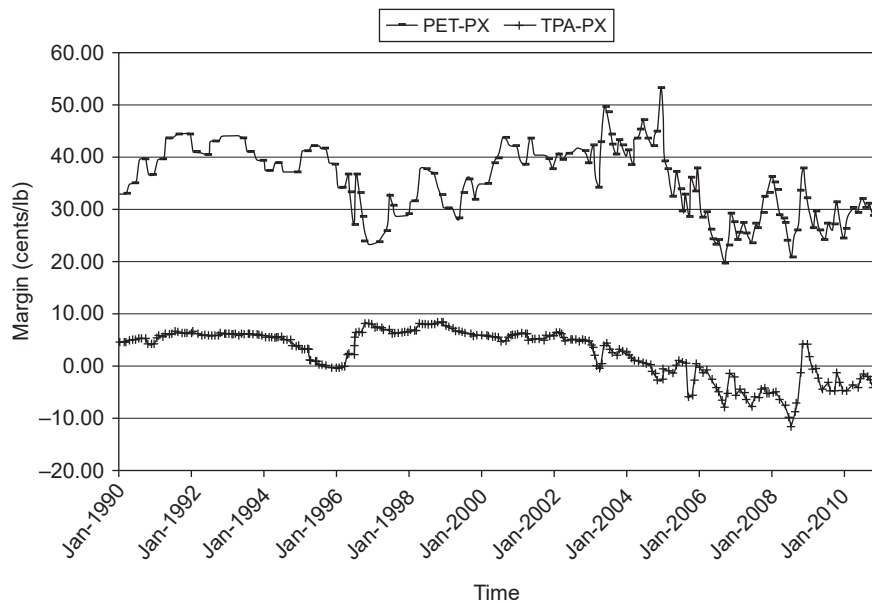


FIG. 8.9 Simple margins for the PET value chain.

1. When plant A produces material that is suitable for internal consumption but does not meet specifications for traded product. In this case, the transfer price to plant B should be discounted to allow for the added costs incurred in plant B from handling the less pure feed.
2. When plant A is underutilized or cannot sell its product and has recovered all of its initial capital investment, then the transfer price to plant B can be set at the cash cost of production of plant A (see [Section 8.2.4](#)).
3. When the pricing of product from the upstream plant is set to drive capacity utilization or conservation, for example, by using a sliding price scale based on the amount of material used.

When transfer pricing is used, it is important to keep in mind which processes actually bring in money from customers and which do not. If unrealistic transfer prices are used, uneconomic projects may seem attractive and poor investment decisions may be made.

Example 8.1

Develop forecasts for the PET-PX margin based on contract pricing using data from (a) Jan. 2000–Oct. 2010, (b) Jan. 2006–Oct. 2010, and (c) Oct. 2009–Oct. 2010. In each case, predict the PET-PX margin in Jan. 2020. Which forecast appears to be most reasonable?

Solution

The forecasts are easily constructed in Microsoft Excel using the data that were used to generate Fig. 8.9. We can make several forecasts from each data set. The simplest forecast of margins would be to assume the margin remains constant at the average value over each time period. This method gives the following results:

Time period	Average margin (cents/lb)
Jan. 2000–Oct. 2010	34.6
Jan. 2006–Oct. 2010	27.8
Oct. 2009–Oct. 2010	27.4

If margins are assumed to remain constant at the average value, then these would be the forecast predictions for Jan. 2020 margins.

Alternatively, we can make a linear regression through the data and see if there is any trend in margins with time. The results are plotted against the data sets for each time period in Figs. 8.10(a), (b), and (c).

None of the correlations are particularly good (R^2 values of 0.50, 0.09, and 0.22, respectively), although case (a) is technically the best correlation. Substituting into the fitted linear regression equations gives the following results for Jan. 2020 margins:

Time period	Linear regression forecast of Jan. 2020 margin (cents/lb)
Jan. 2000–Oct. 2010	10.4
Jan. 2006–Oct. 2010	37.3
Oct. 2009–Oct. 2010	61.6

Looking at Figs. 8.10(a) and 8.9, it seems unreasonable to believe that PET-PX margins will go as low as 10¢/lb or as high as 60¢/lb, which illustrates the difficulty in extrapolating forecasts based on linear trends. The middle forecast had the lowest correlation coefficient, but appears to yield the most reasonable value for the Jan. 2020 margin when viewed in the context of the historic data.

The final method we can consider is to look at the distribution of the data. Fig. 8.11 shows histograms of the data in each time period. For case (a), the data set appears to be bimodal, which inspection of Fig. 8.10(a) reveals to be due to an apparent step-change decrease in margins during 2005. For case (b) the data are approximately normally distributed, whereas the data set for case (c) is too small to give a good distribution.

After looking at the distributions it would seem imprudent to assume that margins would be restored to the levels they enjoyed before 2006, so we can eliminate data set (a). We can also eliminate data set (c) as being too small, so data set (b) is most suitable for a more detailed statistical analysis. For this data set we have an average value of 27.8¢/lb and a standard deviation of 3.8¢/lb. We have 98% confidence that the margin will be within ± 2.05 standard deviations of the mean, so the 98% confidence interval for the 2020 forecast, assuming no time trend in the data, would be from 20.0¢/lb ($= 27.8 - 7.8$) to 35.6¢/lb ($= 27.8 + 7.8$). Note that the forecast predicted by linear regression of the time trend in the same data set lies outside this 98% confidence interval.

Regardless of the level of analysis carried out, none of the forecasts presented is clearly the best. We have no numerical basis to determine whether to believe the downward trend in margins seen during 2000–2010 or the slight upward trend experienced during 2006–2010 or the stronger upward trend during 2009–2010. A marketing group that had experience of the business conditions in this market might have supporting evidence that would bias them towards any of these positions. It would also be just as effective to look at the data in Fig. 8.9 or Fig. 8.10(a) and predict that the PET-PX margin is likely to be about 30¢/lb $\pm$ 10¢.

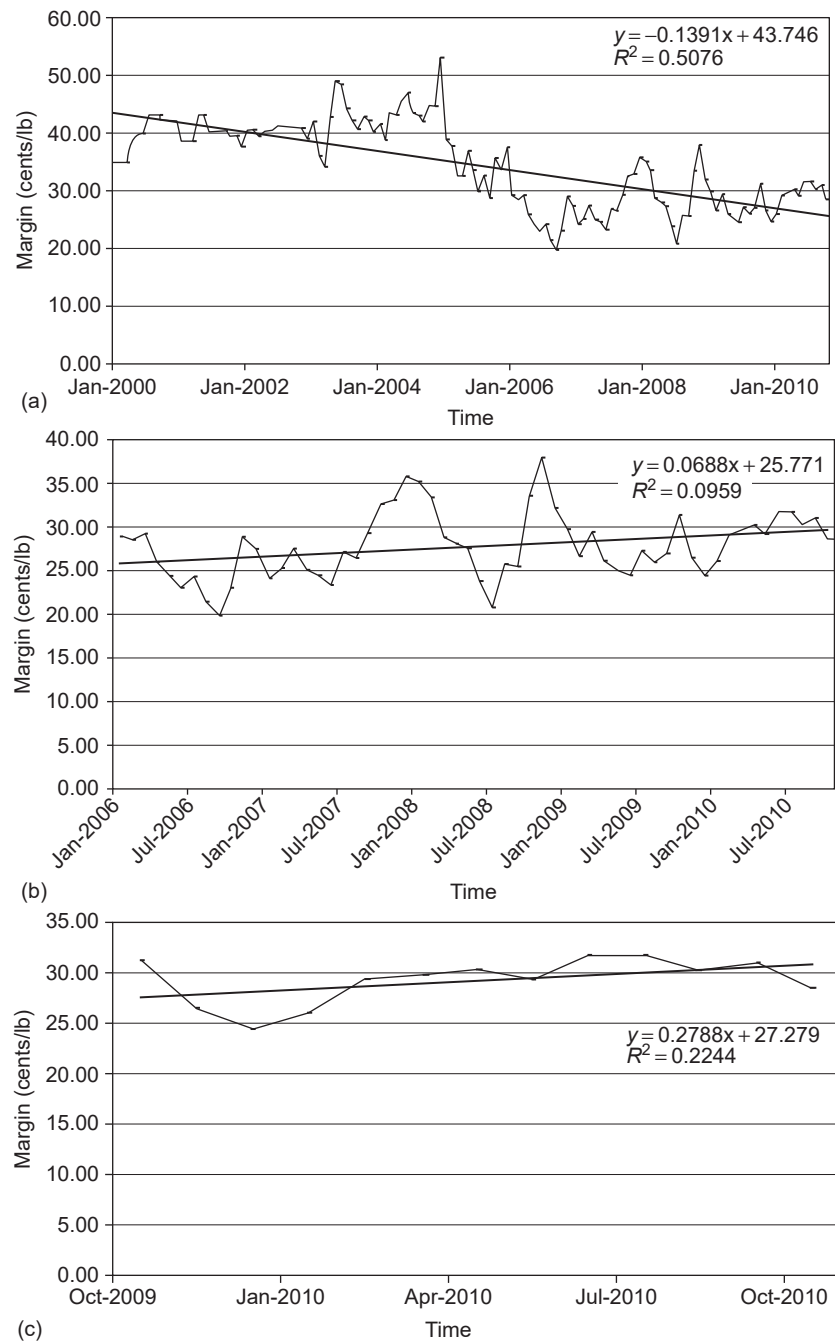


FIG. 8.10 (a) Forecast of PET-PX margins using data from Jan. 2000 to Oct. 2010. (b) Forecast of PET-PX margins using data from Jan. 2006 to Oct. 2010. (c) Forecast of PET-PX margins using data from Oct. 2009 to Oct. 2010.

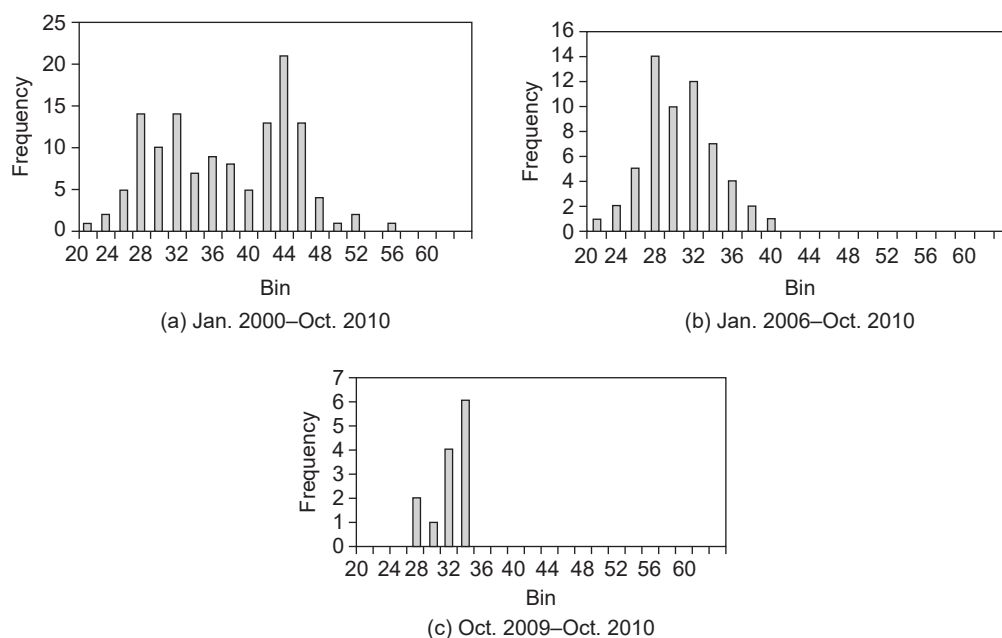


FIG. 8.11 Histograms of PET-PX margin data.

8.4 Estimating variable production costs

Variable costs of production were introduced in [Section 8.2.1](#) and are those production costs that are directly proportional to the rate of production. For most chemicals, the major variable costs are the costs of raw materials and utilities.

8.4.1 Raw materials costs

The annual cost of each raw material is just the annual consumption multiplied by the price. Raw materials prices can be found and forecasted using the sources and methods introduced in [Section 8.3](#).

For an existing plant, the amount of raw materials consumed per kg or lb of product can easily be determined from the plant purchasing records. One of the first steps in a revamp design is usually to assess the actual consumption to see if it is substantially different from what would be expected based on process stoichiometry and expected selectivities. Understanding any offset between the expected and actual raw material consumption can provide insights into how the process should be modified to become more efficient.

In a new design, the amount of raw materials needed to make the desired product is usually estimated using a process model to determine the overall plant material balance; see [Chapter 4](#). Whenever possible, the process model should be benchmarked against an existing plant or pilot plant. Methods for estimating raw material consumption in the early stages of process design, when detailed models are not available, are given in [Section 2.6](#).

Raw materials costs are typically the main contributor to overall cost of production. For most bulk chemicals and petrochemicals, raw materials costs will represent 80% to 90% of the CCOP. For specialty chemicals, raw materials costs can be as much as 95% of the CCOP. Raw materials costs are only low as a fraction of CCOP for processes that recycle waste materials or have very inexpensive feeds such as air or stranded natural gas.

8.4.2 Utilities costs

Plant utilities include fuel, process steam, cooling water, other heating or cooling fluids, electricity, process water, nitrogen, instrument air, and other service streams. Utility streams are described in detail in [Section 3.2](#), and methods are given for determining the cost of each utility. Most utility costs are based on the cost of fuel (typically natural gas) and electricity.

Determination of process utility costs is much more difficult than determination of raw materials costs, although utilities are usually less than 15% of the CCOP and are typically about 5% to 10% of CCOP. The use of energy recovery and heat integration methods means that determination of the utility consumption of a proposed design requires not only a completed mass and energy balance of the process but also at least a preliminary design of the heat recovery network (see [Sections 3.3 and 3.5](#)).

The cost of providing heat to a process is often reduced by using process waste streams as fuel. The recovery of heat from combustion of process wastes is discussed in more detail in [Section 3.4](#). When waste streams are used as fuel, there is a double benefit, as fuel costs are reduced and waste-handling costs are eliminated.

8.4.3 Consumables costs

Consumables include materials such as acids, bases, sorbents, solvents, and catalysts that are used in the process. Over time these become depleted or degraded and require replacement. In some cases a continuous purge and make-up is used (for example, for acids and bases), whereas in other cases an entire batch is periodically replaced (for example, sorbents, chromatography media, and catalysts).

The prices of acids, bases, and solvents can be found from the same sources used for raw materials prices. Whenever possible, the cheapest base (NaOH) or acid (H_2SO_4) would be used in the process, but for neutralizing spent sulfuric acid, lime (CaO) or ammonia (NH_3) is often used, as these bases react with sulfuric acid to form insoluble sulfates that can be recovered and sold as by-products. The cost of the process acid or base must always include the costs of neutralizing the spent stream.

The price of adsorbents and catalysts varies very widely depending on the nature of the material. The cheapest catalysts and adsorbents cost less than \$1/lb, whereas more expensive catalysts containing noble metals such as platinum and palladium have costs that are mainly determined by the amount of precious metal on the catalyst. In some cases, the value of the noble metal on a load of catalyst is so high that the chemical plant rents the catalyst rather than buying it, and when the catalyst is spent it is returned to the manufacturer for precious metal recovery.

Although small in quantity and typically costing less than 3% of CCOP, consumables can add a lot of capital cost and complexity to a plant. The plant must be designed with systems for handling, storing, metering, and disposing of all the consumables used. In many chemicals plants over half of the total pieces of equipment are associated with consumables handling.

8.4.4 Waste disposal costs

Materials produced by the process that cannot be recycled or sold as by-products must be disposed of as waste. In some cases additional treatment is required to concentrate the waste stream before sending it to final disposal.

Hydrocarbon waste streams such as off-spec products, slop oils, spent solvents, and off-gases (including hydrogen-rich gases) can often be incinerated or used as process fuel; see [Section 3.4](#). This allows the fuel value of the stream to be recovered, and the waste stream can be assigned a value based on its heat of combustion:

$$P_{W_{FV}} = P_F \times \Delta H_C^o \quad (8.7)$$

where: $P_{W_{FV}}$ = waste value as fuel (\$/lb or \$/kg)

P_F = price of fuel displaced by burning waste (\$/MMBtu or \$/GJ)

ΔH_C^o = heat of combustion (MMBtu/lb or GJ/kg)

If additional systems such as flue gas scrubbers must be fitted to allow the waste to be combusted, then the waste stream value should be discounted to recover the extra cost.

Dilute aqueous streams are sent to wastewater treatment unless the contaminants are toxic to the bacteria in the wastewater plant. Acidic or basic wastes are neutralized before treatment. Neutralization is usually carried out using a base or acid that will form a solid salt that can be precipitated from the water, so that the total dissolved solids (TDS) load on the wastewater plant is not excessive. The cost of wastewater treatment is typically about \$6 per 1000 gal (\$1.5 per metric ton), but there may also be local charges for spent water discharge.

Inert solid wastes can be sent to landfills at a cost of about \$50/ton, or in some cases used to make roads. Wastes from neutralizing spent sulfuric acid are typically calcium sulfate (gypsum) that can be used as road fill, or ammonium sulfate, which can be sold as fertilizer.

Concentrated liquid streams that cannot be incinerated locally (for example, compounds containing halogens) and noninert solids must be disposed of as hazardous waste. This entails shipping the material to a hazardous waste

company for incineration in a specialized plant or long-term storage in a suitable facility. The costs of hazardous waste disposal depend strongly on the plant location, proximity to waste disposal plants, and the nature of the hazardous waste, and must be evaluated on a case-by-case basis.

Additional information on waste disposal considerations is given in [Chapter 11](#).

8.5 Estimating fixed production costs

Fixed production costs are those costs that do not vary with the rate of production. Fixed costs include labor, maintenance, and overhead charges as well as taxes. Some finance charges are also counted as fixed costs, as described in the following sections.

8.5.1 Labor costs

The wages paid to plant operators and supervisors are treated as a fixed cost of production for almost all chemicals plants, because plant operation requires experience and safety training and it is not practical to ramp the workforce up and down with short-term changes in demand. Almost all plants are operated on a shift-work basis (even batch plants), with typically 4.8 operators per shift position. This gives a four-shift rotation with allowance for weekends, vacations and holidays, and some use of overtime. Most plants require at least three shift positions: one operator in the control room, one outside, and one in the tank farm or other feed/product shipping and receiving area. Plants that use more mechanical equipment, particularly solids-handling plants, typically require more shift positions. More shift positions are also needed when handling highly toxic compounds. In some cases two or more smaller plants may be grouped together with a common control room and tank farm to reduce the number of operators needed. Very few plants run entirely unattended, though, with the exception of gas processing plants, which hold no inventories of feed or product and are usually automated to allow a single control room operator to watch over several plants. A chart for estimating the minimum number of shift positions is given in [Fig. 8.12](#), but it should be emphasized that this only gives a rough guide. The design engineer should always carefully think through the operations required per shift, particularly for processes that handle solids or involve batch operations or frequent sampling.

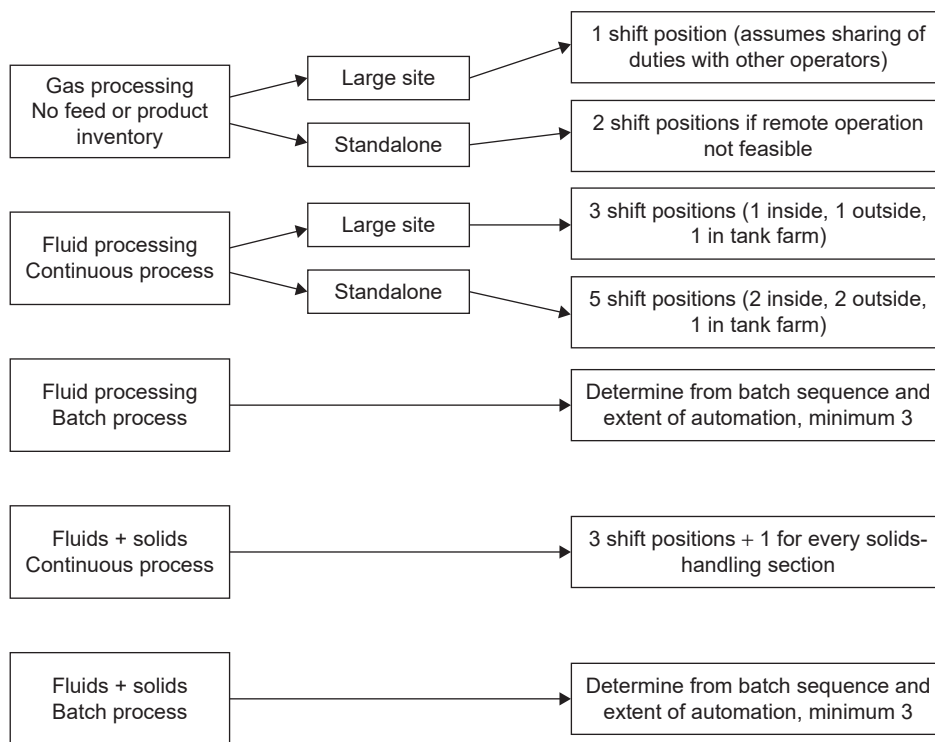


FIG. 8.12 Algorithm for assessing the minimum number of shift positions.

Operator salaries vary by region and experience level. For initial estimates, an average United States Gulf Coast (USGC) salary of \$50,000 per shift position per year, not including overhead, can be used.

Supervision and management costs are typically taken as 25% of operating labor. The actual level of supervision required depends on the extent of the plant or site. In some cases, one shift supervisor can cover several plants.

Direct salary overhead costs are the costs of providing employee benefits and training. These include such nonsalary costs as health insurance, 401k contributions to employee savings plans, training courses, and benefits such as subscriptions and professional society memberships. The level of nonsalary benefits varies widely between sectors and companies, as some companies use benefits packages as a way to attract and retain employees, whereas others place more emphasis on salary alone. For example, historically the oil industry has offered relatively high salaries and low benefits, whereas the pharmaceutical industry paid lower salaries for the same experience level but offered more benefits. Direct salary overhead typically varies between 40% and 60% of labor plus supervision costs.

8.5.2 Maintenance costs

Maintenance is a fixed cost, as the plant must be kept in good repair regardless of the level of production. Operating at less than full capacity can actually increase the rate of maintenance expenditure, as damage to plant equipment is more likely during start-up, shut-down, or turndown than during steady-state operation at design capacity.

Maintenance costs include the cost of replacing or repairing parts and equipment, as well as the cost of labor needed to carry out the maintenance work. Until as recently as the 1990s, many chemical companies employed a large number of trained maintenance technicians to provide maintenance services. Over the past two decades it has become more common for companies to subcontract major maintenance overhauls and keep just a few skilled maintenance workers on site to handle day-to-day repairs.

Annual maintenance costs are typically estimated as a fraction of ISBL investment, ranging from 3%/y for a process that handles liquids and gases to 5%/y for a process that involves solids handling or other large mechanical equipment. If a process is known to require regular equipment replacement, the design engineer should make an estimate of the annualized replacement cost (see [Section 9.7.2](#) and Example 9.6) and add this to the maintenance costs. This is the case when the designer expects the process metallurgy to cause some plant items to have a shorter life than the project life, or when excessive wear and tear is expected on equipment such as grinders, dryers, or conveyors.

8.5.3 Land, rent, and local property taxes

Most chemical plants in North America are constructed on rented land or in rented buildings, as it is usually easier and financially more attractive to lease land or property than to tie up capital in purchasing land, putting in the necessary infrastructure, and constructing buildings. Many new plants are built on so-called “brownfield” sites that were previously occupied by other process operations. Local governments often provide incentives for companies to redevelop abandoned industrial sites, and chemical companies themselves often encourage other companies to locate plants on unoccupied parts of their site so as to defray some of their own utility and infrastructure costs.

The costs of land or building rental vary widely between locations and play a strong role in determining plant location; see [Chapter 11](#). As a first approximation, the annual rental cost of land is typically taken as 1% of the ISBL plus OSBL capital cost for a plant that has few buildings, or 2% of ISBL plus OSBL capital cost if the plant is to be located indoors.

Some local, regional, state, or provincial governments charge property taxes on commercial land or buildings. The level of property tax should be determined in consultation with local authorities. For a preliminary analysis, 1% of ISBL plus OSBL capital cost can be assumed.

If the land for the plant is purchased, then the cost of land is included in the initial capital cost and no land charge is added to the fixed costs. When land is capitalized, it cannot be depreciated, but the cost of the land is recovered at the end of the project, less any site clean-up and remediation costs.

8.5.4 Insurance

All plants require insurance to cover third-party liability as well as potential plant damage. Most chemical companies maintain insurance coverage through insurance brokers, although some choose to self-insure, essentially setting aside a part of their operating income to cover liabilities. Insurance premiums are based on prior performance and risk assessments carried out by specialist risk management companies and are typically about 1% of ISBL plus OSBL capital cost per year.

8.5.5 Interest payments

If the project is financed by bonds or loans, the regular payments of interest (or interest plus amortization of principal) are a fixed cost of the project. Debt financing is discussed in more detail in [Section 9.3.2](#). Creditors have a primary claim on earnings over shareholders, so payments on debt must be made as a cost of production rather than set aside to be paid out of retained earnings.

Most companies do not break out the relative proportion of debt and equity financing on a project-by-project basis and instead evaluate projects using an overall average cost of capital; see [Section 9.3.4](#). Repayment of debts associated with the fixed capital investment is therefore included with the overall expected return on capital of the project. In this case the FCOP does not include an interest charge associated with the fixed capital investment. It is, however, common to assume that working capital will be funded by entirely by debt, as the value of the working capital is always sufficient to repay the principal of the loan, and the cost of maintaining the working capital is then equal to just the annual interest due. The annual interest payment is then equal to the working capital multiplied by the interest rate, which can be estimated from the interest rate paid on corporate bonds by similar companies; see [Section 9.3.2](#).

When a company has only one plant—for example, when a new venture is being considered—it is best to separate debt financing from equity financing and calculate the cost of servicing the debt as a fixed cost of production. This gives a truer picture of the likely return on equity from the project.

8.5.6 Corporate overhead charges

There are three main contributions to corporate overhead charges:

Research and development costs: R&D costs cover the costs of fundamental discovery projects, new product development, scale-up and testing, clinical trials for new medicines, and applications testing to develop new markets for existing products. R&D costs range from less than 1% of revenues for operating companies in the fuels and bulk petrochemicals sectors to as much as 15% of revenues for some biotechnology and pharmaceuticals companies.

Selling and marketing costs: Selling costs include the costs of paying the sales force, advertising costs including promotional materials, travel to visit customers and trade shows, and other costs associated with closing sales. Additional marketing expenses include the costs of market research and analysis, competitive studies, branding and any other costs associated with developing an understanding of customer needs and preferences, and using this understanding to guide new product development and product positioning and pricing. Selling and marketing costs are highly dependent on the type of product. Bulk commodities that are produced to ASTM standards have almost zero selling and marketing costs, whereas consumer and specialty products can have selling costs as high as 5% of total cost of production.

General and administrative costs: G&A costs include all the costs of administering a corporation: general management; human resources; purchasing and procurement; finance; accounting; strategic planning; business development; property management; information technology; health, safety, and environment; corporate communications; and legal services. In smaller companies some of these functions may be outsourced to consultants, but the costs must nonetheless be borne by the company, and a share of these costs is attributed to any new project. G&A costs can be assessed by looking at the annual income statement of a company (see [Section 9.3.1](#)) and give an indication of how bloated the corporate bureaucracy has become. G&A costs are usually

allocated based on either revenue or headcount. A typical initial estimate using a headcount-based approach might be 65% of labor cost plus supervision and overhead, whereas a revenue-based approach would assume the same proportion of revenues as in the overall corporate income statement.

Corporate overhead varies widely depending on the industry sector. Oil refining companies that carry out minimal R&D have much lower overhead than pharmaceuticals manufacturers. Company financial reports can be used to gain insight into the typical distribution of overhead costs in different companies and industries. These reports are freely available and easily obtained over the Internet; see [Section 9.3.1](#).

8.5.7 License fees and royalties

A plant that makes use of proprietary technology must make license or royalty payments to the owner of the technology. If a company practices a technology that is under patent protection without obtaining a license to use the technology, the patent holder can sue them for infringement. If the court finds in favor of the patent holder, then the company can be ordered to cease use of the technology and pay damages.

In the process industries, patents typically cover processes, compositions of matter, chemical and biological routes, process conditions, catalysts, enzymes, genetically modified organisms, processing equipment, and control strategies and algorithms. Patent holders often include royalty fees in the pricing of a product such as a catalyst, enzyme, or piece of equipment, and warrant the customer that their product does not infringe on anyone else's patent.

When the royalty is not included in the price of a product, license fees are negotiated between the patent holder (the licensor) and the company that wants to practice the technology (the licensee). The pricing of the royalty payment depends on the availability of alternatives, the licensee's desire to acquire the technology, and the patent holder's desire to sell. See [Section 8.3.1](#) for a general discussion of pricing. Royalties can be set as a percentage of revenues, but the most common method is a fixed rate per lb or kg of product based on total production capacity. Although the royalty rate is proportional to plant capacity, royalties are a fixed cost and must usually be paid regardless of whether or not the plant actually operates at full capacity.

In some cases, the technology owner will agree to let the licensee capitalize the royalty payments. In this case the licensee pays an up-front amount equal to the sum or net present value of future royalty payments, or some fraction thereof. This arrangement allows the patent holder to recognize revenue sooner and allows the licensee to write off the royalty payments as an initial cost of obtaining the technology rather than a running fixed cost. Both parties are then absolved from the inconvenience of handling regular royalty payments.

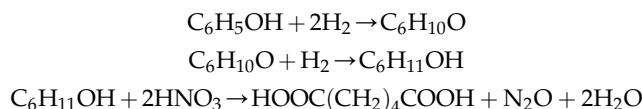
8.6 Summarizing revenues and production costs

It is useful to create a single-page summary of all of the production costs and revenues associated with a project, as this makes it easier to review the project economics and understand the relative contribution of different components to the overall cost of production. The summary sheet usually lists the quantity per year and per unit production of product; the price; the cost per year; and the cost per unit production of product for each of the raw materials, by-products, consumables, and utilities, as well as fixed costs and capital charges.

Most chemical companies have a preferred format for summarizing costs of production and often use standard spreadsheets. Good examples are given in the PERP reports published by Nexant (www.nexant.com/products). A template for summarizing production costs is given in Appendix G and can be downloaded in Microsoft Excel format from the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793. The use of this template is illustrated in Example 8.2.

Example 8.2

Adipic acid is used in the manufacture of nylon 6,6. It is made by hydrogenation of phenol to a mixture of cyclohexanol and cyclohexanone (known as KA oil: ketone and alcohol), followed by oxidation with nitric acid. The overall reaction can be written approximately as:



The actual process requirements of phenol, hydrogen, nitric acid, and utilities and consumables have been determined to be:

Material	Amount	Units
Phenol	0.71572	lb/lb product
Hydrogen	0.0351	lb/lb product
Nitric acid (100% basis)	0.71778	lb/lb product
By-product off-gas	0.00417	lb/lb product
Various catalysts and chemicals	32.85	\$/metric ton product
Electric power	0.0939	kWh/lb product
Cooling water	56.1	gal/lb product
HP steam	0.35	lb/lb product
MP steam	7.63	lb/lb product
Boiler feed water	0.04	gal/lb product

These yields were taken from Chem Systems PERP report 98/99-3 Adipic acid (Chem Systems, 1999). The nitric acid consumption is given on a 100% basis, but 60% nitric acid is used in the process.

Estimate the cash cost of production and total cost of production for a new 400,000 metric ton per year (400 kMTA) adipic acid plant located in northeast Asia (the same scale plant introduced in Example 7.8). The prices of adipic acid, phenol, hydrogen, and nitric acid have been forecasted for northeast Asia as \$1400/MT, \$1000/MT, \$1100/MT, and \$380/MT, respectively. The price of fuel is \$6/GJ (roughly \$6/MMBtu), and electricity costs \$0.05/kWh. Assume a 15% cost of capital and a 10-year project life.

Solution

It is convenient to summarize costs of production in a spreadsheet, as discussed in [Section 8.6](#). The template from Appendix G has been used in this example and is given in [Fig. 8.13](#). In addition to entering the information from the problem statement into the spreadsheet (with any necessary conversion of units), a few more calculations are needed, as described next.

Closing mass balance

The first thing that is apparent when entering the yield data is that the mass balance for the process does not close properly with the information given. This suggests that we still need to account for some waste streams.

The first waste stream is apparent from the process stoichiometry. Nitric acid is recycled in the process until it is eventually converted to N_2O and vented to the atmosphere. The yield of N_2O can therefore be found by a mass balance on nitrogen:

$$\text{Nitrogen fed} = \text{nitrogen purged}$$

$$400,000 \times 0.71778 \times \frac{14}{63} = m_{\text{N}_2\text{O}} \times \frac{2 \times 14}{44}$$

Company Name Address COST OF PRODUCTION Adipic Acid from Phenol		Project Name Adipic acid from phenol				Sheet 1															
Form XXXX-YY-ZZ		Project Number	REV	DATE	BY	APVD	REV														
			1	1.1.07	GPT																
			2	5.6.20	GPT																
Owner's Name Northeast Asia		Capital Cost Basis Year 2006 updated to 2020		Units Metric		On Stream 8,000 hr/yr															
Plant Location Case Description						333.33 day/yr															
YIELD ESTIMATE				CAPITAL COSTS																	
Yield information taken from ChemSystems PERP report 98/99-3, Adipic Acid, p. 89 Yields input for phenol, nitric acid, hydrogen, off-gas, utilities and consumables				<table border="1"> <tr><td></td><td>\$MM</td></tr> <tr><td>ISBL Capital Cost</td><td>247.9</td></tr> <tr><td>OSBL Capital Cost</td><td>99.2</td></tr> <tr><td>Engineering Costs</td><td>34.7</td></tr> <tr><td>Contingency</td><td>52.1</td></tr> <tr><td>Total Fixed Capital Cost</td><td>433.9</td></tr> <tr><td>Working Capital</td><td>60.9</td></tr> </table>					\$MM	ISBL Capital Cost	247.9	OSBL Capital Cost	99.2	Engineering Costs	34.7	Contingency	52.1	Total Fixed Capital Cost	433.9	Working Capital	60.9
	\$MM																				
ISBL Capital Cost	247.9																				
OSBL Capital Cost	99.2																				
Engineering Costs	34.7																				
Contingency	52.1																				
Total Fixed Capital Cost	433.9																				
Working Capital	60.9																				
Scale of production set to 400 t/y = 880 MMb/yr																					
REVENUES AND RAW MATERIAL COSTS																					
MASS BALANCE MB closure 101%																					
Key Products	Units	Units/Unit product	Units/yr	Price \$/unit	\$MM/yr	\$/unit main product															
Adipic acid	t	1	400,000	1400	560.00	1400.00															
Total Key Product Revenues (REV)	t	1	400,000		560.00	1400.00															
By-products & Waste Streams																					
Nitrous oxide (vented)	t		100,261	0	0.00	0.00															
Off-gas	t	0.00417	1,670	700	1.17	2.92															
Organic Waste (Fuel value)	t	0.03072	12,288	300	3.69	9.22															
Aqueous Waste	t		273,440	-1.5	-0.41	-1.03															
Total Byproducts and Wastes (BP)	t	0.0348939	387,659		4.44	11.11															
Raw Materials																					
Phenol	t	0.71572	286,288	1000	286.29	715.72															
Nitric acid 60% (100% basis)	t	0.71778	287,112	380	109.10	272.76															
water with nitric acid	t		191,408	0	0.00	0.00															
Hydrogen, 99%	t	0.0351	14,040	1100	15.44	38.61															
Total Raw Materials (RM)	t	1	778,848		410.83	1027.09															
Gross Margin (GM = REV + BP - RM)					153.61	384.03															
CONSUMABLES																					
Various catalyst and chemicals	kg	32.85	13,138,263	1.00	13.14	32.85															
Other	kg	0	0	0.00	0.00	0.00															
Total Consumables (CONS)					13.14	32.85															
UTILITIES																					
Electric	kWh	206.0	10,300	0.05	4.120	10.30															
HP Steam	t	0.4	18	14.30	2.002	5.01															
MP Steam	t	7.6	382	12.00	36.624	91.56															
LP Steam	t	0.0	0	8.90	0.000	0.00															
Boiler Feed	t	0.3	17	1.10	0.145	0.36															
Condensate	t	0.0	0	0.80	0.000	0.00															
Cooling Water	t	463.0	23,150	0.024	4.445	11.11															
Fuel Fired	GJ	0.0	0	6.00	0.000	0.00															
Total Utilities (UTS)					47.336	118.340															
Variable Cost of Production (VCOP = RM - BP + CONS + UTS)					466.86	1167.16															
FIXED OPERATING COSTS																					
Labor	4.8 Operators per Shift Position				\$MM/yr	\$/unit product															
Number of shift positions	9				30,000 \$/yr each	1.30	3.24														
Supervision	25% of Operating Labor				0.32	0.81															
Direct Ovhd.	45% of Labor & Superv.				0.73	1.82															
Maintenance	3% of Fixed Investment				13.02	32.54															
Overhead Expense	65% of Labor & Maint.				9.99	24.97															
Plant Overhead	2% of Fixed Investment				6.51	16.27															
Tax & Insurance	0% of Fixed Capital				0.00	0.00															
Interest on Debt Financing	6% of Working Capital				3.66	9.14															
Fixed Cost of Production (FOOP)					35.52	88.80															
ANNUALIZED CAPITAL CHARGES																					
Fixed Capital Investment	\$MM	Interest Rate	Life (yr)	ACCR	\$MM/yr	\$/unit product															
Royalty Amortization	433.910	15%	10	0.199	86.46	216.14															
Inventory Amortization	15.000	15%	10	0.199	2.99	7.47															
Catalyst 1	0.000	15%	3	0.438	0.00	0.00															
Catalyst 2	0.000	15%	3	0.438	0.00	0.00															
Adsorbent 1	0.000	15%	3	0.438	0.00	0.00															
Equipment 1	0.000	15%	5	0.298	0.00	0.00															
Equipment 2	0.000	15%	5	0.298	0.00	0.00															
Total Annual Capital Charge					89.45	223.62															
SUMMARY																					
Variable Cost of Production					\$MM/yr	\$/unit product															
466.86					466.86	1167.16															
Fixed Cost of Production					35.52	88.80															
Cash Cost of Production					502.38	1255.96															
Gross Profit					57.62	144.04															
Total Cost of Production					591.83	1479.57															

FIG. 8.13 Cost of production worksheet for Example 8.2.

where m_{N_2O} is the flow rate of N_2O , which can be calculated as 100,261 MT/y. As a first approximation, there is no cost for handling this stream, although we might revisit this at a more detailed design stage if we need to fit vent scrubbers or other equipment to handle this off-gas.

The second waste stream is also apparent from the overall stoichiometry. Phenol has a molecular weight of 100, and adipic acid has a molecular weight of 146, so the stoichiometric requirement of phenol is $= 100/146 = 0.68493$ lb/lb product. The actual process consumption has been estimated as 0.71572 lb/lb product, so the difference ($0.71572 - 0.68493 = 0.03079$ lb/lb) must be converted into organic by-products. It is possible that some of the organic by-product may be material that is lost with the hydrogen-rich fuel gas, but as a first approximation we can assume that we recover an organic liquid waste product from the process. It is also possible (in fact quite likely) that some of the material that we are calling organic by-product is actually losses of organics in the nitrous oxide vent stream. Because this stream probably must be scrubbed before discharge, it is fair to assume as a first approximation that any organic material in it would be collected as an organic waste. This assumption should be revisited at a later stage in the design process when better information on process yields is available. The organic waste stream is priced at a typical fuel value of \$300/MT, assuming that it can be burned as process fuel.

The third waste stream is an aqueous waste. This consists of the water that is brought in with the nitric acid, the water formed by the reaction stoichiometry, and any other water consumed, for example, in vent scrubbers or process water washes.

The water brought in with the nitric acid is easily found by mass balance, because it is equal to the mass flow rate of nitric acid (100% basis) $\times 40/60 = 400,000 \times 0.71778 \times 4/6 = 191,408$ MT/y.

The water formed by reaction stoichiometry can be estimated as 1 mole per mole nitric acid consumed (i.e., 18 MT per 63 MT consumed), giving $400,000 \times 0.71778 \times 18/63 = 82,032$ MT/y. Note that we could also have estimated this as 2 moles per mole product, but that would give an overestimate of the water production, as the amount of nitric acid consumed is less than the apparent stoichiometric requirement. This is because the overall reaction given here is only an approximation and does not include the reaction of cyclohexanone.

The water consumed in process washes and scrubbers is harder to estimate, but because no process water consumption was listed under utilities, we can assume as a first approximation that all the process water needs are met by internal recycles. This gives a total wastewater flow of $191,408 + 82,032 = 273,440$ MT/y. The wastewater stream is assigned a cost of \$1.5/MT (see [Section 8.4.4](#)).

When the values for nitrous oxide, organic waste, and aqueous waste are entered in the spreadsheet, the mass balance shows 101 MT of product for every 100 MT of feed. This is not perfectly closed, but is good enough at this stage in the analysis. The error is most likely in the organic or aqueous waste streams and will have little impact on the economic analysis. This should, of course, be revisited when better process yield data and a converged process simulation are available.

Estimating utility costs

The amounts of utilities consumed are easily estimated from the production rate and the information in the problem statement (with conversion to metric units).

The prices of steam at different levels can be taken from Example 3.2, because the costs of electricity and natural gas are the same. The prices of boiler feed water, condensate, and cooling water are estimated as described in [Section 3.2](#).

The utility cost is about 10% of the variable cost of production. This is typical for many commodity chemical processes.

Estimating fixed costs

This is a relatively complex process and essentially contains two plants: phenol hydrogenation and KA oil oxidation. We should therefore assume at least four shift positions for each plant, say nine total. For a northeast Asian basis we expect that the salary cost per shift position will be lower than the typical \$50,000 per year that we would assume for a USGC plant. As a first approximation, this is estimated as \$30,000/y. The remaining salary and overhead costs are fixed following the assumptions given in [Section 8.5.1](#).

The total fixed capital cost of this plant was estimated in Example 7.8 to be \$433.9MM. Interest charges are not included for the fixed capital (because we will calculate an annualized charge based on overall cost of capital later). An interest charge is included for the working capital, as working capital is recovered at the end of the project and so should not be amortized, as discussed in [Section 9.7.2](#).

The total fixed cost of production is calculated to be \$36 MM/y, which is low compared with the variable cost of production (\$467 MM/y). It is not uncommon for fixed costs to make a relatively minor contribution to the TCOP for a world-scale plant.

Estimating working capital

The working capital is estimated as 7 weeks' cash cost of production minus 2 weeks' feedstock costs plus 1% of the fixed capital investment, as described in [Section 9.2.3](#). Because the CCOP includes the interest payable on the working capital, this sets up a circular reference in the spreadsheet. The spreadsheet options must be adjusted to ensure that the calculation iterates to convergence. The converged result is \$60.9 MM. Note that the value calculated is about 6% less than it would have been had we estimated the working capital as 15% of fixed capital investment.

Estimating annualized capital costs

The fixed capital investment is to be annualized over 10 years at a 15% interest rate. The method for annualization of capital costs is discussed in [Section 9.7.2](#). For this interest rate and recovery period, the annual capital charge ratio is 0.199, so the annual capital charge is $= 0.199 \times 433.9 = \$86.46 \text{ MM/y}$, or \$216.14 /MT of product. As a quick check, we can see that this is roughly 16% of the TCOP, which is in the typical range for commodity chemical processes (10% to 20%).

In addition to the fixed capital investment, we should make an allowance for a process royalty. The problem statement did not specify whether the plant was to be built using proprietary technology, but it is reasonable to assume that a royalty will need to be paid. If a \$15 MM royalty is added, this annualizes to a cost of \$3 MM/y, or roughly 0.5% of revenues, which is a reasonable initial estimate. This should be revisited during more detailed design when discussions with technology vendors take place.

Estimating cost of production

The CCOP is the sum of the fixed and variable production costs (Equation 8.3):

$$\text{CCOP} = \text{VCOP} + \text{FCOP} = 466.86 + 36.94 = \underline{\underline{\$503.81 \text{ MM/y}}}$$

The TCOP is the sum of the CCOP and the annual capital charge ([Equation 8.6](#)):

$$\text{TCOP} = \text{CCOP} + \text{ACC} = 502.38 + 89.45 = \underline{\underline{\$591.83 \text{ MM/y}}}$$

It is worth noting that the calculated TCOP is greater than the projected annual revenue of \$560 MM/y. This suggests that the project would not earn the expected 15% interest rate. This is explored further in Examples 9.5, 9.7, and 9.9 and in Problem 9.6.

8.7 References

Chem Systems. (1999). Adipic acid: PERP report 98/99-3. Chem Systems, Inc.

8.8 Nomenclature

		Dimensions in \$MLT ^θ
ACC	Annual capital charge	\$
CCOP	Cash cost of production	\$M ⁻¹ or \$T ⁻¹
FCOP	Fixed cost of production	\$M ⁻¹ or \$T ⁻¹
P _F	Price of fuel	\$M ⁻¹ L ⁻² T ²
P _{WV}	Value of waste as fuel	\$M ⁻¹
TCOP	Total cost of production	\$M ⁻¹ or \$T ⁻¹
VCOP	Variable cost of production	\$M ⁻¹ or \$T ⁻¹
ΔH ^{°c}	Heat of combustion	L ⁻² T ²

8.9 Problems

8.1. A gas cleaning plant uses a recirculating alkanolamine solvent to remove H_2S and CO_2 from natural gas. The solvent circulation rate has been estimated to be 2.2 gallons per thousand standard cubic feet of natural gas. The solvent is regenerated using low-pressure steam at a rate of 1.3 lb of steam per gallon of solvent. The solvent is pumped from the regenerator pressure of 1.5 bar gauge to the absorber pressure of 60 bar gauge. The annual cost of solvent make-up for a 50 million scf/d plant is \$125,000. If fuel gas costs \$6/MMBtu and electricity costs \$0.05/kWh, estimate the variable cost per scf of natural gas.

8.2. The consumption of feed and utilities for a plant that produces 60 metric tons per year (60 t/y) of nylon 6 are given here. The plant fixed capital cost has been estimated to be \$114 MM. Estimate the cash cost of production.

Raw materials

1. Caprolactam, 1.02 tons per ton product, price \$1700 per metric ton

Utilities

1. Fuel gas, 9 MMBtu/t product, cost \$4.50/MMBtu

2. Cooling water, 32,000 gal/t product

3. Electrical power, 130 kWh/t product

8.3. The production of methyl ethyl ketone (MEK) is described in Appendix F, Problem F.3. A preliminary design has been made for a plant to produce 10,000 metric tons per year. The plant operating rate will be 8000 hours per year. Estimate the cash cost of production. (Note: The capital cost for this process is estimated in Problem 7.10)

Raw materials

1. 2-butanol, 1.045 kg per kg of MEK, price \$800 per metric ton

2. Solvent (trichloroethane) make-up 7000 kg per year, price \$1.0/kg

Utilities

3. Fuel oil, 3000 metric tons per year, heating value 45 GJ/metric ton

4. Cooling water, 120 metric tons per hour

5. Steam, low pressure, 1.2 metric tons per hour

6. Electrical power, 1 MW

The fuel oil is burnt to provide flue gases for heating the reactor feed and the reactor. Some of the fuel requirements could be provided by using the by-product hydrogen. Also, the exhaust flue gases could be used to generate steam. The economics of these possibilities need not be considered.

8.4. Natural gas is widely used as fuel in the process industries, so the price of natural gas is often an important variable in economic analysis. Develop a 15-year forecast of U.S. natural gas prices. Historic natural gas prices are available at www.eia.doe.gov. The “industrial” price data set is most typical of the contract prices paid by large industrial consumers. Consider any structural changes that may have occurred in the natural gas market when making your forecast.

8.5. Estimate the price (\$/metric ton) in 2010, 2020, and 2030 of:

a. Sulfuric acid, 98 wt%

b. Sodium hydroxide

c. Activated carbon adsorbent

d. Deionized water

8.6. Table 8.3 contains 6-month average price data for styrene and its feedstocks ethylene and benzene for South Korea over the period Jan. 1999 to Dec. 2010. Calculate the gross margin for making styrene from benzene and ethylene, and plot how the gross margin varied over this time period. Develop a forecast of the gross margin of styrene from benzene and ethylene for the time period 2010–2030.

8.7. Use the data in Table 8.3 to construct forecasts of the prices of benzene, ethylene, and styrene in South Korea from 2010 to 2030.

TABLE 8.3 South Korean prices for benzene, ethylene, and styrene (all prices in \$/metric ton)

Period	Benzene	Ethylene	Styrene
1H99	218.98	355.67	403.02
2H99	262.86	481.00	603.81
1H00	366.38	520.67	838.58
2H00	404.54	587.83	684.75
1H01	338.19	534.50	529.21
2H01	216.70	467.50	416.08
1H02	293.30	450.67	570.92
2H02	375.11	516.50	659.23
1H03	483.86	580.00	677.75
2H03	415.99	558.17	709.96
1H04	638.07	669.17	870.61
2H04	1055.59	778.83	1278.42
1H05	925.58	829.67	1087.93
2H05	785.97	950.00	1021.90
1H06	815.33	1025.83	1075.45
2H06	931.42	1080.00	1252.43
1H07	1037.46	1122.83	1288.28
2H07	1018.83	1286.83	1341.37
1H08	1145.30	1560.83	1451.25
2H08	846.33	1435.17	1094.41
1H09	539.82	754.83	838.95
2H09	845.93	1096.83	1077.45
1H10	949.62	1252.17	1226.10
2H10	907.75	1221.07	1176.12

Notes:

1. 1H03 denotes the first half of 2003, 2H99 the second half of 1999, etc.

2. Source: CMAI, www.CMAIglobal.com.

Economic evaluation of projects

KEY LEARNING OBJECTIVES

- How corporations finance projects
- How to allow for taxes, depreciation, and other investment incentives
- Different criteria that companies use to compare the financial attractiveness of alternative projects and other factors that are also considered in project selection
- How to allow for error in cost estimates
- How to carry out an economic sensitivity analysis

9.1 Introduction

As the purpose of investing money in a chemical plant is to earn more money, some means of comparing the economic performance of projects is needed. Before a company agrees to spend a large amount of capital on a proposed project, the management must be convinced that the project will provide a sound investment compared with other alternatives. This section introduces the principal methods used for making economic comparisons between projects.

9.2 Cash flows during a project

9.2.1 Cash flow diagrams

During any project, cash initially flows out of the company to pay for the costs of engineering, equipment procurement, plant construction, and plant start-up. Once the plant is constructed and begins operation, revenues from the sale of product begin to flow into the company. The estimation of project capital costs is discussed in Chapter 7 and the estimation of revenues and costs of production is described in Chapter 8.

The “net cash flow” at any time is the difference between the earnings and expenditure. A cash flow diagram, such as that shown in Fig. 9.1, shows the forecast cumulative net cash flow over the life of a project. The cash flows are based on the best estimates of investment, operating costs, sales volume, and sales price that can be made for the project. A cash flow diagram gives a clear picture of the resources required for a project and the timing of the earnings. The diagram can be divided into the following characteristic regions:

A–B The investment required to design the plant.

B–C The heavy flow of capital to build the plant and provide funds for start-up, including working capital.

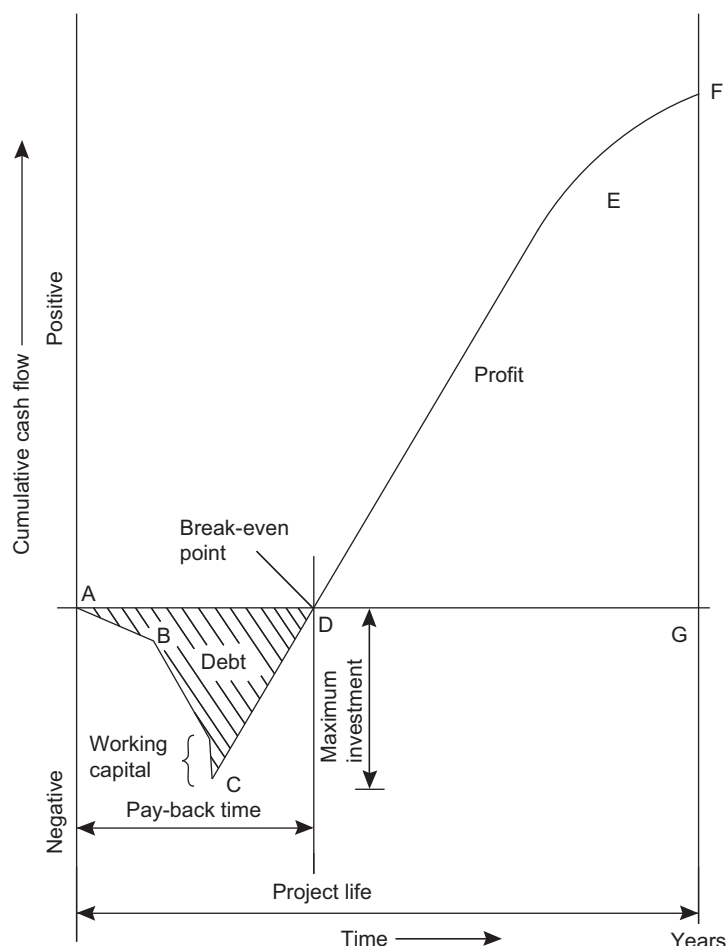


FIG. 9.1 Project cash flow diagram.

C–D The cash flow curve turns up at C, as the process comes on stream and income is generated from sales. The net cash flow is now positive, but the cumulative amount remains negative until the investment is paid off, at point D.

Point D is known as the *break-even point*, and the time to reach the break-even point is called the *pay-back time*. (In a different context, the term “break-even point” is also sometimes used for the percentage of plant capacity at which the income equals the cost of production.)

D–E In this region the cumulative cash flow is positive. The project is earning a return on the investment.

E–F Toward the end of project life the rate of cash flow may tend to fall off because of increased operating costs and falling sales volume and price as a result of obsolescence of the plant, and the slope of the curve changes.

Point F gives the final cumulative net cash flow at the end of the project life.

Net cash flow is a relatively simple and easily understood concept and forms the basis for the calculation of other, more complex, measures of profitability. Taxes and the effect of depreciation are usually not considered in cash flow diagrams.

9.2.2 Cash outflows during design and construction

The rate at which the operating company spends money during the design and construction phase of a project is usually determined by the terms of the contract that they have with an engineering, procurement, and construction (EPC) company. The EPC company, commonly known as a *contractor*, maintains subcontracts with fabricators of pressure vessels, heat exchangers, and other large process equipment, as well as with suppliers of pipe, instruments,

steelwork, wiring, and all the other materials needed to build the plant. The actual purchase (procurement) of individual pieces of equipment is all handled by the contractor.

Most plant construction contracts are divided into several phases based on achievement of defined milestones. A percentage of the total contract amount is payable as each milestone is reached. For example, a common approach is to pay a percentage at signing, at completion of detailed engineering, at groundbreaking, and at mechanical completion and the balance upon successful completion of a performance demonstration run. The exact amount payable at each milestone will be subject to contract negotiations, but the final payment is usually large enough to ensure that contractor will not make any profit until the plant is running properly.

Large projects can take several years to complete. A typical schedule for a single large plant could be a year in detailed engineering and 2 years in construction. Pharmaceutical plants that require regulatory approval can take 5 or 6 years. This is discussed further in Section 9.6.2 (see [Table 9.2](#) for a typical start-up schedule).

9.2.3 Working capital

Working capital is the additional money needed above what it cost to build the plant, start the plant up, and keep the plant running. Working capital is best thought of as the money that is tied up in sustaining plant operation. Working capital typically includes:

1. Value of raw material inventory – usually estimated as 2 weeks' delivered cost of raw materials.
2. Value of product and by-product inventory – estimated as 2 weeks' cost of production.
3. Cash on hand – estimated as 1 week's cost of production.
4. Accounts receivable – products shipped but not yet paid for; estimated as 1 month's cost of production, but could be larger depending on customer payment terms.
5. Credit for accounts payable – feedstocks, solvents, catalysts, packaging, etc., received but not yet paid for; estimated as 1 month's delivered cost, but could be larger depending on terms negotiated with vendors.
6. Spare parts inventory – estimated as 1% to 2% of inside battery limits (ISBL) plus outside battery limits (OSBL) investment cost.

It can be seen that the sum of items 1 through 5 is roughly 7 weeks' cost of production minus 2 weeks' feedstock costs (item 5 is a credit).

Working capital can vary from as low as 5% of the fixed capital for a simple, single-product process, with little or no finished product storage, to as high as 30% for a process producing a diverse range of product grades for a sophisticated market, such as synthetic fibers. A typical figure for petrochemical plants is 15% of the fixed capital (ISBL plus OSBL cost).

Working capital is better estimated from the cost of production rather than the capital investment. The rules of thumb given earlier can be used to make a preliminary estimate of the working capital required.

Working capital is still required as long as the plant is in operation, and it should not be confused with miscellaneous start-up costs. The working capital that is tied up in operating the plant can be reduced by tighter management of inventories and accounts receivable, and these are the goals of lean management methods. Working capital is recovered at the end of the plant life when the remaining inventories are eliminated. Working capital cannot be depreciated because it does not lose value due to "wear and tear."

Companies will sometimes discuss working capital turns, or turnover, defined as the ratio of annual revenues divided by average working capital. Working capital turnover can be increased by reducing working capital. Higher (or increasing) values of working capital turns indicate that the leaders of a business are managing working capital efficiently.

Other methods for estimating the working capital requirement are given by [Bechtel \(1960\)](#), [Lyda \(1972\)](#), and [Scott \(1978\)](#).

9.2.4 Cash flows at the end of the project

If a plant ceases operation, or is "mothballed" (shut down on a semi-permanent basis), then the working capital is recovered, but must be reinvested if the plant is restarted. When a plant is shut down permanently, it can be sold in its entirety or else broken up and sold as scrap. Several companies specialize in buying and reselling secondhand plants, and advertisements for used plants and equipment can usually be found in the classified sections of the trade journals and on resale websites such as www.equipnet.com. The scrap value can be estimated based on the

equipment weight and is usually less than 10% of the ISBL investment. OSBL investments are not recovered unless the entire site is shut down.

If land was purchased for the plant, which is increasingly uncommon, then the land can be sold as an additional end-of-life credit. Site remediation costs must be subtracted from the value of the land, so the additional cash flow from the sale of land is not necessarily positive.

These cash flows at the end of the project are often not included in profitability analysis, as their timing is uncertain and they are often far enough in the future that they have negligible impact on any of the measures of profitability.

9.3 Project financing

The construction and operation of chemical plants require large amounts of capital. Corporations engaged in the production of chemicals must raise the finances to support such investments. Like taxation, corporate financing is a specialized subject with many intricacies that require expert knowledge. The way a corporation raises funds to finance projects determines the cost of capital for that corporation, and hence sets the expected financial rate of return that the projects must achieve. The design engineer therefore needs a superficial awareness of this subject to carry out economic analysis and optimization of the design.

9.3.1 Basics of corporate accounting and finance

The purpose of financial accounting is to report the economic performance and financial condition of a company to its owners (shareholders), lenders, regulatory agencies, and other stakeholders. The primary means for financial reporting is the annual report to shareholders. The annual reports for companies in the chemical, pharmaceutical, and fuels industries generally contain:

1. A letter from the chief executive officer (CEO) describing the past year's operations, significant acquisitions, divestitures and restructuring, and plans for the short and long term
2. Financial information:
 - a. Balance sheet
 - b. Income statement
 - c. Cash flow statement
 - d. Notes to the financial statements
 - e. Comments from the independent auditors
3. Information on the directors and executive management of the company
4. A report on the health, safety, and environmental performance of the company (sometimes published separately)

The annual report of any publicly traded company will usually be available online and can easily be found by visiting the company's website. The site will usually have a prominent link to "information for investors" or something similar. No attempt has been made to create fictitious financial statements for the purposes of this book, as an abundance of real examples is readily available online. The reader is encouraged to search online for real cases to study.

Balance sheet

The balance sheet is a snapshot of the financial condition of the company. It lists all the assets owned by the company and all the liabilities or amounts owed by the company. The difference between assets and liabilities is the stockholder's equity (i.e., notionally the amount of money the stockholders would have available to share out if they decided to liquidate the company).

$$\text{Stockholder's equity} = \text{assets} - \text{liabilities} \quad (9.1)$$

Assets are traditionally listed in order of decreasing liquidity. Liquidity is a measure of how easily the asset could be turned into cash. Assets include:

- Cash and cash equivalents.
- Notes and accounts receivable (i.e., money owed to the company for goods shipped but not yet paid for).
- Inventories of raw materials, products, spare parts, and other supplies.

- Prepaid taxes and expenses.
- Investments such as equity stakes in other companies or joint ventures.
- Property, plant, and equipment. This is listed at book value (i.e., cost less accumulated depreciation). The actual market value of these assets may be considerably higher.
- Intangible assets such as patents, trademarks, goodwill, etc.

Liabilities are usually listed in the order in which they are due, starting with current liabilities. Liabilities include:

- Accounts payable (i.e., payment owed on goods already received by the company).
- Notes and loans due for repayment.
- Accrued liabilities and expenses such as legal settlements, amounts set aside for warranties, guarantees, etc.
- Deferred income taxes.
- Long-term debt.

The difference between assets and liabilities is the shareholder's equity. This consists of the capital paid in by the owners of common and preferred stocks, together with earnings retained and reinvested in the business. The capital paid in by the shareholders is often listed as the par value of the stock (typically 25¢ to \$1 per share) plus the additional capital paid in when the stock was initially sold by the company. Note that this reflects only the capital raised by the company and has no relation to subsequent increases or decreases in the value of the stock that may have resulted from trading on the stock market.

Income statement

The income statement or consolidated statement of operations is a summary of the incomes, expenditures, and taxes paid by the company over a fixed period. Results are usually presented for the past 3 calendar years.

The income statement lists the following items:

1. Sales and operating revenues (positive)
2. Income from equity holdings in other companies (positive)
3. Cost of goods sold (negative)
4. Selling, general, and administrative expenses (negative)
5. Depreciation (negative on the income statement, but will be added back on the cash flow statement)
6. Interest paid on debt (negative)
7. Taxes other than income tax, such as excise duties (negative)
8. Income taxes (negative)

The sum of items 1 through 5 is sometimes listed as earnings before interest and taxes (EBIT). The sum of items 1 through 7 is listed as income before taxes, or taxable income, and is usually positive. The net income is the sum of items 1 through 8 (i.e., income before taxes minus taxes paid). Net income is also usually expressed as earnings per share of common stock.

The fixed and variable costs discussed in [Chapter 8](#) generally fall into the category of cost of goods sold, except that the costs of corporate functions such as research and development, marketing, finance, information technology, legal, and human resources are usually rolled up into the selling, general, and administrative (SG&A) expenses.

The income statement gives a good insight into the overall profitability and margins of a business. It has to be read carefully, though, as several items listed are noncash charges, such as depreciation, that do not affect the cash flow of the business. Corrections for these items are made in the cash flow statement.

Cash flow statement

The cash flow statement gives a summary of overall cash flows into and out of the business as a result of operating activities, investments, and financing activities. It is also usually reported for the past 3 calendar years.

The cash flow from operating activities section starts with the net income. Adjustments are made for noncash transactions (depreciation and deferred taxes are added back in) and changes in assets and liabilities.

The cash flow from investing activities section lists the cash spent on acquiring fixed assets such as property, plant, and equipment, less any revenues from the sale of fixed assets. It also lists acquisitions or divestitures of subsidiary businesses.

The cash flow from financing activities section summarizes changes in the company's long-term and short-term debt, proceeds from issues of common stock, repurchase of stocks, and dividends paid to stockholders.

The sum of cash flows from operations, investments, and financing gives the net change in cash and cash equivalents. This is then added to the cash and cash equivalents from the beginning of the year to give the cash and cash equivalents at the end of the year, which appears on the balance sheet.

Summary

The business and accounting literature contains a wealth of information on how to read and analyze corporate financial statements. Most engineers work for or with corporations and have a direct personal interest in understanding financial performance; however, a detailed treatment of the subject is beyond the scope of this book. Excellent introductions to finance and accounting are given in the books by Spiro (1996), Fields (2016), and Shim and Henteleff (1995).

9.3.2 Debt financing and repayment

Most debt capital is raised by issuing long-term bonds. A mortgage is a bond that is backed by pledging a specific real asset as security against the loan. An unsecured bond is called a *debenture*. The ratio of total debt divided by total assets is known as the *debt ratio* (DR), or leverage of the company.

All debt contracts require payment of interest on the loan and repayment of the principal (either at the end of the loan period or amortized over the period of the loan). Interest payments are a fixed cost, and if a company defaults on these payments, its ability to borrow money will be drastically reduced. Because interest is deducted from earnings, the greater the leverage of the company, the higher the risk to future earnings, and hence to future cash flows and the financial solvency of the company. In the worst case, the company could be declared bankrupt and the assets of the company sold off to repay the debt. Finance managers therefore carefully adjust the amount of debt owed by the company so that the cost of servicing the debt (the interest payments) does not place an excessive burden on the company.

The rate of interest owed on debt depends on the bond markets, government central banks, and the credit worthiness of the company. When new bonds are issued, they must be offered at a competitive interest rate; otherwise, they will not sell. If the bond issuer has a high credit rating, they will be able to issue bonds at low interest rates, close to the interest rates set by the government for government bonds. (Most governments try to maintain a high credit rating and avoid defaulting on their bonds so that they can continue to borrow money at low interest rates.) If the credit rating of the issuer is lower, there is a higher chance that the debt may not be repaid, in which case it must be offered at a higher interest rate to offset this risk. Credit rating services such as Moody's and Standard & Poor's study the finances of governments and corporations and publish credit ratings. These ratings are usually not advertised by issuers unless they are very high, but they are published in the financial papers. The difference in interest rate between low-rated and high-rated bonds issued at the same time is typically 2% to 3%.

Once they have been issued, bonds are traded on the New York Stock Exchange, the American Stock Exchange, or other global financial exchanges. Although the price of the bond in subsequent trading may vary from the offer price (or face value), the interest rate remains fixed. The *Wall Street Journal* reports prices daily for the most actively traded corporate bonds. Bond prices can also be found at the Financial Industry Regulatory Authority (FINRA) website (<http://finra-markets.morningstar.com/MarketData/>), together with a lot of other useful information on bond markets and investing. The interest rate is listed as the "coupon," and the date on which the bond expires is the "maturity." Bonds are also assigned a unique nine-digit identification number by the American Bankers' Association Committee on Uniform Security Identification Procedures (CUSIP). For example, in 2006 Honeywell, Inc., issued a 30-year bond CUSIP #438516AR7 with coupon 5.700 and maturity 03/15/2036.

9.3.3 Equity financing

Equity capital consists of the capital contributed by stockholders, together with earnings retained for reinvestment in the business. Stockholders purchase stocks in the expectation of getting a return on their investment. This return can come from the dividends paid annually to stockholders (the part of earnings returned to the owners) or from growth of the company that is recognized by the stock market and leads to an increase in the price of the stock.

Most stock is usually held by sophisticated institutional investors such as banks, mutual funds, insurance companies, and pension funds. These investors employ expert analysts to assess the performance of companies relative to other companies in the same sector and to the market as a whole. If the management of a company

does not deliver the financial return expected by investors, the stock price will suffer and the management will soon be replaced.

Simple measures of the effectiveness of management are the return on equity (ROE) and earnings per share. ROE is defined as:

$$\text{ROE} = \frac{\text{net annual profit}}{\text{stockholders' equity}} \times 100\% \quad (9.2)$$

The stockholders' expectation of return on their equity can be expressed as an interest rate and is known as the *cost of equity capital*. The cost of equity required to meet the expectations of the market is usually substantially higher than the interest rate owed on debt because of the riskier nature of equity finance (because debt holders are paid first and hence have the primary right to any profit made by the business). For most corporations in the United States at the time of writing, the cost of equity is in the range 25% to 30%.

9.3.4 Cost of capital

Very few companies operate entirely on debt or equity financing alone, and most use a balance of both. The overall cost of capital is simply the weighted average of the cost of debt and the cost of equity.

$$i_c = (DR \times i_d) + ((1 - DR) \times i_e) \quad (9.3)$$

where: i_c = cost of capital

DR = debt ratio

i_d = interest rate due on debt

i_e = cost of equity

For example, if a company were financed 55% with debt at an average 8% interest and 45% with equity that carried an expectation of a 25% return, then the overall cost of capital would be:

$$\begin{aligned} i_c &= (0.55 \times 0.08) + (0.45 \times 0.25) \\ &= \underline{\underline{0.1565}} \end{aligned}$$

Because the equity is by definition (Equation 9.1) the assets minus the liabilities (debt), the overall return on assets (ROA) can be expressed as:

$$\text{ROA} = \frac{\text{net annual profit}}{\text{total assets}} \times 100\% \quad (9.4)$$

It follows that:

$$\frac{\text{ROA}}{\text{ROE}} = \frac{\text{stockholders' equity}}{\text{total assets}} = 1 - DR \quad (9.5)$$

The overall cost of capital sets the interest rate that is used in economic evaluation of projects. The total portfolio of projects funded by a company must meet or exceed this interest rate if the company is to achieve its targeted ROE and hence satisfy the expectations of its owners.

9.4 Taxes and depreciation

The profits generated by most chemical plants are subject to taxation. Taxes can have a significant impact on the cash flows from a project. The design engineer needs to have a basic understanding of taxation and tax allowances such as depreciation to make an economic evaluation of the project.

9.4.1 Taxes

Individuals and corporations must pay income tax in most countries. The details of tax law can be complicated, and governments enact changes almost every year. Companies generally retain tax specialists, either as employees or as consultants, who have deep expertise in the intricacies of the field. Such specialized knowledge is not required

for engineering design projects, which are usually compared on a relatively simple after-tax basis. The design engineer may occasionally need to consult a tax expert, though, particularly when comparing projects in different countries with different tax laws.

Information on corporate taxes in the United States is given on the Internal Revenue Service website at www.irs.gov. At the time of writing, the rate of federal income tax on corporations in the United States is 21% (IRS Publication 542). In many locations corporations must also pay state or local income taxes.

In Canada, corporations pay income tax under the Canadian Income Tax Act (R.S.C., 1985, c. 1 (5th Supp.)). Information on the Canadian Income Tax Act can be obtained from www.laws-lois.justice.gc.ca.

The amount of tax that must be paid in a given year is calculated by multiplying the taxable income by the tax rate. The taxable income is given by:

$$\text{Taxable income} = \text{gross profit} - \text{tax allowances} \quad (9.6)$$

Various types of tax allowances are permitted in the tax laws of different countries, the most common of which is depreciation, discussed in Section 9.4.3. The after-tax cash flow is then:

$$\begin{aligned} CF &= P - (P - D)t_r \\ &= P(1 - t_r) + Dt_r \end{aligned} \quad (9.7)$$

where: CF = after-tax cash flow

P = gross profit

D = sum of tax allowances

t_r = rate of taxation

It can be seen from Equation 9.7 that the effect of tax allowances is to reduce taxes paid and increase cash flow.

In some countries, taxes are paid in a given year based on the previous year's income. This is true for the United States, where corporate taxes are based on a calendar year of operations and are due by March 15 of the following year. This complicates the calculations somewhat, but is easily coded into a spreadsheet.

9.4.2 Investment incentives

National and regional governments often provide incentives to encourage companies to make capital investments, because these investments create employment, generate taxation revenue, and provide other benefits to politicians and the communities they represent.

The most common incentives used are tax allowances. Most countries allow some form of depreciation charge as a tax allowance, by which the fixed capital investment can be deducted from taxable income over a period, as described in Section 9.4.3. Other incentives that are often used include:

1. Tax waivers or vacations, in which no taxes are paid for a fixed period, typically 2 to 5 years after the project begins generating revenue.
2. Investment grants or credits, in which the government makes a cash contribution towards the initial investment.
3. Low-cost loans, in which the government either loans capital directly or else subsidizes the interest due on a commercial loan.
4. Loan guarantees, in which the government agrees to underwrite loans for the project, reducing the risk of the loan and hence making it easier to secure financing on advantageous terms.

Any economic comparison between different process alternatives for the same site should usually be made using the same assumptions on investment incentives. This might not always be the case, though, for example, if one project is eligible for a government grant because of using renewable energy and another project is not. It should also be noted that differences in incentives can have a significant impact on investment decisions when comparing investments at a company-wide level in a global context.

9.4.3 Depreciation charges

Depreciation charges are the most common type of tax allowance used by governments as an incentive for investment. Depreciation is a noncash charge reported as an expense, which reduces income for taxation purposes. There is no cash outlay for depreciation, and no money is transferred to any fund or account, so the depreciation charge is added back to the net income after taxes to give the total cash flow from operations.

$$\begin{aligned}
 CF &= I - (I \times t_r) + D \\
 &= (P - D) - ((P - D) \times t_r) + D \\
 &= P(1 - t_r) + Dt_r
 \end{aligned} \tag{9.8}$$

where: I = taxable income

D = depreciation tax allowance

It can be seen that Equations 9.7 and 9.8 are equivalent.

Depreciation charges can be thought of as an allowance for the “wear and tear, deterioration or obsolescence of the property” as a result of its use (IRS publ. 946).

The book value of an asset is the original cost paid minus the accumulated depreciation charged. The book value has no connection to the resale value or current market value of the asset.

$$\text{Book value} = \text{initial cost} - \text{accumulated depreciation} \tag{9.9}$$

Note that the law usually only allows depreciation of fixed capital investments and not total capital, because working capital is not consumed and can be recovered at the end of the project. If land was purchased for the project, then the cost of the land must be deducted from the fixed capital cost, as land is assumed to retain its value and cannot be depreciated.

Over time the book value of the asset or fixed investment decreases until it is fully “paid off” or “written off,” at which point depreciation can no longer be charged. The schedule of how depreciation charges are taken is set by the tax law. In the United States, most investments are depreciated using either straight-line depreciation or the Modified Accelerated Cost Recovery System (MACRS) described later (IRS publ. 946), but other methods are used internationally, and in a globalized economy it is necessary for design engineers to have familiarity with these also.

Straight-line depreciation

Straight-line depreciation is the simplest method. The depreciable value, C_d , is depreciated over n years with annual depreciation charge D_i in year i , where:

$$D_i = \frac{C_d}{n} \quad \text{and} \quad D_j = D_i \quad \forall j \tag{9.10}$$

The depreciable value of the asset is the initial cost of the fixed capital investment, C , minus the salvage value (if any) at the end of the depreciable life. For chemical plants, the salvage value is often taken as zero, as the plant usually continues to operate for many years beyond the end of the depreciable life.

The book value of the asset after m years of depreciation, B_m is:

$$\begin{aligned}
 B_m &= C - \sum_{i=1}^m D_i \\
 &= C - \frac{mC_d}{n}
 \end{aligned} \tag{9.11}$$

When the book value is equal to the salvage value (or zero), then the asset is fully depreciated and no further depreciation charge can be taken.

Straight-line depreciation must be used in the United States for software (with a 36-month depreciable life), patents (with life equal to the patent term remaining), and other depreciable intangible property (IRS publ. 946).

Straight-line depreciation is allowed under the U.S. tax code for other classes of property, with defined life depending on the type of asset. Although straight-line depreciation is not as favorable as the accelerated methods described in the following sections, it is still the preferred method for most large corporations. This is because it is difficult and costly for an established company to change accounting methods, and most large corporations have used straight-line depreciation in previous years.

Declining-balance depreciation

The declining-balance method is an accelerated depreciation schedule that allows higher charges in the early years of a project. This helps improve project economics by giving higher cash flows in the early years. In the declining-balance method, the annual depreciation charge is a fixed fraction, F_d , of the book value:

$$D_1 = C F_d \quad (9.12)$$

$$B_1 = C - D_1 = C(1 - F_d)$$

$$D_2 = B_1 F_d = C(1 - F_d)F_d$$

$$B_2 = B_1 - D_2 = C(1 - F_d)(1 - F_d) = C(1 - F_d)^2$$

Hence:

$$D_m = C(1 - F_d)^{m-1}F_d \quad (9.13)$$

$$B_m = C(1 - F_d)^m \quad (9.14)$$

The fraction F_d must be equal to or less than $2/n$, where n is the depreciable life in years. When $F_d = 2/n$, this method is known as *double declining-balance depreciation*.

Modified Accelerated Cost Recovery System

The MACRS depreciation method was established by the U.S. Tax Reform Act of 1986. The details of the MACRS depreciation method are given in IRS publication 946, which is available online at www.irs.gov/publications. The method is basically a combination of the declining-balance method and the straight-line method. The declining-balance method is used until the depreciation charge becomes less than it would be under the straight-line method, at which point the MACRS method switches to charge the same amount as the straight-line method.

Under MACRS depreciation, different recovery periods are assigned to different kinds of asset, based on a usable life ("class life") designated by the U.S. Internal Revenue Service. For chemicals manufacture, the latest version of the IRS publication 946 at the time of writing lists a class life of 9.5 years and a recovery period of 5 years (see www.irs.gov/publications/P946 Appendix B). Class life for other process industries ranges from 7.5 years for offshore oil production to 18 years for coal gasification, sugar production, and vegetable oil purification. Always consult the IRS publication to determine the appropriate schedule for a given project. It should be noted, however, that for roads, docks, and other civil infrastructure a 15-year recovery period is used, whereas for cogeneration, power transmission, and natural gas piping the recovery period is 20 years, so some offsite investments are depreciated on a different schedule from that used for the ISBL investment.

Another important convention within MACRS depreciation is that the method assumes that all property is acquired mid-year and hence assigns half of the full year's depreciation in the first and last years of the recovery period. The result is the schedule of depreciation charges given in Table 9.1.

There are other details of MACRS depreciation that are not discussed here, and at the time of writing the tax law also allows assets to be depreciated by the straight-line method (over the class life, not the recovery period, and still

TABLE 9.1 MACRS depreciation charges

Recovery year	Depreciation rate ($F_i = D_i/C_d$)		
	Five-year recovery	Seven-year recovery	Fifteen-year recovery
1	20	14.29	5.00
2	32	24.49	9.50
3	19.2	17.49	8.55
4	11.52	12.49	7.70
5	11.52	8.93	6.93
6	5.76	8.92	6.23
7		8.93	5.90
8		4.46	5.90
9			5.91
10			5.90

TABLE 9.1 MACRS depreciation charges—cont'd

Recovery year	Depreciation rate ($F_i = D_i/C_d$)		
	Five-year recovery	Seven-year recovery	Fifteen-year recovery
11			5.91
12			5.90
13			5.91
14			5.90
15			5.91
16			2.95

following the half-year convention). Many large corporations use straight-line depreciation instead of MACRS, as they have historically used the straight-line method and do not want the added costs and financial uncertainty of changing the accounting approach.

The tax law is revised frequently, and the most recent version of IRS publication 946 should be consulted for the current regulations. Similarly, when analyzing international projects, the appropriate national and regional tax laws must be checked to ensure that the correct depreciation rules are followed. There are several other less widely used depreciation methods that are not discussed here. A good overview of these is given by [Humphreys \(1991\)](#).

Example 9.1

A chemical plant with a fixed capital investment of \$100 million generates an annual gross profit of \$50 million. Calculate the depreciation charge, taxes paid, and after-tax cash flows for the first 10 years of plant operation using straight-line depreciation over 10 years and using MACRS depreciation with a 5-year recovery period. Assume the plant is built at time zero and begins operation at full rate in year 1. Assume the rate of corporate income tax is 21% and taxes must be paid based on the previous year's income.

Solution

The solution is easily coded into a spreadsheet. The results are shown in the following tables:

Solution 9.1a

Year	Gross profit (MM\$)	Depreciation charge (MM\$)	Taxable income (MM\$)	Taxes paid (MM\$)	Cash Flow (MM\$)
0	0	0	0	0	-100
1	50	10	40	0	50
2	50	10	40	8.4	41.6
3	50	10	40	8.4	41.6
4	50	10	40	8.4	41.6
5	50	10	40	8.4	41.6
6	50	10	40	8.4	41.6
7	50	10	40	8.4	41.6
8	50	10	40	8.4	41.6
9	50	10	40	8.4	41.6
10	50	10	40	8.4	41.6

Solution 9.1b

Year	Gross profit (MM\$)	Depreciation charge (MM\$)	Taxable income (MM\$)	Taxes paid (MM\$)	Cash Flow (MM\$)
0	0	0	0	0	-100
1	50	20	30	0	50
2	50	32	18	6.30	43.70
3	50	19.2	30.8	3.78	46.22
4	50	11.52	38.48	6.47	43.53
5	50	11.52	38.48	8.08	41.92
6	50	5.76	44.24	8.08	41.92
7	50	0	50	9.29	40.71
8	50	0	50	10.50	39.50
9	50	0	50	10.50	39.50
10	50	0	50	10.50	39.50

9.5 Simple methods for economic analysis

This section introduces some economic measures that can be estimated quickly if the project investment and cash flows are known. These methods are widely used for preliminary “eyeballing” of project attractiveness, and in some companies they are used for preliminary screening of projects. They are not suitable for detailed project selection, as they involve too many simplifying assumptions. The choice of which method to use in a particular application is addressed in Section 9.9.3.

9.5.1 Pay-back time

A simple method for estimating the pay-back time is to divide the total initial capital (fixed capital plus working capital) by the average annual cash flow:

$$\text{simple pay-back time} = \frac{\text{total investment}}{\text{average annual cash flow}} \quad (9.15)$$

This is not the same pay-back time indicated by the cash flow diagram, as it assumes that all the investment is made in year zero and revenues begin immediately. For most chemical plant projects, this is not realistic, as investments are typically spread over 1 to 3 years and revenues may not reach 100% of design basis until the second year of operation. The simple pay-back time is strictly based on a cash flow, but for simplicity, taxes and depreciation are often neglected and the average annual income is used instead of cash flow.

9.5.2 Return on investment

Another simple measure of economic performance is the return on investment (ROI). The ROI is defined in a similar manner to ROA and ROE:

$$\text{ROI} = \frac{\text{net annual profit}}{\text{total investment}} \times 100\% \quad (9.16)$$

Net annual profit is the same as annual operating income after tax. If ROI is calculated as an average over the whole project, then:

$$\text{ROI} = \frac{\text{cumulative net profit}}{\text{plant life} \times \text{initial investment}} \times 100\% \quad (9.17)$$

Calculation of the after-tax ROI is complicated if the depreciation term is less than the plant life and if an accelerated method of depreciation such as MACRS is used. In such cases, it is just as easy to calculate one of the more meaningful economic criteria such as net present value or discounted cash flow rate of return, described later. Because of this complication, a pretax ROI is often used instead:

$$\text{pretax ROI} = \frac{\text{pretax cash flow}}{\text{total investment}} \times 100\% \quad (9.18)$$

Note that pretax ROI is based on cash flow, not profit or taxable income, and therefore does not include a depreciation charge.

ROI is also sometimes calculated for incremental modifications to a large project, as described in Section 9.9.3.

Example 9.2

Calculate the simple pay-back time, ROI, and pretax ROI for the project introduced in Example 9.1, assuming MACRS depreciation is used.

Solution

If we base the simple pay-back on income instead of cash flow, it is:

$$\text{Simple pay-back (before tax)} = \frac{\$ 100 \text{ MM}}{\$ 50 \text{ MM/yr}} = \underline{\underline{2 \text{ years}}}$$

If, instead, we average the cash flows after allowing for taxes and depreciation, then using the numbers from the solution to Example 9.1b, we obtain average annual cash flow = \$42.65 MM, in which case:

$$\text{Simple pay-back (after tax)} = \frac{\$ 100 \text{ MM}}{\$ 42.65 \text{ MM/yr}} = \underline{\underline{2.34 \text{ years}}}$$

The ROI can be calculated from Equation 9.17:

$$\text{ROI} = \frac{426.5}{10 \times 100} \times 100\% = \underline{\underline{42.65\%}} \quad (9.17)$$

The pretax ROI is just the inverse of the before-tax simple pay-back and is $50/100 = 50\%$.

Note that calculation of the after-tax pay-back time and ROI requires calculation of the schedule of depreciation charges and taxes. Because this usually involves creating a table or spreadsheet, most of the effort required to calculate more useful economic measures has already been invested. Simple pay-back and ROI are therefore often only used on a before-tax basis.

Example 9.3

A plant is producing 10,000 metric tons per year (10 kt/y or 10 kMTA) of a product. The overall yield is 70 wt% (0.7 kg of product per kg raw material). The raw material costs \$500/metric ton, and the product sells for \$900/metric ton. A process modification has been devised that will increase the yield to 75%. The additional investment required is \$1,250,000, and the additional operating costs are negligible. Is the modification worth making?

Solution

There are two ways of looking at the earnings to be gained from the modification:

1. If the additional production given by the yield increase can be sold at the current price, the earnings on each additional ton of production will equal the sales price less the raw material cost.
2. If the additional production cannot be readily sold, the modification results in a reduction in raw material requirements rather than increased sales, and the earnings (savings) are from the reduction in annual raw material costs.

The second way gives the lowest figures and is the safest basis for making the evaluation. At 10 kMTA production:

$$\text{Raw material requirements at 70\% yield} = \frac{10,000}{0.7} = 14,286$$

$$\text{at 75\% yield} = \frac{10,000}{0.75} = 13,333$$

$$\text{Cost savings} = 953 \text{ t/y, which is worth } 953 \times 500 = \$476,500/\text{y}$$

$$\text{Pretax ROI} = \frac{476,500}{1,250,000} = \underline{\underline{38\%}}$$

As the annual savings are constant, the pretax simple pay-back period is the inverse of the pretax ROI:

$$\text{Simple pay - back period} = \frac{1,250,000}{476,500} = 2.62 \text{ years}$$

Based on the attractive ROI and pay-back period, this investment would seem to be worth pursuing further. Whether or not it was implemented would depend on the hurdle rate set for investments by the company.

9.6 Present value methods

The simple economic measures introduced in Section 9.5 are not able to capture the time dependence of cash flows during the project. The timing of cash flows is very important to investors, first because not all of the capital must be

financed immediately, and second because capital that is repaid sooner can be put back to work in another investment. Present value methods account for this *time value of money* and are preferred over simple methods when evaluating large investments; see Section 9.9.3.

9.6.1 Time value of money

In Fig. 9.1 the net cash flow is shown at its value in the year in which it occurred. So the figures on the ordinate show the “future worth” of the project. The cumulative value is the “net future worth” (NFW).

The money earned in any year can be reinvested as soon as it is available and can start to earn a return. So money earned in the early years of the project is more valuable than that earned in later years. This “time value of money” can be allowed for by using a variation of the familiar compound interest formula. The net cash flow in each year of the project is brought to its “present value” at the start of the project by discounting it at some chosen compound interest rate.

Future worth

The future worth of an amount of money, P , invested at interest rate, i , for n years is:

$$\text{Future worth in year } n = P(1 + i)^n$$

Hence the present value of a future sum is:

$$\text{present value of future sum} = \frac{\text{future worth in year } n}{(1 + i)^n} \quad (9.19)$$

The interest rate used in discounting future values is known as the *discount rate* and is chosen to reflect the earning power of money. In most companies the discount rate is set at the cost of capital (see Section 9.3.4).

Inflation

Discounting of future cash flows should not be confused with allowing for price inflation. Inflation is a general increase in prices and costs, usually caused by imbalances between supply and demand. Inflation raises the costs of feed, products, utilities, labor, and parts, but does not affect depreciation charges, which are based on original cost. Discounting, on the other hand, is a means of comparing the value of money that is available now (and can be reinvested) with money that will become available at some time in the future. All of the economic analysis methods can be modified to allow for inflation. See, for example, [Humphreys \(1991\)](#), [Chapter 6](#). In practice, most companies assume that although prices may suffer inflation, margins and hence cash flows will be relatively insensitive to inflation. Inflation can therefore be neglected for the purposes of comparing the economic performance of projects.

9.6.2 Net present value

The net present value (NPV) of a project is the sum of the present values of the future cash flows:

$$\text{NPV} = \sum_{n=1}^{n=t} \frac{CF_n}{(1 + i)^n} \quad (9.20)$$

where: CF_n = cash flow in year n

t = project life in years

i = interest rate (= cost of capital, percent /100)

The NPV is always less than the total future worth of the project because of the discounting of future cash flows. NPV is easily calculated using spreadsheets, and most spreadsheet programs have a NPV function.

The NPV is a strong function of the interest rate used and the period studied. When different periods are analyzed, the period is sometimes denoted by a subscript. For example, NPV_{10} would denote the NPV over a 10-year period.

NPV is a more useful economic measure than simple pay-back and ROI, because it allows for the time value of money and for annual variation in expenses and revenues. Few large projects are completed in a single year and immediately begin production at full capacity. A more typical start-up schedule for a chemical process is given in Table 9.2. For pharmaceutical products, the time scale can be even longer, as the plant must be certified to be in compliance with good manufacturing practices (cGMP) and must be inspected and approved by the Food and Drug Administration (FDA) before production can begin. The time from breaking ground to production for new pharmaceutical products produced by fermentation is typically about 6 years (Lee, 2010). NPV is also a more appropriate method to use when considering after-tax income using an accelerated depreciation method such as MACRS.

9.6.3 Discounted cash flow rate of return

By calculating the NPV at various interest rates, it is possible to find an interest rate at which the cumulative NPV at the end of the project is zero. This particular rate is called the *discounted cash flow rate of return* (DCFROR) and is a measure of the maximum interest rate that the project could pay and still break even by the end of the project life.

TABLE 9.2 Typical start-up schedule

Year	Costs	Revenues	Explanation
1st year	30% of fixed capital	0	Engineering + long lead-time items
2nd year	40%–60% of fixed capital	0	Procurement and construction
3rd year	10%–30% of fixed capital + working capital + FCOP + 30% VCOP	30% of design basis revenue	Remaining construction Initial production
4th year	FCOP + 50%–90% VCOP	50%–90% of design basis revenue	Shake-down of plant
5th year +	FCOP + VCOP	100% of design basis revenue	Full production at design rates

$$\sum_{n=1}^{n=t} \frac{CF_n}{(1 + i')^n} = 0 \quad (9.21)$$

where: CF_n = cash flow in year n

t = project life in years

i' = the discounted cash flow rate of return (percent /100)

The value of i' is found by trial-and-error calculations or by using the appropriate function (e.g., “goal seek”) in a spreadsheet. A more profitable project will be able to pay a higher DCFROR.

DCFROR provides a useful way of comparing the performance of capital for different projects, independent of the amount of capital used, the life of the plant, or the actual interest rates prevailing at any time. DCFROR is a more useful method than NPV when comparing projects of very different sizes. The NPV of large projects is usually greater than that of small projects, but then the investment is also much greater. DCFROR is independent of project size, and the project with the highest DCFROR always provides the best “bang for the buck.” When DCFROR is used as an investment criterion, companies usually expect projects to have a DCFROR greater than the cost of capital.

DCFROR can also be compared directly with interest rates. Because of this, it is sometimes known as the *interest rate of return* or *internal rate of return* (IRR).

Example 9.4

Estimate the NPV at a 12% interest rate and the DCFROR for the project described in Example 9.1, using the MACRS depreciation method.

Solution

Calculating the present values of the cash flows from the previous example requires adding two columns to the spreadsheet. We first calculate the discount factor $(1 + i)^{-n}$ and then multiply this by the cash flow in year n to give the present value of the cash flow. The present values can then be summed to give the NPV:

Year	Gross profit (MM\$)	Depreciation charge (MM\$)	Taxable income (MM\$)	Taxes paid (MM\$)	Cash Flow (MM\$)	Discount factor	Present value of CF (MM\$)
0	0	0	0	0	-100	1	-100
1	50	20	30	0	50	0.893	44.64
2	50	32	18	6.30	43.70	0.797	34.84
3	50	19.2	30.8	3.78	46.22	0.712	32.90
4	50	11.52	38.48	6.47	43.53	0.636	27.67
5	50	11.52	38.48	8.08	41.92	0.567	23.79
6	50	5.76	44.24	8.08	41.92	0.507	21.24
7	50	0	50	9.29	40.71	0.452	18.41
8	50	0	50	10.50	39.50	0.404	15.95
9	50	0	50	10.50	39.50	0.361	14.24
10	50	0	50	10.50	39.50	0.322	12.72
				Interest rate		12.00%	
				Total = Net present value =			146.40

Note that we could also have calculated NPV directly using the NPV function. In Microsoft Excel, the NPV function starts at the end of year 1, so any cash flows in year 0 should be added to or subtracted from the result returned by the function and should not be included in the function range.

The DCFROR can then be found by adjusting the interest rate until the NPV is equal to zero. This is easily accomplished in the spreadsheet using the "Goal Seek" tool, giving DCFROR = 44.35%.

Example 9.5

The adipic acid plant from Examples 7.8 and 8.2 is to be built in 2020 with 30% of the fixed investment in year 1 and 70% in year 2, and the plant will operate at 50% of capacity in year 3 before reaching full capacity in year 4. The plant can be depreciated by the straight-line method over 10 years, and profits can be assumed to be taxed at 21% per year, payable the next year. Assume that losses cannot be offset against revenues from other operations for tax purposes (i.e., no tax credits in years when the plant makes a loss). Estimate the following:

1. The cash flow in each year of the project.
2. The simple pay-back period.
3. The NPV with a 15% cost of capital after 10 years of production and after 15 years of production at full capacity.
4. The DCFROR for 15 years of production at full capacity.

Is this an attractive investment?

Solution

The solution requires calculating the cash flows in each year of the project. This is easily coded into a spreadsheet, as illustrated in Fig. 9.2. A blank template of this spreadsheet is given in Appendix G and is available in Microsoft Excel format in the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793.

Company Name				Project Name				Adipic acid from phenol			
Address				Project Number				Sheet 1			
ECONOMIC ANALYSIS Adipic Acid from Phenol				REV	DATE	BY	APVD	REV	DATE	BY	APVD
				1	1.1.07	GPT					
				2	5.6.20	GPT					
Form XXXXX-YY-ZZ											
Owner's Name				Capital Cost Basis Year				2020			
Plant Location				Units				Metric			
Case Description				On Stream				8,000 hr/yr 333.33 day/yr			
REVENUES AND PRODUCTION COSTS				CAPITAL COSTS				CONSTRUCTION SCHEDULE			
MMM/yr				MMM				Year % FC % WC % FCOP % VCOF			
Main product revenue 560.0				ISBL Capital Cost 247.9				1 30% 0% 0% 0%			
Byproduct revenue 4.4				OSBL Capital Cost 99.2				2 70% 0% 0% 0%			
Raw materials cost 410.8				Engineering Costs 34.7				3 0% 100% 100% 50%			
Utilities cost 47.3				Contingency 52.1				4 0% 0% 100% 100%			
Consumables cost 13.1				Total Fixed Capital Cost 433.9				5 0% 0% 100% 100%			
VCOF 466.8								6 0% 0% 100% 100%			
Salary and overheads 16.6				Working Capital 60.9				7+ 0% 0% 100% 100%			
Maintenance 13.0											
Interest 3.7											
Royalties 3.0											
FCOP 35.5											
ECONOMIC ASSUMPTIONS											
Cost of equity 25%				Debt ratio 0.5				Tax rate 21%			
Cost of debt 5%								Depreciation method Straight-line			
Cost of capital 15%								Depreciation period 10 years			
CASHFLOW ANALYSIS											
All figures in \$MM unless indicated											
Project year		Cap Ex	Revenue	CCOP	Gr. Profit	Deprcn	Taxbl Inc	Tax Paid	Cash Flow	PV of CF	NPV
1		130.2	0.0	0.0	0.0	0.0	0.0	0.0	-130.2	-113.2	-113.2
2		303.7	0.0	0.0	0.0	0.0	0.0	0.0	-303.7	-229.7	-342.9
3		60.9	280.0	268.9	11.1	43.4	-32.3	0.0	-49.9	-32.8	-375.6
4		0.0	560.0	502.3	57.7	43.4	14.3	0.0	57.7	33.0	-342.7
5		0.0	560.0	502.3	57.7	43.4	14.3	3.0	54.7	27.2	-315.5
6		0.0	560.0	502.3	57.7	43.4	14.3	3.0	54.7	23.6	-291.8
7		0.0	560.0	502.3	57.7	43.4	14.3	3.0	54.7	20.6	-271.3
8		0.0	560.0	502.3	57.7	43.4	14.3	3.0	54.7	17.9	-253.4
9		0.0	560.0	502.3	57.7	43.4	14.3	3.0	54.7	15.5	-237.9
10		0.0	560.0	502.3	57.7	43.4	14.3	3.0	54.7	13.5	-224.3
11		0.0	560.0	502.3	57.7	43.4	14.3	3.0	54.7	11.8	-212.6
12		0.0	560.0	502.3	57.7	43.4	14.3	3.0	54.7	10.2	-202.4
13		0.0	560.0	502.3	57.7	0.0	57.7	3.0	54.7	8.9	-193.5
14		0.0	560.0	502.3	57.7	0.0	57.7	12.1	45.6	6.4	-187.0
15		0.0	560.0	502.3	57.7	0.0	57.7	12.1	45.6	5.6	-181.4
16		0.0	560.0	502.3	57.7	0.0	57.7	12.1	45.6	4.9	-176.6
17		0.0	560.0	502.3	57.7	0.0	57.7	12.1	45.6	4.2	-172.3
18		0.0	560.0	502.3	57.7	0.0	57.7	12.1	45.6	3.7	-168.7
19		0.0	560.0	502.3	57.7	0.0	57.7	12.1	45.6	3.2	-165.5
20		-60.9	560.0	502.3	57.7	0.0	57.7	12.1	106.5	6.5	-159.0
ECONOMIC ANALYSIS											
Average cash flow		48.9 \$MM/yr		NPV		10 years		-224.3 \$MM		IRR	
Simple pay-back period		10.1 yrs		15 years		-181.4 \$MM		15 years		5.0%	
Return on investment (10 yrs)		1.37%		20 years		-159.0 \$MM		20 years		7.2%	
Return on investment (15 yrs)		3.63%		NPV to yr		19		-165.5 \$MM			

In the final year of the project, the working capital is released and should be taken as a positive increment to the cash flow. This is shown as occurring in year 20 in Fig. 9.2, but should be adjusted when the length of the project is varied, as described next.

The present value of the cash flow in year n can be found by multiplying by $(1 + i)^{-n}$, as stated in Equation 9.19. The NPV up to year n is the cumulative sum of all the present values of cash flow up to that year.

Simple pay-back period

The simple pay-back is calculated from the fixed investment and the average annual cash flow (Equation 9.15). The average annual cash flow should be based only on the years in which the plant generates revenue (i.e., years 3 to 20) and is found to be \$48.9 MM/y. Note that it does not matter if this range includes the year in which working capital is invested, as long as it also includes the year in which working capital is recovered. The working capital thereby cancels out and is not included in the average cash flow.

The simple pay-back period is then found from:

$$\text{simple pay-back time} = \frac{\text{total investment}}{\text{average annual cash flow}} = \frac{433.9 + 60.9}{48.9} = \underline{\underline{10.1 \text{ years}}} \quad (9.15)$$

Net present value

The NPV with a 15% cost of capital after 10 years of production is the NPV at the end of year 13. This can be looked up in the cash flow table and is \$-193.5 MM. If the plant is closed after 10 years of production and the working capital is released, there would be an additional cash flow of \$60.9 MM in year 13, increasing the NPV to \$-132.6 MM.

The NPV after 15 years of production is the NPV at the end of year 18, which can also be found from the cash flow table, and is \$-168.7 MM. If the plant is closed after 15 years of production and the working capital is released, there would be an additional cash flow of \$60.9 MM in year 18, increasing the NPV to \$-107.7 MM.

In all cases the NPV for this project is negative, so it is not an attractive investment with a 15% cost of capital. We already knew this would be the case based on the cost of production analysis in Example 8.2, which had shown that the total cost of production (TCOP) with capital recovered at a 15% interest rate was greater than the expected revenue.

Internal rate of return (DCFRROR)

The DCFRROR (IRR) of the project after 15 years of production at full capacity can be found by either adjusting the interest rate (manually or using the goal seek function) until the NPV at the end of year 18 is equal to zero, or by using the IRR function in the spreadsheet over the range year 1 to year 18. The working capital should be included as a recovered cost in year 18.

The answer obtained in either case is DCFRROR = 8.03%. This is the maximum interest rate at which this project can be financed to break even in 15 years of production. Note that this is not the same as the IRR at the end of year 15 calculated in Fig. 9.2.

Summary

None of the economic measures indicates that this is an attractive project with the projected costs, revenues, and capital expenses. It should perhaps be noted, though, that this analysis was based on a class 5 estimate of the capital cost ($\pm 50\%$). If we had any technical improvement in mind that could reduce either the capital investment or the cost of production, we might want to develop the design further to assess if the economic analysis was sufficiently improved.

9.7 Annualized cost methods

9.7.1 Amortization charges

An alternative method of comparing the magnitude of a capital investment in current dollars with a revenue stream in the future is to convert the capital cost into a future annual capital charge. A capital cost can be annualized (amortized) by determining the annual payment that is needed to repay the initial investment, together with the expected return on capital in the form of compound interest.

If an amount P is invested at an interest rate i , then after n years of compound interest, it matures to the sum $P(1+i)^n$.

If, instead, an amount A is invested each year, also at interest rate i , then it matures to a sum, S , where:

$$S = A + A(1+i) + A(1+i)^2 + \dots + A(1+i)^{n-1} \quad (9.22)$$

so:

$$S(1+i) = A(1+i) + A(1+i)^2 + \dots + A(1+i)^n \quad (9.23)$$

Hence, subtracting Equation 9.22 from Equation 9.23:

$$Si = A[(1+i)^n - 1] \quad (9.24)$$

If the annual payments A have matured to give the same final sum as would have been obtained by investing the principal P at the same interest rate, then:

$$S = P(1+i)^n = [(1+i)^n - 1]$$

Hence:

$$A = P \frac{[i(1+i)^n]}{[(1+i)^n - 1]} \quad (9.25)$$

A is then the regular annual payment that must be made to generate the same amount of money over n years as would be earned by investing P at interest rate i for n years. A is also the annual payment that would have to be paid to pay off (amortize) the principal and interest on a loan of amount P , borrowed at interest rate i , over a term of n years.

We can define an annual capital charge ratio, $ACCR$, as:

$$ACCR = \frac{A}{P} = \frac{[i(1+i)^n]}{[(1+i)^n - 1]} \quad (9.26)$$

The $ACCR$ is the fraction of the principal that must be paid out each year to fully repay the principal and all accumulated interest over the life of the investment. This is the same formula used for calculating fixed annual payments on home mortgages and other loans where the principal is amortized over the loan period.

TABLE 9.3 Values of annual capital charge ratio ($ACCR$) for different interest rates

Interest rate, i	$ACCR$: 10-year life	$ACCR$: 20-year life
0.1	0.163	0.117
0.12	0.177	0.134
0.15	0.199	0.16
0.2	0.239	0.205
0.25	0.280	0.253
0.3	0.323	0.302

9.7.2 Annualized capital cost and total annualized cost

If the cost of capital is used as the interest rate (see Section 9.3.4), then the $ACCR$ can be used to convert the initial capital expense into an annual capital charge, or annualized capital cost:

$$\text{Annual capital charge (ACC)} = ACCR \times \text{total fixed capital cost} \quad (9.27)$$

The annual capital charge can be added to the operating costs to give a total annualized cost of production (TAC):

$$TAC = \text{operating costs} + ACCR \times \text{total fixed capital cost} \quad (9.28)$$

The TAC can be compared with forecasted future revenues. The TAC is also sometimes referred to as TCOP.

Table 9.3 shows values of ACCR for different values of i and n . For a typical cost of capital of about 15% and a plant life of 10 years, the value of ACCR is 0.199, or about one-fifth of the capital investment.

A few important things should be noted when using the annualized cost method:

1. The method assumes investment and cash flows begin immediately, and so it does not capture information on the timing of early expenditures and revenues. In this respect it is inferior to NPV and DCFROR.
2. The method does not take into account taxes or depreciation and assumes that all of the revenue from the project is available to provide a return on the initial investment. The tax and depreciation schedule is not easily annualized if the MACRS system is used.
3. Working capital is recovered at the end of the project and so strictly only the fixed capital should be annualized. Equations 9.25 and 9.26 can be modified for the case where an additional sum becomes available at the end of the investment term, but this modified version is seldom used in practice, and working capital is often either neglected in the annualized cost method or else (wrongly) thrown in with fixed capital. A simple way around this problem is to assume that the working capital is entirely funded by debt, in which case the cost of carrying the working capital is reduced to an interest payment that appears as part of the fixed costs of production. At the end of the project life the working capital will be released and will be available to repay the principal on the debt.
4. As described in Section 8.5, several of the fixed costs of production are proportional to the fixed capital invested (FC). If we assume annual charges of 3% of FC for maintenance, 2% of FC for property tax, and 65% plant overhead, then the annual capital charge ratio is increased by $0.02 + (1.65 \times 0.03) = 0.07$.
5. If we also assume engineering costs are 10% of (ISBL + OSBL) capital investment and add 15% of (ISBL + OSBL) capital as contingency, then with a 10-year plant life and a 15% interest rate, the annual capital charge ratio is:

$$\begin{aligned} \text{ACCR} &= [0.199 \times (1.0 + 0.1 + 0.15) + 0.07] \times [\text{Installed ISBL} + \text{OSBL capital cost}] \\ &= 0.32 \times [\text{Installed ISBL} + \text{OSBL capital cost}] \end{aligned} \quad (9.29)$$

Equation 9.29 is the basis for the widely used rule of thumb of annualizing capital cost by dividing by 3. When using this rule of thumb, it is important to remember that some, but not all, of the fixed costs have been counted in the annual capital charge.

The annualized cost method involves more assumptions than calculating NPV or DCFROR, but it is widely used as a quick way of comparing investments with the resulting benefits. Annualized cost is also useful as a method for analyzing small projects and modifications that lead to reduced operating costs (for example, heat recovery projects), because the annualized capital outlay can be directly traded off against the expected annual savings, and there is usually no change in working capital, operating labor, or other fixed costs of production. Small projects usually can be executed quickly, so the error introduced by neglecting the timing of investments and revenues is less important than it is when designing a new plant or a major investment project.

The annualized cost method is also used when comparing the costs of equipment with different expected operating lives. Annualization of the costs allows equipment with different service life to be compared on the same annual basis. This is illustrated in the example that follows.

Example 9.6

A carbon steel heat exchanger that costs \$140,000 is expected to have a service life of 5 years before it requires replacement. If type 304 stainless steel is used, then the service life will be increased to 10 years. Which exchanger is the most economical if the cost of capital is 12%?

Solution

With a 12% interest rate and 5-year life, the annual capital charge ratio is

$$\text{ACCR} = \frac{[i(1+i)^n]}{[(1+i)^n - 1]} = \frac{[0.12(1.12)^5]}{[(1.12)^5 - 1]} = 0.277 \quad (9.26)$$

The annualized capital cost of the carbon steel exchanger is then $= \$140,000 \times 0.277 = \underline{\underline{\$38,780/\text{y}}}$.

From Table 7.6, we can estimate the cost of the type 304 stainless steel exchanger to be $\$140,000 \times 1.3 = \$182,000$. From Table 9.3 (or Equation 9.26), with a 10-year life and 12% interest rate, the annual capital charge ratio is 0.177, so the annualized cost of the stainless steel exchanger is:

$$= \$182,000 \times 0.177 = \underline{\underline{\$32,210/\text{y}}}$$

In this case, it would be more economical to buy the stainless steel heat exchanger.

9.8 Sensitivity analysis

9.8.1 Simple sensitivity analysis

The economic analysis of a project can only be based on the best estimates that can be made of the investment required and the cash flows. The actual cash flows achieved in any year will be affected by changes in raw materials costs and other operating costs and will be highly dependent on the sales volume and price. A sensitivity analysis is a way of examining the effects of uncertainties in the forecasts on the viability of a project. To carry out the analysis, the investment and cash flows are first calculated using what are considered the most probable values for the various factors; this establishes the base case for analysis. Various parameters in the cost model are then adjusted, assuming a range of error for each factor in turn. This will show how sensitive the cash flows and economic criteria are to errors in the forecast figures. A sensitivity analysis gives some idea of the degree of risk involved in making judgments on the forecast performance of the project.

The results of a sensitivity analysis are usually presented as plots of an economic criterion such as NPV or DCFROR vs. the parameter studied. Several plots are sometimes shown on the same graph using a scale from $0.5 \times$ base value to $2 \times$ base value as the abscissa, as illustrated in Example 9.7.

9.8.2 Parameters to study

The purpose of sensitivity analysis is to identify those parameters that have a significant impact on project viability over the expected range of variation of the parameter. Typical parameters investigated and the range of variation that is usually assumed are given in Table 9.4.

Varying the production rate (while keeping investment and fixed costs constant) investigates the effects of unexpectedly high downtime due to maintenance or operations problems, as well as unexpected difficulties in selling the full volume of product that could be produced. An increase in production rate beyond the design capacity might also be possible if the plant design margins allow some extra capacity or if the yields can be improved by use of a better catalyst, etc.

TABLE 9.4 Sensitivity analysis parameters

Parameter	Range of variation
Sales price	$\pm 20\%$ of base (larger for cyclic commodities)
Production rate	$\pm 20\%$ of base
Feed cost	-10% to $+30\%$ of base
Fuel cost	-50% to $+100\%$ of base
Fixed costs	-20% to $+100\%$ of base
ISBL capital investment	-20% to $+50\%$ of base
OSBL capital investment	-20% to $+50\%$ of base
Construction time	-6 months to $+2$ years
Interest rate	base to base $+2$ percentage points

The choice of which feed and product prices to use in the sensitivity analysis depends strongly on the method of price forecasting that has been used. Typically, total raw material cost is studied rather than treating each feed separately, but if raw material costs are found to be the dominant factor, they may be broken out into the costs of individual raw materials.

9.8.3 Statistical methods for risk analysis

In a simple sensitivity analysis, each parameter is varied individually and the output is a qualitative understanding of which parameters have the most impact on project viability. In a more formal risk analysis, statistical methods are used to examine the effect of variation in all of the parameters simultaneously and hence quantitatively determine the range of variability in the economic criteria. This allows the design engineer to estimate the degree of confidence with which the chosen economic criterion can be said to exceed a given threshold.

A simple method of statistical analysis was proposed by Piekarski (1984) and is described in Humphreys (2005). Each item in the estimate is expressed as a most likely value, ML ; an upper value, H ; and a lower value, L . The upper and lower values can be estimated using the ranges of variation given in Table 9.4. The mean and standard deviation are then estimated as:

$$\text{mean value, } \bar{x} = \frac{(H + 2ML + L)}{4} \quad (9.30)$$

$$\text{standard deviation, } S_x = \frac{(H - L)}{2.65} \quad (9.31)$$

Note that the mean is not necessarily equal to the most likely value if the distribution is skewed. This is often the case for cost estimates, where the likelihood of costs being underestimated is much greater than the likelihood of them being overestimated.

The mean and standard deviation of other parameters can then be estimated by combination of the individual means and standard deviations using the mathematics of statistics given in Table 9.5.

This allows relatively easy estimation of the overall error in a completed cost estimate, and with a little more difficulty can be extended to economic criteria such as NPV, TAC, or ROI.

Rather than build the simple method noted earlier into a spreadsheet, a more sophisticated approach is to take the economic model and subject it to analysis using Monte Carlo simulation. In Monte Carlo simulation, random numbers are generated and used to establish the value of each parameter within its allowed range. For example, each parameter could be set equal to $L + (R \times (H - L)/10)$, where R is a random number between 1 and 10. The overall probability distribution in the calculated parameter (economic criterion) can be estimated by performing a large number of such simulations. Several commercial programs for Monte Carlo simulation are available, for example REP/PC (Decision Sciences Corp.), @RISK (Palisade Corp.), and CRYSTAL BALL (Decisioneering Corp.).

TABLE 9.5 Mathematics of statistics

If: $y = f(\bar{x}, \bar{z})$, then the standard deviation of y , S_y , is given as a function of S_x and S_z .	
Function y of $\bar{x}$, $\bar{z}$	Standard deviation S_y
$y = a\bar{x} + b\bar{z}$	$S_y = \sqrt{a^2 S_x^2 + b^2 S_z^2}$
$y = \bar{x} \bar{z}$	$S_y = \bar{x} \bar{z} \sqrt{\frac{S_x^2}{\bar{x}^2} + \frac{S_z^2}{\bar{z}^2}}$
$y = \frac{\bar{x}}{\bar{z}}$	$S_y = \frac{\bar{x}}{\bar{z}} \sqrt{\frac{S_x^2}{\bar{x}^2} + \frac{S_z^2}{\bar{z}^2}}$

Notes:

1. These formulae are strictly true only when the covariance of x and z is zero, that is, there is no statistical interrelation between x and z and x and z have been estimated from a small set of data points.
2. For a more general description of the formulae, see Ku (1966).

Care must be taken in formulating Monte Carlo simulation problems. The Monte Carlo method implicitly assumes that all parameters vary randomly and independently. If two parameters are correlated (for example, feedstock and product prices or feedstock and energy prices), they should not be varied independently. The correct approach is to vary one of the parameters and then predict the other by correlation, imposing a smaller random error on the predicted parameter to reflect the accuracy of the correlation.

The cost estimating literature contains a lot of information on risk analysis. Good introductions to the use of statistics in risk analysis are given by [Humphreys \(2005\)](#) and [Sweeting \(1997\)](#). [Anderson \(2010\)](#) gives an overview of the subject and suggests several approaches to presenting the data and results.

9.8.4 Contingency costs

The concept of a contingency charge to allow for variation in the capital cost estimate was introduced in [Section 7.2.1](#), where it was suggested that a minimum contingency charge of 10% of ISBL plus OSBL fixed capital should be used.

If the confidence interval of the estimate is known, the contingency charges can also be estimated based on the desired level of certainty that the project will not overrun the projected cost. For example, if the cost estimate is normally distributed, then the estimator has the following confidence levels:

- 90% confidence that the cost is less than $\bar{x} + 1.3 S_x$
- 95% confidence that the cost is less than $\bar{x} + 1.65 S_x$
- 98% confidence that the cost is less than $\bar{x} + 2.05 S_x$
- 99% confidence that the cost is less than $\bar{x} + 2.33 S_x$

Although many of the components of a cost estimate are skewed distributions, when these are combined, the resulting distribution is often approximately normal. These guidelines can thus be used to determine the amount of contingency charge needed for a given level of confidence.

Note also that a 10% contingency charge gives 98% confidence of the cost coming in under estimate if the estimate has accuracy $\pm 6.5\%$ (using the approximate method of calculating S_x given in [Equation 9.31](#)). This illustrates that a 10% contingency charge should really be viewed as a minimum level and is only appropriate for detailed estimates (Class 1 and Class 2) when the technology is well understood.

Additional guidance on improving estimates of contingency costs is given by [Hollmann \(2014\)](#).

Example 9.7

Carry out a sensitivity analysis on the adipic acid project described in Example 9.5. Investigate the sensitivity of the NPV after 10 years of production to variation in total fixed capital (-20% to $+50\%$), annual gross profit (-20% to $+30\%$), and project construction schedule (-6 months to $+1.5$ years).

Solution

Because the economic analysis is entered in a spreadsheet, it is easy to investigate the effect of varying any parameter. The simplest method is to define a ratio parameter in an empty cell and then multiply the cell or cells of interest by this ratio. The ratio can then be varied and the effect on NPV tabulated. For fixed capital, the ratio varies from 0.8 to 1.5, and for annual gross profit, the ratio varies from 0.8 to 1.3.

The effect of variation in the project schedule is a little more difficult. The easiest way to study this is to change the numbers in the construction schedule section of the worksheet. It is usually a good idea to set up the economic analysis with the construction schedule as a defined parameter (as in [Fig. 9.2](#)) so that the effect of changes in schedule can be studied. For the purposes of this example it was assumed that a 1-year delay in schedule would mean the project would still incur 100% of fixed costs in year 3, but would achieve 50% of revenues and variable costs in year 4 instead of year 3. A 6-month delay would then give 25% of revenues and variable costs in year 3 and 75% in year 4, etc. A 6-month acceleration in schedule was assumed to bring 25% of revenues and variable costs into year 2, 75% of revenues and variable costs in year 3, and 100% of working capital and 50% of fixed costs into year 2 to allow for hiring staff and beginning production sooner.

The results can be plotted on the same graph for all three variables. For a base case schedule of 3 years, a 6-month reduction is $2.5/3 = 83\%$ of base case. A 1.5-year delay would be $4.5/3 = 150\%$ of base case.

The results are shown in [Fig. 9.3](#). It can be seen that none of the parameters studied can cause the project to have positive NPV over the expected range of variation. The project economics are more sensitive to capital investment (CAPEX) and gross profit than to schedule. Gross profit usually depends on market conditions and cannot be controlled for a commodity such as

adipic acid. The investors in this project should therefore concentrate their efforts on driving down capital cost rather than trying to accelerate the schedule.

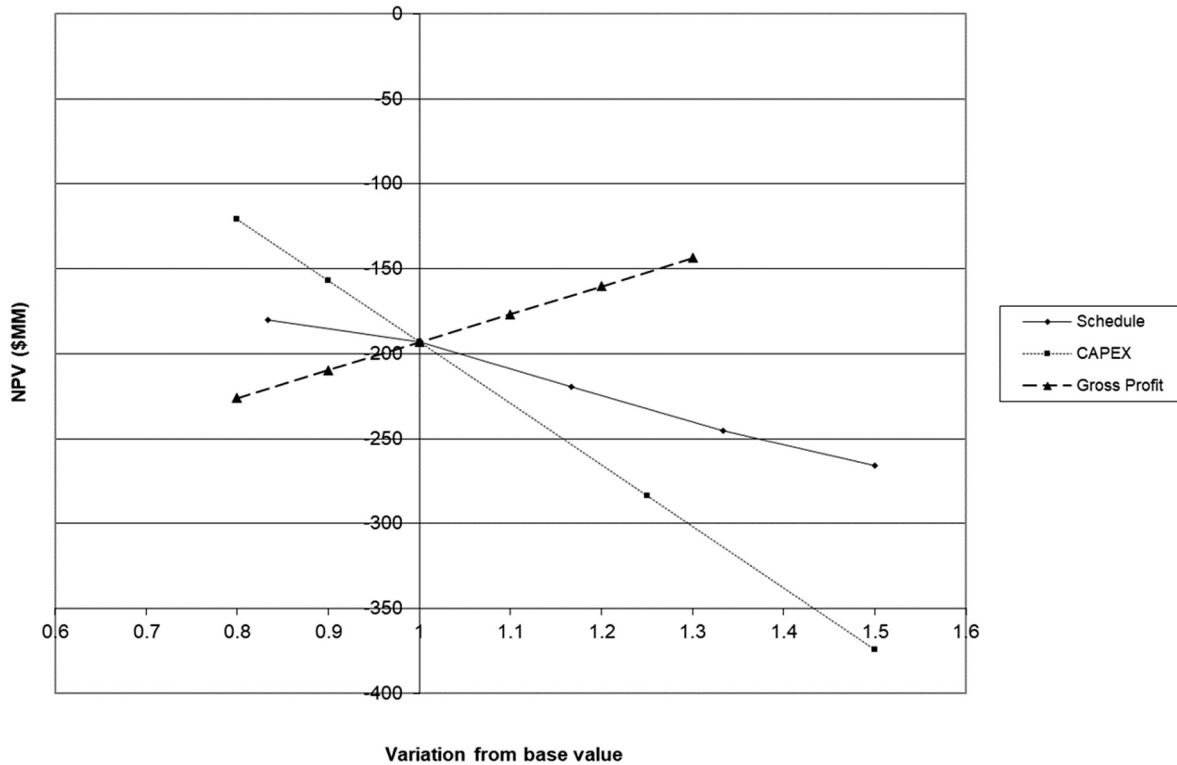


FIG. 9.3 Sensitivity analysis for Example 9.7

Example 9.8

A preliminary (Class 4) estimate of the ISBL capital cost of building a 200,000 ton per year ethanol plant by corn dry milling has been stated as \$130 MM -30% / $+50\%$. The plant is to be built on a greenfield site, and offsite costs are estimated to be between \$40 MM and \$60 MM. Estimate a value for the total project cost that will give 98% confidence that the project can be carried out within the amount estimated.

Solution

For the ISBL cost, $H = \$195$ MM, $L = \$91$ MM, and $ML = \$130$ MM, so:

$$\bar{x}_{ISBL} = \frac{(H + 2ML + L)}{4} = \frac{(195 + 260 + 91)}{4} = \$136.5 \text{ MM} \quad (9.30)$$

$$S_{x,ISBL} = \frac{(H - L)}{2.65} = \frac{195 - 91}{2.65} = \$39.2 \text{ MM} \quad (9.31)$$

Similarly, for the OSBL costs, assuming the most likely value is in the middle of the range given:

$$\bar{x}_{OSBL} = \frac{(H + 2ML + L)}{4} = \frac{(40 + 100 + 60)}{4} = \$50 \text{ MM} \quad (9.30)$$

$$S_{x,OSBL} = \frac{(H - L)}{2.65} = \frac{60 - 40}{2.65} = \$7.55 \text{ MM} \quad (9.31)$$

Both mean values should be increased by 10% to allow for engineering costs, so combining the means gives:

$$\bar{x}_{Total} = 1.1 \bar{x}_{ISBL} + 1.1 \bar{x}_{OSBL} = \$205.2 \text{ MM}$$

$$S_{x,Total} = \sqrt{(1.1 S_{x,ISBL})^2 + (1.1 S_{x,OSBL})^2} = \$43.9 \text{ MM}$$

We have 98% confidence that the cost is less than $\bar{x} + 2.05 S_x = 205.2 + (2.05 \times 43.9) = \295 MM. If we budget the project for this amount (or tender a contract), then we are accepting a 1 in 50 risk that the project will exceed the given budget. Note that this is substantially larger than the most likely value of $1.1 \times 180 = \$198$ MM, showing the impact of the high error in a Class 4 estimate. The required contingency to give 98% confidence is $295 - 198 = \$97$ MM, or 49% of the most likely estimate.

Example 9.9

If the capital costs in the adipic acid example described in Example 9.5 are known to within -10% to $+30\%$ and the cash flow from operations is predicted within $\pm 10\%$, estimate the mean and standard deviation of the NPV after 10 years of production. Is there a greater than 1% likelihood of the NPV being greater than zero?

Solution

Once again, the spreadsheet is easily extended to calculate the statistical parameters.

In years 1 and 2, the cash flows are only associated with capital expense and are all negative. We can therefore use the present value of cash flow as the most likely value (ML) and estimate the lower bound (L) as $1.3 \times ML$ (because we know the numbers are negative) and the upper bound (H) as $0.9 \times ML$.

In years 4 to 13, the cash flows are only positive and associated with operations. We can use the present value of cash flow as the value of ML and then estimate $L = 0.9 \times ML$ and $H = 1.1 \times ML$.

Year 3 is a little more complex, as the cash flow has contributions from both capital outlays and income from operations. The most likely value is still the value previously predicted (-34.0). The low value (L) will be the present value of 90% of the gross profit minus 130% of the capital expense:

$$\frac{(0.9 \times 12.8) - (1.3 \times 59.6)}{(1.15)^3} = -43.3$$

Similarly, the high value (H) will be the present value of 110% of the gross profit minus 90% of the capital expense.

We can then tabulate the values of ML , H , and L and calculate the mean and standard deviation using [Equations 9.30](#) and [9.31](#), as in [Table 9.6](#).

TABLE 9.6 Solution to Example 9.9

Year	ML	L	H	x	S	S^2
1	-113.19	-147.15	-101.87	-118.85	17.09	291.93
2	-229.67	-298.57	-206.70	-241.15	34.67	1201.80
3	-32.59	-45.29	-27.86	-34.58	6.58	43.26
4	33.01	29.71	36.31	33.01	2.49	6.21
5	27.21	24.49	29.93	27.21	2.05	4.22
6	23.66	21.29	26.03	23.66	1.79	3.19
7	20.57	18.52	22.63	20.57	1.55	2.41
8	17.89	16.10	19.68	17.89	1.35	1.82
9	15.56	14.00	17.11	15.56	1.17	1.38
10	13.53	12.17	14.88	13.53	1.02	1.04
11	11.76	10.59	12.94	11.76	0.89	0.79
12	10.23	9.21	11.25	10.23	0.77	0.60
13	8.89	8.01	9.78	8.89	0.67	0.45
			Sum	-212.27		1559.09
			S		39.5	

The overall mean NPV is the sum of the mean values = \$-212.3 MM

The standard deviation is the root of the sum of squares of the S_x values (see Table 9.5) = \$39.5 MM.

The 99% confidence interval is $\bar{x} + 2.33 S_x$, so we have 99% confidence that the NPV will be less than $-212.3 + (2.33 \times 39.5) = \-120.3 MM. The design team can state that there is less than 1% probability of this project delivering an NPV greater than \$-120 million at 15% cost of capital after 10 years of production.

9.9 Project portfolio selection

A typical company involved in the chemical, pharmaceutical, or fuels industries will evaluate many projects each year. Only a few of these projects are selected for implementation. This section discusses some of the criteria and methods used in making that selection.

9.9.1 Types of projects

Investment projects are carried out for a variety of reasons.

Regulatory compliance projects are often required as a result of changes in environmental or other legislation. If the government changes the rules on plant safety, emissions, or product specifications, then unless an exemption can be obtained, the plant must be modified or closed down. Regulatory compliance projects often have poor financial performance unless the costs of going out of business are considered.

Cost reduction projects are aimed at reducing the cost of production of an existing plant. The most common cost reduction investments are for *preventive maintenance*, in which equipment is replaced, repaired, or cleaned after a planned interval and before the equipment deteriorates to the point where it could affect process performance or safety. Most preventive maintenance projects are small and are handled through the plant maintenance budget, but some can be very large, expensive projects requiring a major plant shutdown, for example, replacing the fired tubes in a main plant furnace. Another common type of cost reduction project is *heat recovery* or *heat integration projects*, in which the plant heat exchange network or utility system is upgraded to reduce energy costs.

Whenever possible, companies also seek to fund *growth projects* that can be expected to give high returns on the capital invested. Growth projects include expansions of existing units, often referred to as *debottlenecking* or *revamp projects*, as well as construction of entirely new plants in *grassroots projects*.

In all cases except grassroots projects, a large amount of information about the existing plant, site, and products is usually needed before the project can be designed. Much effort is typically spent on reconciling simulation or other models to the plant performance so as to make the models useful for designing the plant modifications.

Grassroots projects are typically used as undergraduate design projects because they are self-contained and do not require model reconciliation; however, in industrial practice, they make up less than 10% of all projects.

9.9.2 Limits on the project portfolio

The most obvious limit on the portfolio of projects that can be funded is the availability of capital, which is in turn limited by the financing arrangements of the company (see Section 9.3).

Capital spending is often set in proportion to sales, operating profit, or total assets. Table 9.7 shows recent information on capital spending for some of the largest chemical companies in the world. These capital expenditures are plotted against revenues in Fig. 9.4 and against total assets in Fig. 9.5. Although the figures show that the correlation is not particularly strong, it can be seen from Table 9.7 that most of the companies' capital spending was between 5% and 10% of sales (mean 7.7%) and also between 4% and 8% of assets (mean 6.2%).

A second important constraint on the number of projects that can be carried out is the availability of critical resources. A company with a small engineering staff will only be able to carry out a few projects at one time. Even if extensive use is made of EPC contractors, the owners will still need to provide some engineering support to each project. The availability of EPC contractors can also be an issue during times of peak industry construction. Projects that require extensive research and development work may be delayed because of constraints on the availability of researchers and pilot plant facilities.

Often the most important constraint is set by regulatory timelines. Regulatory compliance projects must be completed in time for the plant or product to comply with the new law. This may dictate a narrow window of typically less than 5 years in which the project must be planned, designed, and constructed, giving the company little choice on when the project must be begun.

Regulatory timelines are extremely important for pharmaceutical products. A new drug is protected by patent for 20 years from the date the patent is filed. Beyond that time, competitors are able to sell generic versions of the drug, and the price usually falls significantly. Before a new drug can be marketed, both the product and the manufacturing process must be approved by the FDA. Pharmaceutical manufacturers thus seek to maximize the revenue that they can obtain from a drug between FDA approval and patent expiration. This requires making advance preparations during the approvals process so that the rate of production can be ramped up quickly when final approval is obtained. The portfolio of investment projects for a pharmaceutical company will be strongly influenced by the expected outcomes of the regulatory approval process for new products.

9.9.3 Decision criteria

There is no single best criterion for economic evaluation of projects. Each company uses its own preferred methods and sets criteria for the minimum performance that will allow a project to be funded. The design engineer must be careful to ensure that the method and assumptions used are in accordance with company policy and that projects are compared on a fair basis. Projects should always be compared using the same economic criterion, but do not have to be compared on *the exact same* basis, because in a global economy there may be significant regional advantages in feed and product pricing, capital costs, labor costs, financing, or investment incentives. Different economic criteria may be used depending on the type of project, although the same criterion must always be used when comparing alternative projects of the same type.

As well as economic performance, many other factors have to be considered when evaluating projects, such as:

1. Safety
2. Environmental problems (waste disposal)
3. Political considerations (government policies and incentives)
4. Location of customers and suppliers (supply chain)
5. Availability of appropriately trained and skilled labor and supporting services
6. Corporate growth strategies
7. Company experience in the particular technology

Some of these factors are discussed in more detail in the discussion of site selection and design in [Chapter 11](#).

Different projects will have different levels of risk, and management must decide whether the potential financial reward is sufficient to justify taking on the associated risks. For example, projects that use new technology or projects located in countries that may be politically unstable have higher risk and would normally require a higher economic impact than less risky investments.

At the plant or site scale, management may have been given a small discretionary capital budget that can be used for preventive maintenance and cost reduction projects (if it is not swallowed up by regulatory compliance projects). These projects are often ranked using simple measures such as pay-back, ROI, or TAC. For a project to be considered for funding, it must meet a minimum (or maximum) criterion, known as a *hurdle rate*. For example, a company may dictate that projects should not be funded unless the pay-back period is less than 2 years. Regulatory compliance projects are often evaluated based on minimum incremental TAC, because it is implicitly assumed that there will be no additional revenue. If there is additional revenue, for example, from sale of a by-product, then this can be offset against the costs. If the cost of compliance is excessive, the alternative costs of closing down or selling the site will also be evaluated.

Small projects or modifications to ongoing projects are often evaluated based on an “incremental ROI” defined as:

$$\text{Incremental ROI} = \frac{\text{incremental profit}}{\text{incremental investment}} \times 100\% \quad (9.32)$$

A separate hurdle rate is set for incremental ROI to ensure that modifications to a large project pay out in their own right and do not get funded just because of the attractiveness (or size) of the base project. This helps prevent creep of project expenses.

Major growth and expansion projects that require significant investment are usually evaluated at the corporate level. Most companies look at the IRR or DCFROR, the fixed and working capital, and the NPV with the interest

TABLE 9.7 Capital spending of large chemical companies

Company	Sales (MM\$)	Net profit (MM\$)	Total assets (MM\$)	Capital spending (MM\$)	Capital/sales	Capital/net profit	Capital/assets
DowDuPont	85,977	8412	188,030	3837	0.045	0.456	0.020
BASF	74,066	7462	102,288	4602	0.062	0.617	0.045
Sinopec	69,210	3929	23,735	2962	0.043	0.754	0.125
Sabic	42,120	9518	79,751	3866	0.092	0.406	0.048
ExxonMobil Chemical	32,443	4162	36,035	1494	0.046	0.359	0.041
LyondellBasell Industries	30,783	,615	12,477	1846	0.060	0.329	0.148
Mitsubishi Chemical	28,747	2382	39,915	1883	0.066	0.791	0.047
LG Chem	25,637	2043	26,330	3838	0.150	1.879	0.146
Air Liquide	24,322	2379	45,312	2605	0.107	1.095	0.057
Evonik Industries	17,755	2078	23,968	1220	0.069	0.587	0.051
Covestro	17,273	2982	13,099	835	0.048	0.280	0.064
Bayer	16,859	3708	90,769	1217	0.072	0.328	0.013
Sumitomo Chemical	16,081	1182	18,576	1171	0.073	0.991	0.063
Braskem	15,885	2250	16,212	741	0.047	0.329	0.046
Lotte Chemical	15,051	1790	18,921	1680	0.112	0.939	0.089
Linde PLC	14,900	2244	93,386	1883	0.126	0.839	0.020
Shin-Etsu Chemical	14,439	3657	27,525	2054	0.142	0.562	0.075
Mitsui Chemicals	13,432	846	13,597	482	0.036	0.570	0.035
Solvay	13,353	1606	25,999	817	0.061	0.509	0.031
Yara	12,928	402	16,656	1336	0.103	3.323	0.080
DSM	10,951	1446	16,120	668	0.061	0.462	0.041
Indorama	10,747	903	11,739	506	0.047	0.560	0.043
Asahi Kasei	10,654	1174	13,517	785	0.074	0.669	0.058
Arkema	10,418	1171	11,949	698	0.067	0.596	0.058

Notes:

1. Source: [Tullo \(2019\)](#).
2. Numbers are based on 2018 financial data.

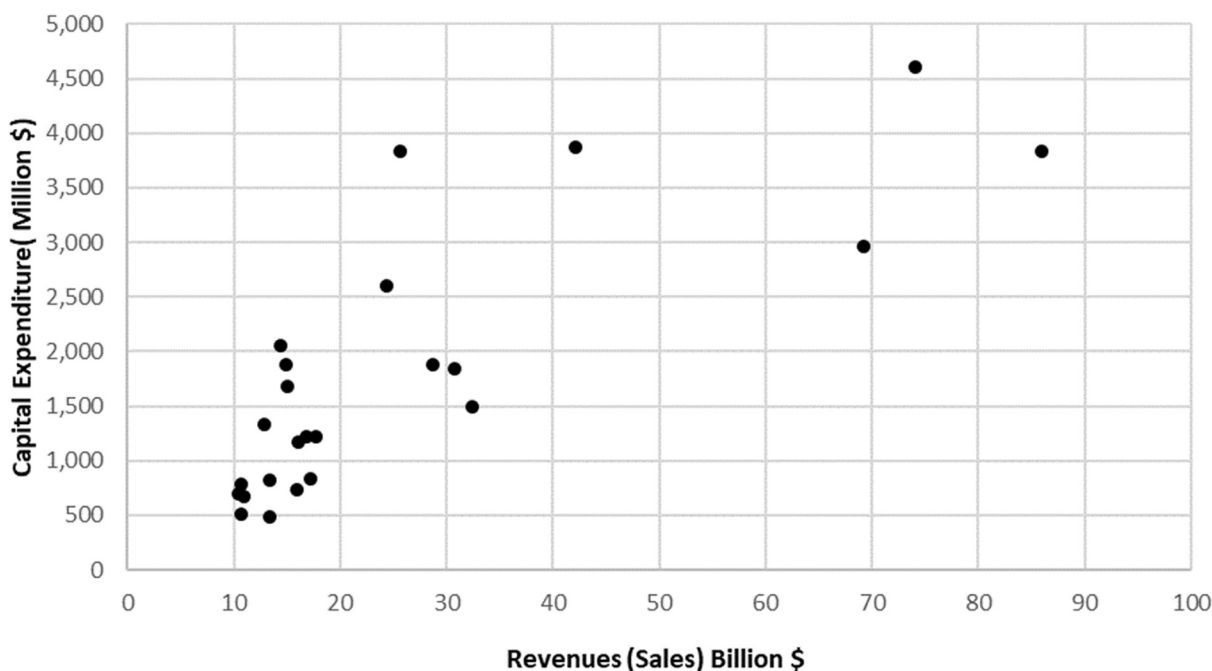


FIG. 9.4 Capital expenditure of major chemical companies versus revenues.

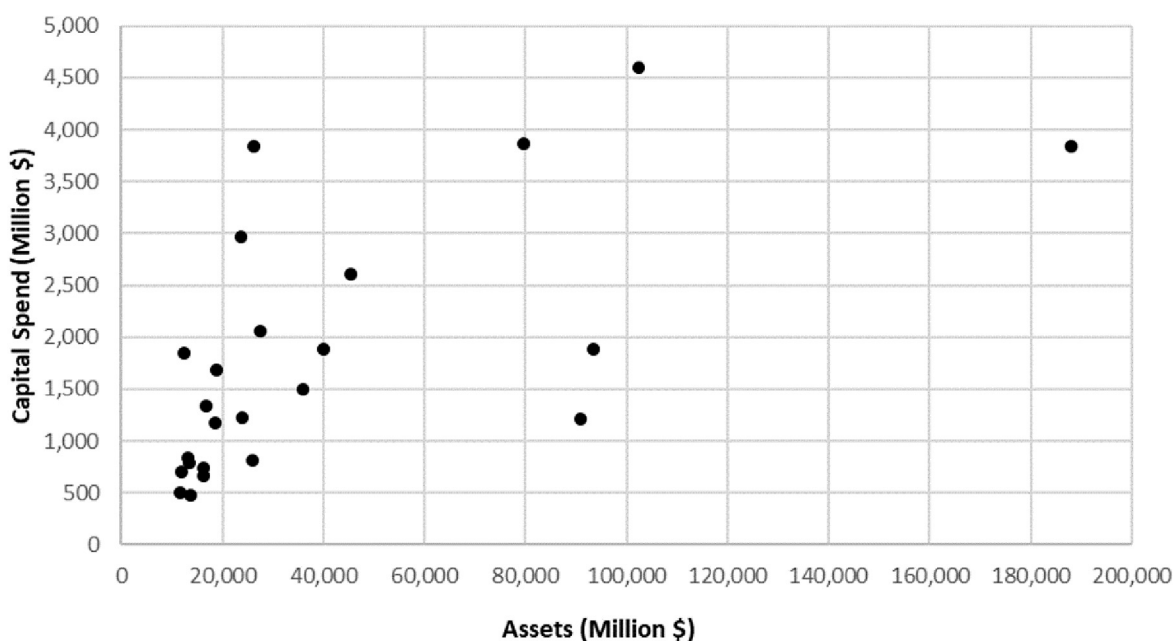


FIG. 9.5 Capital expenditure of major chemical companies versus assets.

rate set equal to the cost of capital. The selection of projects is constrained by the factors described in Section 9.9.2. The set of projects chosen may also be strongly influenced by strategic factors such as the desire to expand a particular business or product line, or a desire to expand the presence of the company in a region that is experiencing rapid economic growth, such as India, China, the Middle East, or Latin America.

Two means of simplifying the selection problem are usually used so that the company's senior management is not faced with a list of thousands of potential projects. The first is to set internal hurdle rates based on simple measures such as IRR or pay-back so that unattractive projects are weeded out at an early stage of the evaluation process. The second method is to divide the available capital budget into categories (sometimes referred to as *buckets*) so as to balance the competing needs of different regions and businesses, growth areas vs. established products, etc. The

various strategic business units or regional subsidiaries (depending on how the company is organized) each submit their proposed capital budgets and a ranked list of projects. Corporate senior management then makes strategic adjustments between the different categories and determines where to draw the line in each list such that the overall portfolio is balanced in accordance with the strategic objectives that they have set for the company. In a large corporation, this process may be repeated at two or more levels of management, with the list of selected projects being passed up to a higher level for further review and approval before the capital is authorized.

The problem of portfolio selection is easily expressed numerically as a constrained optimization: maximize economic criterion subject to constraint on available capital. This is a form of the “knapsack problem,” which can be formulated as a mixed-integer linear program (MILP), as long as the project sizes are fixed. (If not, then it becomes a mixed-integer nonlinear program.) In practice, numerical methods are rarely used for portfolio selection, as many of the strategic factors considered are difficult to quantify and relate to the economic objective function.

9.10 References

- Anderson, J. (2010). Communicating the cost of product and process development. *Chemical Engineering Progress*, 106(2), 46.
- Bechtel, L. B. (1960). *Estimate working capital needs*, 67 p. 127. New York: Chemical Engineering.
- Fields, E. (2016). *The essentials of finance and accounting for nonfinancial managers* (3rd ed.). AMACOM.
- Hollmann, J. K. (2014). Improve your contingency estimates for more realistic project budgets. *Chemical Engineering*, 121(12), 36.
- Humphreys, K. K. (1991). *Jelen's cost and optimization engineering* (3rd ed.). McGraw-Hill.
- Humphreys, K. K. (2005). *Project and cost engineers' handbook* (4th ed.). AACE International.
- Ku, H. (1966). Notes on the use of propagation of error formulas. *Journal of Research of the National Institute of Standards and Technology*, 70C(4), 263–273.
- Lee, A. L. (2010). *Genentech: A rich history of chemical engineering innovation*, 11. Salt Lake City, Nov: AIChE 3rd Corporate Innovation Award Lecture.
- Lyda, T. B. (1972). How much working capital will the new project need? *Chemical Engineering*, 79, 182.
- Piekarski, J. A. (1984). Simplified risk analysis in project economics. *AACE Transactions*, 1984, D5.
- Scott, R. (1978). Working capital and its estimation for project evaluation. *Engineering and Process Economics*, 3, 105.
- Shim, J. K., & Hentleff, N. (1995). *What every engineer should know about accounting and finance*. Marcel Dekker.
- Spiro, H. T. (1996). *Finance for the non-financial manager* (4th ed.). Wiley.
- Sweeting, J. (1997). *Project cost estimating — Principles and practice*. Institution of Chemical Engineers.
- Tullo, A. H. (2019). C&EN's Global Top 50. *Chemical and Engineering News*, 97(30), 30.

American laws and standards

IRS Publication 542. (2019). *Corporations*. United States Department of the Treasury Internal Revenue Service.

IRS Publication 946. (2018). *How to depreciate property*. United States Department of the Treasury Internal Revenue Service.

9.11 Nomenclature

		Dimensions in \$MLT ⁰
A	Annual amount invested	\$
ACC	Annual capital charge	\$
$ACCR$	Annual capital charge ratio	—
B_m	Book value in after m years of depreciation	\$
C	Capital cost	\$
C_d	Depreciable value	\$
CF	Cash flow	\$
CF_n	Cash flow in year n	\$
D	Sum of tax allowances, depreciation	\$

		Dimensions in \$MLT ⁰
D_i	Depreciation charge in year i	\$
DR	Debt ratio (leverage)	—
FC	Fixed capital cost	\$
F_d	Fraction of book value depreciated each year in declining balance method	—
H	High value of range	\$
I	Taxable income	\$
i	Interest rate	—
i'	Discounted cash flow rate of return (internal rate of return)	—
i_c	Cost of capital	—
i_d	Interest rate due on debt	—
i_e	Cost of equity	—
L	Low value of range	\$
ML	Most likely value of range	\$
m	Number of years	T
NPV	Net present value	\$
n	Number of years	T
P	Gross profit, or principle invested in equations 9.22 to 9.26	\$
R	Random number between 1 and 10	—
ROA	Return on assets	—
ROE	Return on equity	—
ROI	Return on investment	—
S	Matured sum	\$
S_x	Standard deviation	\$
TAC	Total annualized cost	\$
TCOP	Total cost of production	\$M ⁻¹ or \$T ⁻¹
t	Time, project life in years	T
t_r	Tax rate	—
$\bar{x}$	Mean value	\$

9.12 Problems

- 9.1. What annual payment is needed to amortize a loan of \$25 million at 7.5% interest over a 20-year period?
- 9.2. A plant is proposing to install a combined heat and power (CHP) system to supply electrical power and process steam. Power is currently taken from a utility company, and steam is generated using on-site boilers. The capital cost of the CHP plant is estimated to be \$23 million. CHP is expected to give net savings of \$10 million per year. The plant is expected to operate for 10 years after the completion of construction. Calculate the cumulative net present value of the project at a discount rate of 12% using MACRS depreciation with a 7-year recovery term. Also calculate the discounted cash flow rate of return.

Construction will take 2 years, and the capital will be paid in two equal increments, at the end of the first and second years. The savings (income) can be taken as paid at the end of each year. Production will start on the completion of construction.

- 9.3. A process heat recovery study identifies five potential modifications, none of which are mutually exclusive, with the costs and energy savings given here.

Project	Capital cost (MM\$)	Fuel savings (MMBtu/hr)
A	1.5	15
B	0.6	9
C	1.8	16
D	2.2	17
E	0.3	8

If fuel costs \$6/MMBtu and the plant operates for 350 days/year, which projects have a simple pay-back period less than 1 year?

What is the maximum 10-year NPV that can be achieved with a 15% interest rate and a 21% tax rate? Assume all the projects can be built immediately, and use MACRS depreciation with a 5-year recovery term. What combination of projects is selected to meet the maximum NPV?

- 9.4. An electronics company wants to fit a solvent recovery system on the vent gas from its circuit board manufacturing line. The solvent recovery system consists of a chiller, a knockout drum, and an adsorbent bed. The adsorbent is periodically regenerated by circulating hot air over the bed and to the chiller and knockout. After consultation with equipment vendors, the following purchased prices are estimated for the major plant equipment:

Item	Cost (\$)
Chiller	4000
Knockout drum	1000
Packaged refrigeration plant	3000
Adsorbent vessel (×2)	1500 each
Air blower	4000
Air heater	3000

Estimate the ISBL cost of the plant and the total project cost. If the annual operating costs are \$38,000 and the annual savings in recovered solvent are \$61,500, what is the IRR of this project?

- 9.5. A shell and tube heat exchanger is to be used in a corrosive service. If carbon steel tubes are used, the cost of the exchanger is estimated to be \$160,000, and it is expected that the exchanger will need to be replaced after 2 years. If stainless steel tubes are used, the service life will be increased to 5 years, but the cost increases to \$400,000. If the cost of capital is 15%, which option should be selected?
- 9.6. The adipic acid plant described in Examples 9.5 and 9.9 is to be built in China, with a location factor of 0.85 and a tax rate of 25%. The cost estimate needs to be updated from 2006 to 2022. Up to 45% of the total investment can be secured as a low-cost loan at an interest rate of 1%.
1. What is the cost of capital if the cost of equity is 40%?
 2. What is the NPV for 15 years of production?
 3. What is the IRR if the debt must be amortized over 15 years as a fixed cost of production?
- 9.7. A Class 2 estimate of the ISBL capital cost of building a 200,000 MMlb/yr ethylene cracking plant has been stated as \$850 MM –5% / +10%. The plant is to be built as an expansion of an existing site, and off-site costs are estimated to be between \$170 MM and \$250 MM. Estimate a value for the total project cost that will give 98% confidence that the project can be carried out within the amount estimated.

- 9.8. Find the most recent annual report for a publicly traded company in the process industries. These are readily available online from company websites. Review the balance sheet, income statement, and cash flow statement and answer the following questions:
1. What fraction of assets is in property plant and equipment?
 2. What fraction of assets is tied up in working capital (inventories, supplies, and receivables)?
 3. What are the relative values of stockholders' equity and the market value (capitalization) of the company? Can you explain the difference?
 4. What fraction of revenues is costs of production? What fraction is selling, general, and administrative (SG&A) overhead? What fraction is R&D costs? What is the overall gross margin of the company, and what do these numbers tell you about the company and the segment of industry it serves?
 5. How much cash came in from operations? How much was invested in property, plant, and equipment? How much cash was brought in from financing (or how much was spent on acquiring shares, paying debt, paying dividends, etc.)? What do these numbers tell you about the financial priorities of the company?

(Note to instructors: This problem can be set as an individual or group exercise. It can be very educational to assign different companies to different students or groups and have them present the results orally.)

Safety and loss prevention

KEY LEARNING OBJECTIVES

- The importance of safety in the design and operation of chemical plants
- Safety legislation with which companies must comply
- Standards and codes of practice that help ensure safer designs
- Process and materials hazards that must be considered in design
- Methods such as HAZOP, FMEA, and quantitative risk analysis that are used to analyze and quantify process hazards
- How relief valves are designed and used to prevent failure of vessels due to overpressure

10.1 Introduction

The safe design and operation of facilities is of paramount importance to every company that is involved in the manufacture of fuels, chemicals, and pharmaceutical products.

Any organization has a legal and moral obligation to safeguard the health and welfare of its employees and the general public. Safety is also good business; the good management practices needed to ensure safe operation also ensure efficient operation.

The term *loss prevention* is an insurance term, the loss being the financial loss caused by an accident. This loss will not only be the cost of replacing the damaged plant, paying fines, and settling third-party claims but also the loss of earnings from lost production and lost sales opportunity. In the event of a major incident, such costs can be large enough to overwhelm a company.

All manufacturing processes are hazardous to some extent, but in chemical processes there are additional, special, hazards associated with the chemicals used and the process conditions. The designer must be aware of these hazards and ensure, through the application of sound engineering practice, that the risks are reduced to tolerable levels.

In this chapter the discussion of safety in process design will of necessity be limited. A more complete treatment of the subject can be found in the books by Wells (1996) (1997), Mannan (2012), Fawcett and Wood (1982), Green (1982), Crowl and Louvar (2019), Cameron and Raman (2005), and Carson and Mumford (1988, 2002) and in the general literature, particularly the publications by the American Institute of Chemical Engineers and the Institution of Chemical Engineers. The proceedings of the symposia on safety and loss prevention organized by these bodies also contain many articles of interest on general safety philosophy, techniques and organization, and the hazards

associated with specific processes and equipment. A good general overview of safety issues in process design is given in the AIChE Center for Chemical Process Safety book *Guidelines for Engineering Design for Process Safety* (CCPS, 2012). The Institution of Chemical Engineers has published a book on safety of particular interest to students of chemical engineering: [Marshall and Ruhemann \(2000\)](#).

Although an effort has been made to provide a summary of the major legal requirements and relevant codes and standards, the area of process safety undergoes continuous improvement and many of the standards are rewritten annually, so the information presented here may not be current by the time of publication. Updated information can be obtained from the standards organizations and regulatory agencies. The design engineer should always consult the most recent version of the laws, regulations, or standards and should always make a thorough check to ensure that the design complies with all local regulations and current best practices.

10.1.1 Safety legislation

Because of the particular hazards associated with processing large quantities of chemicals and fuels, most governments have enacted legislation to ensure that the best safety practices are followed. In the United States, the major federal laws relating to chemical plant safety are:

1. The Occupational Safety and Health Act, 29 U.S.C. 651 *et seq.* (1970): Employers must provide a place of employment free from recognized hazards to safety and health, such as exposure to toxic chemicals, excessive noise, mechanical dangers, heat or cold stress, or unsanitary conditions. Employers must provide personal protective equipment and training, including communication of hazards. Facilities must undergo hazard analysis. The Occupational Safety and Health Administration (OSHA) is established to promote best practices, inspect facilities, set standards, and enforce the law.
2. The Toxic Substances Control Act (TSCA), 15 U.S.C. 2601 *et seq.* (1976): The Environmental Protection Agency (EPA) is required to regulate the 75,000 chemical substances used in industry. Extensive review is required before a new chemical can be manufactured, imported, or sold in the United States. The EPA can ban or restrict the import, manufacture, and use of any chemical. Under TSCA, anyone has a right and an obligation to report information about new or alleged health or environmental effects caused by a chemical. Companies must submit a premanufacture notice (PMN) to the EPA 90 days before manufacturing or importing a new chemical. In 2016 TSCA was amended by the Frank R. Lautenberg Chemical Safety for the 21st Century Act, which introduced additional requirements around risk-based chemical assessments, public transparency for chemical information, and enforceable deadlines for the EPA to evaluate the safety of existing chemicals in commerce.
3. The Emergency Planning and Community Right-to-Know Act (EPCRA), 42 U.S.C. 11001 *et seq.* (1986): Facilities must make plans for major incidents. Plans must be made public to the local community.

In addition, releases of material to the environment as a result of loss of containment during an incident are prohibited by various environmental laws. These are discussed in [Chapter 11](#).

Various states, municipalities, and other bodies may also enact legislation that regulates the safe operation of chemical plants (for example, local fire codes). Local regulations may place stricter requirements on the design and operation of facilities but do not absolve the owner or designer from obligations under national or federal laws.

The most recent version of local, national, and federal laws and standards must always be consulted during design. The requirements for legal compliance may change over time due to revisions of legislation, court actions, or executive orders. The design engineer should be aware of the relevant legislation, but is not usually expected to interpret the requirements of the law and will usually rely upon corporate lawyers and professional safety experts to set company policies, codes, and standards that ensure legal compliance. If the design engineer has any concerns that corporate policies are not meeting regulatory requirements, these concerns should immediately be raised with management, and if there is no satisfactory response, with the regulatory agencies.

In Canada, the major safety legislation is:

1. The Canadian Centre for Occupational Health and Safety Act (R.S.C., 1985, c. C-13) (1978): Created the Canadian Centre for Occupational Health and Safety (CCOHS) to promote health and safety in the workplace, establish and maintain safety standards, develop programs to reduce or eliminate occupational hazards, and collect statistics.
2. The Canadian Oil and Gas Operations Act (R.S.C., 1985, c. O-7): The National Energy Board and chief safety officer shall review installations, equipment, operating procedures, and personnel before authorizing any oil or gas production activity.

3. The Hazardous Products Act (R.S., 1985, c. H-3): Prohibits the advertising, sale, and import of hazardous products. Requires Workplace Hazardous Materials Information System (WHMIS) labels and Material Safety Data Sheets (MSDS) for chemicals. Allows inspection of facilities for compliance.
4. The Canadian Environmental Protection Act (S.C. 1999, c.33): Establishes a list of toxic substances (sec. 90). The list is given as Schedule 1 of the act.

In this book, only the particular hazards associated with chemical and allied processes will be considered. The more general hazards present in all manufacturing processes, such as the dangers from rotating machinery, falls, falling objects, use of machine tools, and of electrocution, will not be considered. General industrial safety and hygiene are covered in several books: [King and Hirst \(1998\)](#), [Asfahl and Rieske \(2018\)](#), and [Channing \(2013\)](#).

10.1.2 Layers of plant safety

Safety and loss prevention in process design can be considered under the following broad headings:

1. Identification and assessment of the hazards.
2. Control of the hazards: for example, by containment of flammable and toxic materials.
3. Control of the process: prevention of hazardous deviations in process variables (pressure, temperature, flow) by provision of automatic control systems, interlocks, alarms and trips, together with good operating practices and management.
4. Limitation of the loss, the damage and injury caused if an incident occurs: pressure relief, plant layout, and provision of firefighting equipment.

Another way of expressing this is in terms of layers of plant safety, illustrated in [Fig. 10.1](#). Each of the layers in [Fig. 10.1](#) can be activated if the lower levels have all failed.

The most basic level of plant safety is safe process and equipment design. If the process is inherently safe (see [Section 10.1.3](#)) incidents are much less likely to occur. The process equipment is the primary means for containing the chemicals that are being processed, as well as keeping out air, and containing high temperatures and pressures. Vessel design codes and standards incorporate safety margins for equipment to reduce the risk of it failing in operation (see [Chapter 14](#)). Most countries require chemical plants to be built and operated in accordance with national or industry standards.

The basic process control system (BPCS) should be designed to maintain the plant under safe conditions of temperature, pressure, flow rates, levels, and compositions. In most continuous plants the process control system will attempt to maintain the process within reasonable bounds of a steady-state condition. In batch or cyclic processes, the variation of process parameters (“ramping”) will be controlled to occur at a safe rate to prevent overshooting.

If a process variable falls outside of the safe operating range, this should trigger an automatic alarm in the plant control room. The purpose of the alarm is to warn the process operators of the triggering condition so that the operators can intervene accordingly. Care should be taken when designing the plant control system not to include too many alarms and to clarify the necessary operator responses, because too many alarms can overwhelm the

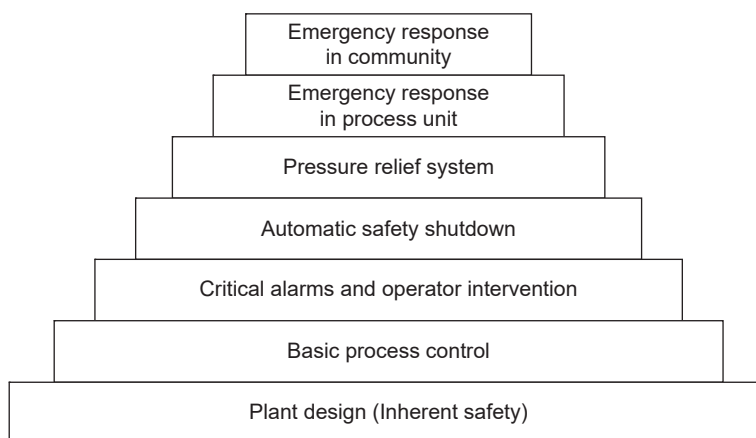


FIG. 10.1 Layers of plant safety.

operators and increase the likelihood of human error (see [Section 10.3.7](#)). Alarms should be set so that they are not frequently triggered by normal process variability (in which case they will tend to be ignored) and to allow time for the operator to respond before the next safety layer is activated. See [Chapter 5](#) for more discussion of process control and instrumentation.

In the event that process operators are unable to bring the process back into control when there is a significant deviation of a variable that indicates a hazardous condition, an automatic shut-down of the process (also known as a *trip*) should be activated. Trip systems are sometimes activated by the plant control system and sometimes self-actuated, as described in [Chapter 5](#). Emergency shut-down will usually involve shutting off feeds and sources of heat, depressurizing the process, and purging the plant with inert material. When designing the emergency shut-down procedures and systems, care must be taken to ensure that unsafe conditions are not created or worsened. For example, in some high-temperature or exothermic processes, it may be safer to continue feed of one of the reagents while shutting off the others, so as to remove heat from the reactor. Closing all the valves in a plant is almost never the safest means of shutting it down. A good guide to the design of control, alarm, and shut-down systems is given in the American Petroleum Institute Recommended Practice API RP 14c. Although intended for offshore production platforms, this standard covers many unit operations found in chemical plants and refineries. Methods for quantifying the required reliability of shut-down systems are described in [Section 10.8](#).

If the plant safety shut-down is not rapid enough and an overpressure situation develops, the pressure relief system is activated. Pressure vessel design codes such as the ASME Boiler and Pressure Vessel Code require relief devices to be fitted on all pressure vessels. If the relief system has been properly designed and maintained, then in the event of an overpressure incident the plant contents will be vented via relief valves or bursting disks into the relief system, where liquids are recovered for treatment and vapors are sent to flare stacks or discharged to the atmosphere if it is safe to do so. The pressure relief system should allow the plant to be relieved of any source of overpressure before damage to process equipment (leaks, bursting, or explosion) can occur. Pressure relief systems are discussed in [Section 10.9](#).

If a loss of containment does occur in a chemical plant, an emergency response is required. A small-scale loss of containment might be a leak or a spill. Leaks of liquids are usually visible and obvious, whereas leaks of vapor can be much harder to detect and require special instrumentation for monitoring. If the material that escapes from the process is flammable, the first manifestation of loss of containment might be a small or localized fire (often smoldering insulation is an early warning). Plant personnel should be trained to respond to such emergencies. Many large sites also have a dedicated emergency response staff to fight fires and clean up chemical spills. The emergency response in the process unit does not always cause the unit to be shut down, depending on the scale of the incident, but the root cause of every incident must always be determined, and any deficiency in the plant must be corrected safely before normal operations are resumed.

In the event that an incident develops into a more serious accident, the resources required will be beyond those available on the plant or site. Local community emergency response providers will be brought into the site, and injured members of the workforce and local population will need treatment at local hospitals. The local community must be able to plan for such events, and local emergency responders must be trained to cope with the hazards associated with the plant. The EPCRA ensures that the local community has access to the necessary information.

10.1.3 Intrinsic and extrinsic safety

Processes can be divided into those that are intrinsically safe and those for which the safety has to be engineered in. An intrinsically safe process is one in which safe operation is inherent in the nature of the process—a process that causes no danger, or negligible danger, under all foreseeable circumstances (all possible deviations from the design operating conditions). The term *inherently safe* is often preferred to intrinsically safe to avoid confusion with the narrower use of the term *intrinsically safe* as applied to electrical equipment (see [Section 10.3.5](#)). In the context of risk management, an inherently safe design is a design that has a very low likelihood of causing injury even in the absence of protective systems; see [Section 10.8](#).

Clearly, the designer should always select a process that is inherently safer whenever it is practical and economic to do so; however, most chemical manufacturing processes are, to a greater or lesser extent, inherently unsafe, and dangerous situations can develop if the process conditions deviate from the design values. The safe operation of such processes depends on the design and provision of engineered safety devices and on good operating practices to prevent a dangerous situation from developing and to minimize the consequences of any incident that arises from the failure of these safeguards.

The term *engineered safety* covers the provision in the design of control systems; alarms; trips; pressure-relief devices; automatic shut-down systems; duplication of key equipment services; and firefighting equipment, sprinkler systems, and blast walls to contain any fire or explosion.

The design of inherently safer process plants is discussed by Kletz in a booklet published by the Institution of Chemical Engineers, Kletz (1984), and in Kletz and Amyotte (2010), CCPS (2012), and Chapter 32 of Mannan (2012). Kletz makes the telling point that material that is not there cannot leak out and thus cannot catch fire, explode, or poison anyone. This is a plea to keep the inventory of dangerous material to the absolute minimum required for the operation of the process. The AIChE Center for Chemical Process Safety (CCPS) has published their guidelines for inherently safer design methods in the CCPS *Guidelines for Inherently Safer Chemical Processes*, CCPS (2019), which includes a checklist for inherently safer chemical reaction process design and operation in Appendix A (available for download in the resource section on the CCPS website).

10.2 Materials hazards

In this section the special hazards of chemicals are reviewed (toxicity, flammability, and reactivity). Hazards arising from process operations will be discussed in Section 10.3.

10.2.1 Toxicity

Most of the materials used in the manufacture of chemicals are poisonous to some extent, and almost every chemical is toxic if someone is exposed to enough of it. The potential hazard will depend on the inherent toxicity of the material and the frequency and duration of any exposure.

It is usual to distinguish between the short-term (acute) effects and the long-term (chronic) effects. Acute effects normally have symptoms that develop rapidly after exposure, for example, burns to the skin after direct contact, respiratory failure, renal failure, cardiac arrest, paralysis, etc. Acute effects are usually associated with a short exposure to a high concentration of toxin (although what constitutes a “high concentration” depends on the toxicity). The chronic symptoms of poisoning develop over a long time, for example, cancer, and often persist or recur frequently. Chronic effects may occur as the result of long-term exposure to low levels of a toxin, but may also occur as a delayed response to a short-term exposure to high levels of a toxin.

Highly toxic materials that cause immediate injury, such as phosgene or chlorine, are usually classified as safety hazards, whereas materials whose effects are only apparent after long exposure at low concentrations—for instance, carcinogenic materials such as vinyl chloride—are usually classified as industrial health and hygiene hazards. The permissible limits and the precautions to be taken to ensure the limits are met will be very different for these two classes of toxic materials. Industrial hygiene is as much a matter of good operating practice and control as of good design.

The inherent toxicity of a material is measured by tests on animals. It is usually expressed as the lethal dose at which 50% of the test animals are killed, the LD₅₀ (lethal dose 50) value. The dose is expressed as the quantity in milligrams of the toxic substance per kilogram of body weight of the test animal.

Some values of LD₅₀ for oral ingestion by rats are given in Table 10.1. Estimates of the LD₅₀ for humans are based on tests on animals. The LD₅₀ measures the acute effects; it gives only a crude indication of the possible chronic effects. The LD₅₀ for humans should always be taken as the lowest measured value for other mammalian species. In some cases, LD₅₀ data are given for different routes of ingestion. For example, ethanol has LD₅₀ values 3450 (oral, mouse), 7060 (oral, rat), and 1440 (intravenous, rat).

There is no generally accepted definition of what can be considered toxic and nontoxic.

A system of classification is given in the *Classification, Packaging and Labelling of Dangerous Substances Regulations*, 1984 (United Kingdom), which is based on European Union (EU) guidelines; for example:

LD ₅₀ , absorbed orally in rats, mg/kg	
≤25	very toxic
25–200	toxic
200–2000	harmful

TABLE 10.1 Toxicity data

Compound	PEL (ppm)	LD ₅₀ (mg/kg)
Carbon monoxide	50	1807
Carbon disulfide	20	3188
Chlorine	1	239
Chlorine dioxide	0.1	292
Chloroform	50	1188
Cyclohexane	300	
Dioxane	100	4200
Ethylbenzene	100	3500
Formic acid	5	1100
Furfural	5	260
Hydrogen chloride	5	4701
Hydrogen cyanide	10	3.7
Isopropyl alcohol	400	5045
Toluene	100	5000
Xylene	100	4300

Source: OSHA.

These definitions apply only to the short-term (acute) effects. In fixing permissible limits on concentration for the long-term exposure of workers to toxic materials, the exposure time must be considered together with the inherent toxicity of the material. The “threshold limit value” (TLV) is a commonly used guide for controlling the long-term exposure of workers to contaminated air. The TLV is defined as the concentration to which it is believed the average worker could be exposed, day by day, for 8 hours a day, 5 days a week, without suffering harm. It is expressed in ppm for vapors and gases and in mg/m³ (or grains/ft³) for dusts and liquid mists. A comprehensive source of data on the toxicity of industrial materials is Sax’s handbook, [Lewis \(2012\)](#), which also gives guidance on the interpretation and use of the data. Recommended TLV values are given by the American Conference of Government Industrial Hygienists (www.acgih.org/home). In the United States, permissible exposure limits (PEL) for known toxins are set by OSHA. Values can be found on the OSHA website at <https://www.osha.gov/dsg/annotated-pels/>.

Fuller details of the methods used for toxicity testing, the interpretation of the results, and their use in setting standards for industrial hygiene are given in the more specialized texts on the subject; see [Carson and Mumford \(1988\)](#) and [Mannan \(2012\)](#).

10.2.2 Flammability

The hazard caused by a flammable material depends on a number of factors:

1. The flash point of the material
2. The autoignition temperature of the material
3. The flammability limits of the material
4. The energy released in combustion

An introduction to the flammability properties of materials is given by [Crowl \(2012\)](#).

Flash point

The flash point is a measure of the ease of ignition of a liquid. It is the lowest temperature at which the material will ignite from an open flame. The flash point is a function of the vapor pressure and the flammability limits of the material. It is measured in standard apparatus, following standard procedures (ASTM D92 and ASTM D93). Both

open- and closed-cup apparatuses are used. Closed-cup flash points are lower than open-cup, and the type of apparatus used should be stated clearly when reporting measurements. Flash points are given in Sax's handbook, [Lewis \(2012\)](#). The flash points of many volatile materials are below normal ambient temperature, for example, ether -45°C , gasoline -43°C (open cup).

Autoignition temperature

The autoignition temperature of a substance is the temperature at which it will ignite spontaneously in air without any external source of ignition. It is an indication of the maximum temperature to which a material can be heated in air, for example, in drying operations.

Flammability limits

The flammability limits of a material are the lowest and highest concentrations in air, at normal pressure and temperature, at which a flame will propagate through the mixture. They show the range of concentration over which the material will burn in air if ignited. At very low concentrations in air a flame will not propagate, as there is insufficient fuel. Similarly, at very high concentrations a flame will not propagate due to insufficient oxidant. Flammability limits are characteristic of the particular material and differ widely for different materials. For example, hydrogen has a lower limit of 4.1 and an upper limit of 74.2% by volume, whereas for gasoline the range is only from 1.3% to 7.0%. The flammability limits for a number of materials are given in [Table 10.2](#). The limits for a wider range of materials are given in Sax's handbook, [Lewis \(2012\)](#).

A flammable mixture may exist in the space above the liquid surface in a storage tank. The vapor space above highly flammable liquids is usually purged with inert gas (nitrogen), or floating-head tanks are used. In a floating-head tank a "piston" floats on top of the liquid, eliminating the vapor space.

10.2.3 Materials incompatibility

Some materials are naturally unstable and can spontaneously decompose, polymerize, or undergo other reactions. These reactions can be initiated or accelerated by promoters such as light, heat, sources of free radicals or ions or catalysts such as metal surfaces. These reactions can sometimes be retarded by adding inhibitors or diluents. Reactions of this kind are usually exothermic and if allowed to proceed will lead to a "runaway" reaction with serious consequences.

Some materials are by their nature highly reactive and will react with many other compounds at low temperatures. Examples include strong oxidizing agents such as peroxides and chlorates, strong reducing agents, strong alkalis, strong acids, and the metallic forms of alkali metals. In addition to being reactive with many other chemicals, these materials can attack the materials from which the plant is constructed.

Other groups of compounds are known to react together rapidly and exothermically. These include mixtures such as acids and bases; acids and metals; fuels and oxidants; and free radical initiators and epoxides, peroxides, or unsaturated molecules.

Another important class of incompatible materials is compounds that become more hazardous when contacted with water. For example, carbonyl sulfide (COS) and calcium sulfide (CaS) both release toxic H_2S on contact with water. Dry powders of sodium or potassium cyanide release toxic HCN in the presence of moisture. Care must be taken to prevent such materials from coming into contact with water during processing and storage. The 1985 Bhopal disaster was started by a runaway reaction involving a water-sensitive chemical.

Materials that are used to construct the process equipment and instrumentation must also be checked for compatibility with the process chemicals. This includes not only the metals or alloys from which the major vessels are built but also welding, brazing, or soldering materials; components of pumps, valves, and instruments; gaskets; seals; linings; and lubricants.

Information on incompatible materials can be found in most safety data sheets (SDSs). Wiley has published a guide to chemical incompatibilities—[Pohanish and Greene \(2009\)](#)—and they are also discussed in [Urban \(2017\)](#). The U.S. National Fire Protection Association (NFPA) also publishes standards NFPA 491 Guide to Hazardous Chemical Reactions and NFPA 49 Hazardous Chemicals Data, both of which provide data on incompatible materials.

Materials incompatibility is one of the most frequent causes of process incidents. Degradation of seals and gaskets that have become softened by solvent effects can lead to minor leaks or major loss of containment, and hence to fires, explosions, or more serious accidents. If seal or gasket leakage is identified in a process, the plant engineer should

TABLE 10.2 Flammability ranges

Material	Lower limit	Upper limit
Hydrogen	4.1	74.2
Ammonia	15.0	28.0
Hydrocyanic acid	5.6	40.0
Hydrogen sulfide	4.3	45.0
Carbon disulfide	1.3	44.0
Carbon monoxide	12.5	74.2
Methane	5.3	14.0
Ethane	3.0	12.5
Propane	2.3	9.5
Butane	1.9	8.5
Isobutane	1.8	8.4
Ethylene	3.1	32.0
Propylene	2.4	10.3
n-Butene	1.6	9.3
Isobutene	1.8	9.7
Butadiene	2.0	11.5
Benzene	1.4	7.1
Toluene	1.4	6.7
Cyclohexane	1.3	8.0
Methanol	7.3	36.0
Ethanol	4.3	19.0
Isopropanol	2.2	12.0
Formaldehyde	7.0	73.0
Acetaldehyde	4.1	57.0
Acetone	3.0	12.8
Methylethyl ketone	1.8	10.0
Dimethylamine (DEA)	2.8	184
Trimethylamine (TEA)	2.0	11.6
Gasoline	1.3	7.0
Kerosene (jet fuel)	0.7	5.6
Gas oil (diesel)	6.0	13.5

Note: Volume percentage in air at ambient conditions.

consult with the manufacturer to confirm the material is suitable for the service. If necessary, all the seals or gaskets of that material should be replaced with something more resilient to the process conditions.

10.2.4 Ionizing radiation

The radiation emitted by radioactive materials is harmful to living matter. Small quantities of radioactive isotopes are used in the process industry for various purposes—for example, in level and density-measuring instruments and for the nondestructive testing of equipment.

The use of radioactive isotopes in industry is covered by government legislation. In the United States this is under OSHA (29 CFR 1910.1096), and in Canada it is governed by the Radiation Emitting Devices Act (R.S., 1985, c. R-1). Low levels of radiation may also be present in natural minerals. Care should be taken if these radioactive materials are concentrated or accumulated in the process or dispersed into the environment.

A discussion of the particular hazards that arise in the chemical processing of nuclear fuels is outside the scope of this book.

10.2.5 Safety Data Sheets

A Safety Data Sheet (SDS) is a document summarizing the hazards and health and safety information for a chemical. In the United States, the OSHA Hazard Communication Standard (29 CFR 1910.1200) requires that chemical manufacturers must make an SDS available to employees and customers for every chemical manufactured or sold. The Canadian Hazardous Products Act (R.S., 1985, c. H-3) has the same requirement. SDSs are compliant with the UN-sponsored Globally Harmonized System (GHS) for hazard communication, as revised into the OSHA Hazard Communication Standard in 2012, and replaced the earlier MSDSs, though they contain largely the same information (Burgess, 2013).

The SDS contains the information needed to begin analyzing materials and process hazards, to understand the hazards to which the workforce is exposed, and to respond to a release of the material or other major incident where emergency response personnel may be exposed to the material.

An SDS usually contains the following sections:

1. Chemical product and company information: chemical name and grade; catalog numbers and synonyms; manufacturer's contact information, including 24-hour contact numbers
2. Composition and information on ingredients: chemical names, CAS numbers, and concentration of major components of the product
3. Hazards identification: summary of the major hazards and health effects
4. First aid measures: procedures for contact with eyes and skin or by ingestion or inhalation
5. Firefighting measures: information on firefighting, extinguishing media, flammability data, NFPA ratings
6. Accidental release measures: procedures for dealing with leaks or spills
7. Handling and storage: procedures for transfer, storage, and general use of the material
8. Exposure controls and personal protection: required engineering controls such as eyewashes, safety showers, ventilation, etc.; OSHA PEL data; required personal protective equipment
9. Physical and chemical properties
10. Stability and reactivity: conditions that cause instability, known incompatible materials, hazardous decomposition products
11. Toxicological information: acute effects, LD₅₀ data, chronic effects, carcinogenicity, teratogenicity, mutagenicity
12. Ecological information: ecotoxicity data for insects and fish, other known environmental impacts
13. Disposal considerations: requirements for disposal under the Resource Conservation and Recovery Act (RCRA—see Chapter 11)
14. Transport information: shipping information required by the U.S. Department of Transport as well as other international bodies
15. Regulatory information: U.S. federal and state, European, Canadian, and international regulations listing the material; includes TSCA listing and Clean Air Act and Clean Water Act limits (see Chapter 11)
16. Additional information: date of creation and revisions, legal disclaimers

Most SDS forms are created by chemical manufacturers. They can be found in libraries, on manufacturer's websites and by contacting manufacturers or suppliers directly. Websites are available that catalogue SDS forms from multiple sources. The Canadian Centre for Occupational Health and Safety (www.msdsdsonline.com) also maintains an extensive database (for purchase) at their website. For legal reasons (limitation of liability), most SDS forms contain a disclaimer stating that the user should also make their own evaluation of compatibility and fitness for use. An example SDS form is given in Appendix I, available in the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793.

10.2.6 Design for materials hazards

Under OSHA hazard communication standard 29 CFR 1910.119, employers are required to carry out an assessment to evaluate the risk to health from any chemicals handled and establish what precautions are needed to protect

employees. A written record of the assessment should be kept and details made available to employees. The design engineer should consider the preventive aspects of the use of hazardous substances. Points to consider are:

1. Substitution: of the processing route with one using less hazardous material, or substitution of toxic process materials with nontoxic or less toxic materials.
2. Containment: sound design of equipment and piping to avoid leaks. For example, specifying welded joints in preference to gasketed flanged joints that are liable to leak or suffer materials incompatibility problems.
3. Prevention of releases: by process and equipment design, operating procedures, and design of disposal systems.
4. Ventilation: use open structures or provide adequate ventilation systems.
5. Disposal: provision of effective vent stacks to disperse material vented from pressure relief devices or use of vent scrubbers. Collection and treatment of sewer and run-off waters and liquids collected from relief systems.
6. Emergency equipment and procedures: automated shut-down systems, escape routes, rescue equipment, respirators, antidotes (if appropriate), safety showers, eye baths, and emergency services.

In addition, good plant operating practice would include:

1. Written instruction in the use of the hazardous substances and the risks involved.
2. Adequate training of personnel.
3. Provision of protective clothing and equipment.
4. Good housekeeping and personal hygiene.
5. Monitoring of the environment to check exposure levels. Consider the installation of permanent instruments fitted with alarms.
6. Regular medical check-ups on employees to check for the chronic effects of toxic materials.
7. Training of local emergency response personnel.

The process design engineer should always collect the SDS of every component used in the process, including solvents, acids, bases, adsorbents, etc., at as early a stage in the design as possible. The information in the SDS can be used to improve the inherent safety of the process, for example, by eliminating incompatible mixtures or substituting less hazardous chemicals as feeds, intermediates, or solvents. The SDS information can also be used to ensure that the design meets regulatory requirements on vapor recovery and other emissions.

10.3 Process hazards

In addition to the hazards caused by chemical or materials properties, hazards can arise from the conditions under which processes are carried out and the equipment that is used.

10.3.1 Pressure

Overpressure, a pressure exceeding the system design pressure, is one of the most serious hazards in chemical plant operation. If the pressure exceeds the maximum allowable working pressure of the vessel by more than the safety margin allowed in the vessel design code, then failure of the vessel can occur, usually at a joint or flange. Failure of a vessel, or the associated piping, can precipitate a sequence of events that culminate in a disaster. Overpressure occurs when mass, moles, or energy accumulate in a contained volume or space with a restricted outflow. Specific causes of overpressure are discussed in [Section 10.9.1](#).

Pressure vessels are required to be fitted with some form of pressure-relief device, set at the maximum allowable working pressure, so that potential overpressure is relieved in a controlled manner (ASME Boiler and Pressure Vessel Code, [Section VIII](#) Division 1, Part UG-125). See [Section 10.9](#) for a more detailed discussion of pressure relief.

Process equipment must also be protected from underpressure (vacuum), as this places compressive stresses on vessel walls that can lead to failure by buckling. Design for underpressure is discussed in [Section 10.9.6](#).

The factors to be considered in the design of relief systems are set out in a comprehensive paper by [Parkinson \(1979\)](#) and by [Moore \(1984\)](#). More extensive references, including design codes and standards and relief system design software, are given in [Section 10.9](#).

10.3.2 Temperature deviations

Excessively high temperature, over and above that for which the equipment was designed, can cause structural failure and initiate a disaster. High temperatures can arise from loss of control of reactors and heaters and, externally, from open fires. In the design of processes where high temperatures are a hazard, protection against high temperatures is provided by:

1. Provision of high-temperature alarms and interlocks to shut down reactor feeds, or heating systems, if the temperature exceeds critical limits.
2. Use of additional temperature detectors to provide highly localized and redundant temperature monitoring. These include detectors on the wall of the vessel ("skin thermocouples") as well as in thermowells exposed to the process fluids. In some cases, temperature-sensitive paint that changes color above a certain threshold is used on the outside of the vessel.
3. Provision of emergency cooling systems for reactors where heat continues to be generated after shut-down, for instance, in some polymerization systems.
4. Provision of quench systems for emergency shut-down, designed to flood the equipment with a cold, inert material.
5. Structural design of equipment to withstand the worst possible temperature excursion.
6. The selection of inherently safe heating systems for hazardous materials.

Steam and other vapor heating systems are inherently safer than fired heat and electric heat, as the temperature cannot exceed the saturation temperature at the supply pressure if the vapor is de-superheated and the supply line has a relief system that prevents it from becoming overpressurized. Other heating systems rely on control of the heating rate to limit the maximum process temperature. Electrical heating systems can be particularly hazardous, because the heating rate is proportional to the resistance of the heating element, which increases with temperature.

Very low temperatures can also be hazardous. Low temperatures can be caused by ambient conditions, operation of cryogenic processes, expansion of gases and vapors, flashing of liquids (autorefrigeration), and endothermic reactions. Low temperatures can cause embrittlement and stress cracking in metals. At very low temperatures some metals undergo microstructural transformations that cause substantial changes in density (for example, brass). When water freezes in a confined volume, the increase in specific volume can cause pipes or vessels to crack. Minimum design metal temperatures are specified in pressure vessel design (see [Chapter 14](#)).

10.3.3 Noise

Excessive noise is a hazard to health and safety. Long exposure to high noise levels can cause permanent damage to hearing. At lower levels, noise is a distraction and causes fatigue. Regulations on noise are set by OSHA (29 CFR 1910.95, 2007).

The unit of sound measurement is the decibel, defined by the expression:

$$\text{Sound level} = 20 \log_{10} \left[\frac{\text{RMS sound pressure (Pa)}}{2 \times 10^{-5}} \right], \text{ dB} \quad (10.1)$$

The subjective effect of sound depends on frequency as well as intensity.

Industrial sound meters include a filter network to give the meter a response that corresponds roughly to that of the human ear. This is termed the "*A*" *weighting network*, and the readings are reported as dB(A). Permanent damage to hearing can be caused at sound levels above about 85 dB(A), and it is normal practice to provide ear protection in areas where the level is above 80 dB(A).

Excessive plant noise can lead to complaints from neighboring factories and local residents. Due attention should be given to noise levels when specifying, and when laying out, equipment that is likely to be excessively noisy, such as compressors, fans, burners, and steam relief valves. This equipment should not be placed near the control room.

Several books are available on the general subject of industrial noise control—[Bies and Hansen \(2009\)](#)—and on noise control in the process industries—[Cheremisinoff \(1996\)](#) and [IMechE \(1993\)](#).

10.3.4 Loss of containment

The primary means for protecting employees and the public from exposure to toxic chemicals is the plant itself. Loss of containment can occur due to:

1. Pressure relief events.
2. Operator errors such as leaving a sample point open or dripping.
3. Poor maintenance procedures, including failure to isolate, drain, and purge properly before maintenance, leading to release when the equipment is opened, or failure to reconnect items properly and close drain valves when maintenance is complete.
4. Leaks from degraded equipment, including damaged seals, gaskets, and packings and corroded or eroded vessels and pipes.
5. Emissions from solid-handling operations (dust).
6. Internal equipment leaks (particularly in heat exchanger tubesheets) that allow utility services such as cooling water to become contaminated with process chemicals.
7. Spills from drum or tanker loading and emptying.

Frequent loss-of-containment incidents are usually an indication that a plant has been poorly maintained and are a leading indicator of major incidents.

If the potential impact of a loss of containment is high, then the design engineer should provide a means of containment or mitigation. These means might include:

1. Secondary containment (bunding) to prevent run-off, but note that this can create a worse hazard if the chemicals are flammable and easily ignited.
2. Contained drainage and sewer systems to collect run-off and rainwater for waste treatment.
3. Use of concrete foundations to protect groundwater.
4. Containment of the plant inside a building with ventilation and vent scrubbers (used for hazardous dusts and highly toxic compounds).

10.3.5 Fires and ignition sources

A fire occurs whenever sufficient amounts of fuel and oxidant are mixed and contacted with an ignition source. If a fuel is above its autoignition temperature, then ignition can occur spontaneously in air. Though precautions are normally taken to eliminate sources of ignition on chemical plants, it is best to work on the principle that a leak of flammable material will ultimately find an ignition source. Guidelines for the control of ignition sources are given in [section 7.9](#) of NFPA 30.

Electrical equipment

The sparking of electrical equipment, such as motors, is a major potential source of ignition, and flameproof equipment is normally specified. Electrically operated instruments, controllers, and computer systems are also potential sources of ignition of flammable mixtures.

The use of electrical equipment in hazardous areas is covered by the National Electrical Code (NFPA standard 70), NFPA standards 496 and 497, and OSHA standard 29 CFR 1910.307. The American Petroleum Institute Recommended Practices API RP 500 and API RP 505 should also be consulted.

Articles 500 and 505 of the National Electrical Code (NFPA 70) define classified areas in which flammable materials may be present at high enough concentrations to be ignitable. In a Class I location, the ignitable material is a gas or vapor. Class I locations are further divided into either divisions or zones as follows:

Class I, Division 1: locations in which ignitable concentrations of flammable gas or vapor can exist under normal operating conditions; or can exist frequently because of repair, maintenance or leakage; or can be released during breakdown or faulty operation of equipment that might also cause simultaneous failure of electrical equipment such that the electrical equipment could become a source of ignition.

Class I, Division 2: locations in which flammable gases, vapors, or volatile liquids are processed but are normally contained within closed systems; or ignitable concentrations are normally prevented by forced ventilation; or the location is adjacent to a Class I, Division 1 location and an ignitable concentration of gas or vapor might occasionally be communicated to it.

Class I, Zone 0: locations in which ignitable concentrations of vapor or gas are present continuously or for long periods.

Class I, Zone 1: locations where ignitable concentrations of flammable gas or vapor are likely to exist under normal operating conditions; or may exist frequently because of repair, maintenance, or leakage; or could be released by equipment breakdown or faulty operation that could simultaneously cause failure of the electrical

equipment in such a way that the electrical equipment could act as a source of ignition; or is adjacent to a Class I, Zone 0 location from which ignitable concentrations of flammable gas or vapor could be communicated.

Class I, Zone 2: locations where ignitable concentrations of flammable gas or vapor are not likely to exist in normal operation, and if they do occur will exist only for short periods; or flammable gases, vapors, or liquids are processed but are normally contained within closed systems; or ignitable concentrations are normally prevented by forced ventilation; or the location is adjacent to a Class I, Zone 1 location and an ignitable concentration of gas or vapor might occasionally be communicated to it.

Guidance on how to classify a location is given in the standards NFPA 30, NFPA 497, API RP 500, and API RP 505.

Various protection techniques can be used to prevent electrical equipment from becoming an ignition source. Explosion-proof enclosures, pressurized and purged equipment, and intrinsically safe circuits can be used in Division 1 and Division 2 locations. Nonsparking equipment can be used in Division 2 locations. Details of the types of equipment permitted in each classified location are given in NFPA 70.

It is also important that electrical equipment in hazardous areas be properly maintained, particularly in older plants. Guidelines for maintenance of electrical equipment are given in standard NFPA 70B.

The standards should be consulted for the full specification before selecting equipment for use in classified locations.

The design and specification of intrinsically safe control equipment and systems is discussed by [MacMillan \(1998\)](#) and [Cooper and Jones \(1993\)](#). Purged and pressurized enclosures for electrical equipment are described in NFPA 496.

Static electricity

The movement of any nonconducting material, powder, liquid, or gas can generate static electricity, producing sparks. Precautions must be taken to ensure that all piping is properly earthed (grounded) and that electrical continuity is maintained around flanges. Escaping steam, or other vapors and gases, can generate a static charge. Gases escaping from a ruptured vessel can self-ignite from a static spark. For a review of the dangers of static electricity in the process industries, see the article by [Napier and Russell \(1974\)](#) and the books by [Pratt \(1997\)](#) and [Britton \(1999\)](#). Protection against static electricity, lightning, and stray currents is discussed in API RP 2003. The NFPA standard NFPA 77 is the national standard on static electricity protection in the United States.

Process flames

Open flames from process furnaces, incinerators, and flare stacks are obvious sources of ignition and must be sited well away from plants containing flammable materials.

Miscellaneous sources

It is the usual practice in plants handling flammable materials to control the entry to the site of obvious sources of ignition, such as matches, cigarette lighters, and battery-operated equipment. The use of portable electrical equipment, welding, cutting, and spark-producing tools and the movement of gasoline engine vehicles would also be subject to strict control. Exhaust gases from diesel engines are also a potential source of ignition.

Flame traps

Flame arresters are fitted in the vent lines of equipment that contains flammable material to prevent the propagation of flame through the vents. Various types of proprietary flame arresters are used. In general, they work on the principle of providing a heat sink, usually expanded metal grids or plates, to dissipate the heat of the flame. Flame arrestors and their applications are discussed by [Rogowski \(1980\)](#), [Howard \(1992\)](#), [Mendoza et al. \(1998\)](#), API RP 2210, and BS EN ISO 16852.

Traps should also be installed in plant ditches to prevent the spread of flame. These are normally liquid U-legs, which block the spread of a flame along ditches.

Fire protection

Recommendations on the fire precautions to be taken in the design of chemical plants are given in the standards NFPA 30, [API RP 2001](#), and API RP 2218. Legal requirements for fire protection are given in the OSHA standard 29 CFR 1910, subpart L. The relevant international standard is BS EN ISO 19353.

To protect against structural failure, water deluge systems are usually installed to keep vessels and structural steelwork cool in a fire. Water-mist fire protection systems are described in NFPA 750 and API RP 2030. The lower sections of structural steel columns are often lagged with concrete or other suitable materials.

Plants that handle flammable liquids are usually designed to have slightly sloping ground or use drainage ditches or trenches to control run-off so that pools do not form. Drainage ditches and slopes should always direct flow away from sources of ignition.

10.3.6 Explosions

An explosion is the sudden, catastrophic release of energy, causing a pressure wave (blast wave). An explosion can occur without fire, such as the failure through overpressure of a steam boiler or an air receiver. A good general introduction to explosions is given by [Crowl \(2003\)](#).

When discussing the explosion of a flammable mixture, it is necessary to distinguish between detonation and deflagration. If a mixture detonates, the reaction zone propagates at a supersonic velocity (above approximately 300 m/s) and the principal heating mechanism in the mixture is shock compression. The pressure wave in a detonation can be up to 20 bar. In a deflagration, the combustion process is the same as in the normal burning of a gas mixture, the combustion zone propagates at a subsonic velocity, and the pressure build-up is slow and usually less than 10 bar. Whether detonation or deflagration occurs in a gas–air mixture depends on a number of factors, including the concentration of the mixture and the source of ignition. Unless confined or ignited by a high-intensity source (a detonator), most materials will not detonate; however, the pressure wave (blast wave) caused by a deflagration can still cause considerable damage. In a confined space, such as a pipe or inside a building, a deflagration can propagate into a detonation.

Certain materials—for example, acetylene and many peroxides—can decompose explosively in the absence of oxygen; such materials are particularly hazardous.

Confined vapor cloud explosion (CVCE)

A relatively small amount of flammable material—a few kilograms—can lead to an explosion when released into the confined space of a building.

Unconfined vapor cloud explosion (UCVCE)

This type of explosion results from the release of a considerable quantity of flammable gas, or vapor, into the atmosphere and its subsequent ignition. Such an explosion can cause extensive damage, such as occurred at Flixborough, [HMSO \(1975\)](#) and BP Texas City, [CSHIB \(2005\)](#). Unconfined vapor explosions are discussed by [Munday \(1976\)](#) and [Gugan \(1979\)](#).

Boiling liquid expanding vapor explosion (BLEVE)

BLEVEs occur when there is a sudden release of vapor containing liquid droplets due to the failure of a vessel. A serious incident involving the failure of a liquefied petroleum gas (LPG) storage sphere occurred at Feyzin, France, in 1966, when the tank was heated by an external fire fueled by a leak from the tank; see [Mannan \(2012\)](#) and [Marshall \(1987\)](#).

Dust explosions

Finely divided combustible solids, if intimately mixed with air, can explode. Several disastrous explosions have occurred in grain silos and sugar factories.

Dust explosions usually occur in two stages: a primary explosion that disturbs deposited dust, followed by the second, more severe, explosion of the dust thrown into the atmosphere. Any finely divided combustible solid is a potential explosion hazard. Particular care must be taken in the design of dryers, conveyors, cyclones, and storage hoppers for polymers and other combustible products or intermediates. The extensive literature on the hazard and control of dust explosions should be consulted before designing powder handling systems: [Field \(1982\)](#), [Cross and Farrer \(1982\)](#), [Barton \(2001\)](#), [Eckhoff \(2003\)](#), NFPA 61, NFPA 654, NFPA 664, and BS EN 1127.

Explosivity properties

Information on explosive materials is given in the standards NFPA 495, NFPA 491, and BS EN 1839. *Sax's Handbook of Hazardous Materials* ([Lewis, 2012](#)) is also a good general reference.

The expansion factor is defined as the molar density of the reagents divided by the molar density of the products in an explosive mixture. The expansion factor is a measure of the increase in volume resulting from combustion. The maximum value of the expansion factor is for adiabatic combustion.

The flame speed is the rate of propagation of a flame front through a flammable mixture with respect to a fixed observer. Materials such as hydrogen and acetylene that have high flame speeds are more prone to detonation.

Values of these properties, autoignition temperature and adiabatic flame temperature are given for hydrogen and some hydrocarbons in Table 10.3, which is based on data from Drysdale (2011).

Design implications

The usual approach in design is to prevent explosions from occurring, for example, by not allowing flammable mixtures to form in the process. If internal explosion is a possibility, then it must be considered as a pressure relief scenario and the pressure relief devices must be sized to prevent detonation. This will usually require the use of large bursting disks; see Section 10.9. Flame arrestors should also be specified on process piping to prevent a deflagration event from propagating into a detonation. Particular care should be taken when designing plants that contain both pressurized fuels and pressurized oxidants. General guidelines on explosion protection are given in NFPA 69 and NFPA 68; see also the European standards BS EN 1127, BS EN 14460, and BS EN 60079 and the references listed in Section 10.3.5. Explosion suppression and venting systems are discussed in BS EN 14373, BS EN 14797, and BS EN 14994. Specific guidelines for various sectors of the process industries are given in other standards, such as offshore production installations (BS EN ISO 13702), agricultural and food processing plants (NFPA 61), wood processing and woodworking facilities (NFPA 664), and plants handling sulfur (NFPA 655).

10.3.7 Human error

The intervention of well-trained process operators is a vital layer in process safety, as it is usually the last opportunity to restore the process to a safe condition before an emergency shut-down or incident occurs (see Fig. 10.1).

Even with computer control of basic plant regulatory control systems and a capable, experienced, and well-trained staff, there is always a possibility of human error. Kletz et al. (1995) describe incidents that have occurred due to human errors in control system design, installation, and operation. The likelihood of operator error is substantially increased if operating procedures are not clearly documented and followed or if there are lapses in training and supervision. Kletz (1999) has suggested the following failure probabilities:

TABLE 10.3 Explosivity properties

Fuel	Formula	Maximum flame speed (m/s)	Adiabatic flame temperature (K)	Expansion factor	Autoignition temperature (°C)
Hydrogen	H ₂	22.1	2318	6.9	400
Methane	CH ₄	2.8	2148	7.5	601
Ethane	C ₂ H ₆	3.4	2168	7.7	515
Propane	C ₃ H ₈	3.3	2198	7.9	450
n-Butane	C ₄ H ₁₀	3.3	2168	7.9	405
Pentane	C ₅ H ₁₂	3.4	2232	8.1	260
Hexane	C ₆ H ₁₄	3.4	2221	8.1	225
Acetylene	C ₂ H ₂	14.8	2598	8.7	305
Ethylene	C ₂ H ₄	6.5	2248	7.8	490
Propylene	C ₃ H ₆	3.7	2208	7.8	460
Benzene	C ₆ H ₆	5	2287	8.1	560
Cyclohexane	C ₆ H ₁₂	4.2	2232	8.1	245

Adapted from Drysdale (2011).

Action required	Probability of failure
A valve to be closed directly below an alarm	0.001
Simple action in a quiet environment	0.01
Simple action in a distracting environment	0.1
Complex and rapid action required	1.0

The Chemical Safety and Hazard Investigation Board (CSHIB) preliminary report on their investigation of the explosion at the BP Texas City refinery on March 23, 2005, in which 15 people were killed and over 170 were injured, describes multiple failures in supervision, operating procedures, and training that contributed to the accident (CSHIB, 2005). One of these was holding a safety training meeting in the control room while the operators were trying to start up the plant.

10.4 Analysis of product and process safety

The analysis of health, safety, and environmental (HS&E) impact of technology is so important that it is carried out at every stage of a project, using the project technical information as it becomes available. As more design detail is developed, more quantitative methods can be used for analyzing safety and environmental impact.

Table 10.4 shows typical steps in the evolution of a new product or process from initial concept to manufacture. In the early stages of process development, the detail of the process has not been established, but qualitative

TABLE 10.4 Health, safety, and environmental impact analysis during the evolution of a project

Stage	Information available	HS&E analysis methods
Research concept	Chemistry	SDS review
	SDS information	major hazard review
Conceptual design	Process flow diagram	Process FMEA/HAZAN
	Equipment list	Pollution prevention analysis
	Vessel designs	Preliminary operating procedures
Reactor models		
Preliminary design	P&I diagrams	HAZOP
	Process control scheme	Emergency procedures
	Metallurgy	Safety indices
	Detailed mass and energy balance	Safety checklists
	Hydraulics	Effluent summary
	Offsites	
Detailed design engineering	Mechanical designs	Quantitative risk analysis
	Instrument specs	Fault tree analysis
	Vendor details	
	Plot plans	
Procurement, construction	Piping isometrics	As built HAZOP
	As-built specs	Operator training
Operation	Commissioning log	Ongoing training
	Operations log	Change management procedures
	Maintenance log	Revised operating procedures

assessments of major hazards can be made by collecting information from the SDS forms for the chemicals involved. Once a conceptual flow scheme has been developed, semi-quantitative methods such as failure mode and effects analysis (FMEA—see [Section 10.5](#)) and systematic procedures for identifying hazards such as HAZAN can be applied. An initial pollution prevention analysis can be made if the major process effluents are known. Some companies also calculate safety indices at this stage to give a semi-quantitative comparison of the safety of the new process compared with existing processes ([Section 10.6](#)). When the process P&I diagram has been established and a full mass and energy balance has been completed, a full hazard and operability study (HAZOP—see [Section 10.7](#)) can be carried out and the operating and emergency procedures can be updated. Safety checklists ([Section 10.4.1](#)) are often completed at this stage and then updated and amended at subsequent stages. During detailed design and procurement, vendor information on instrument reliability becomes available. This information can be used to make a more quantitative analysis of likely failure rates, and hence determine whether duplicate or backup systems are needed ([Section 10.8](#)). When the plant begins operation, any changes or modifications made during commissioning or in operation must also go through a detailed hazard analysis.

In the United States, the OSHA standard 29 CFR 1910.119 *Process Safety Management of Highly Hazardous Chemicals* requires that a hazard analysis must be carried out for any process involving certain listed chemicals (see Appendix A of the standard) or involving more than 10,000 lb (4535.9 kg) of flammable gas or liquid. Employers must involve employees in the hazard analysis, and it must be made available to employees and updated at least every 5 years. Employees and contractors must be trained in safe work practices associated with the identified process hazards. Full details of these and other legal requirements, as well as descriptions of the information that must be included in the analysis, can be found in the standard. The most recent version of this standard and all other OSHA regulations are available at www.osha.gov. The regulations do not specify the method of hazard analysis that must be used, and most employers use several or all of the methods listed in the following sections, increasing the complexity of the analysis as more information is developed during the course of the project.

When the product that will be produced by the plant is a food, vitamin, cosmetic, medical implant, or human or veterinary drug additional safety analysis must be carried out to comply with regulations set by the Food and Drug Administration (FDA). The FDA requires that plants that come under its jurisdiction must follow Current Good Manufacturing Practice (CGMP) regulations. Additional documentation of the design and operation of such plants is required, and these documents are subject to inspection and certification by the FDA. Details of FDA regulations can be found in the FDA Compliance Policy Guides, which are available at www.fda.gov. GMP guidelines are addressed in more detail in the discussion of bioreactor quality control in [Section 15.9.8](#).

Overviews of the different methods used for safety analysis are given in [Crowl and Louvar \(2019\)](#), [Mannan \(2012\)](#), [CCPS \(2008\)](#), and [BS EN ISO 17776 \(2000\)](#).

10.4.1 Safety checklists

Checklists are useful aids to memory. A checklist that has been drawn up by experienced engineers can be a useful guide for the less experienced; however, too great a reliance should never be put on the use of checklists to the exclusion of other considerations and techniques. No checklist can be completely comprehensive, covering all the factors to be considered for any particular process or operation.

A short safety checklist, covering the main items that should be considered in process design, is given here. More detailed checklists are given by [Carson and Mumford \(1988\)](#) and [Wells \(1980\)](#). [Balemans \(1974\)](#) gives a comprehensive list of guidelines for the safe design of chemical plants drawn up in the form of a checklist. A loss prevention checklist is included in *Dow's Fire and Explosion Index Hazard Classification Guide* ([Dow, 1994](#)).

Design safety checklist

Materials

- a. flash point
- b. flammability range
- c. autoignition temperature
- d. composition
- e. stability (shock sensitive?)
- f. toxicity, TLV
- g. corrosion
- h. physical properties (unusual?)
- i. heat of combustion/reaction

Process

1. Reactors
 - a. exothermic – heat of reaction
 - b. temperature control – emergency systems
 - c. side reactions – dangerous?
 - d. effect of contamination
 - e. effect of unusual concentrations (including catalyst)
 - f. corrosion
2. Pressure systems
 - a. need?
 - b. design to current codes
 - c. materials of construction – adequate?
 - d. pressure relief – adequate?
 - e. safe venting systems
 - f. flame arresters

Control systems

- a. fail safe
- b. backup power supplies
- c. high/low alarms and trips on critical variables
 - i. temperature
 - ii. pressure
 - iii. flow
 - iv. level
 - v. composition
- d. backup/duplicate systems on critical variables
- e. remote operation of valves
- f. block valves on critical lines
- g. excess-flow valves
- h. interlock systems to prevent misoperation
- i. automatic shut-down systems

Storage

- a. limit quantity
- b. inert purging/blanketing
- c. floating roof tanks
- d. dykeing
- e. loading/unloading facilities – safety
- f. earthing
- g. ignition sources – vehicles

General

- a. inert purging systems needed
- b. compliance with electrical codes
- c. adequate lighting
- d. lightning protection
- e. sewers and drains adequate, flame traps
- f. dust-explosion hazards
- g. buildup of dangerous impurities – purges
- h. plant layout
 - i. separation of units
 - ii. access
 - iii. siting of control rooms and offices
 - iv. services
- i. safety showers, eye baths

Fire protection

- a. emergency water supplies
- b. fire mains and hydrants
- c. foam systems
- d. sprinklers and deluge systems
- e. insulation and protection of structures
- f. access to buildings
- g. firefighting equipment

The checklist is intended to promote thought and to raise questions such as: Is it needed? What are the alternatives? Has provision been made for...? Has it been checked for...? Has it been provided?

10.5 Failure mode effect analysis

Failure mode and effects analysis (FMEA) is a method originally developed in manufacturing, which is used to determine the relative importance of different component failures within an overall system or product. It can be applied to analysis of chemical plant safety (OSHA 29 CFR 1910.119), as well as to design of products and even business plans and commercial projects. The method is semi-quantitative. It assigns numerical rankings to different failure modes based on the (qualitative) perceptions of the participants. Different groups or individuals will not necessarily reach the same conclusions, so the method is best used in the early stages of design as a means of brainstorming for safety issues. More rigorous methods such as HAZAN and HAZOP should be applied when more design details are available.

10.5.1 FMEA procedure

An FMEA should ideally be carried out as a group brainstorming exercise. The group should include a diverse set of experts. When an FMEA is used for process safety analysis, these should include:

1. An expert on process chemistry
2. An expert on process equipment
3. An expert on process control
4. An expert on process operations
5. An expert on safety analysis
6. The process design engineer

The analysis then proceeds as follows:

1. The group begins by reviewing the process and defining a set of process steps or key inputs.
2. For each step or input, they then brainstorm for failure modes (i.e., ways in which the step or input might not perform its desired function).
3. For each failure mode, the group brainstorms possible consequences. There may be multiple consequences for a given failure mode.
4. For each failure mode (and consequences) the group lists possible causes. Once again, several causes may trigger the same failure mode.
5. For each cause, the team lists the systems *that are currently in place* to prevent the cause from happening or allow the cause to be detected in time for operators to respond before the failure mode occurs. At this step, it is very important that the team considers the design as it currently exists. They must not assume that something will be added later to take care of any identified problem.
6. Once the brainstorming phase is completed (usually after several sessions), the team reviews the list of consequences and assigns each of them a “severity” number (SEV). The severity is a measure of the impact of the consequence. Different scales can be used for severity, as discussed in [Section 10.5.2](#).
7. The team then assigns a “likelihood of occurrence” number (OCC) to each of the causes. The occurrence number is a measure of either the probability or frequency of the cause occurring.
8. For each of the current control methods or systems, the team assigns a “detection” number (DET) that rates the probability that the existing systems will prevent the cause or failure mode from happening or detect the cause and allow an operator response before the failure mode occurs.

9. The three numbers, SEV, OCC, and DET, are multiplied together to give an overall risk probability number (RPN).
10. Based on the RPN values, actions are assigned to each item in the FMEA. Low RPN items may require no action, whereas high RPN issues may require major changes to the process design and instrumentation.

An FMEA should always be associated with a particular revision of a design. Whenever a new revision is released, the FMEA should be updated.

10.5.2 FMEA rating scales

The numbers assigned to the FMEA SEV, OCC, and DET parameters are only a qualitative indication of the probability or impact. Because of this (and to reduce the time spent arguing over whether an item deserves a 4 or a 5 rating) most experienced practitioners use a 1, 4, 7, 10 scale to increase the granularity of the responses.

It is important that the team agree on the meaning of each rating in the context of each of the FMEA parameters before starting to assign ratings. A suggested rating scale is given in [Table 10.5](#), but other scales may be more appropriate in other cases.

It should be noted that the DET scale is inverse to the OCC scale. A high value of the DET number corresponds to a low probability of detection, whereas a high value of the OCC number corresponds to a high probability of occurrence.

When assigning ratings to the different FMEA parameters, it is good practice to attempt to reach a consensus within the team. If no consensus can be developed (usually between a pair of values), then the best practice is to choose the higher value of the pair.

10.5.3 Interpretation of FMEA scores

Once the RPN values have been calculated, the list should be ranked by RPN and should be checked for consistency. This is particularly necessary when the FMEA has been completed during several sessions. FMEA is essentially a qualitative method, and the rankings based on RPN are, at best, only an indication of the team's assessment of the relative risk of the different failure modes. The team should not be overly concerned with the relative ranking of two issues as long as both are ranked appropriately high or low in the overall list.

Every item in the list should be reviewed to determine what follow-up action is required. If a 1, 4, 7, 10 scale is used, then specific actions leading to changes in design or operating procedures are usually required for every item with an RPN score greater than 100. This ensures that any item that scores 7 or 10 in one rating leads to an action unless it scores 1 in one of the other ratings.

Because FMEA is a qualitative method, it is difficult to draw comparisons between FMEA studies of different processes. If a team studying process A identifies 70 items in an FMEA, whereas a team studying process B identifies 200 items, then either process B has more associated risks or the team assigned to process B made a more thorough analysis. A short list of FMEA items (fewer than 50) usually indicates an incomplete analysis rather than a safe process.

TABLE 10.5 Suggested rating scale for FMEA

Rating	SEV	OCC	DET
1	Effect is insignificant	Failure is very unlikely	Current safeguards will always prevent failure mode
4	Minor disruption, possible loss of production	Occasional failure possible	High probability that current safeguard will detect or prevent
7	Major disruption, possible damage to local equipment	Infrequent failure is likely	Low probability that current safeguard will detect or prevent
10	Severe disruption, major damage to plant, possible injury to personnel	Failure is very likely or frequent	No current method of detection

10.5.4 Tools for FMEA

Failure mode and effects analysis (FMEA) is easily carried out using spreadsheets. A Microsoft Excel template is available in Appendix G in the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793. Additional information on FMEA is given by Birolini (2017), Dodson and Nolan (1999), and Stamatis (2003, 2019).

10.6 Safety indices

Some companies make use of safety indices as a tool for assessing the relative risk of a new process or plant. The most widely used safety index is the Dow Fire and Explosion Index, developed by the Dow Chemical Company and published by the American Institute of Chemical Engineers (Dow, 1994) (www.aiche.org). A numerical “fire and explosion index” (F&EI) is calculated, based on the nature of the process and the properties of the process materials. The larger the value of the F&EI, the more hazardous the process (Table 10.6).

To assess the potential hazard of a new plant, the index can be calculated after the piping and instrumentation and equipment layout diagrams have been prepared. In earlier versions of the guide, the index was then used to determine what preventive and protection measures were needed; see Dow (1973). In the current version, the preventive and protection measures that have been incorporated in the plant design to reduce the hazard are taken into account when assessing the potential loss in the form of loss control credit factors.

It is worthwhile to estimate the F&EI at an early stage in the process design, as it will indicate whether alternative, less hazardous process routes should be considered.

Only a brief outline of the method used to calculate the Dow F&EI will be given in this section. The full guide should be studied before applying the technique to a particular process. Judgment, based on experience with similar processes, is needed to determine the magnitude of the various factors used in the calculation of the index and the loss control credit factors.

10.6.1 Calculation of the Dow F&EI

The procedure for calculating the index and the potential loss is set out in Fig. 10.2.

The first step is to identify the units that would have the greatest impact on the magnitude of any fire or explosion. The index is calculated for each of these units.

The basis of the F&EI is a *material factor* (MF). The MF is then multiplied by a *unit hazard factor*, F_3 , to determine the F&EI for the process unit. The unit hazard factor is the product of two factors that take account of the hazards inherent in the operation of the particular process unit: the general and special process hazards (Fig. 10.3).

Material factor

The material factor is a measure of the intrinsic rate of energy release from the burning, explosion, or other chemical reaction of the material. Values for the MF for over 300 of the most commonly used substances are given in the guide. The guide also includes a procedure for calculating the MF for substances not listed: from knowledge of the flash points (for dusts, dust explosion tests) and a reactivity value, N_r . The reactivity value is a qualitative description of the reactivity of the substance and ranges from 0 for stable substances to 4 for substances that are capable of unconfined detonation. Some typical material factors are given in Table 10.7.

In calculating the F&EI for a unit, the value for the material with the highest MF, which is present in significant quantities, is used.

TABLE 10.6 Assessment of hazards

Fire and Explosion Index range	Degree of hazard
1–60	Light
61–96	Moderate
97–127	Intermediate
128–158	Heavy
>159	Severe

Adapted from Dow (1994).

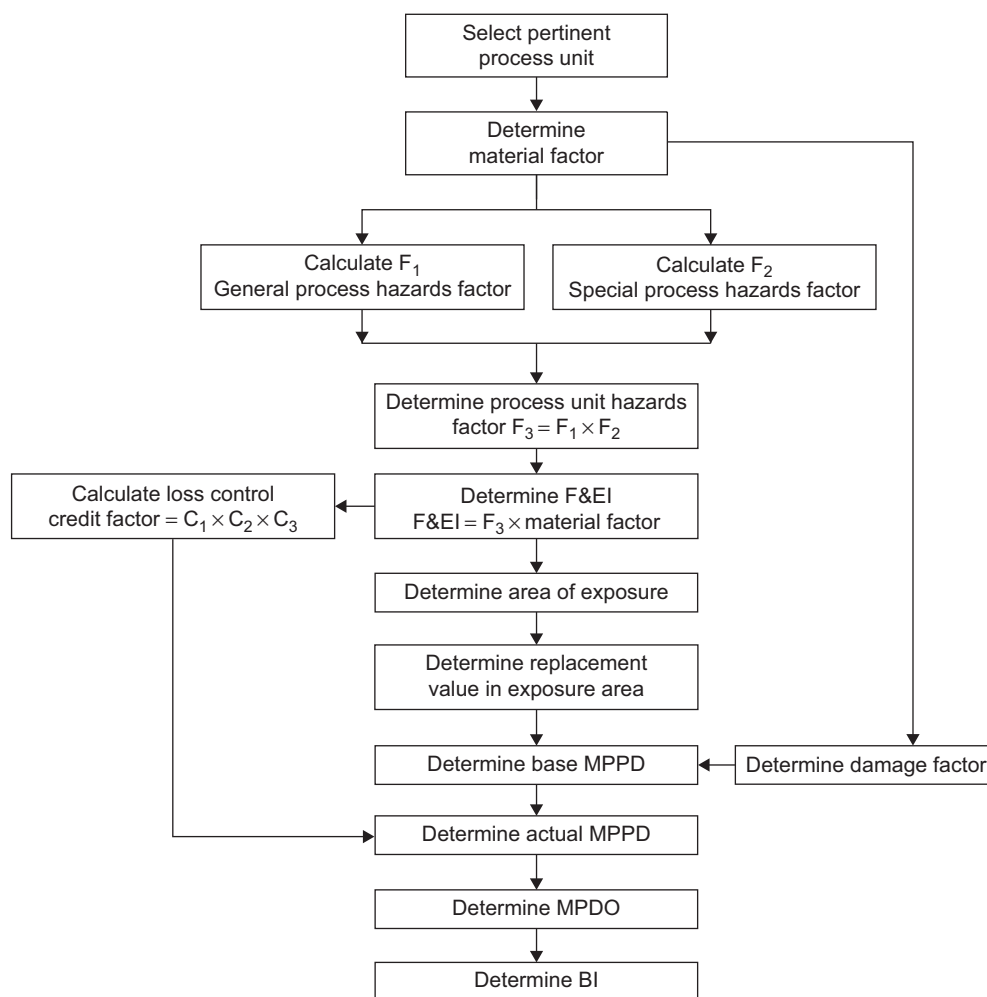


FIG. 10.2 Procedure for calculating the Fire and Explosion Index and other risk analysis information. From Dow [1994] reproduced by permission of the American Institute of Chemical Engineers. © 1994 AIChE. All rights reserved.

General process hazards

The general process hazards are factors that play a primary role in determining the magnitude of the loss after an incident.

Six factors are listed on the calculation form (see Fig. 10.3).

- A. *Exothermic chemical reactions*: The penalty varies from 0.3 for a mild exotherm, such as hydrogenation, to 1.25 for a particularly sensitive exotherm, such as nitration.
- B. *Endothermic processes*: A penalty of 0.2 is applied to reactors only. It is increased to 0.4 if the reactor is heated by the combustion of a fuel.
- C. *Materials handling and transfer*: This penalty takes account of the hazard involved in the handling, transfer, and warehousing of the material.
- D. *Enclosed or indoor process units*: Accounts for the additional hazard where ventilation is restricted.
- E. *Access of emergency equipment*: Areas not having adequate access are penalized. Minimum requirement is access from two sides.
- F. *Drainage and spill control*: Penalizes design conditions that would cause large spills of flammable material adjacent to process equipment, such as inadequate design of drainage.

Special process hazards

The special process hazards are factors that are known from experience to contribute to the probability of an incident involving loss.

Twelve factors are listed on the calculation form (see Fig. 10.3).

FIRE & EXPLOSION INDEX

AREA/COUNTRY	DIVISION	LOCATION	DATE
SITE	MANUFACTURING UNIT	PROCESS UNIT	
PREPARED BY:	APPROVED BY:(Superintendent)	BUILDING	
REVIEWED BY:(Management)	REVIEWED BY:(Technology Center)	REVIEWED BY:(Safety & Loss Prevention)	
MATERIALS IN PROCESS UNIT			
STATE OF OPERATION — DESIGN — START UP — NORMAL OPERATION — SHUTDOWN		BASIC MATERIAL(S) FOR MATERIAL FACTOR	
MATERIAL FACTOR (See Table 1 or Appendices A or B) Note requirements when unit temperature over 140 °F (60 °C)			
1. General Process Hazards		Penalty Factor Range	Penalty Factor Used(1)
Base Factor		1.00	1.00
A. Exothermic Chemical Reactions		0.30 to 1.25	
B. Endothermic Processes		0.20 to 0.40	
C. Material Handling and Transfer		0.25 to 1.05	
D. Enclosed or Indoor Process Units		0.25 to 0.90	
E. Access		0.20 to 0.35	
F. Drainage and Spill Control _____ gal or cu.m.		0.25 to 0.50	
General Process Hazards Factor (F₁)			
2. Special Process Hazards			
Base Factor		1.00	1.00
A. Toxic Material(s)		0.20 to 0.80	
B. Sub-Atmospheric Pressure (< 500 mm Hg)		0.50	
C. Operation In or Near Flammable Range _____ Inerted _____ Not Inerted			
1. Tank Farms Storage Flammable Liquids		0.50	
2. Process Upset or Purge Failure		0.30	
3. Always in Flammable Range		0.80	
D. Dust Explosion (See Table 3)		0.25 to 2.00	
E. Pressure (See Figure 2) Operating Pressure _____ psig or kPa gauge Relief Setting _____ psig or kPa gauge			
F. Low Temperature		0.20 to 0.30	
G. Quantity of Flammable/Unstable Material: Quantity _____ lb or kg H _C = _____ BTU/lb or kcal/kg			
1. Liquids or Gases in Process (See Figure 3)			
2. Liquids or Gases in Storage (See Figure 4)			
3. Combustible Solids in Storage, Dust in Process (See Figure 5)			
H. Corrosion and Erosion		0.10 to 0.75	
I. Leakage – Joints and Packing		0.10 to 1.50	
J. Use of Fired Equipment (See Figure 6)			
K. Hot Oil Heat Exchange System (See Table 5)		0.15 to 1.15	
L. Rotating Equipment		0.50	
Special Process Hazards Factor (F₂)			
Process Unit Hazards Factor (F₁ x F₂) = F₃			
Fire and Explosion Index (F₃ x MF = F&EI)			

(a)

(1) For no penalty use 0.00.

FIG. 10.3 Dow Fire and Explosion Index calculation form. Note: The figure numbers refer to the Dow guide. Gallons are U.S. gallons. Note: 1 m³ = 264.2 U.S. gal; 1 kN/m² = 0.145 psi; 1 kg = 2.2 lb; 1 kJ/Kg = 0.43 BTU/lb.

From Dow [1994] reproduced by permission of the American Institute of Chemical Engineers. © 1994 AIChE. All rights reserved.

TABLE 10.7 Some typical material factors

	MF	Flash point °C	Heat of combustion MJ/kg
Acetaldehyde	24	−39	24.4
Acetone	16	−20	28.6
Acetylene	40	gas	48.2
Ammonia	4	gas	18.6
Benzene	16	−11	40.2
Butane	21	gas	45.8
Chlorine	1	—	0.0
Cyclohexane	16	−20	43.5
Ethyl alcohol	16	13	26.8
Hydrogen	21	gas	120.0
Nitroglycerine	40	—	18.2
Sulfur	4	—	9.3
Toluene	16	40	31.3
Vinyl chloride	21	gas	18.6

- A. Toxic materials:** The presence of toxic substances after an incident will make the task of emergency personnel more difficult. The factor applied ranges from 0 for nontoxic materials to 0.8 for substances that can cause death after short exposure.
- B. Subatmospheric pressure:** Allows for the hazard of air leakage into equipment. It is only applied for pressure less than 500 mmHg (0.66 bara).
- C. Operation in or near flammable range:** Covers the possibility of air mixing with material in equipment or storage tanks under conditions where the mixture will be within the explosive range.
- D. Dust explosion:** Covers the possibility of a dust explosion. The degree of risk is largely determined by the particle size. The penalty factor varies from 0.25 for particles above 175 μm to 2.0 for particles below 75 μm .
- E. Relief pressure:** This penalty accounts for the effect of pressure on the rate of leakage should a leak occur. Equipment design and operation become more critical as the operating pressure is increased. The factor to apply depends on the relief device setting and the physical nature of the process material. It is determined from Fig. 2 in the Dow Guide.
- F. Low temperature:** This factor allows for the possibility of brittle fracture occurring in carbon steel or other metals at low temperatures (see [Chapter 6](#) of this book).
- G. Quantity of flammable material:** The potential loss will be greater the greater the quantity of hazardous material in the process or in storage. The factor to apply depends on the physical state and hazardous nature of the process material and the quantity of material. It varies from 0.1 to 3.0 and is determined from Figs. 3, 4, and 5 in the Dow Guide.
- H. Corrosion and erosion:** Despite good design and materials selection, some corrosion problems may arise, both internally and externally. The factor to be applied depends on the anticipated corrosion rate. The severest factor is applied if stress corrosion cracking is likely to occur (see [Chapter 6](#) of this book).
- I. Leakage—joints and packing:** This factor accounts for the possibility of leakage from gaskets, pumps and other shaft seals, and packed glands. The factor varies from 0.1, where there is the possibility of minor leaks, to 1.5 for processes that have sight glasses, bellows, or other expansion joints.
- J. Use of fired heaters:** The presence of boilers or furnaces heated by the combustion of fuels increases the probability of ignition should a leak of flammable material occur from a process unit. The risk involved will depend on the siting of the fired equipment and the flash point of the process material. The factor to apply is determined with reference to Fig. 6 in the Dow Guide.
- K. Hot oil heat exchange system:** Most special heat exchange fluids are flammable and are often used above their flash points, so their use in a unit increases the risk of fire or explosion. The factor to apply depends on the quantity and whether the fluid is above or below its flash point; see Table 5 in the Guide.
- L. Rotating equipment:** This factor accounts for the hazard arising from the use of large pieces of rotating equipment: compressors, centrifuges, and some mixers.

10.6.2 Potential loss

The procedure for estimating the potential loss that would follow an incident is set out in Table 10.8. The first step is to calculate the *damage factor* for the unit. The damage factor depends on the value of the material factor and the process unit hazards factor (F_3 in Fig. 10.3). It is determined using Fig. 8 in the Dow Guide.

An estimate is then made of the area (radius) of exposure. This represents the area containing equipment that could be damaged following a fire or explosion in the unit being considered. It is evaluated from Fig. 7 in the Guide and is a linear function of the F&EI.

An estimate of the replacement value of the equipment within the exposed area is then made and combined with the damage factor to estimate the *base maximum probable property damage* (base MPPD).

The *maximum probable property damage* (MPPD) is then calculated by multiplying the base MPPD by a *loss control credit factor*. The loss control credit factors (Table 10.9) allow for the reduction in the potential loss given by the

TABLE 10.8 Loss control credit factors

Feature	Credit factor range	Credit factor used (2)
1. Process Control Credit Factor (C1)		
a. Emergency Power	0.98	
b. Cooling	0.97–0.99	
c. Explosion Control	0.84–0.98	
d. Emergency Shut-down	0.96–0.99	
e. Computer Control	0.93–0.99	
f. Inert Gas	0.94–0.96	
g. Operating Instructions/Procedures	0.91–0.99	
h. Reactive Chemical Review	0.91–0.98	
i. Other Process Hazard Analysis	0.91–0.98	
C ₁ Value(3) : _____		
2. Material Isolation Credit Factor (C2)		
a. Remote Control Valves	0.96–0.98	
b. Dump/Blowdown	0.96–0.98	
c. Drainage	0.91–0.97	
d. Interlock	0.98	
C ₂ Value(3) : _____		
3. Fire Protection Credit Factor (C3)		
a. Leak Detection	0.94–0.98	
b. Structural Steel	0.95–0.98	
c. Fire Water Supply	0.94–0.97	
d. Special Systems	0.91	
e. Sprinkler Systems	0.74–0.97	
f. Water Curtains	0.97–0.98	
g. Foam	0.92–0.97	
h. Hand Extinguishers/Monitors	0.93–0.98	
i. Cable Protection	0.94–0.98	
C ₃ Value(3) : _____		
Loss Control Credit Factor = $C_1 \times C_2 \times C_3 =$ _____ Enter on Line 7 in Table 10.9		

(2) For no credit factor enter 1.00.

(3) Product of all factors used.

From Dow (1994). Reproduced by permission of the American Institute of Chemical Engineers. © 1994 AIChE. All rights reserved.

TABLE 10.9 Process unit risk analysis summary

1. Fire & Explosion Index (F&EI)	
2. Radius of Exposure	(Fig. 7)* ft or m
3. Area of Exposure	ft ² or m ²
4. Value of Area of Exposure	\$MM
5. Damage Factor	(Fig. 8)*
6. Base Maximum Probable Property Damage — (Base MPPD) [4 × 5]	\$MM
7. Loss Control Credit Factor	(See earlier)
8. Actual Maximum Probable Property Damage — (Actual MPPD) [6 × 7]	\$MM
9. Maximum Probable Days Outage — (MPDO)	(Fig. 9)* days
10. Business Interruption — (BI)	\$MM
(2) For no credit factor enter 1.00.	
(3) Product of all factors used.	

* Refer to Fire & Explosion Index Hazard Classification Guide for details. From Dow (1994).
 Reproduced by permission of the American Institute of Chemical Engineers. © 1994 AIChE. All rights reserved.

preventive and protective measures incorporated in the design. The Dow Guide should be consulted for details of how to calculate the credit factors.

The MPPD is used to predict the maximum number of days that the plant will be down for repair, the *maximum probable days outage* (MPDO). The MPDO is used to estimate the financial loss due to the lost production: the *business interruption* (BI). The financial loss due to lost business opportunity can often exceed the loss from property damage.

10.6.3 Basic preventive and protective measures

The basic safety and fire protective measures that should be included in all chemical process designs are listed here. This list is based on that given in the Dow Guide, with some minor amendments.

1. Adequate and secure water supplies for firefighting.
2. Correct structural design of vessels, piping, and steel work.
3. Pressure-relief devices.
4. Corrosion-resistant materials and/or adequate corrosion allowances.
5. Segregation of reactive materials.
6. Grounding of electrical equipment.
7. Safe location of auxiliary electrical equipment, transformers, and switch gear.
8. Provision of backup utility supplies and services.
9. Compliance with national codes and standards.
10. Fail-safe instrumentation.
11. Provision for access of emergency vehicles and the evacuation of personnel.
12. Adequate drainage for spills and firefighting water.
13. Insulation of hot surfaces.
14. No glass equipment used for flammable or hazardous materials unless no suitable alternative is available.
15. Adequate separation of hazardous equipment.
16. Protection of pipe racks and cable trays from fire.
17. Provision of block valves on lines to main processing areas.
18. Protection of fired equipment (heaters, furnaces) against accidental explosion and fire.
19. Safe design and location of control rooms.

Note: The design and location of control rooms, particularly with regard to protection against an unconfined vapor explosion, is covered in a publication of the Chemical Industries Association (CIA, 1979).

10.6.4 Mond fire, explosion, and toxicity index

The Mond index was developed from the Dow F&EI by personnel at the ICI Mond division. The third edition of the Dow index, Dow (1973), was extended to cover a wider range of process and storage installations, the processing of chemicals with explosive properties, and the evaluation of a toxicity hazards index. Also included was a

procedure to allow for the offsetting effects of good design and of control and safety instrumentation. Their revised Mond fire, explosion, and toxicity index was discussed in a series of papers by Lewis (1979a, 1979b), which included a technical manual setting out the calculation procedure. An extended version of the manual was issued in 1985, and an amended version published in 1993 (ICI, 1993).

Procedure

The basic procedures for calculating the Mond indices are similar to those used for the Dow index.

The process is first divided into a number of units that are assessed individually.

The dominant material for each unit is then selected and its material factor determined. The material factor in the Mond index is a function of the energy content per unit weight (the heat of combustion).

The material factor is then modified to allow for the effect of general and special process and material hazards, the physical quantity of the material in the process step, the plant layout, and the toxicity of process materials.

Separate fire and explosion indices are calculated. An aerial explosion index can also be estimated to assess the potential hazard of aerial explosions. An equivalent Dow index can also be determined.

The individual fire and explosion indices are combined to give an overall index for the process unit. The overall index is the most important in assessing the potential hazard.

The magnitude of the potential hazard is determined by reference to rating tables, similar to that shown for the Dow index in Table 10.6.

After the initial calculation of the indices (the initial indices), the process is reviewed to see what measures can be taken to reduce the rating (the potential hazard).

The appropriate offsetting factors to allow for the preventive features included in the design are then applied, and final hazard indices are calculated.

Preventive measures

Preventive measures fall into two categories:

1. Those that reduce the number of incidents, such as sound mechanical design of equipment and piping, operating and maintenance procedures, and operator training.
2. Those that reduce the scale of a potential incident, such as measures for fire protection and fixed firefighting equipment.

Many measures will not fit neatly into individual categories but will apply to both.

Implementation

The Mond technique of hazard evaluation is fully explained in the ICI technical manual (ICI 1993), to which reference should be made to implement the method. The calculations are made using a standard form, similar to that used for the Dow index.

10.6.5 Summary

The Dow and Mond indices are useful techniques that can be used in the early stages of a project design to evaluate the hazards and risks of the proposed process.

Calculation of the indices for the various sections of the process will highlight any particularly hazardous sections and indicate where a detailed study is needed to reduce the hazards.

Example 10.1

Evaluate the Dow F&EI for the nitric acid plant illustrated in Chapter 2, Fig. 2.8.

Solution

The calculation is set out on the special form shown in Fig. 10.3(a). Notes on the decisions taken and the factors used are given here.

Unit: Consider the total plant, no separate areas, but exclude the main storages.

Material factor: For ammonia, from the Dow Guide and Table 10.6.

$$MF = 4.0$$

Note: Hydrogen is present and has a larger material factor (21), but the concentration is too small for it to be considered the dominant material.

General process hazards

- A. Oxidizing reaction, factor = 0.5
- B. Not applicable
- C. Not applicable
- D. Not applicable
- E. Adequate access would be provided, factor = 0.0
- F. Adequate drainage would be provided, factor = 0.0

FIRE & EXPLOSION INDEX

AREA/COUNTRY -	DIVISION -	LOCATION SLIGO	DATE 20 JAN 1997
SITE -	MANUFACTURING UNIT NITRIC ACID	PROCESS UNIT COMPLETE PLANT	
PREPARED BY: RKS	APPROVED BY: (Superintendent) ANOTHER	BUILDING -	
REVIEWED BY: (Management) -	REVIEWED BY: (Technology Center) -	REVIEWED BY: (Safety & Loss Prevention) -	
MATERIALS IN PROCESS UNIT AMMONIA, AIR, OXIDES OF NITROGEN, WATER			
STATE OF OPERATION — DESIGN — START UP ☒ NORMAL OPERATION — SHUTDOWN		BASIC MATERIAL(S) FOR MATERIAL FACTOR AMMONIA	
MATERIAL FACTOR (See Table 1 or Appendices A or B) Note requirements when unit temperature over 140 °F (60 °C)			4
1. General Process Hazards		Penalty Factor Range	Penalty Factor Used⁽¹⁾
Base Factor		1.00	1.00
A. Exothermic Chemical Reactions		0.30 to 1.25	0.50
B. Endothermic Processes		0.20 to 0.40	
C. Material Handling and Transfer		0.25 to 1.05	
D. Enclosed or Indoor Process Units		0.25 to 0.90	
E. Access		0.20 to 0.35	
F. Drainage and Spill Control _____ gal or cu.m.		0.25 to 0.50	
General Process Hazards Factor (F₁)			1.50
2. Special Process Hazards			
Base Factor		1.00	1.00
A. Toxic Material(s)		0.20 to 0.80	0.60
B. Sub-Atmospheric Pressure (< 500 mm Hg)		0.50	
C. Operation In or Near Flammable Range _____ Inerted _____ Not Inerted			
1. Tank Farms Storage Flammable Liquids		0.50	
2. Process Upset or Purge Failure		0.30	0.80
3. Always in Flammable Range		0.80	
D. Dust Explosion (See Table 3)		0.25 to 2.00	
E. Pressure (See Figure 2) Operating Pressure <u>103</u> psig or kPa gauge Relief Setting <u>125</u> psig or kPa gauge			0.35
F. Low Temperature		0.20 to 0.30	
G. Quantity of Flammable/Unstable Material: Quantity _____ lb or kg H _C = _____ BTU/lb or kcal/kg			
1. Liquids or Gases in Process (See Figure 3)			
2. Liquids or Gases in Storage (See Figure 4)			
3. Combustible Solids in Storage, Dust in Process (See Figure 5)			
H. Corrosion and Erosion		0.10 to 0.75	0.10
I. Leakage – Joints and Packing		0.10 to 1.50	0.10
J. Use of Fired Equipment (See Figure 6)			
K. Hot Oil Heat Exchange System (See Table 5)		0.15 to 1.15	
L. Rotating Equipment		0.50	0.50
Special Process Hazards Factor (F₂)			3.45
Process Unit Hazards Factor (F₁ x F₂) = F₃			5.20
Fire and Explosion Index (F₃ x MF = F&EI)			21

(b)

(1) For no penalty use 0.00.

FIG. 10.3 (A) Fire and Explosion Index calculation form, Example 10.1. From Dow [1994] reproduced by permission of the American Institute of Chemical Engineers. © 1994 AIChE. All rights reserved.

Special process hazards

- A. Ammonia is highly toxic and likely to cause serious injury, factor = 0.6.
- B. Not applicable.
- C. Operation always is within the flammable limits, factor = 0.8.
- D. Not applicable.
- E. Operation pressure $8 \text{ atm} = 8 \times 14.7 - 14.7 = 103 \text{ psig}$. Set relief valve at 20% above the operating pressure (see [Chapter 13](#) of this book) = 125 psig.
From Fig. 2 in the guide, factor = 0.35.
Note: psig = pounds force per square inch, gauge.
- F. Not applicable.
- G. The largest quantity of ammonia in the process will be the liquid in the vaporizer, say around 500 kg.
Heat of combustion (see [Table 10.3](#)) = 18.6 MJ/kg
Potential energy release = $500 \times 18.6 = 9300 \text{ MJ}$
 $= 9300 \times 10^6 / (1.05506 \times 10^3) = 8.81 \times 10^6 \text{ Btu}$
which is too small to register on Fig. 3 in the Guide, factor = 0.0.
- H. Corrosion-resistant materials of construction would be specified, but external corrosion is possible due to nitric oxide fumes, allow minimum factor = 0.1.
- I. Welded joints would be used on ammonia service and mechanical seals on pumps. Use minimum factor, as full equipment details are not known at the flowsheet stage, factor = 0.1.
- J. Not applicable.
- K. Not applicable.
- L. Large turbines and compressors used, factor = 0.5.

The index works out at 21: classified as “Light.” Ammonia would not normally be considered a dangerously flammable material; the danger of an internal explosion in the reactor is the main process hazard. The toxicity of ammonia and the corrosiveness of nitric acid would also need to be considered in a full hazard evaluation. The process unit risk analysis would be completed when the site for the plant had been determined.

10.7 Hazard and operability studies

A HAZOP study is a systematic procedure for critical examination of the operability of a process. When applied to a process design or an operating plant, it indicates potential hazards that may arise from deviations from the intended design conditions. The technique was developed by the Petrochemicals Division of ICI (see [Lawley, 1974](#)) and is now in general use in the chemical and process industries.

The term *operability study* should more properly be used for this type of study, though it is usually referred to as a *hazard and operability study* (HAZOP study). This can cause confusion with the term *hazard analysis* or *process hazard analysis* (PHA), which is a similar but somewhat less rigorous method. Numerous books have been written illustrating the use of HAZOP. Those by [Hyatt \(2003\)](#), [CCPS \(2000\)](#), [Kletz \(1999\)](#), and [Crawley and Tyler \(2015\)](#) give comprehensive descriptions of the technique, with examples.

A brief outline of the technique is given in this section to illustrate its use in process design. It can be used to make a preliminary examination of the design at the flowsheet stage and for a detailed study at a later stage, when a full process description, final flowsheets, P&I diagrams, and equipment details are available. An “as-built” HAZOP is often carried out after construction and immediately before commissioning a new plant.

10.7.1 Basic principles

A formal operability study is the systematic study of the design, vessel by vessel and line by line, using “guide words” to help generate thought about the way deviations from the intended operating conditions can cause hazardous situations.

The seven guide words recommended are given in [Table 10.10](#). In addition to these words, the following words are used in a special way and have the precise meanings given here:

TABLE 10.10 A list of guide words

Guide words	Meanings	Comments
No or Not	The complete negation of these intentions	No part of the intentions is achieved and nothing else happens.
More Less	Quantitative increases or decreases	These refer to quantities and properties such as flow rates or temperatures, as well as activities like "Heat" or "React."
As well as	A qualitative increase	All the design and operating intentions are achieved, together with some additional activity.
Part of	A qualitative decrease	Only some of the intentions are achieved; some are not.
Reverse	The logical opposite of the intention	This is most applicable to activities, for example, reverse flow or chemical reaction. It can also be applied to substances (e.g., "Poison" instead of "Antidote" or "D" instead of "L" optical isomers).
Other than	Complete substitution	No part of the original intention is achieved. Something quite different happens.

Intention: The intention defines how the particular part of the process was intended to operate; the intention of the designer.

Deviations: These are departures from the designer's intention that are detected by the systematic application of the guide words.

Causes: Reasons why, and how, the deviations could occur. Only if a deviation can be shown to have a realistic cause is it treated as meaningful.

Consequences: The results that follow from the occurrence of a meaningful deviation.

Hazards: Consequences that can cause damage (loss) or injury.

The use of the guide words can be illustrated by considering a simple example. Fig. 10.4 shows a chlorine vaporizer, which supplies chlorine at 2 bar to a chlorination reactor. The vaporizer is heated by condensing steam.

Consider the steam supply line and associated control instrumentation. The designer's intention is that steam shall be supplied at a pressure and flow rate to match the required chlorine demand.

Apply the guide word No:

Possible deviation – no steam flow.

Possible causes – blockage, valve failure (mechanical or power), failure of steam supply (fracture of main, boiler shut-down).

Clearly this is a meaningful deviation, with several plausible causes.

Consequences – The main consequence is loss of chlorine flow to the chlorination reactor.

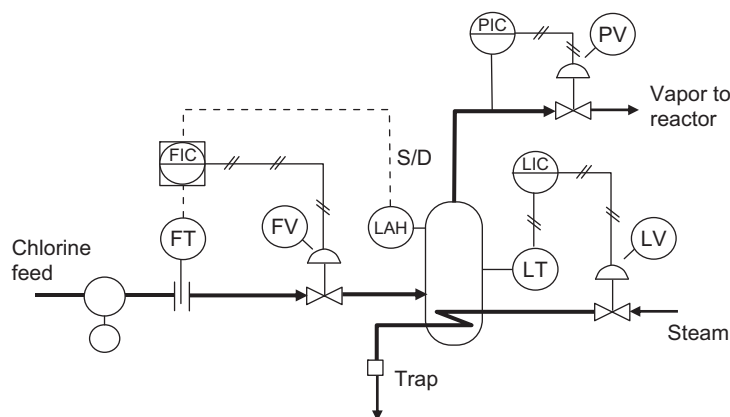


FIG. 10.4 Chlorine vaporizer instrumentation.

The effect of this on the reactor operation would have to be considered. This would be brought out in the operability study on the reactor; it would be a possible cause of no chlorine flow. Because the flow controller does not know that steam flow has been lost, chlorine will continue to be pumped into the vessel until the high-level alarm sounds and the high-level shut-down closes the control valve. A secondary consequence is that the vessel is now filled with liquid chlorine that must be drained to a safe level before operation can be resumed. The operating procedures must include instructions on how to deal with this scenario.

Apply the guide word MORE:

Possible deviation – More steam flow.

Possible cause – Valve stuck open.

Consequences – Low level in vaporizer (this should activate the low-level alarm), higher rate of flow to the reactor.

Note: To some extent the level will be self-regulating, because as the level falls, the heating surface is uncovered.

Hazard – Depends on the possible effect of high flow on the reactor.

Possible deviation – More steam pressure (increase in mains pressure).

Possible causes – Failure of pressure-regulating valves.

Consequences – Increase in vaporization rate. Need to consider the consequences of the heating coil reaching the maximum possible steam system pressure.

Hazard – Rupture of lines (unlikely), effect of sudden increase in chlorine flow on reactor.

A more detailed illustration of the HAZOP method is given in Example 10.2.

10.7.2 Explanation of guide words

It is important to understand the intended meaning of the guide words in Table 10.10. The meaning of the words NO/NOT, MORE, and LESS are easily understood; the NO/NOT, MORE, and LESS could, for example, refer to flow, pressure, temperature, level, and viscosity. All circumstances leading to No flow should be considered, including reverse flow.

The other words need some further explanation:

AS WELL AS: Something in addition to the design intention, such as impurities, side reactions, ingress of air, or extra phases present.

PART OF: Something missing, only part of the intention realized, such as the change in composition of a stream or a missing component.

REVERSE: The reverse of, or opposite to, the design intention. This could mean reverse flow if the intention were to transfer material. For a reaction, it could mean the reverse reaction. In heat transfer, it could mean the transfer of heat in the opposite direction to what was intended.

OTHER THAN: An important and far-reaching guide word, but consequently more vague in its application. It covers all conceivable situations other than that intended, such as start-up, shut-down, maintenance, catalyst regeneration and charging, and failure of plant services.

When referring to time, the guide words SOONER THAN and LATER THAN can also be used.

10.7.3 Procedure

An operability study would normally be carried out by a team of experienced people, who have complementary skills and knowledge, led by a team leader who is experienced in the technique. The team would include a similar set of experts to an FMEA team, as described in Section 10.5. The team examines the process vessel by vessel, and line by line, using the guide words to detect any hazards.

The information required for the study will depend on the extent of the investigation. A preliminary study can be made from a description of the process and the process flow diagrams. For a detailed, final study of the design, the flowsheets, piping and instrument diagrams, equipment specifications, and layout drawings would be needed. For a batch process, information on the sequence of operation will also be required, such as that given in operating instructions, logic diagrams, and flow charts.

A typical sequence of events is shown in Fig. 10.5. After each line has been studied it is marked on the flowsheet as checked.

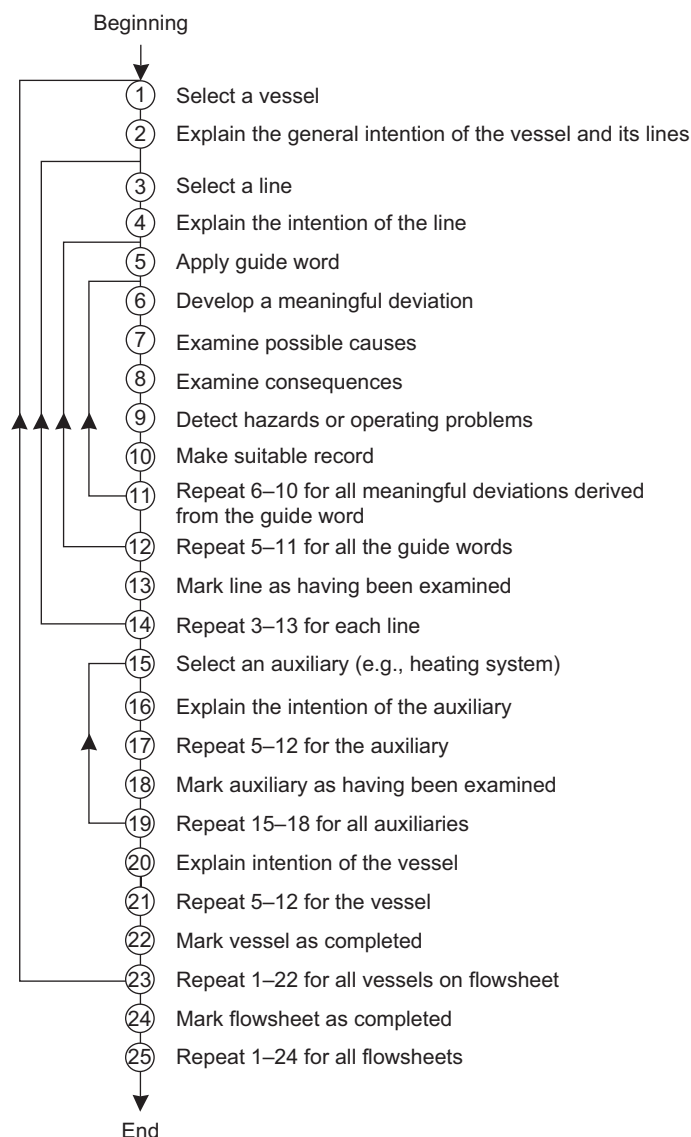


FIG. 10.5 Detailed sequence of an operability study.

A written record is not normally made of each step in the study—only those deviations that lead to a potential hazard are recorded. If possible, the action needed to remove the hazard is decided by the team and recorded. If more information, or time, is needed to decide the best action, the matter is referred to the design group for action or taken up at another meeting of the study team.

When using the operability study technique to vet a process design, the action to be taken to deal with a potential hazard will often be modifications to the control systems and instrumentation: the inclusion of additional alarms, trips, or interlocks. If major hazards are identified, major design changes may be necessary; alternative processes, materials, or equipment should be considered.

Example 10.2

This example illustrates how the techniques used in an operability study can be used to decide the instrumentation required for safe operation. Fig. 10.6(a) shows the basic instrumentation and control systems required for the steady-state operation of the reactor section of the nitric acid process introduced in Fig. 2.8. Fig. 10.6(b) shows the additional instrumentation and safety trips added after making the operability study set out next. The instrument symbols used are explained in Chapter 5.

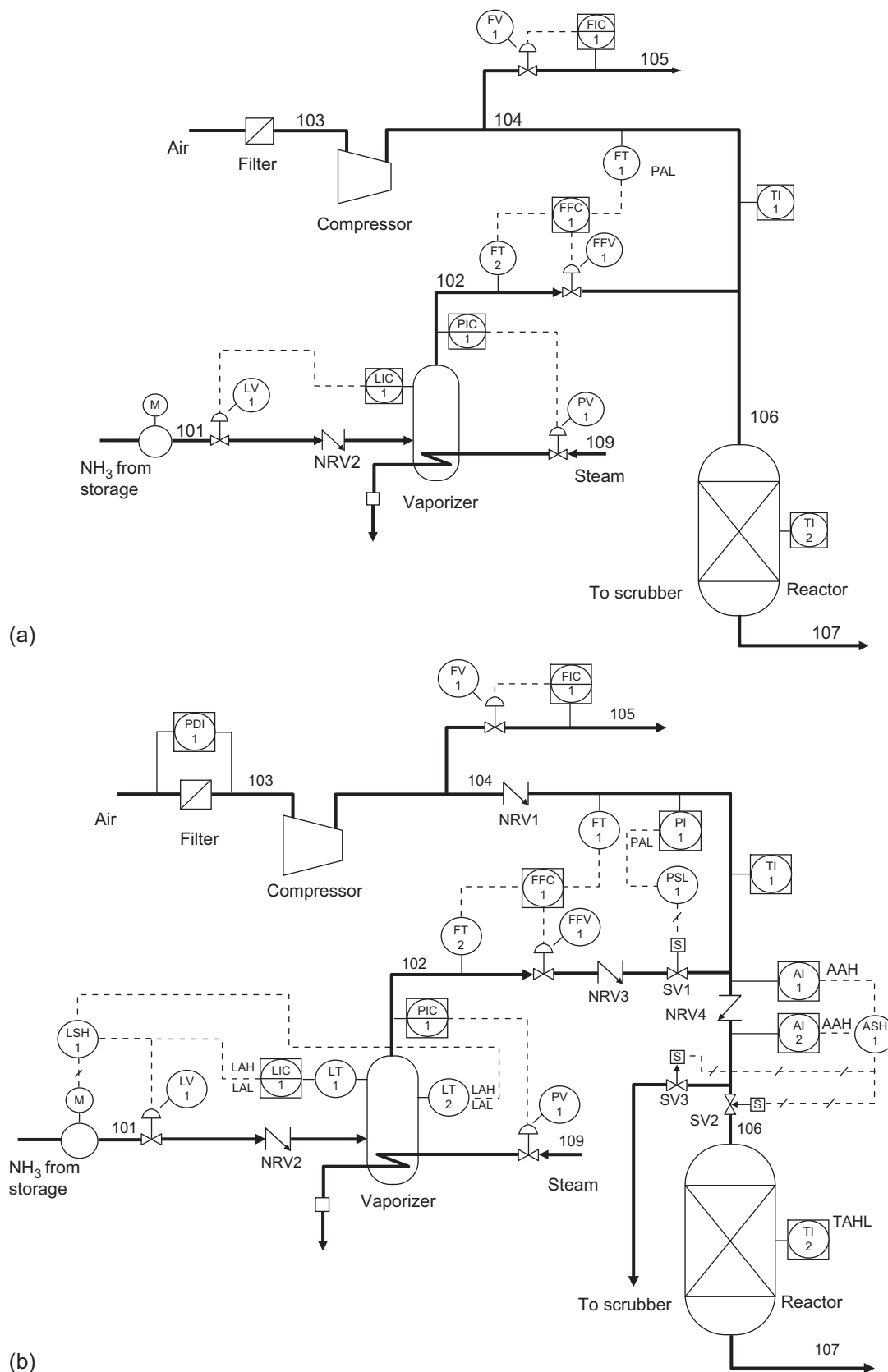


FIG. 10.6 Nitric acid plant reactor section (a) before HAZOP and (b) after HAZOP.

Vessel – Air Filter

Intention – to remove particles that would foul the reactor catalyst

Guide Word	Deviation	Cause	Consequences and Action
Line No. 103			
Intention – transfers clear air at atmospheric pressure and ambient temperature to compressor			
LESS OF	Flow	Partially blocked filter	Possible dangerous increase in NH ₃ concentration; measure and log pressure differential
AS WELL AS	Composition	Filter damaged, incorrectly installed	Impurities, possible poisoning of catalyst; proper maintenance

Vessel – Compressor

Intention – to supply air at 8 bar, 12,000 kg/h, 250 °C, to the mixing tee

Line No. 104

Intention – transfers air to reactor (mixing tee)

NO/NONE	Flow	Compressor failure	Possible dangerous NH ₃ concentration: pressure indicator with low pressure alarm (PI1) interlocked to shut down NH ₃ flow
MORE	Flow	Failure of compressor controls	High rate of reaction, high reactor temperature: high-temperature alarms added to TI2
REVERSE	Flow	Fall in line press. (compressor fails) high pressure at reactor	NH ₃ in compressor – explosion hazard: fit nonreturn valve (NRV1); hot wet acid reactor gas-corrosion; fit second valve (NRV4)

Line No. 105

Intention – transfer secondary air to absorber

NO	Flow	Compressor failure FV1 failure	Incomplete oxidation, air pollution from absorber vent: operating procedures
LESS	Flow	FV1 plugging failure, FIC1	As no flow

Vessel – Ammonia vaporizer

Intention – evaporate liquid ammonia at 8 bar, 25 °C, 731 kg/h

Guide Word	Deviation	Cause	Consequences and Action
Line No. 101			
Intention – transfer liquid NH ₃ from storage			
NO	Flow	Pump failure LV1 fails	Level falls in vaporizer: fit low-level alarm on LIC1
LESS	Flow	Partial failure pump/valve	LIC1 alarms
MORE	Flow	LV1 sticking, LIC1 fails	Vaporizer floods, liquid to reactor: fit high-level alarm on LIC1 with automatic pump shut-down. Add independent level transmitter and alarm LT2.
AS WELL AS	Water brine	Leakage into storages from refrigeration	Concentration of NH ₄ OH in vaporizer: routine analysis, maintenance
REVERSE	Flow	Pump fails, vaporizer press, higher than delivery	Flow of vapor into storages: LIC1 alarms; fit nonreturn valve (NRV2)

—cont'd

Line No. 102

Intention — transfers vapor to mixing tee

NO	Flow	Failure of steam flow, FFV1 fails closed,	LIC1 alarms, reaction ceases: considered low flow alarm, rejected — needs resetting at each rate
LESS	Flow	Partial failure or blockage FFV1	As no flow
	Level	LIC1 fails	LT2 backup system alarms
MORE	Flow	FT2/ratio control misoperation	Danger of high ammonia concentration: fit alarm, fit analyzers (duplicate) with high alarm 12% NH ₃ (AI1, AI2)
	Level	LIC1 fails	LT2 backup system alarms
REVERSE	Flow	Steam failure	Hot, acid gases from reactor — corrosion: fit nonreturn valve (NRV3)

Line 109 (auxiliary)

No	Flow	PV1 fails, trap frozen	High level in vaporizer: LIC1 actuated
----	------	------------------------	--

*Vessel — Reactor*Intention — oxidizes NH₃ with air, 8 bar, 900 °C

NO	Flow	NRV4 stuck closed	Fall in reaction rate: fit low temp, alarm on TI2
LESS	Flow	NRV4 partially closed	As NO
	NH ₃ conc.	Failure to ratio control	Temperatures fall: TI2 alarms (consider low conc. Alarm on AI1, AI2)
MORE	NH ₃ conc.	Failure of ratio control, air flow restricted	High reactor temp.: TI2 alarms 14% explosive mixture enter reactor — potential for disaster; include automatic shut-down bypass actuated by AI1, AI2, SV2 closes, SV3 opens
	Flow	Control systems failure	High reactor temp.: TI2 alarms

Line No. 107

Intention — transfer reactor products to waste-heat boiler

AS WELL AS	Composition	Refractory particles from reactor	Possible plugging of boiler tube: install filter upstream of boiler
------------	-------------	-----------------------------------	---

The most significant hazard of this process is the probability of an explosion if the concentration of ammonia in the reactor is inadvertently allowed to reach the explosive range, >14%. Note that this is a simplified flow diagram, and a HAZOP based on the full P&I diagram would go into considerably more detail.

Operability study

The sequence of steps shown in Fig. 10.4 is followed. Only deviations leading to action and those having consequences of interest are recorded.

10.8 Quantitative hazard analysis

Methods such as FMEA, HAZOP, and the use of safety indices will identify potential hazards but give only qualitative guidance on the likelihood of an incident occurring and the loss suffered; these are left to the intuition of the team members. In a quantitative hazard analysis, the engineer attempts to determine the probability of an event occurring and the potential cost in terms of injuries, financial loss, etc. The international standard IEC 61508 (1998) defines

the risk of a hazard as the probable rate of occurrence (typically expressed as events per year) multiplied by the degree of severity of the harm caused. If there are no protective systems in place, the inherent risk, R_{np} , is:

$$R_{np} = F_{np} \times C \quad (10.1)$$

where: F_{np} is the inherent frequency of the hazard with no protective system (number of events per year)

C is the impact of the hazard (impact per loss event)

The impact can be stated in terms of injuries, serious injuries, emissions, financial loss, or other measures. The analysis is sometimes repeated with different measures of impact, as the organization may have a different tolerance for risk of injuries than for financial risk.

In most designs, protective systems are added to reduce the risk of a hazard to a tolerable level. The tolerable risk is defined as:

$$R_t = F_t \times C \quad (10.2)$$

where: F_t is the tolerable frequency of the hazard (number of events per year)

R_t is the tolerable risk, also sometimes called the *acceptable risk*

The risk reduction factor of the protective system, ΔR , is defined as the ratio of inherent frequency to tolerable frequency:

$$\Delta R = \frac{F_{np}}{F_t} \quad (10.3)$$

It can be seen that the risk reduction factor is the inverse of the average probability that the protective system will fail when it is called upon to operate (average probability of failure on demand), PFD_{av} :

$$PFD_{av} = \frac{F_t}{F_{np}} = \frac{1}{\Delta R} \quad (10.4)$$

The quantitative hazard analysis can thus be used to set targets for the reliability of the protective system. The reliability of the protective system can then be increased until the desired risk reduction factor is attained. Methods that are used to improve the reliability of the protective system are discussed in [Section 10.8.2](#).

The design of safety instrumented systems for maintaining safe operation of processes is discussed in the international standard IEC 61511, which is adopted as ANSI/ISA-84.00.01-2004 in the United States. For a safety system that must operate on demand (i.e., in response to an initiating event), IEC 61511 states that each safety instrumented function (SIF) must have a safety integrity level (SIL) specified to give the required risk reduction or average probability of failure on demand shown in [Table 10.11](#). A good introduction to the application of the functional safety standards IEC 61508 and 61511 and the development of safety instrumented systems is given by [King \(2007\)](#). The book by [Cameron and Raman \(2005\)](#) is an extensive guide to risk management systems.

10.8.1 Fault trees

Incidents usually occur through the coincident failure of two or more items: failure of equipment, control systems and instruments, and misoperation. The sequence of events leading to a hazardous incident can be shown as a fault tree (logic tree), such as that shown in [Fig. 10.7](#). This figure shows the set of events that could lead to a pressure vessel rupture. The AND symbol is used where all the inputs are necessary before the system fails, and the OR symbol where failure of any input, by itself, would cause failure of the system. A fault tree is analogous to the type of logic

TABLE 10.11 Safety integrity levels in demand mode of operation (IEC 61511)

Safety integrity level (SIL)	Target average probability of failure on demand (PFD_{av})	Target risk reduction factor (ΔR)
4	$\geq 10^{-5}$ to $< 10^{-4}$	$> 10,000$ to $\leq 100,000$
3	$\geq 10^{-4}$ to $< 10^{-3}$	> 1000 to $\leq 10,000$
2	$\geq 10^{-3}$ to $< 10^{-2}$	> 100 to ≤ 1000
1	$\geq 10^{-2}$ to $< 10^{-1}$	> 10 to ≤ 100

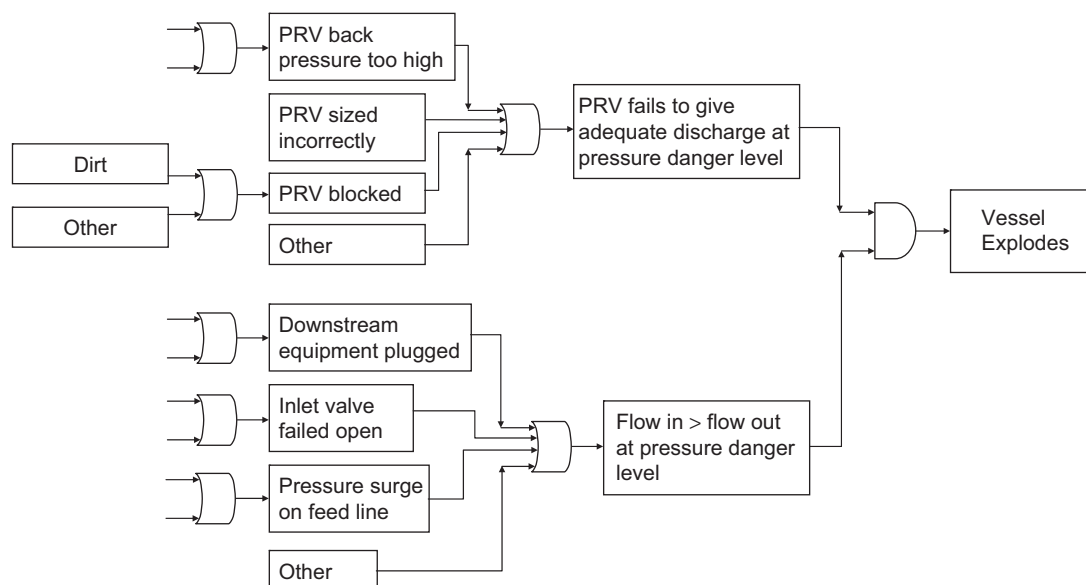


FIG. 10.7 Fault tree for failure of a pressure vessel.

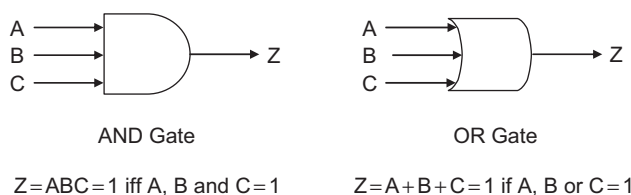


FIG. 10.8 Logic symbols for AND and OR.

diagram used to represent computer operations, and the symbols are analogous to logic AND and OR gates (Fig. 10.8). It can be seen from Fig. 10.7 that failure of the vessel will only occur if there is a cause of overpressure AND a failure of the pressure relief valve (PRV) to respond adequately. These in turn have several possible causes, which may also have possible causes. Each chain of causality should be pursued to the root cause, and the diagram in Fig. 10.7 is incomplete.

The fault trees for even a simple process unit will be complex, with many branches. Fault trees are used to make a quantitative assessment of the likelihood of failure of a system using data on the reliability of the individual components of the system.

Once the fault tree for a subsection of the process has been developed, it can be used to improve the reliability of the design by introducing additional redundant instrumentation. Because a hazardous condition usually requires the failure of one or more devices, introducing additional parallel devices reduces the likelihood of a failure as long as the devices do not have a common mode of failure. The quantitative analysis of the likelihood of an event can be used to determine the level of system redundancy that is required to reduce the likelihood to an acceptably low value.

Event trees are a similar way of representing the same information, but there is not sufficient space to present both methods here; see Mannan (2012) or CCPS (2008).

10.8.2 Equipment reliability

When a fault tree has been constructed, it can be used to estimate the probability of the system failing if the probabilities of the events in the fault tree can be estimated. In most cases, this requires a good understanding of the reliability of instruments, alarms, and safety devices, because these devices would be expected to maintain the process in a safe condition.

If the failure rate, λ , is the number of occasions per year that a protective system develops a fault (yr^{-1}) and the time interval between tests of the device is τ years, then intuitively, the device on average fails halfway between tests. The probability that the device is inactive and will fail on demand (also known as the *fractional dead time*) is then approximately:

$$\phi = \frac{\lambda \tau}{2} \quad (10.5)$$

If the demand rate, δ , is the number of occasions per year that the protective device is actuated then the hazard rate, F , is:

$$F = \delta \phi = \frac{\delta \lambda \tau}{2} \quad (10.6)$$

The intuitive result in Equation 10.6 is true if and only if $\delta\lambda$, $\lambda\tau$, and $\delta\tau$ are all $\ll 1$. For a more rigorous analysis of reliability see Chapters 7 and 13 of Mannan (2012). It can be seen that for a simple system with a single device, the demand rate is the inherent frequency of the hazard, $\delta = F_{np}$, and the probability of the device being inactive is the average probability of failure on demand, $\Phi = PFD_{av}$. Equations 10.6 and 10.4 are therefore equivalent.

The hazard rate can be reduced by using more reliable equipment (lower value of λ), conducting more frequent testing (lower τ), or making improvements that lead to steadier operation (lower δ). Alternatively, two protective systems in parallel can be installed, in which case the hazard rate becomes:

$$F = \frac{4}{3} \delta \phi_A \phi_B \quad (10.7)$$

where: Φ_A is the fractional dead time for system A

Φ_B is the fractional dead time for system B and is subject to the same conditions listed earlier

Example 10.3

Laboratory test data for a trip system shows a failure rate of 0.2 per year. If the demand rate is once every 2 years and the test interval is 6 months, what is the hazard rate? Should a parallel system be installed?

$$\text{Fractional dead time, } \phi = \frac{\lambda \delta}{2} = \frac{0.2 \times 0.5}{2} = 0.05 \quad (10.5)$$

Hazard rate for single system, $F = \delta \Phi = 0.5 \times 0.05 = 0.025$ (i.e., once in every 40 years).

Many plants operate for more than 20 years, so this is probably too high a failure rate to be acceptable. If two systems are used in parallel then:

$$F = \frac{4}{3} \delta \phi_A \phi_B = \frac{4}{3} \times 0.5 \times 0.05 \times 0.05 = \underline{1.67 \times 10^{-3}} \text{ or once in 600 years.}$$

Two systems in parallel should be used, or alternatively, the test frequency could be increased to, say, once every 2 months, giving a more acceptable failure rate of once in every 120 years. Whether the test frequency could be increased will depend on the extent to which testing the device disrupts plant operations. On a large plant with many safety trips and interlocks, it may not be possible to test every system on a frequent basis.

The data on probabilities given in this example are for illustration only and do not represent actual data for these components. Some quantitative data on the reliability of instruments and control systems are given by Mannan (2012). Examples of the application of quantitative hazard analysis techniques in chemical plant design are given by Wells (1996) and Prugh (1980).

The CCPS of the American Institute of Chemical Engineers has published a comprehensive and authoritative guide to quantitative risk analysis (CCPS, 1999). The CCPS has also collected extensive data on device reliability (see CCPS, 1989).

Several other texts are available on the application of risk analysis techniques in the chemical process industries; see CCPS (2000), Frank and Whittle (2001), Cameron and Raman (2005), Crowl and Louvar (2019), Arendt and Lorenzo (2000), Kales (1997), Dodson and Nolan (1999), Green (1983), Kletz (1999), and Crawley and Tyler (2015).

10.8.3 Tolerable risk and safety priorities

If the consequences of an incident can be predicted quantitatively (property loss and the possible number of fatalities), then a quantitative assessment can be made of the risk using [Equation 10.1](#).

If the loss can be measured in money, the cash value of the risk can be compared with the cost of safety equipment or design changes to reduce the risk. In this way, decisions on safety can be made in the same way as other design decisions: to give the best return on the money invested.

Hazards invariably endanger life as well as property, and any attempt to make cost comparisons will be difficult and controversial. It can be argued that no risk to life should be tolerated; however, resources are always limited, and some way of establishing safety priorities is needed.

One approach is to compare the risks, calculated from a hazard analysis, with risks that are generally considered acceptable, such as the average risks in the particular industry and the kind of risks that people accept voluntarily. One measure of the risk to life is the “Fatal Accident Frequency Rate” (FAFR), defined as the number of deaths per 10^8 working hours. This is equivalent to the number of deaths in a group of 1000 people over their working lives. The FAFR can be calculated from statistical data for various industries and activities; some of the published values are shown in [Tables 10.12 and 10.13](#). [Table 10.12](#) shows the relative position of the chemical industry compared with other industries; [Table 10.13](#) gives values for some of the risks that people accept voluntarily.

In the chemical process industries, it is generally accepted that risks with an FAFR greater than 0.4 (one-tenth of the average for industry) should be eliminated as a matter of priority, with the elimination of lesser risks depending on the resources available; see [Kletz \(1977a\)](#). This criterion is for risks to employees; for risks to the general public (undertaken involuntarily), a lower criterion must be used. In the UK, the Health and Safety Executive (HSE) has developed the “as low as reasonably practicable” (ALARP) principle, under which owners can operate a plant in a region defined as tolerable risk as long as they can demonstrate that they have achieved the lowest risk possible, taking into account cost versus risk reduction. The tolerable risk region is defined as a fatality frequency of 10^{-3} to 10^{-6} per person, per year for workers and 10^{-4} to 10^{-6} per person, per year for the general public; see [Schmidt \(2007\)](#). The level of risk to which the public outside the factory gate should be exposed by plant operations will always be a matter of debate and controversy. [Kletz \(1977b\)](#) suggested that a hazard can be considered acceptable if the average risk is less than 1 in 10 million per person, per year. This is an order of magnitude lower than the UK HSE guideline and is equivalent to a FAFR of 0.001—about the same as the chance of being struck by lightning. Additional guidelines on

TABLE 10.12 FAFR for some industries for the period 1978–1990

Industry	FAFR
Chemical industry	1.2
UK manufacturing	1.2
Deep sea fishing	4.2

TABLE 10.13 FAFR for some nonindustrial activities

Activity	FAFR
Staying at home	3
Traveling by rail	5
Traveling by bus	3
Traveling by car	57
Traveling by air	240
Traveling by motorcycle	660
Rock climbing	4000

Source: [Brown \(2004\)](#).

setting tolerable risk are given by [Schmidt \(2007\)](#). There is currently no accepted national guideline on tolerable risk in the United States or Canada.

For further reading on the subject of tolerable risk and risk management, see [Cox and Tait \(1998\)](#) and [Lowrance \(1976\)](#).

10.8.3 Computer software for quantitative risk analysis

The assessment of the risks and consequences involved in the planning and operation of a major plant site is a daunting task.

In industrial practice, the safety instrumented systems that are used are more complex than the simple systems described earlier. If two instruments are used in parallel, with either instrument able to activate a shut-down (a “one out of two” system, denoted 1oo2), then the probability of failure on demand is reduced, as illustrated in Example 10.3, but the likelihood of a spurious shut-down due to instrument failure is doubled. The instrument engineer can overcome this problem by using three instruments with a voting policy that requires two out of three of the instruments to activate before a shut-down is caused (2oo3 voting). The use of programmable logic controllers, distributed control systems, and device communication methods such as Fieldbus means that the reliability of many electrical, electronic, and software components must also be considered in the analysis. Credit can also be taken for the performance of the BPCS as a layer of safety protection (IEC 61511). The calculations soon become too complex to be carried out without using a computer.

The methodology of the classical method of quantitative risk analysis is shown in [Fig. 10.9](#). First, the likely frequency of failure of equipment, pipelines, and storage vessels must be predicted using the techniques described earlier. The probable magnitude of any discharges must then be estimated and the consequences of failure evaluated: fire, explosion, or toxic fume release. Other factors, such as site geography, weather conditions, site layout, and safety management practices, must be taken into consideration. The dispersion of gas clouds can be predicted using suitable models. This methodology enables the severity of the risks to be assessed. Limits have to be agreed on the acceptable risks, such as the permitted concentrations of toxic gases. Decisions can then be made on the location of plant equipment (see [Chapter 11](#)), on the suitability of a site location, and on emergency planning procedures.

The comprehensive and detailed assessment of the risks required for a “safety case” can only be satisfactorily carried out for major installations with the aid of computer software. Programs for quantitative risk analysis have been developed by consulting firms that specialize in safety and environmental protection. Typical of the software available is the SAFETI (Suite for Assessment of Flammability Explosion and Toxic Impact) suite of programs developed by DNV GL (www.dnvgl.com). These programs were initially developed for the authorities in the Netherlands as a response

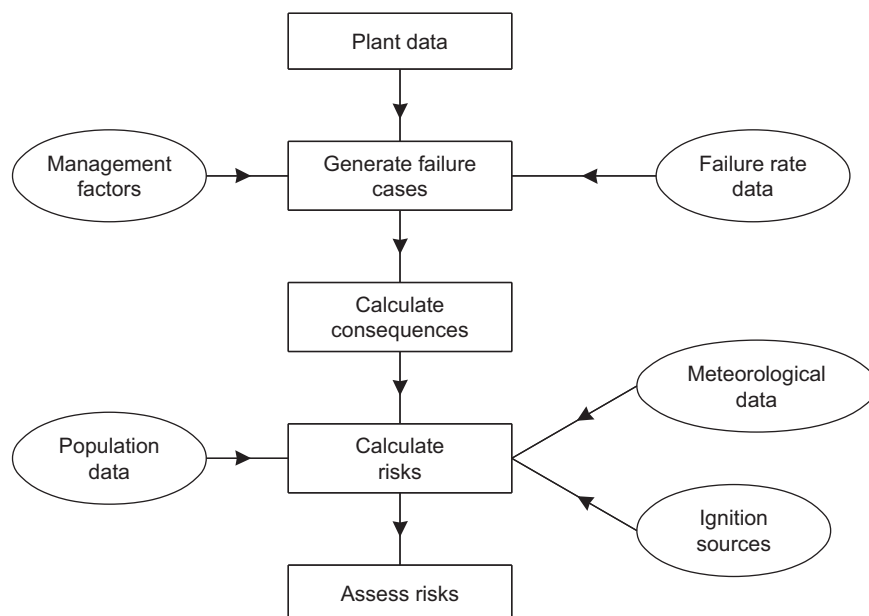


FIG. 10.9 Quantitative risk assessment procedure.

to the Seveso Directives of the EU. The programs have subsequently been developed further and extended and are widely used in the preparation of safety cases; see [Pitblado et al. \(1990\)](#). Other examples include PHAWorks and FaultREASE from Chempute Software (www.chempute.com), FTA-Pro and PHA-Pro from Sphera (www.sphera.com/pha-pro-software), RENO and BlockSim7 from Reliasoft (www.reliasoft.com), and LOGAN from Reliass (www.reliability-safety-software.com). These and other programs for fault tree analysis, HAZOP, process hazard analysis, and quantitative risk analysis are easily found by searching online, and most chemical companies license one of these programs. Some of the programs offer free trial versions and reduced-cost student subscriptions.

Computer programs can be used to investigate a range of possible scenarios for a site; but as with all computer software used in design, they should not be used without caution and judgment. They would normally be used with the assistance and guidance of the consulting firm supplying the software. With intelligent use, guided by experience, such programs can indicate the magnitude of the likely risks at a site and allow sound decisions to be made when licensing a process operation or granting planning permission for a new installation.

10.9 Pressure relief

Pressure-relief devices are an essential requirement for the safe use of pressure vessels. Pressure-relief devices provide a mechanical means of ensuring that the pressure inside a vessel cannot rise to an unsafe level. All pressure vessels within the scope of Section VIII of the ASME Boiler and Pressure Vessel Code must be fitted with a pressure-relief device. The purpose of the pressure-relief device is to prevent catastrophic failure of the vessel by providing a safe means of relieving overpressure if the pressure inside the vessel exceeds the maximum allowable working pressure.

Three different types of relief devices are commonly used:

- *Directly actuated valves*: Weight or spring-loaded valves that open at a predetermined pressure and that normally close after the pressure has been relieved. The system pressure provides the motive power to operate the valve.
- *Indirectly actuated valves*: Pneumatically or electrically operated valves that are activated by pressure-sensing instruments.
- *Bursting discs*: Thin discs of material that are designed and manufactured to fail at a predetermined pressure.

Relief valves are normally used to regulate minor excursions of pressure; bursting discs are used as safety devices to relieve major overpressure. Bursting discs are often used in conjunction with relief valves to protect the valve from corrosive process fluids during normal operation. The design and selection of relief valves is discussed by [Morley \(1989a,b\)](#) and is also covered by the pressure vessel standards (see later). Bursting discs are discussed by [Mathews \(1984\)](#), [Asquith and Lavery \(1990\)](#), and [Murphy \(1993\)](#). The discs are manufactured in a wide range of common engineering steels and alloys as well as a variety of materials for use in corrosive conditions, such as impervious carbon, gold, and silver, and suitable discs can be found for use with all process fluids. Bursting discs and relief valves are proprietary items, and the vendors should be consulted when selecting suitable types and sizes.

Selection and sizing of the relief device are the responsibility of the end user of the pressure vessel. Rules for the selection and sizing of pressure-relief devices are given in the ASME BPV Code Sec. VIII D.1 Parts UG-125 to UG-137 and D.2 Part AR.

Under the rules given in ASME BPV Code Sec. VIII D.1, the primary pressure-relief device must have a set pressure not greater than the maximum allowable working pressure of the vessel. The primary relief device must be sized to prevent the pressure from rising 10% or 3 psi (20 kPa), whichever is greater, above the maximum allowable working pressure (MAWP). If secondary relief devices are used, their set pressure must not be greater than 5% above the maximum allowable working pressure. When multiple relief devices are used, their combined discharge must be adequate to prevent the vessel pressure from rising more than 16% or 4 psi (30 kPa), whichever is greater, above the maximum allowable working pressure. In a relief scenario where the pressure vessel is exposed to an external fire, the relief device or devices must prevent the vessel pressure from increasing to more than 21% above the maximum allowable working pressure.

Pressure-relief devices must be constructed, located, and installed such that they can be easily inspected and maintained. They are normally located at the top of a vessel in a clean, free-draining location. They must be located on or close to the vessel that they are protecting.

10.9.1 Pressure-relief scenarios

Overpressure will occur whenever mass, moles, or energy accumulate in a contained volume or space with a restricted outflow. The rate at which material or energy accumulates determines the pressure rise. If the process control system is not able to respond quickly enough, the pressure-relief device must be activated before the vessel ruptures, explodes, or suffers some other catastrophic loss of containment.

The first step in designing a pressure-relief system is to evaluate the possible causes of overpressure to determine the rate of pressure accumulation associated with each and hence estimate the relief load (the flow rate that must be discharged through the relief device). The API Standard 521 suggests the following causes:

Blocked outlet	Chemical reaction	Electric power loss
Utility failure	External fire	Accumulation of noncondensable species
Cooling or reflux failure	Abnormal heat input	Failure of automatic controls
Inadvertent valve opening	Operator error	Loss of heat in series fractionation
Loss of fans	Check valve failure	Volatile material entering system
Steam or water hammer	Internal explosion	Heat exchanger tube failure
Adsorbent flow failure	Overheating a liquid full system	

This list is not exhaustive, and the design engineers should always brainstorm for additional scenarios and review the results of FMEA, HAZOP, HAZAN, or other process safety analyses.

In evaluating relief scenarios, the design engineer should consider sequential events that result from the same root-cause event, particularly when these can increase the relief load. For example, the loss of electric power in a plant that carries out a liquid-phase exothermic reaction could have the following impacts:

1. Failure of all or part of the automatic control system.
2. Loss of cooling due to failure of cooling water pumps or air coolers.
3. Loss of mixing in the reactor due to failure of the stirrer, leading to localized runaway reaction.

Because these have a common cause, they should be considered as simultaneous events for that cause. If two events do not share a common cause, then the probability that they will occur simultaneously is remote and is not usually considered (API STD 521, 3.2). Root-cause events such as power loss, utility loss, and external fire will often cause multiple other events and hence large relief loads.

The rate at which pressure accumulates is also affected by the response of the process control system. API STD 521 recommends that instrumentation be assumed to respond as designed if it increases the relieving requirement, but no credit should be taken for instrumentation response if it reduces the relieving requirement. For example in Fig. 10.10(a), if the outlet control valve becomes blocked and the pressure in the vessel rises, the flow from the pump will initially decrease because of the higher back-pressure. The flow controller will compensate for this by

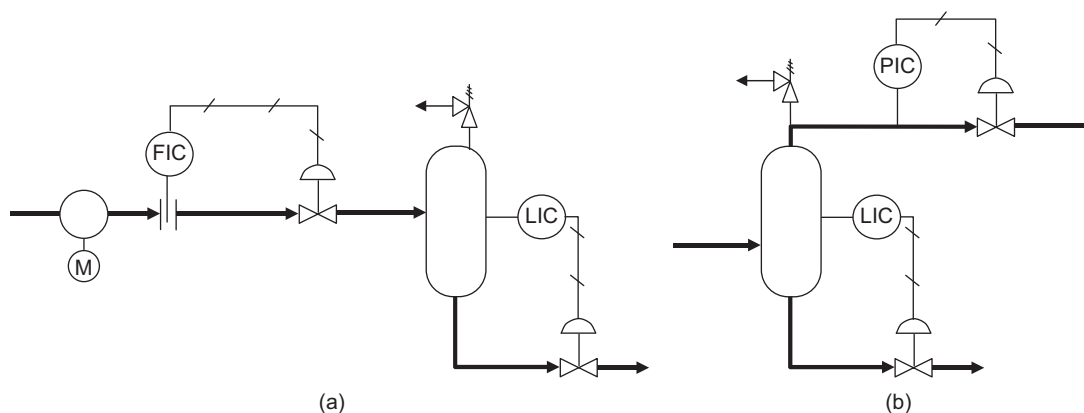


FIG. 10.10 Instrumentation response to pressure-relief scenarios: (a) Instrumentation response increases relieving load. (b) Instrumentation response would reduce relieving load, but API RP 521 recommends taking no credit for instrumentation response.

opening the flow control valve to try to maintain a constant flow rate and will consequently increase the relieving load. The design engineer should assume that the instrumentation responds as designed and the flow rate remains constant. In Fig. 10.10(b), if the outlet control valve becomes blocked, the pressure controller will continue opening the pressure control valve until it is fully open. This provides an alternative outflow and reduces the relieving load, but according to API STD 521 this response should not be considered.

Heat exchangers and other vessels with internal compartments must also be protected from overpressure in the case of an internal failure. This is of particular importance for shell and tube type exchangers, as the common design practice is to put the higher-pressure fluid on the tube side. This saves costs in constructing the shell and obviates sizing the tubes to withstand a high compressive load due to external pressure. If the tube side is at higher pressure, then in the event of a tube or tubesheet failure, the shell will be exposed to the higher tube-side pressure.

Both API STD 521 and ASME BPV Code Section VIII allow multiple vessels connected together to be considered as a single unit for relief scenarios, provided that there are no valves or restriction orifices between the vessels and that the design considers the full relieving load of the system (ASME BPV Code Sec. VIII D.1 UG-133).

10.9.2 Pressure-relief loads

The rate at which pressure accumulates can be estimated by making non-steady-state mass, mole, and energy balances around the vessel or system:

$$\text{in} + \text{formed by reaction} = \text{out} + \text{accumulation} \quad (10.8)$$

Because liquids have very low compressibility, pressure vessels are seldom operated entirely filled with liquid, because small accumulations of material would cause large surges in pressure. Instead, it is common practice to operate with a “bubble” of vapor (often nitrogen) at the top of the vessel. The mass balance equation can then be rearranged into an equation for the rate of change of pressure of this gas with time.

For example, consider a vessel of total volume $V \text{ m}^3$ that is normally operated 80% full of liquid on level control, as in Fig. 10.10(a), and is fed with a flow rate $v \text{ m}^3/\text{s}$ of liquid. If the volume of liquid in the vessel is V_L , then if the outlet becomes blocked and the liquid is assumed to be incompressible, then the change in the volume of the liquid is:

$$\frac{dV_L}{dt} = v \quad (10.9)$$

where t = time, s

The volume occupied by vapor, $V_G = V - V_L$, so:

$$\frac{dV_G}{dt} = -\frac{dV_L}{dt} = -v \quad (10.10)$$

If there is no vapor flow in or out of the vessel, then assuming the vapor behaves as an ideal gas:

$$V_G = nRT/P \quad (10.11)$$

where n = number of moles in of gas in the vessel, mol

R = ideal gas constant, J/mol.K

T = temperature, K

P = pressure, N/m²

If the temperature is constant (which is valid for a blocked outlet relief scenario), then until the relief valve opens:

$$\frac{dP}{dt} = nRT \frac{d}{dt} \left(\frac{1}{V_G} \right) = -\frac{nRT}{V_G^2} \frac{dV_G}{dt} = \frac{P^2 v}{nRT} \quad (10.12)$$

Equation 10.12 can be used to estimate the rate of pressure accumulation.

When the relief valve opens, it allows vapor to discharge at a flow rate, $w \text{ kg/s}$. The number of moles of vapor in the vessel is then given by:

$$\frac{dn}{dt} = -\frac{1000 w}{M_w} \quad (10.13)$$

where M_w is the average molecular weight of the vapor, g/mol

The equation for the rate of change of pressure becomes:

$$\begin{aligned}\frac{dP}{dt} &= RT \frac{d}{dt} \left(\frac{n}{V_G} \right) = \frac{RT}{V_G^2} \left(V_G \frac{dn}{dt} - n \frac{dV_G}{dt} \right) \\ &= \frac{P^2}{nRT} \left(v - \frac{1000 RTw}{M_w P} \right)\end{aligned}\quad (10.14)$$

If the relief valve is sized correctly, then the maximum pressure that can accumulate is 110% of the maximum allowable working pressure, P_m (ASME BPV Code Sec. VIII D.1 UG-125). At this point there is no further accumulation of pressure and $dp/dt = 0$; hence:

$$\frac{1000 RTw}{M_w \times 1.1P_m} = v \quad (10.15)$$

and the required relief load is:

$$w = \frac{1.1P_m M_w v}{1000 RT} \quad (10.16)$$

Equation 10.16 applies as long as only vapor is vented from the vessel. Once the vapor has been displaced by liquid, then the relief load must be the liquid flow rate. If a two-phase mixture is vented, then the calculation becomes more complex.

In most cases the governing relief scenario includes both material and heat input into the system and typically also includes vaporization of material, reaction, and two-phase flow. Such systems are much more difficult to describe using simple differential algebraic models, and the current industrial practice is to use dynamic simulation models for these cases. Dynamic models can be built in any of the commercial process simulators that have this capability. The AIChE Design Institute for Emergency Relief Systems (DIERS) also licenses software called SuperChems (formerly SAFIRE) that is written specifically for pressure-relief system design and incorporates the DIERS-recommended methods and research findings for multiphase, reacting, and highly nonideal systems.

For some relief scenarios, correlations have been established for the relieving load. For the external fire case, API STD 521 (Section 3.15.2) gives:

$$Q = 21000 F_e A_w^{0.82} = w_f \Delta H_{vap} \quad (10.17)$$

where Q = heat input due to fire, BTU/hr

F_e = environmental factor

A_w = internal wetted surface area, ft²

w_f = fire case relieving load, lb/hr

ΔH_{vap} = heat of vaporization, BTU/lb

The environmental factor F_e allows for insulation on the vessel. It is equal to 1.0 for a bare vessel or if the insulation can be stripped off by a liquid jet. The correlation in Equation 10.17 assumes good general design practice and site layout, including use of sewers and trenches or the natural slope of the land to control run-off so that pools do not form. Other formulae for the rate of heat input and relief load are given by ROSPA (1971) and NFPA 30. Local safety regulations and fire codes should be consulted to determine the appropriate method to use in any particular design.

Design codes and standards such as API STD 521 and the DIERS Project Manual (Fisher et al., 1993) should be consulted for other correlations and recommended methods for calculating relief loads. The DIERS Project Manual also discusses calculation of relief loads for underpressure scenarios (see Section 10.9.5).

10.9.3 Design of pressure-relief valves

Spring-loaded relief valves

The most commonly used relief device is the conventional spring-loaded relief valve shown in Fig. 10.11. This design of valve is available in the widest range of sizes and materials (API STD 526, BS EN ISO 4126-1).

In a conventional relief valve the pressure force acts on a disk that is held against a seating surface by a spring. The compression of the spring can be adjusted using an adjusting screw so that the spring force is equal to the pressure force at the valve set pressure.

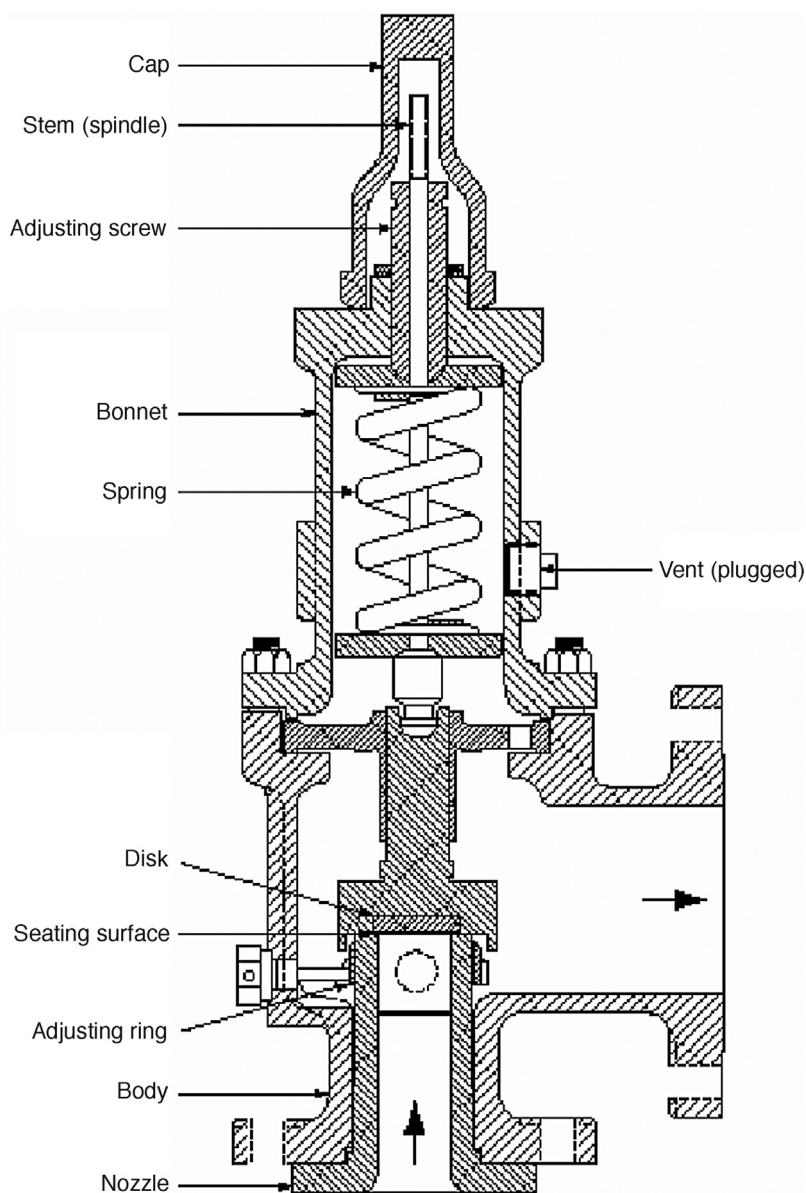


FIG. 10.11 Conventional spring-loaded relief valve. Reproduced with permission from API Recommended Practice 520.

The pressure flow response of a conventional relief valve is illustrated schematically in Fig. 10.12. When the pressure in the vessel reaches 92% to 95% of the set pressure, a spring-loaded relief valve in a gas or vapor service begins to “simmer” and leak gas. Leakage can be reduced by lapping the disk and seating surface to a high degree of polish, using elastomeric seals (at low temperatures only), or using a high-pressure differential between the operating pressure and set pressure. When the set pressure is reached, the valve “pops” and the disk lifts from the seat. The disk and seat are shaped such that the force on the disk continues to increase until the valve is fully open, at which point the flow rate is limited only by the bore area of the seating surface and not by the gap between the seating surface and the disk. At this point the design flow rate is achieved and there should be no further pressure accumulation.

When the pressure falls sufficiently, the spring force can overcome the forces due to the flowing fluid, and the valve reseats. Reseating usually occurs at a lower pressure than the set pressure, giving a different curve for blowdown.

The capacity and lift pressure of a conventional spring-loaded relief valve are affected by the back-pressure in the downstream relief system. The back-pressure exerts forces that are additive to the spring force. Where back-pressure is known to fluctuate or accumulate, balanced pressure relief valves that incorporate a bellows or other means of

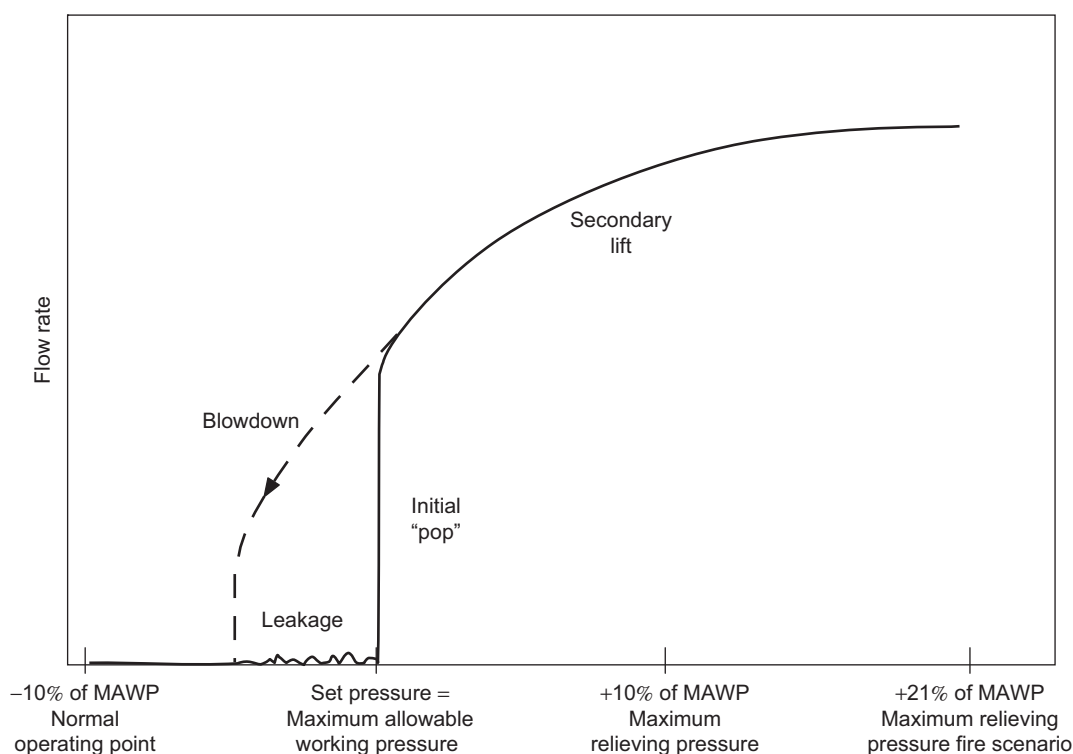


FIG. 10.12 Pressure–flow response of a conventional spring-loaded relief valve.

compensating for back-pressure should be used (see API STD 520 for details). This is particularly important when multiple devices are relieved into the same vent or flare system, as common-cause relief scenarios such as power loss can trigger multiple relief events and send a lot of material into the vent or flare system, increasing the back-pressure acting on the relief valves.

Pilot-operated relief valves

Pilot-operated relief valves are designed to overcome some of the major drawbacks of conventional spring-loaded relief valves. In a pilot-operated relief valve, the spring and disk are replaced by a piston, as shown in Fig. 10.13. A narrow-bore pipe known as a *pilot supply line* connects from the top of the piston to the relief valve inlet via a secondary (pilot) valve of the spring-loaded type. In normal operation both sides of the valve see the same pressure, but because the top surface area of the piston is greater than the area of the seat, the downward force is greater and the valve remains closed. When the pressure exceeds the set pressure, the pilot valve opens and pressure above the piston is lost. This causes the piston to lift, and the valve opens. The pilot valve vent can be exhausted to atmosphere or to the main valve outlet, depending on the containment requirements for the process fluid.

The pressure-flow response of a pilot-operated relief valve is illustrated schematically in Fig. 10.14. Leakage is eliminated, and there is no blowdown.

Pilot-operated relief valves are used in applications that require a low differential between operating pressure and set pressure (for example, revamps where the vessel is now operated closer to the maximum allowable working pressure or vessels operating below 230 kPa or 20 psig): high-pressure services (above 69 bara or 1000 psig) and cases where low leakage is required. They are not available in as broad a range of metallurgies as spring-loaded relief valves. Pilot-operated relief valves are also restricted to lower-temperature applications, as they typically use elastomeric materials to make a seal between the piston and its housing. More details of pilot-operated relief valves are given in BS EN ISO 4126-4.

Sizing relief valves

Guidelines for sizing relief valves are given in API STD 520 and BS EN ISO 4126. Different design equations are recommended for vapor, liquid, steam, or two-phase flows. Sizing methods are also discussed in the DIERS Project Manual (Fisher et al., 1993) and the book by CCPS (2017).

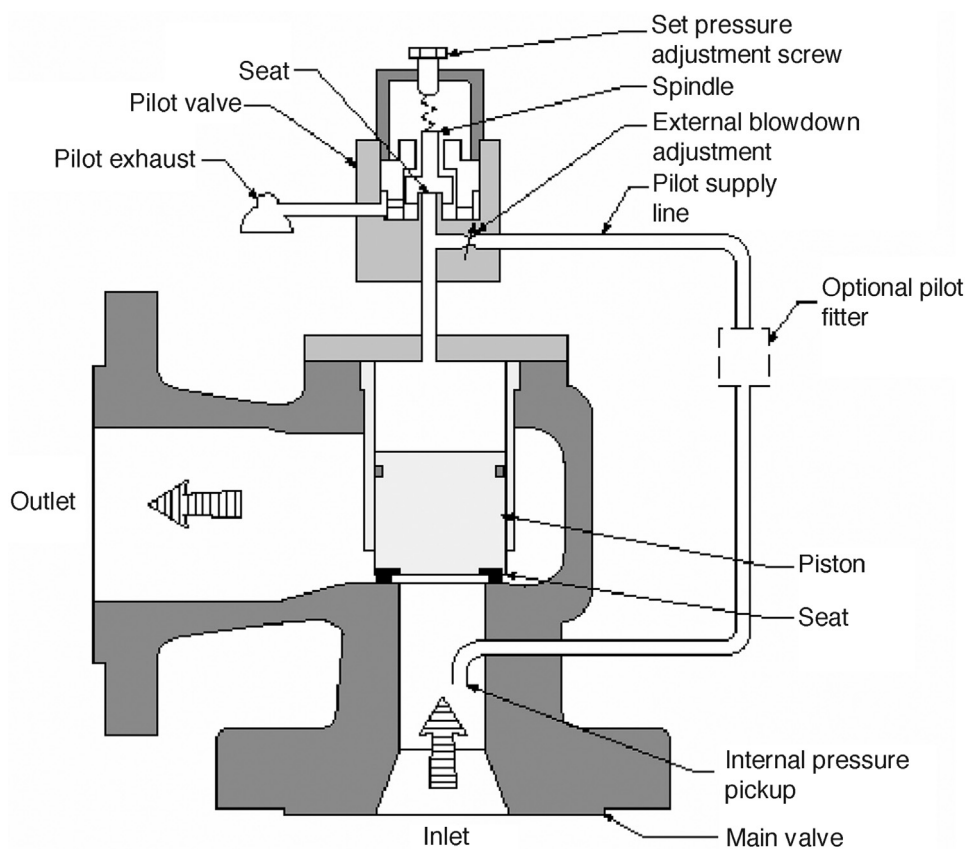


FIG. 10.13 Pop-action pilot-operated relief valve. Reproduced with permission from API Recommended Practice 520.

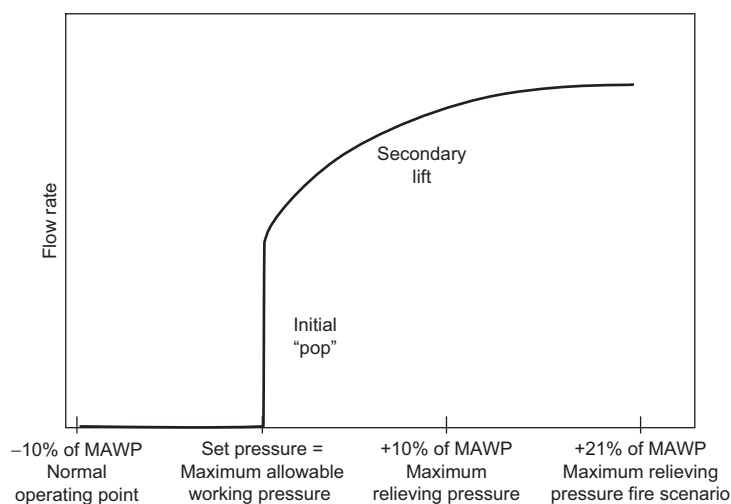


FIG. 10.14 Pressure-flow response of a pilot-operated relief valve.

When the fluid flowing through the valve is a compressible gas or a vapor, the design must consider whether critical flow is achieved in the nozzle of the valve. The critical flow rate is the maximum flow rate that can be achieved and corresponds to a sonic velocity at the nozzle. If critical flow occurs, the pressure at the nozzle exit cannot fall below the critical flow pressure P_{cf} , even if a lower pressure exists downstream. The critical flow pressure can be estimated from the upstream pressure for an ideal gas using the equation:

$$\frac{P_{cf}}{P_1} = \left[\frac{2}{\gamma + 1} \right]^{\gamma/(\gamma-1)} \quad (10.18)$$

where γ = ratio of specific heats = C_p/C_v

P_1 = absolute pressure upstream

P_{cf} = critical flow pressure

Any consistent set of units may be used for pressure as long as the absolute pressure is used, not the gauge pressure. The ratio P_{cf}/P_1 is called the *critical pressure ratio*. Typical values of this ratio are given in [Table 10.14](#). If the downstream pressure is less than the critical flow pressure, critical flow will occur in the nozzle. It can be seen from the table that this will be the case whenever the upstream pressure is more than two times the downstream pressure. Because most relief systems are operated close to atmospheric pressure, critical flow is the usual case.

For critical flow, API STD 520 ([Section 3.6.2](#)) gives the following equation for valve area, A_d :

$$A_d = \frac{13,160}{C K_d P_1 K_b K_c} \frac{w}{\sqrt{M_w}} \quad (10.19)$$

where A_d = discharge area, mm²

w = required flow rate, kg/hr

C = coefficient = $520 \sqrt{\gamma \left(\frac{2}{\gamma+1} \right)^{(\gamma+1)/(\gamma-1)}}$

K_d = coefficient of discharge

P_1 = absolute pressure upstream, kPa

K_b = back-pressure correction factor

K_c = combination correction factor

TABLE 10.14 Critical flow pressure ratios

Gas	Specific heat ratio $\gamma = C_p/C_v$ at 60 °F, 1 atm	Critical flow pressure ratio at 60 °F, 1 atm
Hydrogen	1.41	0.52
Air	1.40	0.53
Nitrogen	1.40	0.53
Steam	1.33	0.54
Ammonia	1.3	0.54
Carbon dioxide	1.29	0.55
Methane	1.31	0.54
Ethane	1.19	0.57
Ethylene	1.24	0.57
Propane	1.13	0.58
Propylene	1.15	0.58
n-Butane	1.19	0.59
n-Hexane	1.06	0.59
Benzene	1.12	0.58
n-Decane	1.03	0.60

Adapted from API RP 520.

Notes:

1. Taken from API RP 520, Table 7.

2. Some values of critical flow pressure ratio have been determined experimentally and do not necessarily agree with predictions from [Equation 10.18](#).

T = relieving temperature, K

Z = compressibility at the inlet condition

M_w = molecular weight, g/mol

For preliminary estimates, the coefficient K_d can be taken as 0.975 for a relief valve and 0.62 for a bursting disk. The back-pressure correction factor, K_b , can initially be assumed to be 1.0 for critical flow. The combination correction factor, K_c , is used when a rupture disk is used upstream of the relief valve (see next section), in which case it is 0.9. If no rupture disk is used, then K_c is 1.0. For vessels designed in accordance with ASME BPV Code Sec. VIII, $P_1 = 1.1$ times the maximum allowable working pressure.

The relief valve selected should be one with equal or greater area than calculated using Equation 10.19. Relief valve sizes are given in API STD 526 and BS EN ISO 4126. Sizing equations for subcritical flow of vapors, liquids, steam, and two-phase mixtures are given in API STD 520.

10.9.4 Design of non-reclosing pressure relief devices

Two types of non-reclosing pressure relief devices are used: rupture disks and breaking-pin devices.

A rupture disk device consists of a rupture disk and a clamp that holds the disk in position. The disk is made from a thin sheet of metal and is designed to burst if a set pressure is exceeded. Some rupture disks are scored so that they can burst without forming fragments that might damage downstream equipment.

Rupture disks are often used upstream of relief valves to protect the relief valve from corrosion or to reduce losses due to relief valve leakage. Large rupture disks are also used in situations that require very fast response time or high relieving load (for example, reactor runaway and external fire cases). They are also used in situations where pressure is intentionally reduced below the operating pressure for safety reasons. The use of bursting disk devices is described in BS EN ISO 4126-2 and BS EN ISO 4126-6.

If a rupture disk is used as the primary pressure-relief device, then when it bursts, the operators have no option but to shut down the plant so that the disk can be replaced before the vessel is repressured. Rupture disks are therefore most commonly used at the inlets of relief valves or as secondary relief devices. Rupture disks can be sized using Equation 10.19 for compressible gases in sonic flow, with a value of $K_d = 0.62$. The combination of safety valves and rupture discs is discussed in BS EN ISO 4126-3.

Breaking-pin devices have a similar construction to spring-loaded relief valves, except the valve disk is held against the seat by a pin that is designed to buckle or break when the set pressure is reached, as illustrated in Fig. 10.15. Once the valve has opened, the pin must be replaced before the valve can be reset.

Both rupture disks and breaking-pin devices are sensitive to temperature. The manufacturer should always be consulted for applications that are not at ambient conditions. Because non-reclosing pressure-relief devices can only be used once, the set pressure is determined by testing a sample of the devices out of each manufactured batch. Pressure-relief valve test methods are specified in ASME PTC 25.

10.9.5 Design of pressure-relief discharge systems

When designing relief venting systems, it is important to ensure that flammable or toxic gases are vented to a safe location. This will normally mean venting at a sufficient height to ensure that the gases are dispersed without creating a hazard. For highly toxic materials, it may be necessary to provide a scrubber to absorb and “kill” the material, for instance, the provision of caustic scrubbers for chlorine and hydrochloric acid gases. If flammable materials have to be vented at frequent intervals, as, for example, in some refinery operations, flare stacks are used.

The rate at which material can be vented will be determined by the design of the complete venting system: the relief device and the associated piping. The maximum venting rate will be limited by the critical (sonic) velocity, whatever the pressure drop. The vent system must be designed such that sonic flow can only occur at the relief valve and not elsewhere in the system; otherwise, the design relief load will not be attained. The design of venting systems to give adequate protection against overpressure is a complex and difficult subject, particularly if two-phase flow is likely to occur. When two-phase flow can occur, the relief system must provide for disengagement of liquid from the vapor before the vapor is vented or sent to flare.

Under normal closed conditions, the leakage from a properly designed and maintained spring-operated relief valve should be minimal; however, valves that open and close frequently or that are exposed to corrosive fluids or dust can suffer damage leading to increased leakage and may require more frequent inspection and maintenance. Leakage from relief valves can be a significant source of plant volatile organic compound (VOC) emissions if the

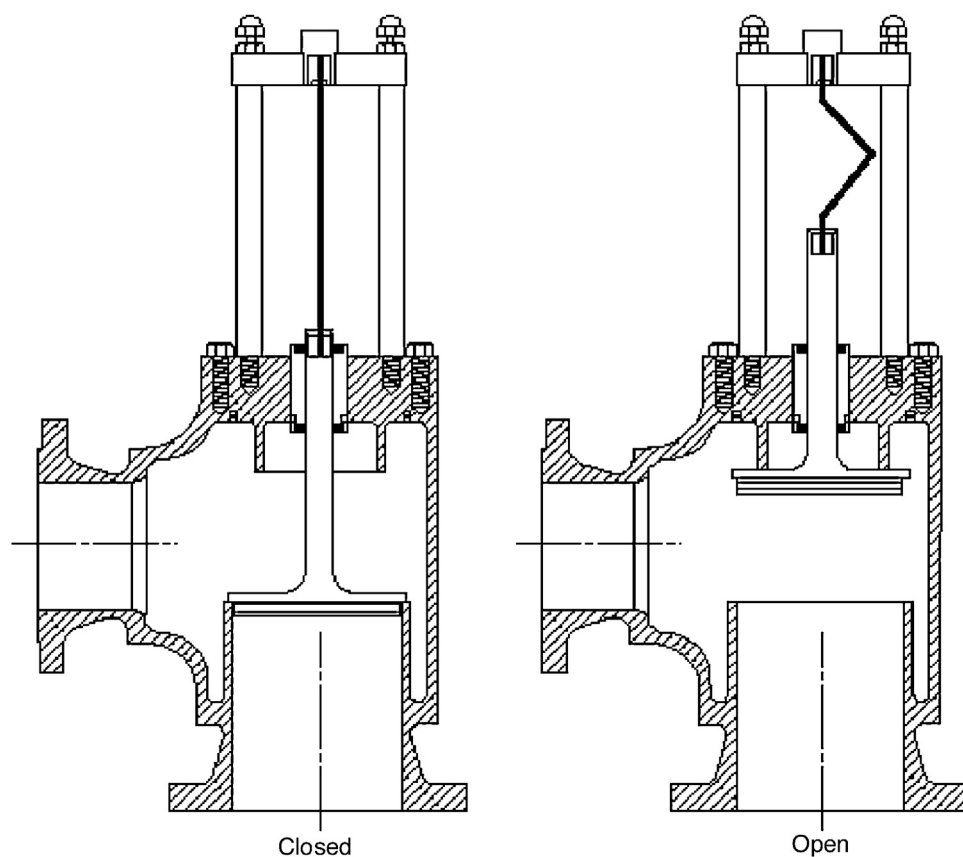


FIG. 10.15 Buckling-pin relief valve. Reproduced with permission from API Recommended Practice 520.

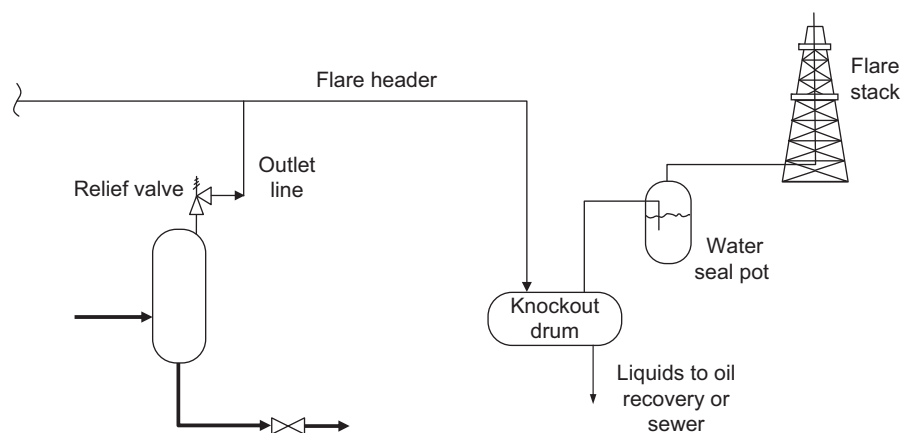


FIG. 10.16 Typical relief system design.

relief system does not vent via a flare stack. Methods for determining leakage from relief valves are given in API STD 527.

Guidelines for relief valve installation and relief systems design are given in API STD 520 Part II, API STD 521 Sections 4 and 5 and the DIERS Project Manual (Fisher et al., 1993). API STD 521 also gives design methods for blow-down drums and flare systems. A typical relief system is shown in Fig. 10.16. For a comprehensive discussion of the problem of vent system design and the design methods available, see the papers by Duxbury (1976, 1979) and the guidelines by CCPS (2017).

10.9.6 Protection from underpressure (vacuum)

Unless designed to withstand external pressure (see [Section 14.7](#)), a vessel must be protected against the hazard of underpressure, as well as overpressure. Underpressure will normally mean vacuum on the inside with atmospheric pressure on the outside. It requires only a slight drop in pressure below atmospheric pressure to collapse a storage tank. Though the pressure differential may be small, the force on the tank roof will be considerable. For example, if the pressure in a 10-m-diameter tank falls to 10 millibar below the external pressure, the total load on the tank roof will be around 80,000 N (equivalent to 8 metric tons weight). It is not an uncommon occurrence for a storage tank to be sucked in (collapsed) by the suction pulled by the discharge pump due to the tank vents having become blocked. Where practical, vacuum breakers (valves that open to atmosphere when the internal pressure drops below atmospheric) should be fitted.

Example 10.4

A gasoline surge drum has capacity 4 m³ (1060 gal) and is normally operated 50% full at 40 °C (100 °F) under 20 bar absolute pressure (280 psig) of hydrogen in the head space and using a level controlled outflow, as shown in [Fig. 10.10\(a\)](#). Gasoline of specific gravity 0.7 is pumped into the surge drum at a normal flow rate of 130 m³/hr.

Assuming the aspect ratio of the vessel (ratio length/diameter) is 3.0 and the heat of vaporization of gasoline is 180 BTU/lb, evaluate the relief loads for the blocked outflow and external fire cases and determine the relief valve size. (In practice, gasoline contains many components that boil over a wide range of temperatures at the design pressure, and a more complex calculation is needed than is given here.)

Blocked outlet case

$$w = \frac{1.1 P_m M_w v}{1000 RT} = \frac{1.1 \times \left(\frac{130}{3600}\right) \times \left(\frac{20 \times 10^5}{0.9}\right) \times 2}{1000 \times 8.314 \times 313} = \underline{\underline{67.8 \text{ g/s}}} \quad (10.16)$$

External fire case

If the vessel has a hemispherical head, then:

$$\text{volume} = \pi \left(\frac{D^2 L}{4} + \frac{D^3}{6} \right) = \frac{11\pi D^3}{12}$$

$$\text{so } D = 1.12 \text{ m}$$

$$\begin{aligned} \text{wetted area} &= \pi (DL + D^2)/2 = 2\pi D^2 \\ &= 7.82 \text{ m}^2 = 84.2 \text{ ft}^2 \end{aligned}$$

Assume $F_e = 1$:

$$\begin{aligned} w_f &= \frac{21000 F_e A_w^{0.82}}{\Delta H_{vap}} = \frac{21000 \times 1 \times 84.2^{0.82}}{180} \\ &= 4423 \text{ lb/hr} \\ &= \underline{\underline{0.56 \text{ kg/s}}} \end{aligned} \quad (10.17)$$

So the external fire case has the higher relieving load and governs the design.

If the vent line discharges to a flare system at atmospheric pressure, then:

$$\frac{P_{outlet}}{P_1} = \frac{1}{20} \ll 0.52$$

so flow in the nozzle is critical.

For hydrogen:

$$C = 520 \sqrt{1.41 \left(\frac{2}{2.41} \right)^{(2.41/0.41)}} \\ = 356.9$$

Assume the valve will lift when the temperature reaches 60 °C (333K), $Z = 1.02$ for hydrogen:

$$A_d = \frac{13,160 w}{C K_d P_1 K_b K_c} \sqrt{\frac{TZ}{M_w}} = \frac{13160 \times 0.56 \times 3600}{356.9 \times 0.975 \times 2000 \times 1.0 \times 1.0} \sqrt{\frac{333 \times 1.02}{2}} \\ = 496.8 \text{ mm}^2 \text{ or } 0.77 \text{ in}^2 \quad (10.19)$$

From API STD 526, we would select an “H” orifice relief valve with effective orifice area 0.785 in². A size 2H3 carbon steel relief valve will allow a set pressure up to 740 psig in the expected range of operation temperature. In practice, however, we would have to consider two-phase flow due to entrainment of boiling liquid with the vapor and might select a larger orifice size after more detailed design.

10.10 References

- Arendt, J. S., & Lorenzo, D. K. (2000). *Evaluating process safety in the chemical industry: A user's guide to quantitative risk analysis*. Wiley – AIChE.
- Asquith, W., & Lavery, K. (1990). 15. Bursting discs – the vital element in relief.
- Asfahl C. R., & Rieske, D.W. (2018.) *Industrial safety and health management* (7th ed.). Pearson.
- Balemans, A. W. M. (1974). Check-lists: Guide lines for safe design of process plants. In C. H. Bushmann (Ed.), *Loss prevention and safety promotion in the process industries* Elsevier.
- Barton, J. (2001). *Dust explosion, prevention and protection – A practical guide*. Gulf.
- Bies, D., & Hansen, C. (2009). *Engineering noise: Theory and practice* (4th ed.). CRC Press.
- Birolini, A. (2017). *Reliability engineering theory and practice* (8th ed.). Springer.
- Britton, L. G. (1999). *Avoiding static ignition hazards in chemical processes*. AIChE.
- Brown, D. (2004). It's a risky business. *Chemical Engineering London*, 758, 42.
- Burgess, P. (2013). The globally harmonized system. *Chemical Engineering New York*, 120(12), 42.
- Cameron, I. T., & Raman, R. (2005). *Process systems risk management, vol.6. (process systems engineering)*. Academic Press.
- Carson, P. A., & Mumford, C. J. (1988). *Safe handling of chemicals in industry, 2 vols*. Longmans.
- Carson, P. A., & Mumford, C. J. (2002). *Hazardous chemicals handbook* (2nd ed.). Newnes.
- CCPS. (1989). *Guidelines for process equipment reliability data, with data tables*. Wiley – AIChE.
- CCPS. (1999). *Guidelines for chemical process quantitative risk analysis* (2nd ed.). Wiley – AIChE.
- CCPS. (2000). *Guidelines for hazard evaluation procedures – With worked examples* (2nd ed.). Wiley – AIChE.
- CCPS. (2008). *Guidelines for hazard evaluation procedures* (3rd ed.). Wiley – AIChE.
- CCPS. (2012). *Guidelines for engineering design for process safety* (2nd ed.). Wiley – AIChE.
- CCPS. (2017). *Guidelines for pressure relief and effluent handling systems* (2nd ed.). Wiley – AIChE.
- CCPS. (2019). *Guidelines for inherently safer chemical processes: A life cycle approach* (3rd ed.). Wiley – AIChE.
- Channing, J. (Ed.). (2013). *Safety at work* (8th ed.). Routledge.
- Cheremisinoff, N. P. (1996). *Noise control in industry: A practical guide*. Noyes.
- CIA. (1979). *Process plant hazards and control building design*. Chemical Industries Association.
- Cooper, W. F., & Jones, D. A. (1993). *Electrical safety engineering* (3rd ed.). Butterworth-Heinemann.
- Cox, S., & Tait, R. (1998). *Safety, reliability and risk management – An integrated approach*. Elsevier.
- Crawley, F., & Tyler, B. (2015). *HAZOP: Guide to best practice* (3rd ed.). Elsevier – IChemE.
- Cross, J., & Farrer, D. (1982). *Dust explosions*. Plenum Press.
- Crowl, D. A. (2003). *Understanding explosions*. Wiley – AIChE.
- Crowl, D. A. (2012). Minimize the risks of flammable materials. *Chemical Engineering Progress*, 108(4), 28.
- Crowl, D. A., & Louvar, J. F. (2019). *Chemical process safety: Fundamentals with applications* (4th ed.). Prentice-Hall.
- CSHIB. (2005). Chemical Safety and Hazard Investigation Board preliminary findings on the BP Americas Texas City explosion. Oct. 27, available from www.csb.gov.
- Dodson, B., & Nolan, D. (1999). *Reliability engineering handbook*. CRC Press.
- Dow. (1973). *Fire and explosion index hazard classification guide* (3rd ed.). Dow Chemical Company.
- Dow. (1994). *Dow's fire and explosion index hazard classification guide*. American Institute of Chemical Engineers.
- Drysdale, D. (2011). *An Introduction to fire dynamics* (3rd ed.). Wiley.
- Duxbury, H. A. (1976). Gas vent sizing methods. *Loss Prevention*, 10, 147.
- Duxbury, H. A. (1979). Relief line sizing for gases. *Chemical Engineering London*, 350, 783.

- Eckhoff, R. K. (2003). *Dust explosions in the process industries: Identification, assessment and control of dust hazards* (3rd ed.). Gulf.
- Fawcett, H. H., & Wood, W. S. (1982). *Safety and accident prevention in chemical operations*. Wiley.
- Field, P. (1982). *Dust explosions*. Elsevier.
- Fisher, H. G., Forrest, H. S., Grossel, S. S., Huff, J. E., Muller, A. R., Noronha, J. A., Shaw, D. A., & Tilley, B. J. (1993). *Emergency relief system design using DIERS technology – The Design Institute for Emergency Relief Systems (DIERS) project manual*. Wiley – AIChE.
- Frank, W. I., & Whittle, D. K. (2001). *Revalidating process hazard analysis*. Wiley – AIChE.
- Green, A. E. (Ed.). (1982). *High risk technology*. Wiley.
- Green, A. E. (Ed.). (1983). *Safety system reliability*. Wiley.
- Gugan, K. (1979). *Unconfined vapor cloud explosions*. Gulf Publishing.
- HMSO. (1975). *The Flixborough disaster, report of the court of enquiry*. Stationery Office.
- Howard, W. B. (1992). Use precautions in selection, installation and operation of flame arresters. *Chemical Engineering Progress*, 88, 69.
- Hyatt, N. (2003). *Guidelines for process hazard analysis (PHA, HAZOP), hazard identification and risk analysis*. CRC Press.
- ICI. (1993). *Mond index: How to identify, assess and minimise potential hazards on chemical plant units for new and existing processes* (2nd ed., p. ICI).
- IMechE. (1993). *Noise control in the process industries*. PEP.
- Kales, P. (1997). *Reliability for technology, engineering and management*. Prentice Hall.
- King, A. (2007). Functional safety standards. *The Chemical Engineer*, 790, 46.
- King, R., & Hirst, R. (1998). *King's safety in the process industries* (2nd ed.). Elsevier.
- Kletz, T. A. (1977a). What risks should we run. *New Sci.*, (May 12th), 320.
- Kletz, T. A. (1977b). Evaluate risk in plant design. *Hydrological Processes*, 56, 207.
- Kletz, T. A. (1984). *Cheaper, safer plants or wealth and safety at work*. Institution of Chemical Engineers.
- Kletz, T. A. (1991). *Plant design for safety: A user friendly approach*. Hemisphere Books.
- Kletz, T. A. (1999). *HAZOP and HAZAN: Identifying process industry hazards* (4th ed.). Taylor & Francis.
- Kletz, T. A., & Amyotte, P. (2010). *Process plants: A handbook for inherently safer design* (2nd ed.). CRC Press.
- Kletz, T. A., Chung, P., Broomfield, E., & Shen-Orr, C. (1995). *Computer control and human error*. Gulf.
- Lawley, H. G. (1974). Operability studies and hazard analysis. *Loss Prevention*, 8, 105.
- Lewis, D. J. (1979a). *The Mond fire, explosion and toxicity index: A development of the Dow index*. Houston: AIChE Loss Prevention Symposium.
- Lewis, D. J. (1979b). The Mond fire, explosion and toxicity index applied to plant layout and spacing. *Loss Prevention*, 13, 20.
- Lewis, R. J. (2012). *Sax's dangerous properties of hazardous materials* (12th ed.). Wiley.
- Lowrance, W. W. (1976). *Of acceptable risk: Science and the determination of safety*. W. Kaufmann.
- Macmillan, A. (1998). *Electrical installations in hazardous areas*. Butterworth-Heinemann.
- Mannan, S. (Ed.). *Lees' loss prevention in the process industries* (4th ed.), 3 vols. Butterworth-Heinemann.
- Marshall, V. C. (1987). *Major chemical hazards*. Ellis Horwood.
- Marshall, V. C., & Ruhemann, S. (2000). *Fundamentals of process safety*. Institution of Chemical Engineers.
- Mathews, T. (1984). Bursting discs for over-pressure protection. *Chemical Engineering London*, 406, 21.
- Mendoza, V. A., Smolensky, V. G., & Straitz, J. F. (1998). Do your flame arrestors provide adequate protection? *Hydrocarbon Processing*, 77(10), 63.
- Moore, A. (1984). Pressure relieving systems. *Chemical Engineering*, 407, 13. London.
- Morley, P. G. (1989a). Sizing pressure safety valves for gas duty. *Chemical Engineering*, 463, 21. London.
- Morley, P. G. (1989b). Sizing pressure safety valves for flashing liquid duty. *Chemical Engineering*, 465, 47. London.
- Munday, G. (1976). Unconfined vapor explosions. *Chemical Engineering*, 308, 278. London.
- Murphy, G. (1993). *Quiet life ends in burst of activity*. Processing. 6.
- Napier, D. H., & Russell, D. A. (1974). Hazard assessment and critical parameters relating to static electrification in the process industries. Proc. First Int. Sym. on Loss Prevention (Elsevier).
- Parkinson, J. S. (1979). Assessment of plant pressure relief systems. *Inst. Chem. Eng. Sym. Design*, 79, K1.
- Pitblado, R. M., Shaw, S. J., & Sevens, G. (1990). The SAFETI risk assessment package and case study application. *Inst. Chem. Eng. Sym. Ser.*, 120, 51.
- Pohanish, R. P., & Greene, S. A. (2009). *Wiley guide to chemical incompatibilities* (3rd ed.). Wiley.
- Pratt, T. H. (1997). *Electrostatic ignitions of fires and explosions*. AIChE.
- Prugh, R. N. (1980). Applications of fault tree analysis. *Chemical Engineering Progress*, 76, 59.
- Rogowski, Z. W. (1980). Flame arresters in industry. *Inst. Chem. Eng. Sym. Ser.*, 58, 53.
- ROSPA. (1971). *Liquid flammable gases: Storage and handling*. Royal Society for the Prevention of Accidents.
- Schmidt, M. (2007). Tolerable risk. *Chemical Engineering New York*, 114(9), 69.
- Stamatis, D. H. (2003). *Failure mode and effect analysis: FMEA from theory to execution* (2nd ed.). ASQ Quality Press.
- Stamatis, D. H. (2019). *Risk management using failure mode and effect analysis (FMEA)*. ASQ Quality Press.
- Urban, P. G. (Ed.). (2017). *Bretherick's handbook of reactive chemical hazards* (8th ed.). Elsevier.
- Wells, G. L. (1980). *Safety in process plant design*. Institution of Chemical Engineers.
- Wells, G. L. (1996). *Hazard identification and risk assessment*. Institution of Chemical Engineers.
- Wells, G. L. (1997). *Major hazards and their management*. Institution of Chemical Engineers.

Bibliography

Further reading on process safety:

- Croner's Dangerous Substances. Croner.
- Fingas, M. (Ed.). (2002). *Handbook of hazardous materials spills technology*. McGraw-Hill.
- Tolley's Health and Safety at Work Handbook 2020 (32nd ed.). LexisNexis.

- Johnson, R. W., Rudy, S. W., & Unwin, S. D. (2003). *Essential practices for managing chemical reactivity hazards*. CCPS and American Institute of Chemical Engineers.
- Martel, B. (2000). *Chemical risk analysis (English translation)*. Penton Press.
- Redmill, F., Chudleigh, M., & Catmur, J. (1999). *Systems safety: HAZOP and software HAZOP*. Wiley.
- Richardson, M., & Gangolli, S. (Eds.). (1995). *Dictionary of substances and their effects*, 7 vols. Royal Society of Chemistry.
- Smith, D. J. (2017). *Reliability, maintainability and risk – Practical methods for engineers* (9th ed.). Butterworth-Heinemann.

American standards and laws

- ANSI/ISA – 84.00.01-2004. (2004) (3 parts) (IEC 61511 Mod). Functional safety: Safety instrumented systems for the process industry sector. American National Standards Institute/Instrumentation, Systems and Automation Society.
- API Recommended Practice 2030. (2014). *Application of water spray systems for fire protection in the petroleum industry* (4th ed.). American Petroleum Institute.
- API Recommended Practice 2218. (2013). *Fireproofing practices in petroleum and petrochemical processing plants* (3rd ed.). American Petroleum Institute.
- API Recommended Practice 14c. (2017). *Recommended practice for analysis, design, installation and testing of safety systems for offshore production facilities*. American Petroleum Institute.
- API Recommended Practice 500. (2012). *Recommended practice for classification of locations for electrical installations at petroleum facilities classified as Class I Division 1 and Division 2* (3rd ed.). American Petroleum Institute.
- API Recommended Practice 505. (2018). *Recommended practice for classification of locations for electrical installations at petroleum facilities classified as Class I Zone 0, Zone 1 and Zone 2* (2nd ed.). American Petroleum Institute.
- API Standard 520 P1. (2014). Sizing, selection, and installation of pressure-relieving devices in refineries. *Part I: sizing and selection*. American Petroleum Institute.
- API Standard 520 P2. (2015). Sizing, selection, and installation of pressure-relieving devices in refineries. *Part II: installation*. American Petroleum Institute.
- API Standard 521. (2014). *Pressure-relieving and depressuring systems*. American Petroleum Institute.
- API Standard 526. (2017). *Flanged steel pressure relief valves* (7th ed.). American Petroleum Institute.
- API Standard 527. (2014). *Seal tightness of pressure relief valves* (4th ed.). American Petroleum Institute.
- API Recommended Practice 2001. (2019). In *Fire protection in refineries* (10th ed.). American Petroleum Institute.
- API Recommended Practice 2003. (2015). *Protection against ignitions arising out of static, lightning and stray currents* (8th ed.). American Petroleum Institute.
- API Recommended Practice. 2210. (2000). *Flame arrestors for vents of tanks storing petroleum products* (3rd ed.). American Petroleum Institute.
- Boiler, A. S. M. E., & Pressure Vessel Code Section VIII. (2019). Rules for the construction of pressure vessels. *ASME International*.
- ASME PTC 25-. (2019). Pressure relief devices – performance test codes. *ASME International*.
- ASTM D92. (2018). *Standard test method for fire and flash points by Cleveland open cup tester*. ASTM International.
- ASTM D93. (2019). *Standard test methods for flash point by Pensky-Martens closed cup tester*. ASTM International.
- NFPA 30. (2018). *Flammable and combustible liquids code*. National Fire Protection Association.
- NFPA 49. (1994). *Hazardous chemicals data*. National Fire Protection Association.
- NFPA 61. (2020). *Standard for the prevention of fires and dust explosions in agricultural and food processing facilities*. National Fire Protection Association.
- NFPA 68. (2017). *Standard on explosion protection by deflagration venting*. National Fire Protection Association.
- NFPA 69. (2019). *Standard on explosion prevention systems*. National Fire Protection Association.
- NFPA 70. (2019). *National electrical code*. National Fire Protection Association.
- NFPA 77. (2019). *Recommended practice on static electricity*. National Fire Protection Association.
- NFPA 491. (1997). *Guide to hazardous chemical reactions*. National Fire Protection Association.
- NFPA 495. (2018). *Explosive materials code*. National Fire Protection Association.
- NFPA 496. (2017). *Standard for purged and pressurized enclosures for electrical equipment*. National Fire Protection Association.
- NFPA 497. (2017). *Recommended practice for the classification of flammable liquids, gases or vapors and of hazardous (classified) locations for electrical installations in chemical process areas*. National Fire Protection Association.
- NFPA 654. (2020). *Standard for the prevention of fire and dust explosions from the manufacturing, processing and handling of combustible particulate solids*. National Fire Protection Association.
- NFPA 655. (2017). *Standard for the prevention of sulfur fires and explosions*. National Fire Protection Association.
- NFPA 664. (2017). *Standard for the prevention of fires and dust explosions in wood processing and woodworking facilities*. National Fire Protection Association.
- NFPA 750. (2019). *Standard on water mist fire protection systems*. National Fire Protection Association.
- OSHA Standard 29 CFR 1910.119. (2008). *Process safety management of highly hazardous chemicals*.
- OSHA Standard 29 CFR 1910.307. (2008). Subpart S. Electrical.
- OSHA Standard 29 CFR 1910.1200. (2012). *Hazard communication*.

European and international standards

- BS EN 1127-1:2019. (2019). *Explosive atmospheres. Explosion prevention and protection. Part 1: Basic concepts and methodology*.
- BS EN 1839:2017. (2017). *Determination of explosion limits and the limiting oxygen concentration (LOC) for gases and vapors*.
- BS EN ISO 4126-1:2013. (2013). *Safety devices for protection against excessive pressure. Part 1: Safety valves*.
- BS EN ISO 4126-2:2019. (2019). *Safety devices for protection against excessive pressure. Part 2: Bursting disc safety devices*.
- BS EN ISO 4126-3:2009. (2009). *Safety devices for protection against excessive pressure. Part 3: Safety valves and bursting disc safety devices in combination*.
- BS EN ISO 4126-4:2013. (2013). *Safety devices for protection against excessive pressure. Part 4: Pilot operated safety valves*.
- BS EN ISO 4126-5:2013. (2013). *Safety devices for protection against excessive pressure. Part 5: Controlled safety pressure relief systems*. CSPRS.

- BS EN ISO 4126-6:2014. (2014). *Safety devices for protection against excessive pressure. Part 6: Application, selection and installation of bursting disc safety devices.*
- BS EN ISO 4126-7:2013. (2013). *Safety devices for protection against excessive pressure. Part 7: Common data.*
- BS EN ISO 13702:2015. (2015). *Petroleum and natural gas industries — control and mitigation of fires and explosions on offshore production installations — requirements and guidelines.*
- BS EN 14373:2005. (2005). *Explosion suppression systems.*
- BS EN 14460:2018. (2018). *Explosion resistant equipment.*
- BS EN 14797:2007. (2007). *Explosion venting devices.*
- BS EN 14994:2007. (2007). *Gas explosion venting protective systems.*
- BS EN ISO 16852:2016. (2016). *Flame arrestors — performance requirements, test methods and limits for use.*
- BS EN ISO 17776:2017. (2017). *Petroleum and natural gas industries — Offshore production installations — Major accident hazard management during the design of new installations.*
- BS EN 19353:2019. (2019). *Safety of machinery — Fire prevention and fire protection.*
- BS EN IEC 60079-0:2018. (2018). *Explosive atmospheres. Part 0: Equipment - General requirements.*
- BS EN 60079-1:2014. (2014). *Explosive atmospheres. Part 1: Equipment protection by flameproof enclosures “d.”*
- BS EN 60079-2:2015. (2015). *Explosive atmospheres. Part 2: Equipment protection by pressurized enclosure.*
- BS EN 60079-10:2016. (2016). *Explosive atmospheres. Part 10: Classification of areas — explosive gas atmospheres.*
- BS EN 60079-11:2012. (2012). *Explosive atmospheres. Part 11: Equipment protection by intrinsic safety “i.”*
- BS EN 60079-13:2017. (2017). *Explosive atmospheres. Part 13: Equipment protection by pressurized room “p” and artificially ventilated room “v.”*
- BS EN 60079-25:2010. (2010). *Explosive atmospheres. Part 25: Intrinsically safe electrical systems.*
- BS EN 60079-29:2017. (2017). *Explosive atmospheres. Part 29: Gas detectors.*
- BS EN 61508:2010. (2010). *Functional safety of electrical/electronic/programmable electronic safety-related systems (7 parts).*
- BS EN 61511:2017. (2017). *Functional safety: Safety instrumented systems for the process industry sector (3 parts).*
- BS EN 62305:2011. (2011). *Protection against lightning (4 parts).*
- IEC 61508. (2010). *Functional safety of electrical/electronic/programmable electronic safety-related systems (2nd ed., 7 parts).*
- IEC 61511. (2016). *Functional safety: Safety instrumented systems for the process industry sector (3 parts).*

10.11 Nomenclature

		Dimensions in MLT
A_d	Discharge area	L^2
A_w	Internal wetted surface area	L^2
C	Impact of a hazard (impact per loss event) in Equation 10.1	—
C	Constant in Equation 10.19	—
C_p	Specific heat capacity at constant pressure	$L^2 T^{-2} \theta^{-1}$
C_v	Specific heat capacity at constant volume	$L^2 T^{-2} \theta^{-1}$
D	Diameter	L
F	Hazard rate (frequency of a hazard in events per year)	T^{-1}
F_e	Environmental factor for external fire	—
F_{np}	Inherent frequency of a hazard with no protective system (number of events per year)	T^{-1}
F_t	Tolerable frequency of a hazard (number of events per year)	T^{-1}
F_3	Unit hazard factor in Dow Fire & Explosion Index	—
K_b	Back-pressure correction factor	—
K_c	Combination correction factor for a relief valve	—
K_d	Coefficient of discharge for a relief valve	—
L	Length of vessel	L
M_w	Molecular weight	—
N_r	Reactivity in the Dow Fire & Explosion Index	—

		Dimensions in MLT
n	Number of moles of gas	M
P	Pressure	$\text{ML}^{-1}\text{T}^{-2}$
P_{cf}	Critical flow pressure	$\text{ML}^{-1}\text{T}^{-2}$
P_m	Maximum allowable working pressure	$\text{ML}^{-1}\text{T}^{-2}$
P_1	Upstream pressure	$\text{ML}^{-1}\text{T}^{-2}$
PFD_{av}	Average probability of failure on demand of a protective system	—
Q	Heat input due to fire	ML^2T^{-3}
R	Ideal gas constant	$\text{L}^2\text{T}^{-2}\theta^{-1}$
R_{np}	Inherent risk of a hazard with no protective system in place	
R_t	Tolerable risk, also sometimes called the <i>acceptable risk</i>	
T	Temperature	θ
t	Time	T
V	Vessel volume	L^3
V_G	Volume occupied by vapor	L^3
V_L	Volume occupied by liquid	L^3
v	Liquid volumetric flow rate	L^3T^{-1}
w	Relieving mass flow rate	MT^{-1}
w_f	Fire case relieving mass flow rate	MT^{-1}
Z	Vapor compressibility	—
γ	Ratio of specific heat capacities	—
ΔH_{vap}	Heat of vaporization	L^2T^{-2}
ΔR	Risk reduction factor, defined in Equation 10.3	
δ	Demand rate (number of occasions per year that a safety system is actuated)	T^{-1}
Φ	Fractional dead time (probability that a safety system is inactive)	—
λ	Failure rate (number of occasions per year that a safety system develops a fault)	T^{-1}
τ	Test interval	T

10.12 Problems

- 10.1.** In the storage of flammable liquids, if the composition of the vapor–air mixture above the liquid surface falls within the flammability limits, a floating roof tank would be used or the tank blanketed with inert gas. Check if the vapor composition for liquids listed here will fall within their flammability range both at atmospheric pressure and 25 °C.
1. Toluene
 2. Acrylonitrile
 3. Nitrobenzene
 4. Acetone
- 10.2.** Complete a failure mode effects analysis for the nitric acid plant reactor section described in Example 10.2. (This is best carried out as a group activity with a group size of three to six.)

- 10.3. Estimate the Dow Fire and Explosion Index and determine the hazard rating for the processes listed here. Use the process descriptions given in Appendix F and develop the designs as needed to estimate the index.
1. Ethylhexanol from propylene and synthesis gas, F.1.
 2. Chlorobenzenes from benzene and chlorine, F.2.
 3. Methyl ethyl ketone from 2-butanol, F.3.
 4. Acrylonitrile from propylene and ammonia, F.4.
 5. Aniline from nitrobenzene and hydrogen, F.8.
- 10.4. Devise a preliminary control scheme for the sections of the nitric acid plant described in [Chapter 2](#), flowsheet Fig. 2.8, which are listed here. Make a practice HAZOP study of each section and revise your preliminary control scheme.
1. Waste heat boiler (WHB)
 2. Condenser
 3. Absorption column
- 10.5. A distillation column separates benzene from toluene using a control scheme similar to that shown in Fig. 5.17(a). Make a practice HAZOP study of the plant section and add any instrumentation that is needed to develop the full P&I diagram.
- 10.6. Develop a fault tree for events that could lead to release of benzene to the atmosphere for the distillation section designed in Problem 10.5.
- 10.7. List the materials incompatibility problems you should be aware of in plants that handle:
1. Hydrogen peroxide
 2. Ethylene oxide
 3. Chlorine
 4. Ammonium nitrate
 5. Styrene
- 10.8. List possible relief scenarios for the vessel designed in Problem 14.1.
- 10.9. Estimate the relieving load for the reactor designed in Problem 14.5 for a blocked outlet scenario, and size the relief valve for this case.
- 10.10. A toluene surge drum has capacity 500 gal and is normally operated 60% full at 100 °F under 300 psig of hydrogen in the head space using a level controlled outflow. The normal flow rate into the vessel is 30,000 lb/hr. Determine the vessel dimensions if the vessel is vertically mounted. Evaluate the relief loads for the blocked outflow and external fire cases and determine the relief valve size.

General site considerations

KEY LEARNING OBJECTIVES

- Factors that are considered in site selection and plant layout
- Environmental legislation that governs chemical plant operations
- Waste minimization methods that can be used to reduce the environmental impact of a chemical plant

11.1 Introduction

When a new chemical plant is being designed, one of the first decisions must be the plant location. If the project is for a new grassroots facility, a suitable site must be found and the site and equipment layout planned. If the plant is to go into an existing site, the impact on the existing infrastructure must be evaluated so that necessary modifications can be made to accommodate the new plant. In either case, provision must be made for the ancillary buildings and services needed for plant operation and for the environmentally acceptable disposal of effluent. These subjects are discussed briefly in this chapter.

11.2 Plant location and site selection

The location of the plant can have a crucial effect on the profitability of a project and the scope for future expansion. Many factors must be considered when selecting a suitable site, and only a brief review of the principal factors will be given in this section. Site selection for chemical process plants is discussed in more detail by [Merims \(1966\)](#) and [Mecklenburgh \(1985\)](#); see also [CCPS \(2003\)](#). The principal factors to consider are:

1. Location with respect to the marketing area
2. Raw material supply
3. Transport facilities
4. Availability of labor
5. Availability of utilities: water, fuel, and power
6. Availability of suitable land
7. Environmental impact, including effluent disposal
8. Local community considerations
9. Climate
10. Political, economic, and strategic considerations

Marketing area

For materials that are produced in bulk quantities, such as cement, mineral acids, fuels, and fertilizers, where the cost of the product per metric ton is relatively low and the cost of transport is a significant fraction of the sales price, the plant should be located close to the primary market. This consideration is much less important for low-volume production, high-priced products, such as pharmaceuticals.

Raw materials

The availability and price of suitable raw materials will often determine the site location. Plants that produce bulk chemicals are best located close to the source of the major raw material, as long as the costs of shipping product are not greater than the cost of shipping feed. For example, at the time of writing much of the new ethylene capacity that is being added worldwide is being built in the Middle East, close to supplies of cheap ethane from natural gas. Oil refineries, on the other hand, tend to be located close to major population centers, as an oil refinery produces many grades of fuel, which are expensive to ship separately over long distances.

Transport

The transport of materials and products to and from the plant can be an overriding consideration in site selection. If practicable, a site should be selected that is close to at least two major forms of transport: road, rail, waterway (canal or river), or a sea port. Road transport is increasingly used, and is suitable for local distribution from a central warehouse. Rail transport is usually cheaper for the long-distance transport of bulk chemicals. Industrial gases and some bulk fuels are shipped by pipeline.

Air transport is convenient and efficient for the movement of personnel and essential equipment and supplies, and the proximity of the site to a major airport should be considered. Air freight can also be used for small volumes of high-value products such as pharmaceuticals, as long as the products are safe to transport by air and meet aviation regulations.

Availability of labor

Labor will be needed for construction of the plant and its operation. Skilled construction workers are often brought in from outside the site area, but there should be an adequate pool of unskilled labor available locally and labor suitable for training to operate the plant. Skilled craft workers such as electricians, welders, and pipe fitters will be needed for plant maintenance. Local labor laws, trade union customs, and restrictive practices must be considered when assessing the availability and suitability of the local labor for recruitment and training.

Utilities (services)

Chemical processes invariably require large quantities of water for cooling and general process use, and the plant must be located near a source of water of suitable quality. Process water may be drawn from a river, from wells, or purchased from a local supplier.

At some sites, cooling water can be taken from a river or lake or from the sea; at other locations cooling towers will be needed. In locations where the ambient humidity is high, use of cooling water may be unattractive compared with air cooling; see [Section 3.2.5](#).

Electrical power is needed at all sites. Electrochemical processes (for example, chlorine manufacture or aluminum smelting) require large quantities of power and must be located close to a cheap source of power.

A competitively priced fuel must be available on site for steam and power generation.

Environmental impact and effluent disposal

All industrial processes produce waste products, and full consideration must be given to the difficulties and cost of their disposal. The disposal of toxic and harmful effluents will be covered by local regulations, and the appropriate authorities must be consulted during the initial site survey to determine the standards that must be met.

An environmental impact assessment should be made for each new project or major modification or addition to an existing process; see [Section 11.5.6](#).

Local community considerations

The proposed plant must fit in with and be acceptable to the local community. Full consideration must be given to the safe location of the plant so that it does not impose a significant additional risk to the local population. Plants should generally be sited so as not to be upwind of residential areas under the prevailing wind.

On a new site, the local community must be able to provide adequate facilities for the plant personnel: schools, banks, housing, and recreational and cultural facilities.

The local community must also be consulted about plant water consumption and discharge and the effect of the plant on local traffic. Some communities welcome new plant construction as a source of new jobs and economic prosperity. More affluent communities generally do less to encourage the building of new manufacturing plants, and in some cases may actively oppose chemical plant construction.

Land (site considerations)

Sufficient suitable land must be available for the proposed plant and for future expansion. The land should ideally be flat, well drained, and have suitable load-bearing characteristics. A full site evaluation should be made to determine the need for piling or other special foundations. Particular care must be taken when building plants on reclaimed land near the ocean in earthquake zones because of the poor seismic character of such land.

Climate

Adverse climatic conditions at a site will increase costs. Abnormally low temperatures require the provision of additional insulation and special heating for equipment and pipe runs. Stronger structures are needed at locations subject to high winds (cyclone/hurricane areas) or earthquakes.

Political and strategic considerations

Capital grants, tax concessions, and other inducements are often given by governments to direct new investment to preferred locations, such as areas of high unemployment. The availability of such grants can be the overriding consideration in site selection.

In a globalized economy, there may be an advantage to be gained by locating the plant within an area with preferential tariff agreements, such as the European Union (EU).

Corporations make strategic decisions on plant location, for example, trading off the desire to satisfy the needs of emerging markets against the risk of locating capital assets in countries that do not have strong enforcement of intellectual property rights. These factors are discussed in more detail in [Section 9.9](#).

11.3 Site layout

The process units and ancillary buildings should be laid out to give the most economical flow of materials and personnel around the site. Hazardous processes must be located at a safe distance from other buildings. Consideration must also be given to the future expansion of the site. The ancillary buildings and services required on a site, in addition to the main processing units (buildings), include:

1. Storage for raw materials and products: tank farms and warehouses
2. Shipping and receiving areas: road or rail tanker filling and unloading facilities, truck loading
3. Maintenance workshops
4. Stores for maintenance and operating supplies
5. Laboratories for process quality control
6. Fire stations and other emergency services
7. Utilities: steam boilers, compressed air, power generation, refrigeration, transformer stations
8. Effluent disposal plant: waste water treatment, solid and or liquid waste collection
9. Offices for general administration
10. Canteens and other amenity buildings, such as medical centers
11. Parking lots

When roughing out the preliminary site layout, the process units are normally sited first and arranged to give a smooth flow of materials through the various processing steps, from receiving of raw material to final product storage and shipping. Process units are normally spaced at least 30 m apart; greater spacing may be needed for hazardous processes.

The location of the principal ancillary buildings should then be decided. They should be arranged so as to minimize the time spent by personnel traveling between buildings. Administration offices and laboratories, in which a relatively large number of people will be working, should be located well away from potentially hazardous

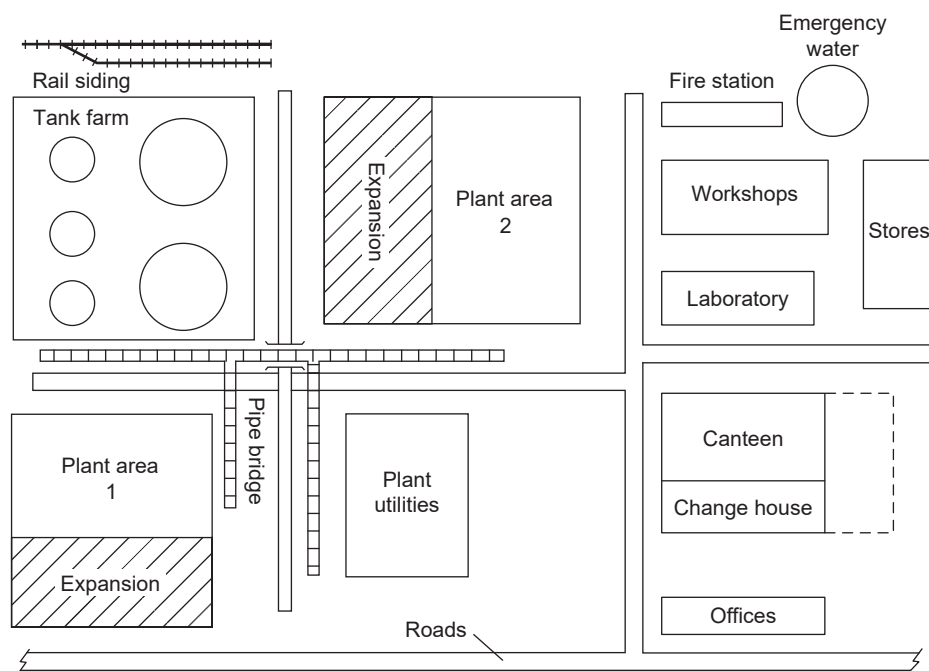


FIG. 11.1 Typical site plan.

processes. Control rooms are normally located adjacent to the processing units, but with potentially hazardous processes may have to be sited at a safer distance.

The siting of the main process units determines the layout of the plant roads, pipe alleys, and drains. Access roads to each building are needed for construction and for operation and maintenance.

Utility buildings should be sited to give the most economical run of pipes to and from the process units.

Cooling towers should be sited so that under the prevailing wind the plume of condensate spray drifts away from the plant area and adjacent properties.

The main storage areas should be placed between the loading and unloading facilities and the process units they serve. Storage tanks containing hazardous materials should be sited at least 70 m (200 ft) from the site boundary.

A typical plot plan is shown in Fig. 11.1.

A comprehensive discussion of site layout is given by Mecklenburgh (1985); see also House (1969), Kaess (1970), and Meissner and Shelton (1992).

11.4 Plant layout

The economic construction and efficient operation of a process unit will depend on how well the plant and equipment specified on the process flowsheet is laid out.

A detailed account of plant layout techniques cannot be given in this short section. A fuller discussion can be found in the book edited by Mecklenburgh (1985) and in articles by Kern (1977a–f, 1978a–f), Meissner and Shelton (1992), Brandt et al. (1992), and Russo and Tortorella (1992).

The principal factors to be considered are:

1. Economic considerations: construction and operating costs
2. The process requirements
3. Convenience of operation
4. Convenience of maintenance
5. Safety
6. Future expansion
7. Modular construction